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New thermo-mechanically processed wire rods for drawing and cold heading applications (Thermorod)

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ABSTRACT

This project was aimed to:

- the assessment of the TMCP potential for achieving microstructures suitable for simplification/elimination of softening annealing of wire rods made of low alloy steels.
- the design and development of steel chemistry, mill lay-out and hot rolling schedules for TMCP of wire rods, in order to improve metallurgical properties.

The investigation on the effect of thermo-mechanical treatment on microstructure evolution and final properties was performed on selected commercial steels (i.e. steels 10MnSi7, 38Cr4, 30MnB4, and 54SiCrV6), by laboratory tests, pilot and industrial trials, and application of thermal and metallurgical models.

Significant effects of temperature and deformation rate on deformation resistance were seen by hot torsion tests. The flow-stress generally increases with increasing strain rate and decreasing deformation temperature. A reduction of deformation temperature from 1000 °C to 850 °C leads to an increase of flow stress from 60% to 100% at a strain value of 0.3 (from 110-130 MPa to 175-240 MPa).
This behaviour was confirmed by the pilot mill trials at TUBEF, where an increase of the rolling force at the last stand of 50-60 % was observed for the various steels when the finish rolling temperature was reduced from 1000 to 850 °C. In this case of multi-pass rolling, also the recrystallization behaviour before the last pass affects the rolling force.

In order to study the austenite evolution during rolling, empirical equations describing changes of austenite (e.g. recrystallisation fraction, austenite grain size) were selected and applied.
The mean austenite grain sizes at the entry of each stand were calculated by various equations and approaches by Sidenor and CSM. These calculated data were compared with the values measured on 38Cr4 steel specimens taken at various locations along the mill and rapidly quenched (< 3 s) to freeze the austenitic structure. The agreement was good, given the nature of the measurements, the accuracy of the process model used to calculate average temperature of the stock, and the assumptions made in the recrystallization model.
The model was applied to assess the effect of chemical composition and rolling temperatures on the austenite grain size (AGS) at finishing.
In the case of a finish rolling temperature of 1040 °C the calculated AGS of steel 38Cr4 is about 60 μm after 12 s from finishing, if water boxes are not activated. The behaviour of steel 10MnSi7 is predicted very similar, whilst the steel 30MnB4 is finer (27 μm) because its grain growth is inhibited by the presence of Ti (C, N) particles. Introducing final water boxes, that in 2 s are able to reduce the average temperature from 1040 to 870 °C, allows to maintain AGS lower than 30 μm.

When the average finish rolling temperature is decreased of 180 °C (from 1040 to 860 °C), introducing water boxes between intermediate and finishing mill, in addition to those just after finishing, an average AGS of less than 15 μm is predicted by the model. However, local microstructure becomes progressively more heterogeneous through the diameter of the wire-rod if adequate equalization is not allowed.

The evolution of austenite grain size was also investigated by the pilot mill trials at TUBEF.
Generally, the AGS is refined reducing the finish rolling temperature from 980 °C to 850 °C, whilst the effect of lying head temperature from 850 °C to 750 °C is often poor.
Initial AGS (i.e. holding time in the reheating stage) shows also some influence; coarser initial AGS causes in general larger AGS at the laying head. Final AGS is dependent on rolling temperatures, especially that at the last stand. In the cases in which this temperature is around 850 ºC the AGS is refined and reaches values of ASTM No.11 (7 μm).

Also hot compression trials were performed in order to study the austenitic grain growth after recrystallisation. The results indicate that for steel 38Cr4, starting from a 40 μm recrystallized austenite grain size, after a holding of 12 s, the grain grows up to 150 μm at 1000 °C. The grain growth is strongly reduced for the same holding period at 850 °C, being the average grain size below 70 μm.

Very effective cooling devices are required to reduce the stock temperature for controlling grain growth after hot rolling as the usual distances between finish roll and cooling conveyor are rather small and wire-rod speed is very high.

The coarse AGS obtained for high laying head temperatures promotes the formation of mixed bainite-martensite microstructures due to the increased hardenability.

The phase transformation characteristics of the investigated steels for different AGS values (fine and coarse) were determined by dilatometry. Both isothermal and continuous cooling conditions were used. Both experimental tests and mathematical simulations indicate that a reduction of AGS is beneficial in reducing the formation of hard and brittle phases such as bainite and martensite on the final microstructure.
Austenite refinement accelerates the ferrite formation and promotes the completion of the phase transformation under the insulated zones (very slow cooling). In this respect, simulations by model clarified that the success in avoiding bainite and martensite formation in steels with very high hardenability such as boron steels for fasteners, is related to the following factors:

- low cooling rates during phase transformation, obtained using insulated covers and low conveyor speeds which give dense loops;
- adequate laying head temperatures.

Low laying head temperatures, for a given conveyor speed, favour the completion of the phase transformation in the insulated zones (formation of ferrite and pearlite). If low laying head temperatures are difficult to be achieved, the AGS refinement helps to avoid bainite and martensite formation, included the segregated regions which result more hardenable due to local enrichment in alloy elements and carbon.

Behind the laying head different cooling rates were achieved during the pilot mill trials at Freiberg. A complete microstructural analysis, was carried out and relations among main

mechanical parameters (ultimate tensile strength, UTS, reduction of area, ROA, hardness, HV), hot rolling and cooling parameters were given. Independently of the steel chemistry, mechanical properties such as UTS, ROA and HV_{30} appeared more homogeneous (especially in the case of 54SiCrV6 steel) in the II trial. In general, the following rule holds: a rigid control of all process parameters and a reduction of temperature values at finishing block (T2-T3) and laying head (T4) lead to a greater microstructural homogeneity in wire-rods, with improvements in terms of reproducibility and narrower dispersion of mechanical properties. If steel is rolled at low temperatures and the process control is good, higher cooling rates can give benefits on the final product properties.

The following considerations were drawn for cold formability by compression test of prototype TMCP wire-rods:

- ***Steel 54SiCrV6***

  An influence of the rolling schedule can be hardly seen. Only for low and nearly constant T1-T4 rolling temperatures it seems possible to avoid cracks during cold deformation.

- ***Steel 30MnB4***

  After machining (i.e. elimination of potential sites for crack initiation) no failure appeared.

- ***Steel 10MnSi7***

  After machining, no failure appeared in air cooled materials, except that rolled at very low finishing temperature (690 °C). In this condition, hot rolling was carried out in the mixed ferrite-austenite range. Therefore ferrite was formed during rolling and work-hardened. This causes a reduction in ductility.

- ***Steel 38Cr4***

  This steel showed some cracks in machined condition; all defects appeared after air cooling treatment. Slow cooled wire rods (cooling from finish temperature to 600 °C and holding 600 s) were crack-free after the compression test. Hence, a slow cooling treatment (such as 600 °C, for 600 s) is the adequate cooling strategy.

Usually, cold drawing trials of prototype wire rods gave good results.

The influence of thermo-mechanical processing on the response to spheroidising annealing was specifically studied in 38Cr4 steel grade. A clear relationship between the degree of spheroidisation and interlamellar spacing was observed: the smaller the interlamellar spacing the better the response to annealing. The interlamellar spacing is a key factor concerning the spheroidisation.
Higher cooling rates decrease interlamellar spacing (improving degree of spheroidisation), but increase the risk of bainite and martensite formation what, in general, gives higher tensile strengths in the as-treated conditions. The best balance of - tensile strength was given by a material which exhibits small interlamellar spacing and a high degree of , due to the air (no retarded) cooling procedure, and is free of bainite and martensite due to the thermo-mechanical treatment (at temperatures below 800ºC).
It was proven that thermo-mechanical processing, at temperatures below 800ºC, allows to shorten subsequent spheroidising heat treatments. In the trials performed in this project a reduction of spheroidising time of 30% was achieved. This means saving up to 40 Euro/ton.

Guidelines and improvements related to the introduction of thermo-mechanical processing process of long products, in particular wire-rods were provided.

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1. INTRODUCTION

Wire rods are used in a wide range of industrial fields. Therefore they have to meet given combinations of properties such as strength, ductility, cold formability, drawability, hardenability. In addition, current market demands closer tolerances and restricted ovality of wire rods. To comply with these requirements, the steel used may need sufficient alloying elements which can give austenite strengthening during hot rolling and deterioration of both cold-heading formability and drawing limit, due to the increase of as-rolled rod hardness. Therefore, expensive heat treatments (e.g. annealing) are carried out to soften the hot rolled wire rod.
In order to reduce production costs, through a simplification of the whole process, on-line controlled cooling systems are applied, but further improvements are necessary to satisfy the customers. [1]

In Japan, when Nippon Steel and Kawasaki Steel Corporation moved to more sophisticated rod products, in-line heat treatment as well as precision rolling technologies based on tension control plus AGC were developed and introduced in 1990's. [2-5]
Uniform heating of billets and tension control during hot rolling were used to achieve the required dimensional accuracy, but in practice it is difficult to guarantee closer tolerances by conventional rod mill over extended production periods.

Advances in wire rod manufacture technologies (e.g. interstand cooling, addition of sizing equipment after conventional finishing-block, improved controlled cooling facilities) can allow the introduction of precision rolling as well as thermo-mechanical control process (TMCP), expanding the range of low and medium carbon steel wire rods, suitable for simplified production routes or direct processing.

By high precision rolling technologies developed by SMS [6-9] and Morgan Construction Company [10, 11] it is possible to achieve low finishing rolling temperatures, in addition to improved product dimensions (tolerances). Location of the compact sizing mill after the finishing block is advantageous not only for obtaining increased productivity, but also for improving metallurgical properties. The ability to roll products at low temperatures should make it possible to use the benefits of thermo-mechanical rolling in grain refinement in order to increase ductility and shorten subsequent thermal treatment processes.

However, for long steel products there are several parameters which must be considered in the thermo-mechanical process: location of rolling equipment, location of water cooling, rolling rates, finish rolling temperatures.

For instance the introduction of a mini-block in existing rod mill installations that have limitations in separating force and power, hence finish rolling temperatures higher than 1000 °C, helps in reducing temperatures below 950 °C, but an adequate distance between the no-twist mill and the sizing mill should be left to use water cooling and reduce the stock temperature.

The location of the sizing mill is important to allow time for the rod to be controlled cooled and equalised prior to the final reduction, thereby reducing the risk of over-heating the rod core. This is crucial for special steels which are sensitive to grain coarsening at high finishing temperatures. Also a critical combination of strain and finishing temperature is needed to promote a uniform microstructure. Static and (meta)-dynamic recrystallisation processes are strongly dependent on temperature and accumulated strain. During rod sizing, without a sufficient amount of strain, either the recrystallisation process is not adequately promoted or a low number of recrystallised sites is nucleated. Consequently, localised coarse grains can occur.

The introduction of thermo-mechanical processing of wire rods leads to higher finishing speed and finer austenite grain size. The consequence is that a different cooling pattern should be used along the conveyor due to a different transformation behaviour and wrap overlapping (coil build-up at the loop layer). Hence, the present cooling practices must be revised.
In this respect, mathematical models able to predict austenite evolution during hot rolling and controlled cooling of wire rods, can be effectively applied, together with present plant and technological data, to predict the effect of plant configuration, rolling and cooling practices on evolution and uniformity of wire rod microstructure. [12-14]

Various ECSC projects have been focused on this aspect and empirical models are already available to optimise important hot rolling and cooling parameters. [15, 16]
New approaches have been developed to calculate phase transformation of medium/high carbon steels in another ECSC Project. [17]

In principle, the computer models that reflect the physics of the process involved can be more easily applied to new operating conditions, albeit still not investigated. On the other hand, the empirical models describing the austenite recrystallisation and grain growth phenomena cannot be applied outside the limits for which they were derived. [12]
This means that specific experiments by laboratory equipment and pilot mill, working at low temperatures, are needed to calibrate these models for the steels of interest.
There is an acknowledged lack of information on recrystallisation kinetics of austenite when deformed in conditions suitable for simulating thermo-mechanically processed wire rods (strain rates > 50 s^{-1}, temperatures from 750 to 900 °C), especially for low alloy steels employed for manufacturing wires for gas shielded arc welding, fasteners, and springs.

In the field of 0.5% C steel for suspension springs, the introduction of TMCP to increase strength and toughness has been investigated by Japanese and MPI. [18-20] In this respect, the effect of vanadium and niobium microadditions on the softening behaviour and fatigue resistance was studied, together with the influence of the austenite deformation on the transformation behaviour.
Also recrystallisation kinetics and grain growth of boron steels used in the production of fasteners, Ti-microalloyed to bind nitrogen and leave boron in solution (effective for hardenability improvement), have not been analysed in detail.

2. OBJECTIVES

Recent advances in wire rod manufacture technologies allow the introduction of thermo-mechanical control process (TMCP) as well as precision rolling, expanding the range of steel wire rods, suitable for simplified production routes or direct processing.
The development of TMCP wire rods which exhibit excellent capability to be directly drawn or cold formed without annealing was investigated for the following steels which are attractive in various sectors:

- High Si – high Mn steel wires for CO_2 gas shielding arc welding.
- High strength steel fasteners of property class 8.8 and above, produced from a variety of steels including medium C – low alloy and Mn-Cr-boron steels.
- High C, Cr-V and Cr-Si steel springs.

Aims of this project are:

- Assessment of the TMCP potential for achieving microstructures suitable for simplification/elimination of softening annealing of wire rods made of low alloy steels.
- Design and development of steel chemistry, mill lay-out and hot rolling schedules for TMCP of wire rods, in order to improve metallurgical properties, without detrimental effects on dimensional accuracy.

The main expected results are:

- Feasibility of TMCP wire rods.
- Quantification of the improvements associated with the introduction of TMCP, including final properties of cold formed products (e.g. fasteners and springs).
- Understanding of metallurgical and process factors that affect recrystallisation kinetics and austenite grain growth during hot rolling and cooling of low alloy steel wire rods at low temperatures, through laboratory tests and pilot mill trials.
- Analysis of the evolution of work hardened austenite during TMCP of wire rods in order to improve both the predicting capability of existing models and operating practices.

The programme foresees a strict co-operation between Centro Sviluppo Materiali (CSM), Lucchini, Sidenor, and TU Bergakamie Freiberg (TUBEF).

3. DESCRIPTION OF ACTIVITIES

The work has been focused on 4 steel grades (Table 1).

- 54SiCrV6: Cr-Si-V spring steel grade for suspensions and automotive applications.
- 30MnB4: B-Mo medium C steel for high strength fasteners by cold heading operations;
- 38Cr4: medium carbon, high Cr and low silicon, normally annealed for the spheroidisation of pearlite;
- 10MnSi7: high Si – high Mn variant of the common low-alloy low C steel for gas shielded arc welding.

The detailed chemical compositions of selected steels is reported below (Table 2):

Code	Steel Type		Distributed by
0 and 1	High C-Si-Cr-V	54SiCrV6	Sidenor & Lucchini
2	Mn-Cr-B	30MnB4	Lucchini
3	Medium C – Low Alloy	38Cr4	Sidenor
4	High Si-Mn	10MnSi7	Lucchini

Table 1 Investigated steels.

STEEL	**54SiCrV6 SIDENOR**	**54SiCrV6 LUCCHINI**	**30MnB4**	**38Cr4**	**10MnSi7**
Code	0	1	2	3	4
% C	0.52	0.54	0.30	0.35	0.062
% Mn	0.64	0.76	0.92	0.70	1.61
% Si	1.48	1.40	0.24	0.22	0.87
% P	< 0.025	0.01	0.009	< 0.025	0.011
% S	< 0.015	0.003	0.003	< 0.015	0.011
% Cr	0.61	0.66	0.226	1.02	0.046
% Ni	0.12	0.019	0.013	0.12	0.010
% Mo	0.04	0.003	0.09	0.02	0.002
% V	0.11	0.11	0.003	0.007	0.003
% Cu	< 0.020	0.012	0.008	< 0.020	0.021
% Al	< 0.020	0.01	0.031	< 0.020	0.002
% Ti	< 0.005	0.002	0.046	< 0.005	0.001
%Nb	< 0.005	< 0.005	< 0.005	< 0.005	0.001
B ppm	< 5	3	50	< 5	5
N ppm	n.d.	107	108	n.d.	94
O ppm	n.d.	12	42	n.d.	7

Table 2 Chemical compositions of selected steels.

The following Work Packages (WPs) were identified:

WP 1 Analysis of process data from wire rod rolling mills.
Lead site: Lucchini
Additional site: Sidenor.

The present performances of existing wire rod mills (Lucchini and Sidenor works) were assessed and a general frame of actual potentials of conventional hot rolling techniques were given. Thermal evolution of both rolling stock during hot rolling and wire-rod during controlled cooling, as a function of steel composition and process conditions (i.e. rolling speed, cooling conditions), was measured (Annex 1) and used to specialise the mathematical models for the prediction of thermal evolution. In particular, the heat transfer coefficient along either the Stelmor conveyor or the EDC were determined, using infrared pyrometers, and thermography over the loops.
Microstructures, mechanical properties of as-rolled wire rods were related to hot working and cooling conditions for a representative range of steel compositions and rod diameters.

WP 2 Set-up and validation of mathematical models
Lead site: CSM
Additional site: Sidenor, TUBEF, Lucchini.

Models taken from the literature (e.g. Maccagno, Jonas, Hodgson et al.), selected by Sidenor on the basis of their predicting capability, and in-house models developed by CSM were revised and specialised to describe the following features for present and future wire rod mill configurations:

- the austenite evolution during hot and warm rolling (Annex 2)
- austenite phase transformation during controlled cooling (Annex 4).

The calibration of the constants of the models were done by means of the data acquired in the WP 1 and WP 3.
The various modelling approaches were tested comparing calculated values and carefully monitored results from laboratory tests, pilot trials and industrial mills.

WP 3 Experimental tests
Lead site: TUBEF
Additional site: CSM, Sidenor, Lucchini.

For the development of thermo-mechanical control process (TMCP) the knowledge of austenite recrystallisation, phase-transformation and precipitation behaviour of selected steel grades is fundamental.
When possible, the experience from former investigations was used.

Task 3.1 Recrystallisation kinetics and grain growth

Work hardening and recrystallisation processes during hot deformation were investigated by hot torsion (Annex 3.1) and compression (Annex 3.2) tests. Austenitizing temperatures were varied in order to simulate different austenite grains. Maximum flow stress and the peak strain were determined at different temperatures and strain rates.

Rolling forces and torques were measured by TUBEF on experiments carried out on its four stand high speed wire mill, using very high strain rates and all the steels (Annex 5.1).
By freezing the samples immediately after hot deformation/rolling, the determination of the recrystallised fraction and the austenite grain size were carried out.
Specific tests were carried out by Sidenor (Gleeble) and by CSM (deformation dilatometry) in order to investigate grain growth after recrystallisation at various temperatures and strains.

Task 3.2 Precipitation

In some cases, where precipitation played a role in determining mechanical properties and drawing behaviour, TEM/STEM-EDS investigation was carried out by CSM in order to assess the precipitation by extraction method.

Task 3.3 Phase transformation kinetics

Dilatometry (CSM) and Gleeble (Sidenor) were used to simulate relevant cooling conditions and construct TTT and CCT diagrams (Annex 3.3).

WP 4 Application of the mathematical models
Lead site: Sidenor
Additional site: CSM, Lucchini.

Validated models (Annex 4) were applied by Sidenor and CSM, referring to existing wire rod mills, respectively by Sidenor and Lucchini, for the acquisition of valuable information on TMCP introduction:

- suitable arrangement of interstand cooling and equalising sections (number and position of cooling devices) in order to ensure that the rolling stock achieves the desired finishing rolling temperature for TMCP with a well uniform temperature profile without inadmissible undercooling of the surface layers and excessive increase in rolling forces;

- selection of alternative controlled cooling programmes in order to promote the formation of given microstructures.

WP 5 Assessment of new TMCP wire rods
Lead site: Lucchini
Additional site: CSM, Sidenor, TUBEF.

Task 5.1 Effect of composition and hot/warm rolling conditions on final microstructures

TUBEF explored different TMCP practices using a four stand high speed mill characterized by great flexibility (Annex 5.1). By means of its pilot mill, TUBEF, together with the other partners, investigated the effect of steel composition and of new thermo-mechanical conditions on microstructure.
The austenitizing temperature was changed in order to have a given content of micro-alloying elements in solution. Usually, in the first part of the mill (stands 1 and 2) the material temperature was above and in the second part (stands 3 and 4) below the recrystallisation temperature. By changing rolling speed and cooling intensity, several temperature evolutions

were obtained inside the mill. Also the cooling strategy after rolling was adjusted using several cooling media, changing their location and intensity.

For these experiments Lucchini and Sidenor prepared the initial material to be rolled in the form of round bars of size 12 mm.
The microstructures and mechanical properties of experimental wire rods were characterised by CSM, Sidenor and Lucchini (Annex 5.2 and Annex 5.3).

Task 5.2 Response of TMCP rods to direct processing (cold heading, drawing).

Compression testing
Compression tests were carried out on cylindrical specimens, measuring load and strain (Annex 5.4).

Drawing experiments
The drawability of the experimental wire rods manufactured by the four stand high speed mill using TMCP and special cooling strategies was tested by TUBEF using drawing machines. The test programme included the determination of the stress-strain curves (Annex 5.5).

Task 5.3 Influence of as-rolled microstructures on annealing response.

Sidenor investigated the evolution of microstructure (spheroidisation) and properties during annealing.

WP 6 Guidelines for the application of TMCP to wire rod production.
All partners.

The whole results were analysed and practicable industrial routes were defined.

4. RESULTS AND DISCUSSION

A full description of the whole work carried out and the results obtained are detailed in the Annexes, which cover all the WPs. In the following sections the main results are examined and discussed in terms of process parameters and metallurgical factors which affect the effectiveness of TMCP for long products for controlling final microstructures, mechanical properties and further processability.

The TMCP for long products means a technology which allows to control the austenite and its subsequent structural changes, not necessarily through the unrecrystallization of austenite as in plate production.

In particular, the following aspects were analysed using data from laboratory tests, pilot and industrial trials and mathematical modelling:

- Flow-stress in low temperature rolling
- Recrystallization and grain growth of austenite

- Phase transformation of austenite
- Final microstructures and mechanical properties of TMCP wire-rods
- Cold formability and drawability of TMCP wire-rods
- Response to spheroidizing annealing of TMCP wire-rods.

4.1. FLOW STRESS

During hot rolling of wire-rod, metallurgical phenomena such as recrystallization and grain growth, influence the state of austenite. Factors which affect the evolution of austenite include thermo-mechanical conditions and steel chemistry. In turn, the microstructure affects metallurgical responses such as flow-stress and the ability to withstand deformation at high temperature without fracture.

Flow-stress curves for a material during a hot working operation are an essential ingredient in calculating the rolling load, which is usually based on a knowledge of the mean flow-stress. The flow-stress is strongly dependent on strain rate and temperature, but is also influenced by austenite grain size and recrystallized fraction, as well as the degree of recovery in any unrecrystallized region.

In this project the effects of deformation temperature and strain rate were investigated by hot torsion tests (Annex 3.1).

Preliminary hot torsion tests in TUBEF showed that the flow curves can be significantly effected by small differences in the chemical composition. Therefore, all steel grades were tested. The program of hot torsion tests was:

- Austenitisation temperature: 1150°C, holding 10 minutes, cooling to deformation temperature
- Deformation temperature:1050°C, 1000°C, 950°C, 900°C, 850°C, 800°C, 750°C
- Deformation rate: 0.2, 2.0 5.0 s^{-1}.

The results of the investigation and evaluation of torsion tests were factors which can be used for the calculation of rolling force and torque. The hardening and softening behaviour of each steel grade was also assessed. All the stress–strain curves of the investigated steels are reported in Annex 3.1.

The peak strain as a function of deformation temperature is shown for the various strain rate values and steels in Fig. 1 to Fig.5.

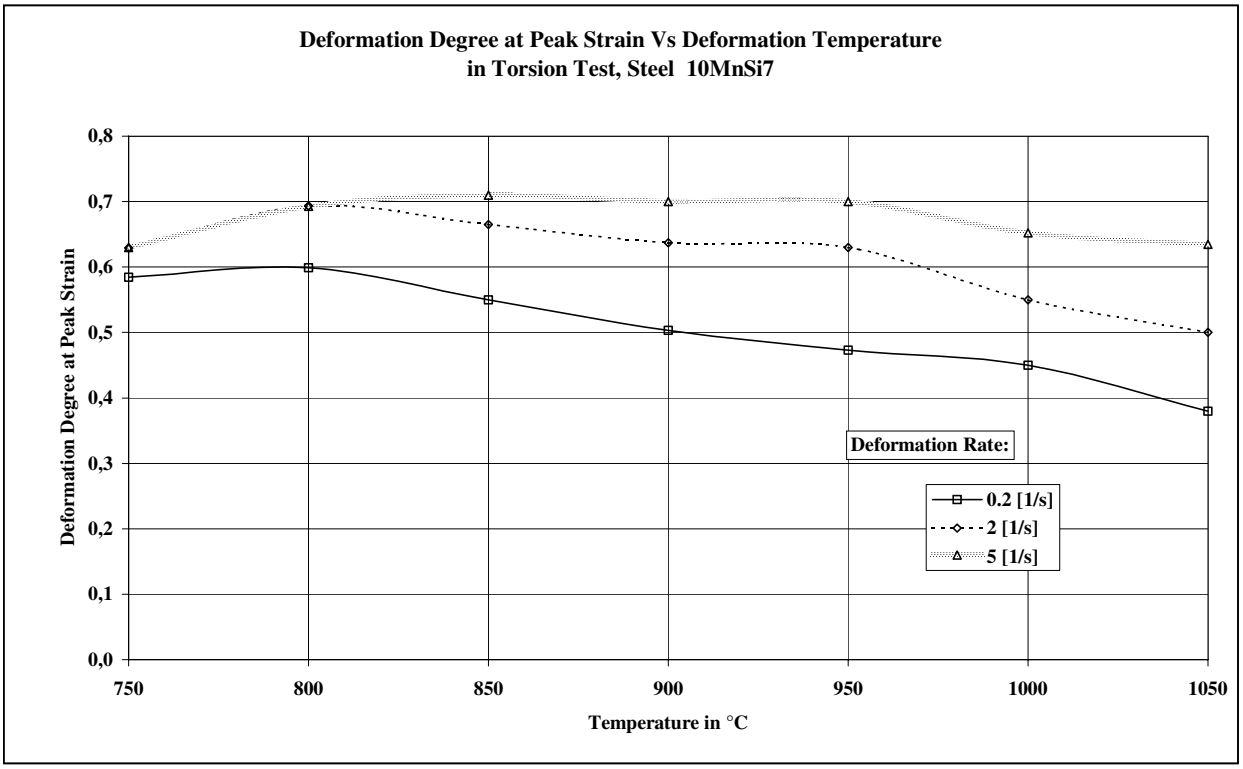


Fig.1 Relationship between peak strain and temperature for 10MnSi7 steel.

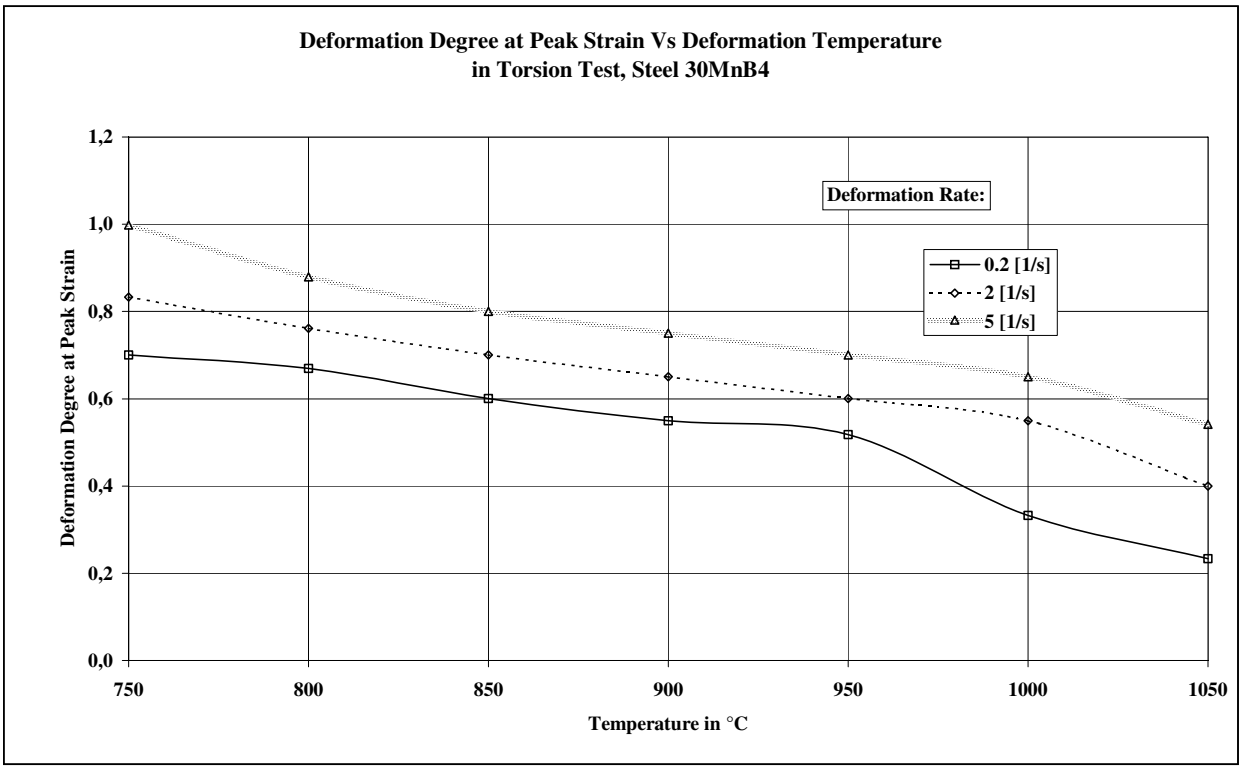


Fig.2 Relationship between peak strain and temperature for 30MnB4 steel.

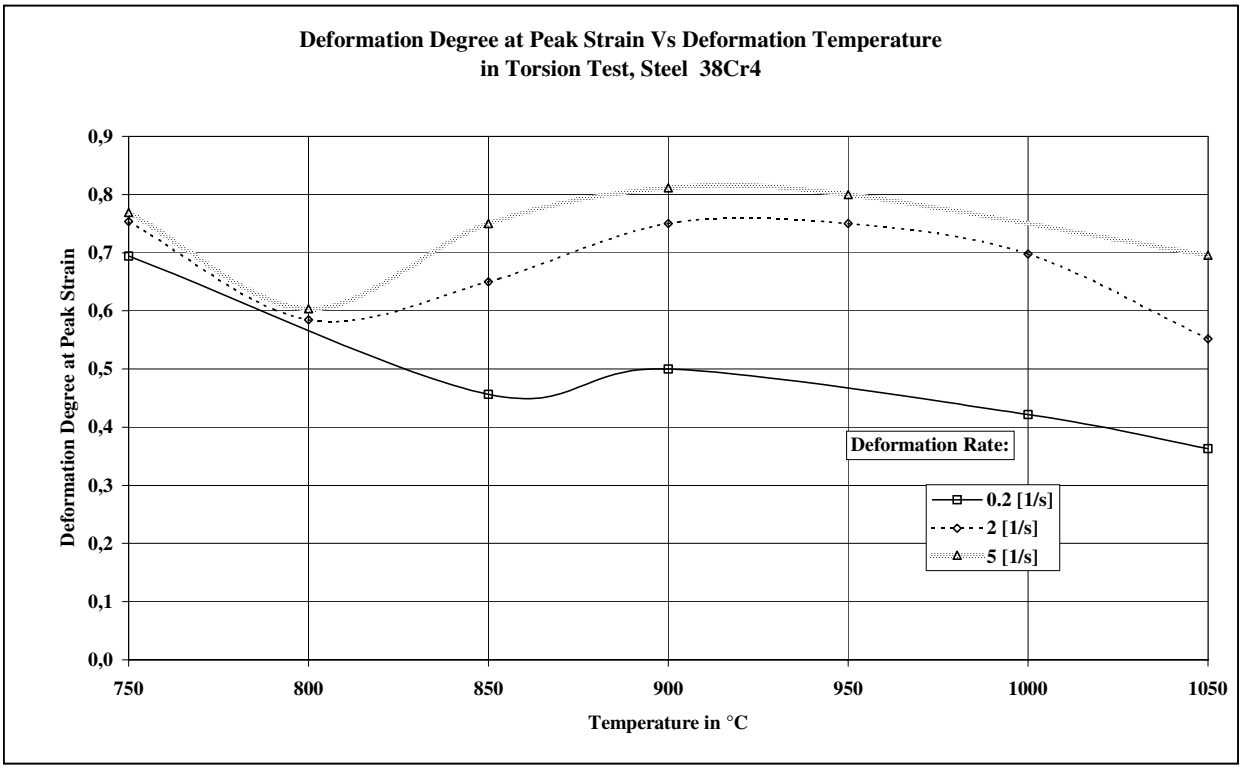


Fig.3 Relationship between peak strain and temperature for 38Cr4 steel.

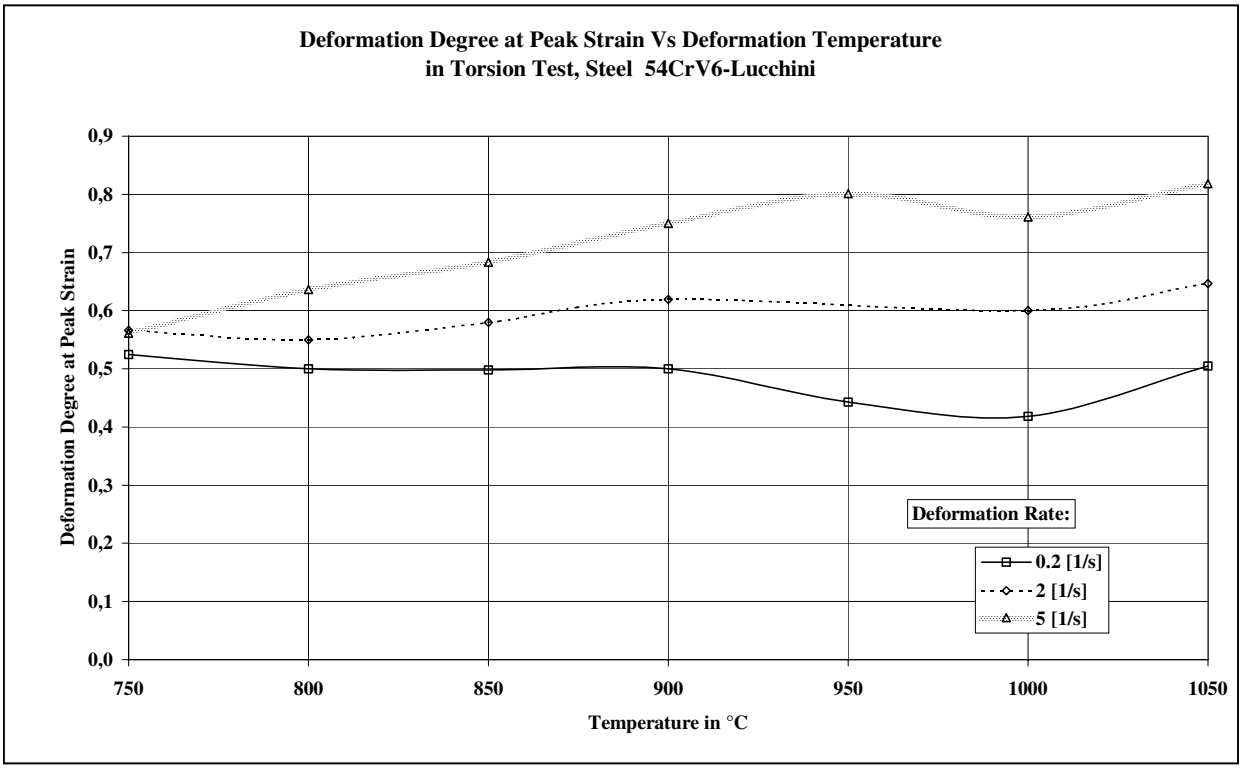


Fig.4 Relationship between peak strain and temperature for 54SiCrV6 steel (Lucchini).

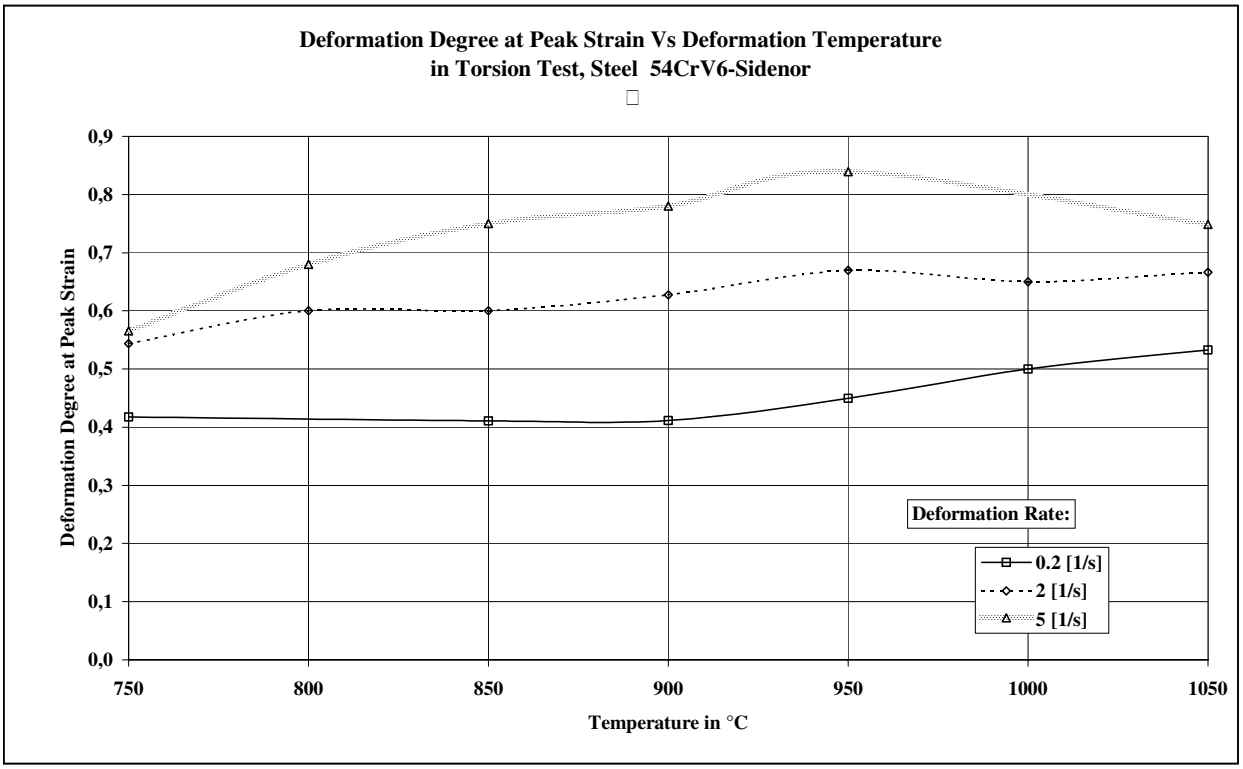


Fig.5 Relationship between peak strain and temperature for 54SiCrV6 steel (Sidenor).

The peak strain gives the magnitude of the critical strain for the initiation of dynamic recrystallization, being:

Critical strain for dynamic recrystallization = constant (Peak strain)

The constant is about 0.85.

A critical strain higher than 0.4 is needed to activate dynamic recrystallization in the case of austenite with grain size of 40 to 60 μm, measured for the re-heating conditions used in the torsion tests.
The peak strain usually increases with increasing the strain rate and reducing the deformation temperature. This behaviour is confirmed by steels 10MnSi7 and 30MnB4 for which calculated and experimental peak strain values are in good agreement (Annex 4), whilst the other steels, with higher carbon and alloy contents, exhibit peak strain values lower than those expected at temperatures lower than 900 °C, especially at high strain rates. This can be related to a heating-up effect which appears during torsion testing, as in any other deformation process, especially at low deformation temperatures and for steels with a high deformation resistance. Therefore for these steels the real deformation temperature differs from the nominal test temperature. Although temperature corrections were made, based on previous experiences using thermocouples and different torsion conditions and steel grades, some discrepancies still remain for steels 54SiCrV6 and 38Cr4.

Significant effects of temperature and deformation rate on deformation resistance were seen. The flow-stress generally increases with increasing strain rate and decreasing deformation temperature.
A reduction of deformation temperature from 1000 °C to 850 °C leads to an increase of flow stress from 60% to 100% at a strain value of 0.3 (from 110-130 MPa to 175-240 MPa).

This behaviour was confirmed by the pilot mill trials at TUBEF, where an increase of the rolling force at the last stand of 50-60 % was observed for the various steels when the finish rolling temperature was reduced from 1000 to 850 °C (Annex 5.1). In this case of multi-pass rolling, also the recrystallization behaviour before the last pass affected the rolling force. Likely, steel 30MnB4, characterised by a more refined initial austenitic grain, exhibits a higher recrystallized fraction of austenite at the entry of the last stand, as indicated by the model simulations (Annex 2), hence a lower increase in rolling force, compared with that of the other steels processed with the same conditions. A slower recrystallization kinetics of the other steels is also expected, for a given prior austenite grain size, due to a higher alloy content.

4.2. RECRYSTALLISED AUSTENITE GRAIN

4.2.1. Model Application

The main microstructural processes of interest during hot rolling prior to transformation are recrystallization and grain growth of austenite. This processes were studied and modelled (Annex 2).

With reference to the average thermo-mechanical evolution of 13 mm diameter wire rods processed by Sidenor mill in Vitoria (Annex 1), the mean austenite grain sizes at the entry of each stand were calculated by various equations and approaches by Sidenor and CSM (Annex 4). These calculated data were compared with the values measured on 38Cr4 steel specimens taken at various locations along the mill and rapidly quenched (< 3 s) to freeze the austenitic structure. The agreement was good, given the nature of the measurements, the accuracy of the process model used to calculate average temperature of the stock, and the assumptions made in the recrystallization model.

The model was applied to assess the effect of chemical composition and rolling temperatures on the austenite grain size (AGS) at finishing (Table 3).

Simulation Code	Steel	Intermediate Water Boxes	Mean Finish Rolling Temperature (°C)	Final Water Boxes	Average AGS (μm)
A	38Cr4	No	1040	No	61
B	10MnSi7	No	1040	No	63
C	30MnB4	No	1040	No	27
D	10MnSi7	No	1040	Yes	29
E	10MnSi7	Yes	860	Yes	13

Table 3 Calculated average AGS for various rolling conditions.

In the case of a finish rolling temperature of 1040 °C the calculated AGS of steel 38Cr4 is about 60 μm after 12 s from finishing, if water boxes were not activated. The behaviour of steel 10MnSi7 is predicted very similar, whilst the steel 30MnB4 is finer (27 μm) because its grain growth is inhibited by the presence of Ti (C, N) particles.

Simulation Code	Intermediate Water Boxes	Finish Rolling Temperature (°C)		Final Water Boxes	AGS (μm)	
		Surface	Centre		Surface	Centre
F	Yes	800	880	Yes	6	16
G	Yes + Equalization	845	870	Yes	12	14

Table 4 Calculated average AGS for various rolling conditions of steel 10MnSi7.

Introducing final water boxes that in 2 s are able to reduce the average temperature from 1040 to 870 °C allowed to maintain AGS lower than 30 μm.

With a reduction of the average finish rolling temperature of 180 °C (from 1040 to 860 °C), introducing intermediate water boxes between intermediate and finishing mill in addition to those just after finishing, an average AGS of less than 16 μm is predicted by the model. However, local microstructure becomes progressively more heterogeneous through the diameter of the wire-rod if adequate equalization is not allowed (Table 4).

4.2.2. Trials by Freiberg Pilot Rolling Mill

The evolution of austenite grain size was also investigated by the pilot mill trials at TUBEF (Annex 5.1).

Objective of the rolling trials was to achieve both a selected start temperature in the first pass as well as defined temperatures at the last stand and laying head.

The desired roughing temperature was adjusted by a precise temperature control at the rod heating stage, before rolling.
The temperature at the laying head was controlled by the following parameters: rolling speed, interstand cooling and number of finish cooling pipes. The schematic layout of wire rod rolling mill is reported in Fig.6.

All relevant data as temperatures, forces and torques were measured and summarised in Annex 5.1.

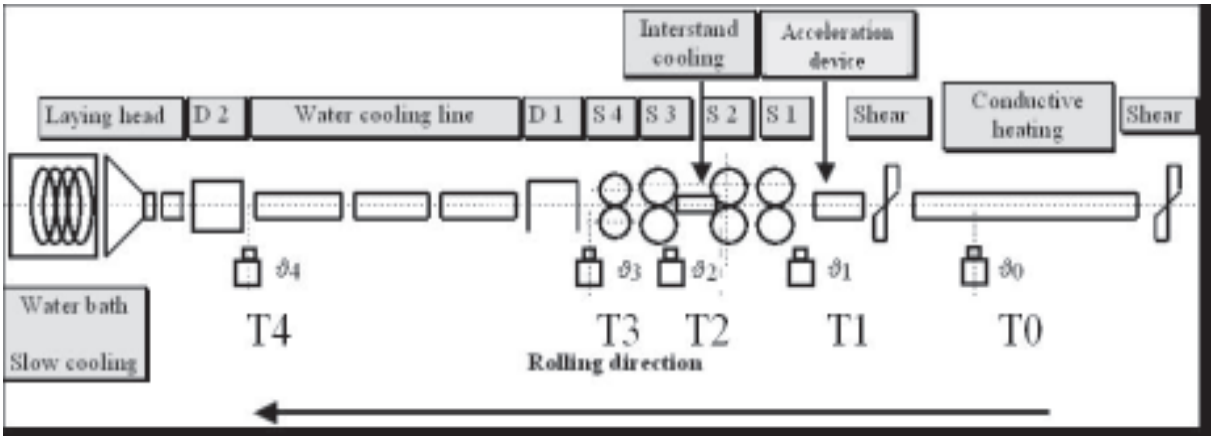


Fig.6 Schematic layout of hot rolling mill in Freiberg (TUBEF).

The AGS at the entry of the first stand, just after the fast reheating of the initial steel rods, was determined for various conditions in terms of holding temperature and time (Table 5).

Test No.	Steel	Treatment
0001	54CrSiV6 (SIDENOR)	Heating to 1100°C → quenching
0002	54CrSiV6 (SIDENOR)	Heating to 1100°C → holding 150s → quenching
1001	54CrSiV6 (LUCCHINI)	Heating to 1100°C → quenching
1002	54CrSiV6 (LUCCHINI)	Heating to 1100°C → holding 150s → quenching
2001	30MnB4 (LUCCHINI)	Heating to 1100°C → quenching
2002	30MnB4 (LUCCHINI)	Heating to 1100°C → holding 150s → quenching
3001	38Cr4 (SIDENOR)	Heating to 1100°C → quenching
3002	38Cr4 (SIDENOR)	Heating to 1100°C → holding 150s → quenching
4001	10MnSi17 (LUCCHINI)	Heating to 1100°C → quenching
4002	10MnSi17 (LUCCHINI)	Heating to 1100°C → holding 150s → quenching

Table 5 Heating up conditions before hot rolling trials at TUBEF.

The austenite grain boundaries were revealed by the Bechet Beaujard's etch with a prior heat treatment at 450 °C for 1 hour. The average AGS was measured by the mean linear intercept method and/or comparison charts (Table 6).
The resulting austenitic grain sizes are given in Table 6. Examples of obtained microstructures are showed in Fig.7.

Sample		Sidenor		CSM		Lucchini	
No.	Steel	AGS (µm)	AGS (ASTM)	AGS (µm)	AGS (ASTM)	AGS (µm)	AGS (ASTM)
0001 0002	54SiCrV6 (SIDENOR)	20 38	8.0 6.1	19 31	8.2 6.8	20 27	8.0 7.5
1001 1002	54SiCrV6 (LUCCHINI)	23 35	7.6 6.4	21 34	7.9 6.4	20 26	8.0 7.5
2001 2002	30MnB4 (LUCCHINI)	18 22	8.3 7.7	n.d. n.d.	n.d. n.d.	18 18	8.3 8.3
3001 3002	38Cr4 (SIDENOR)	18 39	8.3 6.1	16 37	8.7 6.2	20 33	8.0 6.5
4001 4002	10MnSi7 (LUCCHINI)	20 41	8.0 5.9	n.d. n.d.	n.d. n.d.	22 36	7.9 6.4

Table 6 Prior AGS measurements after fast reheating before hot rolling.

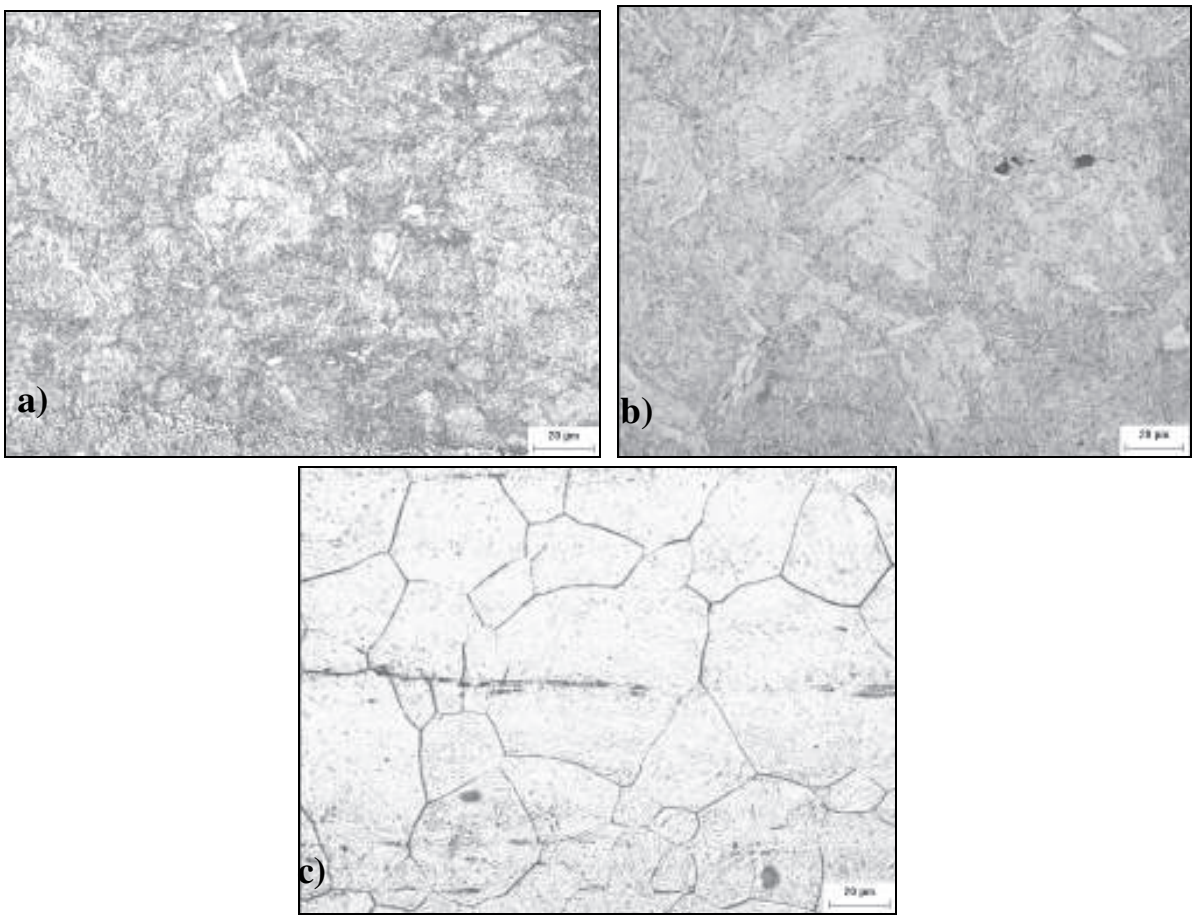


Fig.7 Examples of AGS: (a) 54SiCrV6 steel (SIDENOR 0002), (b) 54SiCrV6 steel (LUCCHINI 1002), (c) 38Cr4 steel (SIDENOR 3002).

The initial AGS is usually in the range 16 to 40 μm. The refined austenitic microstructure corresponds to reheating without holding.

Other investigations on AGS were carried out on the wire-rods manufactured by the pilot mill of TUBEF and water quenched after laying head.
Some examples of AGS changes due to various thermo-mechanical cycles are shown in Fig.8.
Other ones are reported in Annex 5.2, whilst their measurements of average AGS are summarised in Tables 7 and 8.
Generally, the AGS was refined reducing the finish rolling temperature from 980 °C to 850 °C, whilst the effect of lying head temperature from 850 °C to 750 °C was often poor.

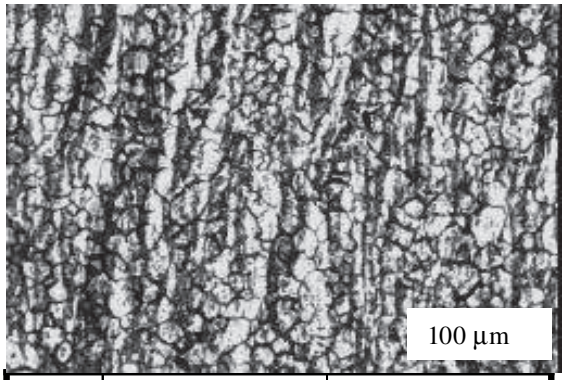
Steel	Specimen Code	T3, Finish Rolling Temperature (°C)	T4, Laying Head Temperature (°C)	AGS (ASTM No.)	
				Centre	Surface
30MnB4	205	990	840	9-10	9
	209	870	770	11	11
38Cr4	302	980	850	10	10-11
	315	840	760	11	10
10MnSi7	401	980	800	10	10-11
	410	900	750	11	10

Table 7 AGS of selected wire-rods manufactured by pilot mill and water quenched after laying head. Comparison between wire-rod surface and centre.

TRIAL I (NO HOLDING TIME)						
Nº	T1 [ºC]	T2 [ºC]	T3 [ºC]	T4 [ºC]	AGS (μm)	AGS (ASTM)
302	970	990	980	850	11	9.7
304	990	1000	980	750	14	9.0
306	910	940	910	800	10	10.0
308	930	860	850	820	7.4	10.9
315	830	865	840	760	7.2	10.9

TRIAL III (150 s HOLDING TIME)						
Nº	T1 [ºC]	T2 [ºC]	T3 [ºC]	T4 [ºC]	AGS (μm)	AGS (ASTM)
362	970	1070	980	740	13	9.2
365	1020	1080	990	870	18	8.3
373	890	860	840	820	7.5	10.8
376	930	990	870	760	9.1	10.3

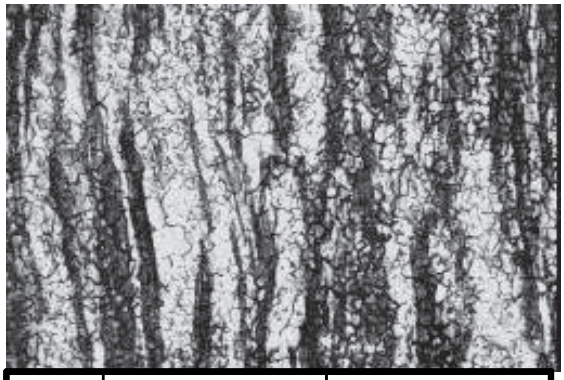
Table 8 AGS of 38Cr4 steel wire-rods manufactured by pilot mill and water quenched after laying head. Effect of initial austenite grain size. T1 to T4 temperatures measured by pyrometers (see Annex 5.1).



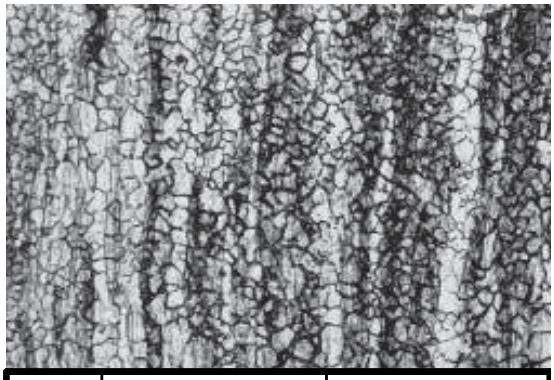
No.	$T_{END\ ROLLING}$	$T_{LAYING\ HEAD}$
302	980	850



No.	$T_{END\ ROLLING}$	$T_{LAYING\ HEAD}$
304	980	750



No.	$T_{END\ ROLLING}$	$T_{LAYING\ HEAD}$
308	850	820



No.	$T_{END\ ROLLING}$	$T_{LAYING\ HEAD}$
315	840	760



No.	$T_{END\ ROLLING}$	$T_{LAYING\ HEAD}$
306	910	800

Fig.8 Steel 38Cr4. Examples of AGS after various thermo-mechanical cycles at pilot mill (magnification: 200X).

The thermo-mechanical cycle of sample 365 (Table 8) attempted to reproduce the industrial rolling conditions (starting AGS and rolling temperatures). The AGS obtained in this case is slightly smaller than the industrial one measured for that steel, i.e. ASTM No. 8.

Initial AGS (i.e. holding time) showed also some influence (Table 8): coarser initial AGS causes in general larger AGS at the laying head.

Final AGS was dependent on rolling temperatures, especially that at the last stand (T3). In the cases in which this temperature was around 850 ºC (samples 308, 315 and 373) the AGS was refined and reached values of ASTM No.11 (7 μm).

4.2.3. Hot Compression Tests

Also hot compression trials were performed in order to study the austenitic grain growth after recrystallisation. The results shown in detail in Annex 3.2 indicate that for steel 38Cr4, starting from a 40 μm recrystallised austenite grain size, after a holding of 12 s, the grain grows up to 150 μm at 1000 °C.

The grain growth is strongly reduced for the same holding period at 850 °C, being the average grain size below 70 μm. A similar behaviour is expected for the other steels, except the steel 30MnB4 for which the movement of recrystallised grain boundaries can be inhibited by very stable and fine Ti(C,N) particles.

Very effective cooling devices are required to reduce the stock temperature for controlling grain growth after hot rolling as the usual distances between finish roll and cooling conveyor are rather small and wire-rod speed is very high. However, if a very severe continuous cooling is used, surface temperatures can reach values lower than 400 °C and martensite formation occurs. Therefore, the cooling system is constituted of a sequence of water cooling boxes separated by some equalizing zones in air where the wire-rod surface temperature can increase. The length of cooling and equalizing zones is designed in order to promote adequate fast cooling of the wire-rod without inadmissible undercooling, avoiding the risk to form undesired brittle microstructural constituents.

4.2.4. Industrial trials at Sidenor on the Effect of cooling devices after finishing block

The effect of cooling devices placed between finishing block and laying head of Sidenor wire rod mill on AGS (Fig.9) and on final microstructure (Table 9 and Fig.10) were specifically studied for 38Cr4 steel grade (13 mm diameter wire rods) by industrial trials.
Two different cooling patterns have been used (Table 9).

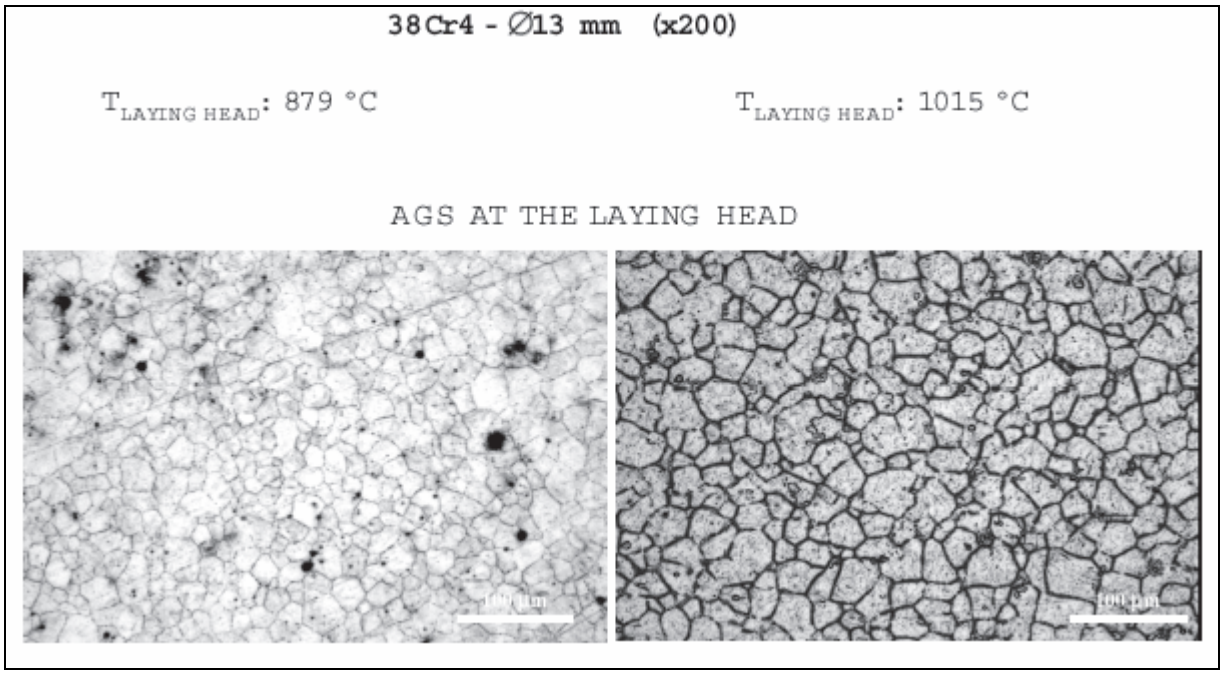


Fig.9 Effect of water cooling devices: AGS of 38Cr4 steel wire-rods for high and low laying head temperatures.

Steel 38Cr4 Cooling degree	Temperature in cooling devices [ºC]		AGS [ASTM No.]		UTS [MPa]	ROA [%]	Bain+Mart.
	Entry	Exit	Centre	Surface			
High	1085	879	8-9	8-10	829	53	0%
Low	1079	1015	6-7	7-8	962	33	90%

Table 9 Influence of cooling profile after finish rolling on microstructural characteristics for industrial wire-rods in steel 38Cr4 (13 mm diameter).

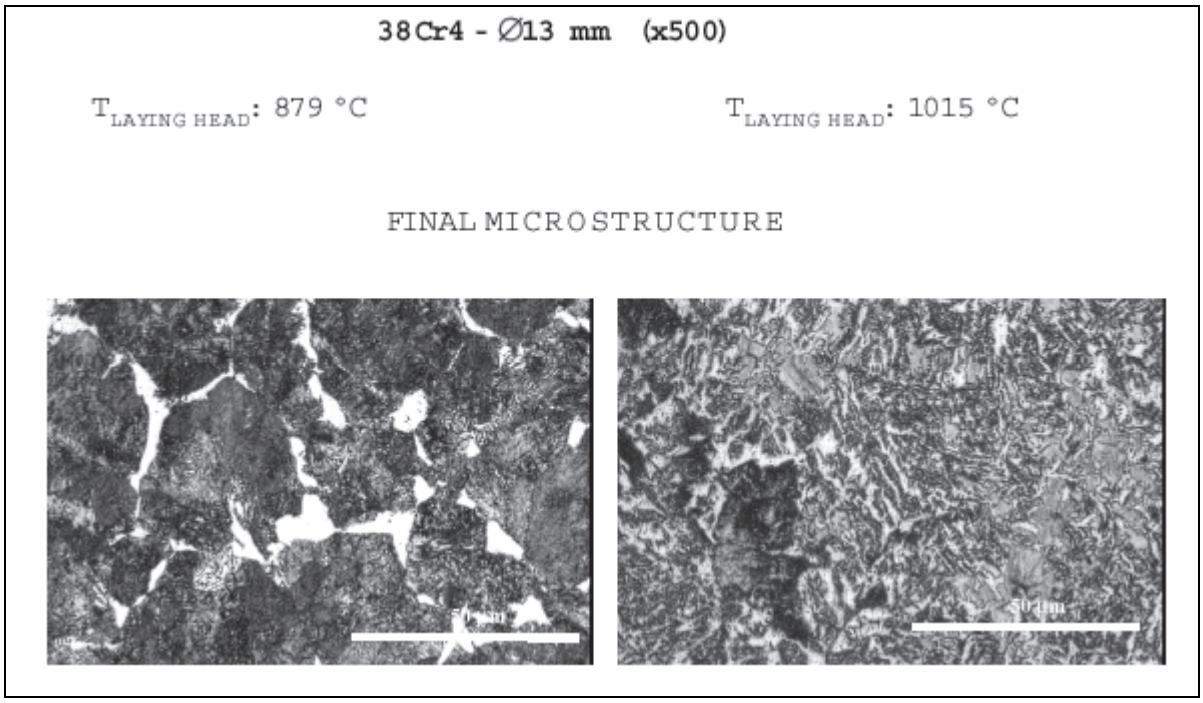


Fig.10 Effect of water cooling devices: microstructure of 38Cr4 steel wire-rods for high and low laying head temperatures.

The first one corresponds to a typical cooling process for this steel grade (high water flow rate), the second one was performed with minimum water flow rate. In both cases a retarded cooling practice was used in the EDC conveyor.

As it was expected, significant differences in both AGS and final microstructure were observed (Figs.9 and 10). The coarse AGS obtained for high laying head temperatures promoted the formation of mixed bainite-martensite microstructures due to the increased hardenability (Table 8).

4.3. PHASE TRANSFORMATION

The phase transformation characteristics of the investigated steels for different AGS values (fine and coarse) were determined by dilatometry. Both isothermal and continuous cooling conditions were used (Annex 3.3).

With the austenitic grain size and cooling rates calculated by modelling (Annex 4) the phase transformation evolution along the conveyor can be predicted using the approach reported in Annex 4.

Both experimental tests and mathematical simulations indicate that a reduction of austenite grain size is beneficial in reducing the formation of hard and brittle phases such as bainite and martensite on the final microstructure.

Conveyor Speed (m/s)	Laying Head Temperature (°C)	AGS (µm)	Transformed austenite Under insulated covers (%)	B+M (%)
0.10	970	29	60	40
0.07	940	29	95	5
0.07	940	15	100	0
0.10	860	29	98	2
0.10	820	29	100	0
0.10	860	15	100	0

Table 10 Calculated effect of process conditions and prior AGS on phase transformation of steel 30MnB4 processed by insulated Stelmor conveyor.

4.3.1. Model Application

Austenite refinement accelerates the ferrite formation and promotes the completion of the phase transformation under the insulated zones (very slow cooling). In this respect, simulations by model (Table 10) clarified that the success in avoiding bainite and martensite formation in steels with very high hardenability such as boron steels for fasteners, is related to the following factors:

- low cooling rates during phase transformation, obtained using insulated covers and low conveyor speeds which give dense loops

- adequate laying head temperatures.

In particular, low laying head temperatures, for a given conveyor speed, favour the completion of the phase transformation in the insulated zones (formation of ferrite and pearlite). If low laying head temperatures are difficult to be achieved, the AGS refinement helps to avoid bainite and martensite formation, included the segregated regions which results more hardenable due to local enrichment in alloy elements and carbon.

4.3.2. Pilot Mill Trials

Behind the laying head different cooling rates were achieved during the pilot mill trials at Freiberg:

- Water quenching to freeze the microstructure after rolling,
- Slow cooling in a thermo-pot with different thermo-pot temperatures to simulate an insulated Stelmor/EDC conveyor.

In the case of quenching the thermo-pot behind the laying head was filled with water and the rods (final diameter of 8 mm) were laid directly into the water bath, so immediately after rolling quenching occurred. By this way a very fast freezing of the microstructure was achieved after rolling.

Slow cooling took place with the same thermo-pot used for quenching, but in this case a lead bath at controlled temperature was employed. The lead was used to obtain a stable temperature in the thermo-pot and a good heat transfer.
In order to calibrate the cooling conditions in the thermo-pot, preliminary tests took place with specimens of steel 10MnSi7, 8 mm diameter. These specimens were equipped with thermocouples, heated in a soaking furnace and cooled down in the thermo-pot.

Because during rolling experiments the loops often do not lay completely on the bottom of the thermo-pot, these preliminary experiments took place with two arrangements:

- Specimen placed on the bottom of the thermo-pot (directly on the swimming steel plate),
- Specimen placed about 50 mm above the bottom.

Fig.11 shows the cooling curves according to both configurations. These tests showed that the differences of cooling rates between the two adjustments were very small. Within 10 min the wire-rods have reached the thermo-pot temperature.

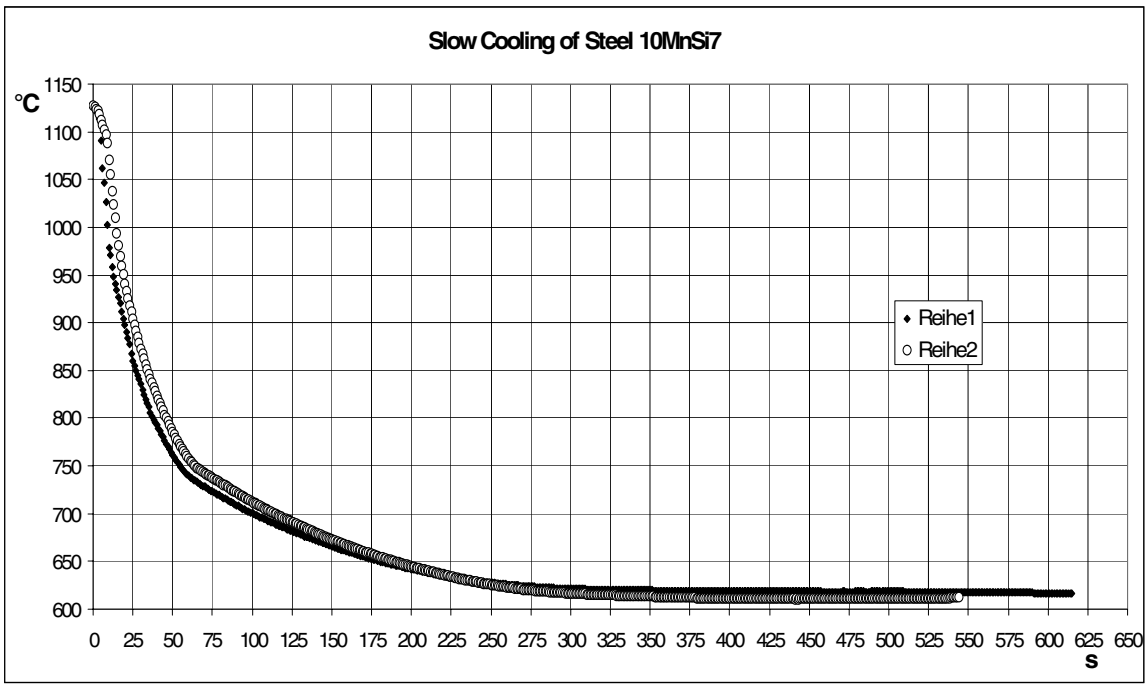


Fig.11 Cooling of 8 mm diameter wire-rods in steel 10MnSi7 in the thermo-pot simulating insulated cooling, Zone 1 (Reihe 1): bottom, Zone 2 (Reihe 2): 50 mm above bottom.

Practically the temperature in the thermo-pot can be varied between 100 °C and 700 °C, to obtain different cooling rates (Table 11).

Lead Temperature of Thermopot	CR between 800-600 °C
	°C/s
600 °C	0.38
180 °C	4.65
120 °C	5.26
20 °C	5.77

Table 11 Cooling rates for different lead temperature in the thermo-pot.

During the experiments (Fig.12) the following temperatures were chosen: 600 °C, 180 °C, 120 °C and 20 °C.
The rolled loops were held in the thermo-pot, after the planned holding time they were removed and cooled in still air.

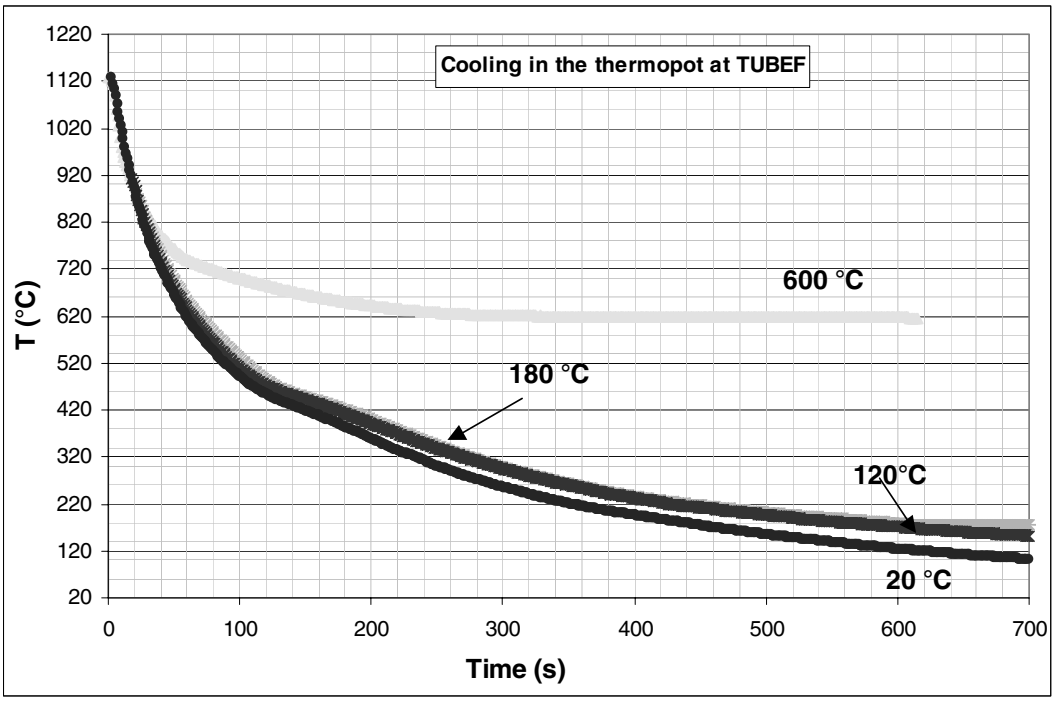


Fig.12 Temperatures evolution inside thermo-pot at different lead temperatures.

In Fig.13 the temperature evolution inside thermo-pot is reported as a function of log time.

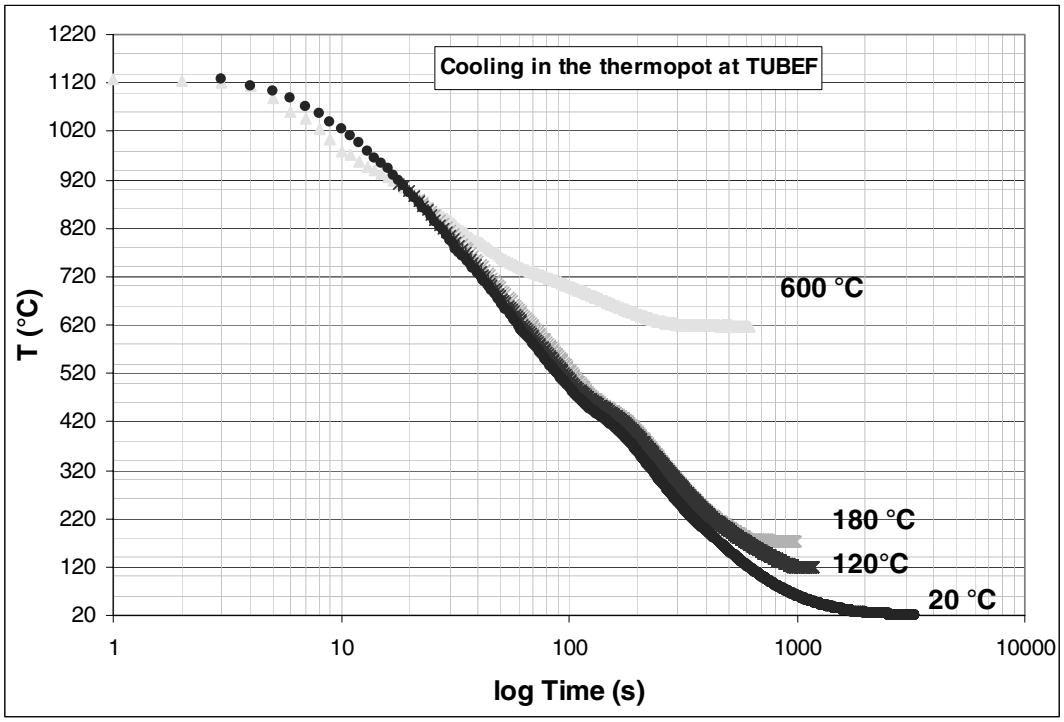


Fig.13 Temperatures evolution vs. log time inside thermo-pot.

A complete microstructural analysis, was carried out (Annex 5.2).
Relations among main mechanical parameters (UTS, ROA, hardness) and hot rolling and cooling parameters were given in Annex 5.3.

Outputs of the sensitivity analysis performed on trials I and II for Lucchini steels 54SiCrV6, 30MnB4 and 10MnSi7 are given in Table 12 and Table 13.

Trial	UTS MPa	ROA %	Hardness HV_{30}	T1 °C	T2 °C	T3 °C	T4 °C
I	1148	55	325	906	930	911	814
II	1204	59	373	988	877	928	801
Δ	+ 5%	+ 7%	+ 15%	+82	- 53	+ 17	- 13

Table 12 Mechanical properties variation between rolling trials I and II for steel 54SiCrV6.

TRIAL	UTS vs. T4	ROA vs. T4	UTS vs. T(1,2,3)	ROA vs. T(1,2,3)	HV_{30} vs. T4
I	YES, parabolic	YES, decreasing	Constant	Constant	YES, parabolic
II	YES, complex	YES, complex	YES, asympt.	NO	YES, complex

TRIAL	HV_{30} vs. UTS	HV_{30} vs. T(1,2,3)	UTS vs. ROA
I	YES, increasing	NO	YES, complex
II	YES, increasing	YES, asympt.	Constant

Table 13 Main correlation laws for steel 54SiCrV6.

The Cr-V steel (54SiCrV6) showed light dependence of mechanical properties from rolling temperatures T1, T2 and T3. The laying head temperature (T4) seemed to be the most influencing factor. In general lower T4 means more ductile material (ROA increase). It is not rare that an increase of ROA corresponds to an increase also in UTS (and HV_{30}) values.

Lower T2 (and T1) values gave a better control in martensite bands formation and more homogeneous structures (measured also by less scattered mechanical parameters). This result was obtained for all steels, not only for 54SiCrV6.

The Ti-B steel (30MnB4) seems more sensitive to T1 and T2 (also T3) changes. Less benefits seemed to derive from fine T4 tuning. As seen for steel 54SiCrV6, lower T2 (and T1) values gave a better control in martensite bands formation and more homogeneous structures (measured also by less scattered mechanical parameters).

TRIAL	UTS MPa	ROA %	Hardness HV_{30}	T1 °C	T2 °C	T3 °C	T4 °C
I	622	65	193	915	962	932	792
II	641	67	220	952	898	931	803
Δ	+ 3%	+ 3%	+ 14%	+ 37	- 64	~ 0	+ 11

Table 14 Mechanical properties variation between rolling trials I and II for 30MnB4 steel.

TRIAL	UTS vs. T4	ROA vs. T4	UTS vs. T(1,2,3)	ROA vs. T(1,2,3)	HV_{30} vs. T4
I	YES, linear decreas.	constant	constant	Constant	YES, lin. decrease
II	YES, asinpth. decrease	constant	YES, parabolic	YES, asinpth. increase	YES, parabolic

TRIAL	HV_{30} vs. UTS	HV_{30} vs. T(1,2,3)	UTS vs. ROA
I	YES, linear increas.	NO	YES, complex
II	YES, asinpth. increase	NO (T1,T2)	constant

Table 15 Main correlation laws for 30MnB4 steel.

TRIAL	UTS MPa	ROA %	Hardness HV_{30}	T1 °C	T2 °C	T3 °C	T4 °C
I	503	78	155	951	965	929	778
II	475	77	155	986	928	945	802
Δ	- 6%	- 1%	0	+ 35	- 37	+ 16	+ 24

Table 16 Mechanical property variation between rolling trials I and II for 10MnSi7 steel.

TRIAL	Rm vs. T4	ROA vs. T4	Rm vs. T(1,2,3)	ROA vs. T(1,2,3)	HV_{30} vs. T4
I	constant	constant	constant	not defined	YES, lin decrese
II	constant	constant	constant (T3,T2), NO (T1)	constant (T3,T2), NO (T1)	constant

TRIAL	HV_{30} vs. Rm	HV_{30} vs. T(1,2,3)	Rm vs. ROA
II	constant	constant	constant
II	constant	NO	constant

Table 17 Main correlation laws for 10MnSi7 steel.

The C-Mn (high Si) steel exhibited a quite flat behaviour for all temperature ranges.
Ultimate tensile strength decreased in rolling trial II, while the same ductility levels were maintained. This behaviour could give some benefits in industrial processing, especially in terms of cold drawing behaviour.
Again, lower T2 (and T1) values gave a better control in martensite bands formation and more homogeneous structures (measured also by less scattered mechanical parameters).

Summarising:

- Independently of the steel chemistry, mechanical properties such as UTS, ROA and HV_{30} appeared more homogeneous (especially in the case of 54SiCrV6 steel) in the II trial. In general, the following rule holds: a rigid control of all process parameters and a reduction of temperature values at finishing block (T2-T3) and laying head (T4) lead to a greater microstructural homogeneity in wire-rods, with improvements in terms of reproducibility and narrower dispersion of mechanical properties.

- If steel is rolled at low temperatures and the process control is good, higher cooling rates can give benefits on the final product properties. This strategy not only reduces the formation of brittle and hard microstructures, but is able to reduce and/or increase UTS at the same ductility (ROA) level. The final choice of what condition is the most suitable for the steel producer should be tuned in relation to evaluations in terms of finishing routes and final applications.

- The presence of brittle structures (bainite, high carbon martensite) originated from cooling and not from segregation regions (i.e. dispersed islands instead of localized bands) seems not to affect ductility of as-rolled products assessed by tensile test. In fact, TMCP wire rods showing dual or multi-phase structures have a relatively low yield stress maintaining high work-hardening rate during cold deformation.

4.4. COLD FORMABILITY AND DRAWABILITY

4.4.1. Cold compression tests on prototype rods

Selected wire rods rolled in Freiberg (trial I and trial II) were cold compression tested to have information on cold headability. The detailed testing conditions and results in terms of visual inspection and stress-strain curves are given in Annex 5.4.

The following considerations were drawn for each steel:

- ***Steel 54SiCrV6 (Lucchini and Sidenor)***

Both these materials showed after compression test (as-rolled and machined conditions) cracks. An influence of the rolling technology can be hardly seen. Only for low and nearly constant T1-T4 rolling temperatures it seems possible to avoid failures during cold deformation (i.e. specimen 121 (Lucchini) and 024 and 025 (Sidenor). An influence of cooling rate after rolling on the crack behaviour was not found.

- ***Steel 30MnB4 (Lucchini)***

In as-rolled conditions several specimens showed cracks after the compression tests. For slow cooling rate (cooling from finish temperature to 600 °C, holding 600 s) all specimens in the as-rolled condition exhibited defects. A few air cooled specimens were "askew", only one cracked. A correlation between start or finishing rolling temperature and cooling rate could not be clearly identified.
Apparently, some micro-cracks formed during rolling and led to cracks in the compression test. After machining (i.e. elimination of potential sites for crack initiation) no failure appeared.

- ***Steel 10MnSi7 (Lucchini)***

Steel 10MnSi7 exhibited the same situation of steel 30MnB4. The 50% of as-rolled specimens cracked during compression test, especially after slow cooling. Air cooled specimens were not affected by severe cracking. It is not possible to argue if this is a consequence of the cooling strategy. After machining, no failure appeared. Only the specimen 424 showed cracks after turning (probably due to its very low finishing rolling temperature (690 °C). In this condition, hot rolling was carried out in the mixed ferrite-austenite range. In fact this high Si low C steel exhibits the highest Ac_3 temperature (Annex 3.3). Therefore ferrite was formed during rolling and work-hardened. This causes a reduction in ductility.

- ***Steel 38Cr4 (Sidenor)***

This steel showed some cracks both in as-rolled and in machined condition; all defects appeared after air cooling treatment. Slow cooled wire rods (cooling from finish temperature to 600 °C and holding 600 s) were crack-free after the compression test. Hence, a slow cooling treatment (such as 600 °C, for 600 s), as performed in the Freiberg rolling trials, is the adequate cooling strategy.

4.4.3. Cold drawing trials of prototype wire rods

For this trial a drawing bench by the Institute for Metal Forming in Freiberg was used. The cold drawing schedule was in two steps: from ∅ 8.0 mm to ∅ 7.62 mm ($\varepsilon = 0.097$) and from ∅7.62 mm to ∅ 7.18 mm ($\varepsilon = 0.119$). A total cold reduction of $\varepsilon = 0.216$ was obtained (Annex 5.5).

Steels 54SiCrV6 (Sidenor), 30MnB4 (Lucchini), 38Cr4 (Sidenor) and 10MnSi7 (Lucchini) showed good drawability.

Steel 54SiCrV6 (Lucchini) had a rupture during the 2nd drawing step.

Metallographic examinations, hardness measurements and tensile tests were performed on selected specimens (Annex 5.5).
The wire from Lucchini steel 54SiCrV6 exhibited more evident bands due to segregated regions in the industrial heat, compared with Sidenor steel 54SiCrV6.

Failures in cold drawing processes could be the consequence of a variety of reasons:

- work hardening induced by the drawing process,

- an as rolled microstructure with low ductility (coarse grained structure, presence of hard and brittle phases),

- the presence of surface cracks and/or embedded scale, which act as crack nucleation sites,

- the presence of inclusions beneath the surface.

Information on the microstructure and mechanical properties of as-rolled wire-rods (Table 18) seems to exclude an intrinsic lack of ductility of the matrix in the sample of Lucchini steel 54SiCrV6, when submitted to tensile testing.

Test No.	**T1 [ºC]**	**T2 [ºC]**	**T3 [ºC]**	**T4 [ºC]**	**UTS [MPa]**	**ROA [%]**	**CR [°C/s]**	**Bainite + Martensite**	**Pearlite + Ferrite**
061 Sidenor	1070	1100	1000	750	1243	19	1	70%	30%
176 Lucchini	950	985	870	770	1271	60	0.45	30%	70%

Table 18 Comparison of process data, microstructure and tensile properties of as-rolled wire-rod samples 061 and 176 in 54SiCrV6 steel.

Investigation by electron transmission microscopy (TEM) showed that the material 176 exhibited a more marked precipitation of cementite at grain boundaries. This fact and the possible presence of a surface defect could have caused the anticipated failure.

4.5. RESPONSE TO ANNEALING

The influence of thermo-mechanical processing on the response to spheroidising annealing was studied in 38Cr4 steel grade by Sidenor.

This steel was rolled in TUBEF pilot mill under different rolling conditions (thermo-mechanical processing) obtaining different mechanical and microstructural characteristics (see Annex 5.3). Selected samples underwent subsequent spheroidising heat treatments and the response to annealing, in terms of strength, ductility and degree of spheroidisation, was studied. This study was completed undertaking a similar investigation in other two samples coming from industrial trials by Sidenor wire rod rolling mill.

The spheroidising annealing was carried out under controlled atmosphere in a laboratory muffle furnace, available in the laboratory of Sidenor I+D. In order to better detect the structure with the best response to annealing, two different spheroidising cycles were performed: the first one simulates industrial times (11.5 hours), the other a shorter cycle (8 hours). A scheme of the cycles with the times performed is shown in Fig.14.

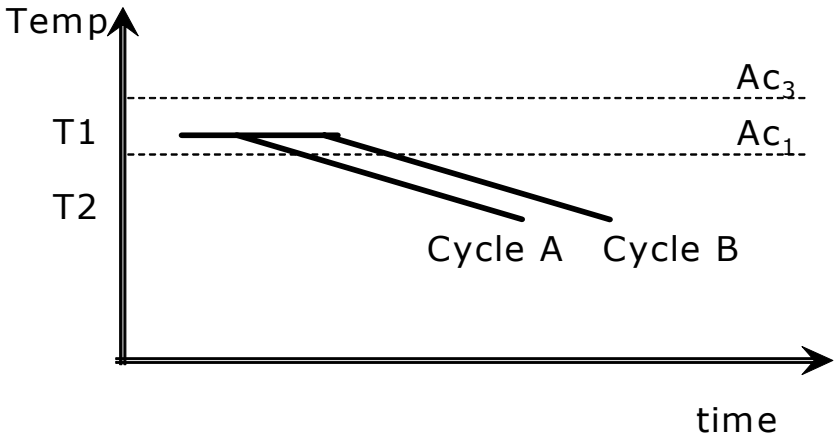


Fig.14 Schematic illustration of the spheroidising cycles.

An important aspect concerning response to annealing is the as-rolled microstructure prior heat treatment. The as-rolled microstructure was studied not only in terms of percentage of each microstructural constituent (ferrite, pearlite, bainite and/or martensite), but also in terms of interlamellar spacing of pearlite, since it is well documented there is a significant influence of this parameter on the response to annealing.

The main characteristics of these materials before and after the spheroidising heat treatments are presented in Table 19.

	As-rolled				**Spheroidised (cycle A - 8 h)**			**Spheroidised (cycle b – 11.5 h)**		
No.	**B+M[1)]**	**IS [2)] [nm]**	**UTS [MPa]**	**ROA [%]**	**UTS [MPa]**	**ROA [%]**	**% Spher**	**UTS [MPa]**	**ROA [%]**	**% Spher**
323	0%	139	740	70	548	70.2	50	504	71.1	95
325	0%	136	720	69	537	71.6	60	502	72.0	95
331	0%	112	750	70	535	72.4	75	508	73.0	95
332	0%	115	738	70	528	72.4	75	506	72.9	100
344	75%	112	991	48	582	64	80	534	72.9	100
347	70%	99	970	47	572	64	85	527	71.1	100
348	85%	97	954	49	574	66	85	536	71.1	100
351	0%	53	857	69	551	71.1	90	557	72.4	100
363	75%	115	1006	34	569	66	85	544	67.8	90
364	80%	110	996	36	572	66	85	542	69	90
374	55%	88	985	42	563	67.8	90	548	68.7	95
375	65%	124	990	63	569	68.7	95	544	71.1	100
IND ret	0%	210	829	53	625	61	25	540	64	50
IND air	75%	149	962	33	576	62	75	530	66.7	90

[1)] Percentage of bainite and martensite
[2)] Interlamellar spacing of pearlite

Table 19 Microstructural and mechanical characterisation on samples of 38Cr4 steel grade before and after the spheroidising cycles.

Taking into account that a proper spheroidising structure must have a high degree of spheroidising (> 90%) to improve cold formability, and a minimum tensile strength (<550 MPa) to minimise wear of tooling during drawing, following considerations can be established from the response to annealing of the studied samples:

All the samples, except the industrial retarded cooling, showed satisfactory response under the treatment B (11.5 hours), so the comparison must be performed on the results of cycle A (8 hours).

Analysing cycle A, a clear relationship between the degree of spheroidisation and interlamellar spacing was observed: the smaller the interlamellar spacing the better the response to annealing, as represented in Fig. 15.

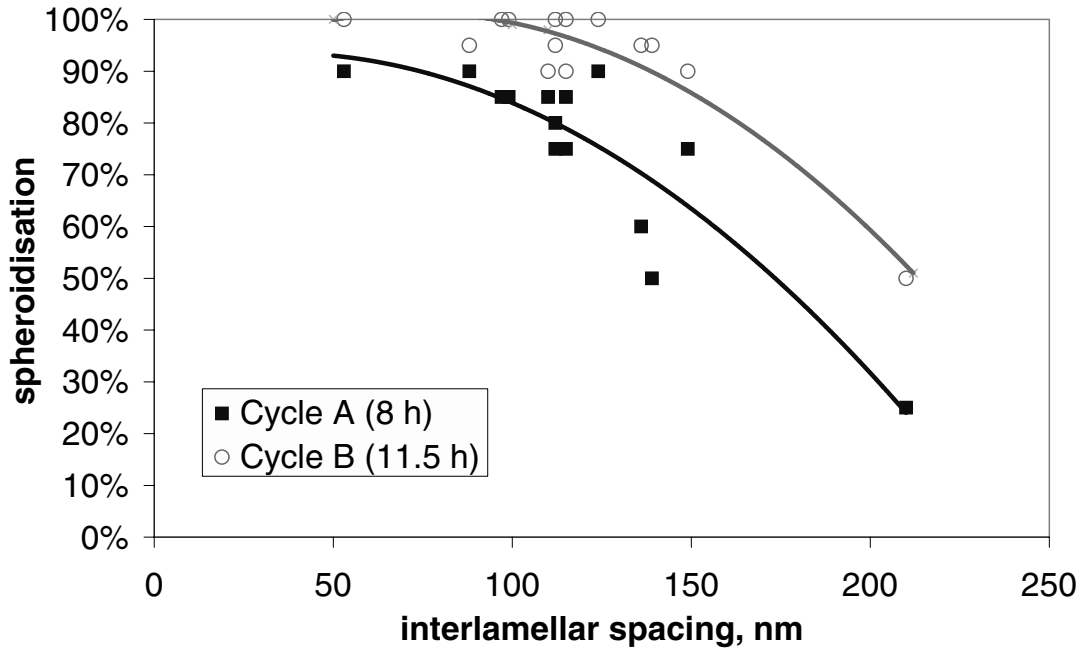


Fig. 15 Effect of interlamellar spacing of pearlite on the degree of spheroidisation after heat treatment.

The interlamellar spacing is a key factor concerning the spheroidisation.

Higher cooling rates decrease interlamellar spacing (improving degree of spheroidisation), but increase the risk of bainite and martensite formation what, in general, gives higher tensile strengths in the as-treated conditions.

The best balance of spheroidisation - tensile strength is given by the sample 351. This sample exhibits small interlamellar spacing and a high degree of spheroidisation due to the air (no retarded) cooling procedure, and on the other hand, due to the thermo-mechanical treatment (at temperatures below 800ºC), acicular structures (bainite and martensite) are avoided, obtaining adequate tensile strength values with high ductility.

In conclusion, it has been proven that thermo-mechanical processing, at temperatures below 800ºC, allows to shorten subsequent spheroidising heat treatments. In the trials performed in this project on steel 38Cr4, a reduction of spheroidising time of 30% was achieved.

5. EVALUATION

The thermo-mechanical process, through the control of recrystallization, grain growth, and phase transformation, can improve mechanical properties, e.g. strength-ductility combination.

Lowering the rolling temperature, but avoiding to deform in the austenite-ferrite range, the austenite grains become finer and ferrite formation is promoted. Therefore, for a given cooling pattern, the control-rolled material exhibits a fully ferrite-pearlite microstructure, whilst the material conventionally rolled can show a mixed microstructure with bainite and possible martensite. Alternatively, thermo-mechanically processed wire-rods can be cooled at higher cooling rates reducing the risk of bainite-martensite formation.

By controlling hot rolling and in-line cooling conditions, the desired microstructure and grade can be achieved.

This work clearly demonstrated that materials with shorter annealing times for spheroidisation can be produced adopting lower rolling temperatures (below 800 °C) in the case of 34Cr4 steel.

However, higher rolling forces are developed during low temperature rolling of long products and high precision rolling technologies such as those developed by SMS and Morgan Construction Company are needed.

The introduction of compact sizing mill is advantageous not only for obtaining increased productivity, but also for improving metallurgical properties.

To ensure that the material quality is uniform, there must be sufficient equalization time after intermediate cooling to reduce the thermal gradient through the wire-rod diameter before finishing.

In this respect, the concept of a cooling and equalization loop between intermediate mill and the finishing block is gaining importance. This loop can be equipped with additional water cooling lines and pre-finishing stands when required.

The ability of new plants to roll at lower temperatures, with a good equalization of temperatures gives cost savings by reducing the time of annealing in the case of cold heading steels (Cr and Cr-B grades). The soft annealing can be shortened more than 30%, saving up to 40 Euro/ton.

6. CONCLUSIONS

The investigation on the effect of thermo-mechanical treatment on microstructure evolution and final properties was performed on selected commercial steels (i.e. steels 10MnSi7, 38Cr4, 30MnB4, 54SiCrV6), by laboratory tests, pilot and industrial trials, and application of thermal and metallurgical models.

Hot torsion tests were performed on all materials. The result was a series of factors which allow the calculations of rolling force and torque. A good agreement between the experimental values of peak strain by torsion test and those calculated by the CSM model was found.
Significant effects of temperature and deformation rate on deformation resistance were seen. The flow-stress generally increases with increasing strain rate and decreasing deformation temperature.
A reduction of deformation temperature from 1000 °C to 850 °C leads to an increase of flow stress from 60% to 100% at a strain value of 0.3 (from 110-130 MPa to 175-240 MPa).

This behaviour was confirmed by the pilot mill trials at TUBEF, where an increase of the rolling force at the last stand of 50-60 % was observed for the various steels when the finish rolling temperature was reduced from 1000 to 850 °C. In this case of multi-pass rolling, also the recrystallization behaviour before the last pass affect the rolling force. Likely, steel 30MnB4, characterised by a more refined initial austenitic grain, exhibits a higher recrystallized fraction of austenite at the entry of the last stand, as indicated by the model simulations, hence a lower increase in rolling force, compared with the other steels processed with the same conditions. A slower recrystallization kinetics of the other steels is also expected, for a given prior austenite grain size, due to a higher alloy content.

In order to study the austenite evolution during rolling, empirical equations describing changes of austenite (e.g. recrystallisation fraction, AGS) were selected and applied.
The mean austenite grain sizes at the entry of each stand were calculated by various equations and approaches by Sidenor and CSM. These calculated data were compared with the values measured on 38Cr4 steel specimens taken at various locations along the mill and rapidly quenched (< 3 s) to freeze the austenitic structure. The agreement was good, given the nature of the measurements, the accuracy of the process model used to calculate average temperature of the stock, and the assumptions made in the recrystallization model.

The model was applied to assess the effect of chemical composition and rolling temperatures on the austenite grain size (AGS) at finishing.
In the case of a finish rolling temperature of 1040 °C the calculated AGS of steel 38Cr4 is about 60 μm after 12 s from finishing, if water boxes are not activated. The behaviour of steel 10MnSi7 is predicted very similar, whilst the steel 30MnB4 is finer (27 μm) because its grain growth is inhibited by the presence of Ti (C, N) particles.
Introducing final water boxes that in 2 s are able to reduce the average temperature from 1040 to 870 °C allow to maintain AGS lower than 30 μm.

With a reduction of the average finish rolling temperature of 180 °C (from 1040 to 860 °C), introducing intermediate water boxes between intermediate and finishing mill in addition to those just after finishing, an average AGS of less than 15 μm is predicted by the model.

However, local microstructure becomes progressively more heterogeneous through the diameter of the wire-rod if adequate equalization is not allowed.

The evolution of AGS was also investigated by the pilot mill trials at TUBEF.
Generally, the AGS is refined reducing the finish rolling temperature from 980 °C to 850 °C, whilst the effect of lying head temperature from 850 °C to 750 °C is often poor.
Initial AGS (i.e. holding time) shows also some influence (Table 8); coarser initial AGS causes in general larger AGS at the laying head.
Final AGS is dependent on rolling temperatures, especially that at the last stand (T3). In the cases in which this temperature is around 850 ºC the AGS is refined and reaches values of ASTM No.11 (7 μm).

Also hot compression trials were performed in order to study the austenitic grain growth after recrystallisation. The results indicate that for steel 38Cr4, starting from a 40 μm recrystallized austenite grain size, after a holding of 12 s, the grains grow up to 150 μm at 1000 °C.
The grain growth is strongly reduced for the same holding period at 850 °C, being the average grain size below 70 μm.

Very effective cooling devices are required to reduce the stock temperature for controlling grain growth after hot rolling as the usual distances between finish roll and cooling conveyor are rather small and wire-rod speed is very high.
The effect of cooling devices placed between finishing block and laying head of Sidenor wire rod mill on AGS and on final microstructure were specifically studied for 38Cr4 steel grade (13 mm diameter wire rods) by industrial trials. Two different cooling patterns have been used. The first one corresponds to a typical cooling process for this steel grade (high water flow rate), the second one was performed with minimum water flow rate. In both cases a retarded cooling practice was used in the EDC conveyor.
The coarse AGS obtained for high laying head temperatures promotes the formation of mixed bainite-martensite microstructures due to the increased hardenability.

The phase transformation characteristics of the investigated steels for different AGS values (fine and coarse) were determined by dilatometry. Both isothermal and continuous cooling conditions were used.

Both experimental tests and mathematical simulations indicate that a reduction of austenite grain size is beneficial in reducing the formation of hard and brittle phases such as bainite and martensite on the final microstructure.
Austenite refinement accelerates the ferrite formation and promotes the completion of the phase transformation under the insulated zones (very slow cooling). In this respect, simulations by model clarified that the success in avoiding bainite and martensite formation in steels with very high hardenability such as boron steels for fasteners, is related to the following factors:

- low cooling rates during phase transformation, obtained using insulated covers and low conveyor speeds which give dense loops
- adequate laying head temperatures.

Low laying head temperatures, for a given conveyor speed, favour the completion of the phase transformation in the insulated zones (formation of ferrite and pearlite). If low laying head temperatures are difficult to be achieved, the AGS refinement helps to avoid bainite and

martensite formation, included the segregated regions which results more hardenable due to local enrichment in alloy elements and carbon.

Behind the laying head different cooling rates were obtaind during the pilot mill trials at Freiberg. A complete microstructural analysis, was carried out and relations among main mechanical parameters (UTS, ROA, Hardness), hot rolling and cooling parameters were given:

- Independently of the steel chemistry, mechanical properties such as UTS, ROA and HV_{30} appeared more homogeneous (especially in the case of 54SiCrV6 steel) in the II trial. In general, the following rule holds: a rigid control of all process parameters and a reduction of temperature values at finishing block (T2-T3) and laying head (T4) lead a greater microstructural homogeneity in wire-rods, with improvements in terms of reproducibility and narrower dispersion of mechanical properties.
- If steel is rolled at low temperatures and the process control is good, higher cooling rates can give benefits on the final product properties. This strategy not only reduces the formation of brittle and hard microstructures, but is able to reduce and/or increase UTS at the same ductility (ROA) level. The final choice of what condition is the most suitable for the steel producer should be tuned in relation to evaluations in terms of finishing routes and final applications.
- The presence of brittle structures (bainite, high carbon martensite) originated from cooling and not from segregation regions (i.e. dispersed islands instead of localized bands) seems not to affect ductility of as-rolled products assessed by tensile test.

The following considerations were drawn for cold formability by compression test foreach steel:

- ***Steel 54SiCrV6 (Lucchini and Sidenor)***

  Both these materials showed after compression test (as-rolled and machined conditions) cracks or fractures. An influence of the rolling technology can be hardly seen. Only for low and nearly constant T1-T4 rolling temperatures it seems possible to avoid failures during cold deformation. An influence of cooling rate after rolling on the crack behaviour was not found.

- ***Steel 30MnB4 (Lucchini)***

  In as-rolled conditions several specimens showed cracks after the compression tests. For slow cooling rate (cooling from finish temperature to 600 °C, holding 600 s) all specimens in the as-rolled condition exhibited defects. Some micro-cracks formed during rolling and led to cracks in the compression test. After machining (i.e. elimination of potential sites for crack initiation) no failure appeared.

- ***Steel 10MnSi7 (Lucchini)***

  Air cooled specimens were not affected by severe cracking. After machining, no failure appeared, except for a material was rolled at very low finishing temperature (690 °C). In this condition, hot rolling was carried out in the mixed ferrite-austenite range. Therefore ferrite was formed during rolling and work-hardened. This causes a reduction in ductility.

- ***Steel 38Cr4 (Sidenor)***

  This steel showed some cracks both in as-rolled and in machined condition; all defects appeared after air cooling treatment. Slow cooled wire rods (cooling from finish temperature to 600 °C and holding 600 s) were crack-free after the compression test. Hence, a slow cooling treatment (such as 600 °C, for 600 s), as performed in the Freiberg rolling trials, is the adequate cooling strategy.

Cold drawing trials of prototype wire rods gave good results, except steel 54SiCrV6 (Lucchini) which exhibited a rupture during the 2nd drawing step.

Metallographic examinations, hardness measurements and tensile tests were performed on selected specimens. The wire from Lucchini steel 54SiCrV6 exhibited more evident bands due to segregated regions in the industrial heat, compared with Sidenor steel 54SiCrV6.
Investigation by electron transmission microscopy (TEM) showed that this material exhibited a more marked precipitation of cementite at grain boundaries. These facts and the possible presence of a surface defect could have caused the anticipated failure.

The influence of thermo-mechanical processing on the response to spheroidising annealing was studied in 38Cr4 steel grade. Taking into account that a proper spheroidising structure must present a high degree of spheroidising (>90%) to improve cold formability, and a low tensile strength (< 550 MPa) to minimise wear of tooling during drawing, following considerations can be established from the response to annealing of studied samples:

- All the samples, except the industrial retarded cooling, present satisfactory response to annealing.

- A clear relationship between the degree of spheroidisation and interlamellar spacing was observed: the smaller the interlamellar spacing the better the response to annealing.

- The interlamellar spacing is a decisive factor concerning the spheroidisation.

- Higher cooling rates decrease interlamellar spacing (improving degree of spheroidisation), but increase the risk of bainite and martensite formation what, in general, gives higher tensile strengths in the as-treated conditions.

- The best balance of spheroidisation - tensile strength was given by a material which exhibits small interlamellar spacing and a high degree of spheroidisation, due to the air (no retarded) cooling procedure, and is free of bainite and martensite due to the thermo-mechanical treatment (at temperatures below 800ºC).

- It was proven that thermo-mechanical processing, at temperatures below 800ºC, allows to shorten subsequent spheroidising heat treatments. In the trials performed in this project a reduction of spheroidising time of 30% was achieved. This means saving up to 40 Euro/ton.

Guidelines and improvements related to the introduction of thermo-mechanical processing process of long products, in particular wire-rods were provided.

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KEYWORDS

Wire-rod, thermo-mechanical process, recrystallisation, grain growth, phase transformation, strength, ductility, microstructure, spheroidisation, cold formability, drawing.

ANNEX 1

Analysis of process data from wire rod rolling mills (WP1)

1. Description of the wire rod mill and collected data in SIDENOR

The industrial rolling mill that SIDENOR used in the research program is located in Vitoria.
The main features of this mill and the parameters which can be recorded, together with the sampling position along the mill are presented in Table 1.1 and Fig.1.1.
Rolling parameters as strain, strain rate and interpass time at each stand, have been calculated for the rolling process of a 13 mm diameter rod. They are shown in Table 1.2.
The time from the last stand to the laying head is 2.5 seconds.

	CHARACTERISTICS	PYROMETERS	SHEARS
REHEATING FURNACE	- Billet: 155x155x8400 mm^3 - Walking beam - 70Tn/h		
ROUGHING STANDS	- Three-high - Nº passes = 7 - Rolling speed = 3.4 m/s - Di: 85 mm	T9 (after pass 6) T10 (before Ind. Furn)	
INDUCTION FURNACE	- Nº coils = 6 - Inner diameter = 165 mm		
INTERMEDIATE MILL	- H-V-H arrangement - Nº stands = 6 - Rolling speed = 1.9 m/s (∅ 5.5mm) - Rolling speed = 2.8 m/s (∅ 18mm) - Di = 34 mm (∅ 5.5mm) - Di = 38 mm (∅ 18mm)	T11 (after Ind Furn)	
CONTINUOUS MILL	- H-V-H with loop control - Nº stands = 7 - Rolling speed = 13.2 m/s (∅ 5.5mm) - Rolling speed = 9.9 m/s (∅ 18mm) - Di = 13 mm (∅ 5.5mm) - Di = 21 mm (∅ 18mm)	 T13 (after pass 18)	X (after pass 13) X (after pass 18)
FINISHING MILL	- H-V arrangement - Nº stands = 2	 T14 (before Finish Bl)	 X (before Finish Bl)
FINISHING BLOCK	- Nº stands = 10 - Rolling speed = 60 m/s (∅ 5.5mm) - Rolling speed = 13.5 m/s (∅ 18mm)		
COOLING DEVICES	- Length = 27000 mm. - 3 zones: - Cooling zone: 16 nozzles - Equalisation - Cooling zone: 6 nozzles + 2 (air) - Water filled.	T15 (after Finis Bl)	
EDC CONVEYOR	- Length = 62000 mm - Nº sections = 6 - Covers over 2, 3 and 4th sections - Several controlled cooling · Cooling in boiling water · Direct quenching after rolling · Retarded cooling (covers) · Air cooling	T16 (after Laying Head)	

Table 1.1 Characteristics of the facilities of the wire rod mill in Vitoria. Position of the pyrometers and shears.

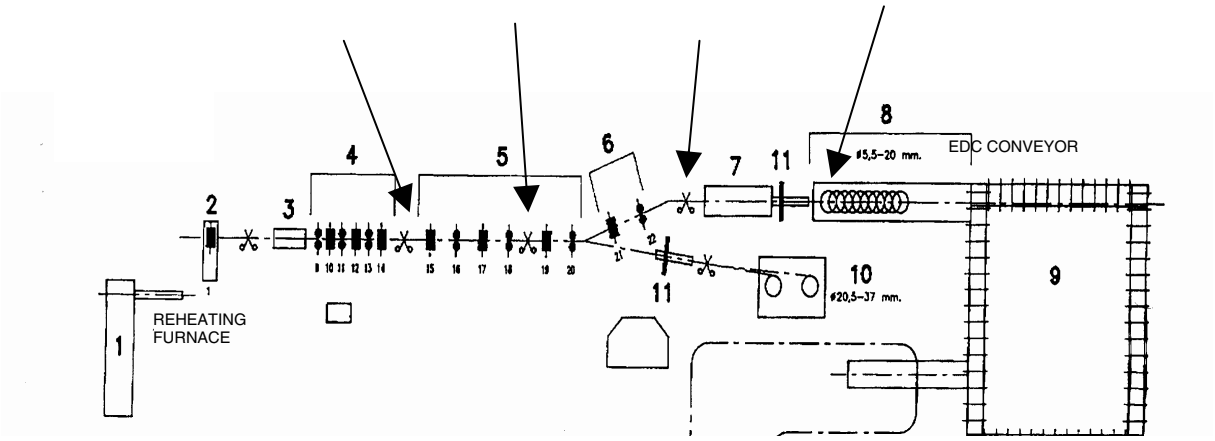


Fig.1.1 Vitoria wire-rod mill.

Pass	Interpass time [s]	Strain rate [s^{-1}]	Strain
1 D	6.50	7.55	0.164
2 D	6.50	8.26	0.191
3 D	6.50	9.42	0.217
4 D	6.50	10.33	0.233
5 D	6.50	7.07	0.105
6 D	6.50	12.48	0.294
7 D	14.24	10.15	0.205
8C	8.33	2.79	0.367
9C	6.25	3.16	0.280
10C	9.71	5.10	0.326
11C	5.68	5.79	0.258
12C	4.17	11.26	0.313
13C	5.75	10.73	0.213
14C	2.09	15.76	0.220
15C	1.69	14.73	0.157
16C	1.03	22.45	0.184
17C	1.13	28.30	0.191
18C	0.58	45.81	0.198
19C	1.33	57.47	0.260
20C	0.54	73.46	0.148
21C	1.60	133.33	0.332
22 B	0.06	134.06	0.118
23 B		193.55	0.206

Table 1.2 Rolling parameters calculated for the rolling process of a 13 mm diameter rod.

The temperatures recorded during rolling for the steel wire rods considered by SIDENOR for this project are shown in Table 1.3.

The controlled cooling was performed by an EDC conveyor that is divided in 6 sections. There are covers in the 2nd, 3rd and 4th section. The loop density in each section in the cooling process for the two steels considered is shown in table 1.4.

Examples of the thermal evolution of steel wire rods along the EDC conveyor are shown in Tables 1.5 and 1.6. They were obtained by infrared thermography.

		38Cr4 (∅ 13 mm)	54SiCrV6 (∅ 13 mm)
PYROMETERS	T9	1064 ºC	1048 ºC
	T10	1070 ºC	1062 ºC
	T14	1098 ºC	1126 ºC
	T15	1088 ºC	1106 ºC
	T16	879 ºC	890 ºC

Table 1.3 Examples of measured temperatures during rolling.

SECTION No.	LOOP DENSITY [loops/m]	COVERS
1	19.4	NO
2	19.4	YES
3	15.0	YES
4	13.9	YES
5	13.9	NO
6	13.9	NO

Table 1.4 Characteristics of the cooling process in the EDC conveyor for both steels.

SECTION Nº	TIME [s]	TEMPERATURE [ºC]	
		Centre of the loop	Edge of the loop
1	0	850	879
2	56	736	778
3	99	687	728
4	120	665	707
5	138	643	690
6	156	619	675

Table 1.5 Thermal evolution for 38 Cr4 (∅13). Covers were used.

SECTION Nº	TIME [s]	TEMPERATURE [ºC]	
		Centre of the loop	Edge of the loop
1	0	872	890
2	56	743	788
3	99	691	734
4	120	669	712
5	138	647	694
6	156	623	678

Table 1.6 Thermal evolution for 54SiCrV6 (∅ 13 mm). Covers were used.

In order to determine the microstructural and mechanical characteristics after cooling of these products, metallographic observations have been made, and hardness has been measured. In table 1.7 the results obtained in these studies are summarised.

STEEL	38Cr4		54SiCrV6	
	Centre of the loop	Edge of the loop	Centre of the loop	Edge of the loop
Microstructure	10%F – 40%P – 50%B	20%F – 80%P	20%F – 80%P	15%F – 85%P
Hardness	310 HB	238 HB	33.5 HRc	32 HRc

Table 1.7 Microstructure and hardness of wire rods produced by SIDENOR (∅ 13 mm).

Due to the variability of some process conditions: segregation, differences in cooling between centre and edge of the loop, there is a scatter in the final product characteristics that sometimes produces, as in the case of 38Cr4 steel, undesired acicular structures in the as-rolled product.
In order to avoid these undesirable structures, the effect of cooling parameters, mainly loop speed at the conveyor, was investigated.

Wire rod thermal evolution in the EDC conveyor for 38Cr4 steel grade (∅13 mm) and for 54SiCrV6 (∅13 mm) were also obtained for other cooling practices (see the scheme shown in table 1.8). Again the temperatures at the centre and edges of the loop were measured by infrared thermography at different points along the conveyor (Fig.1.2).

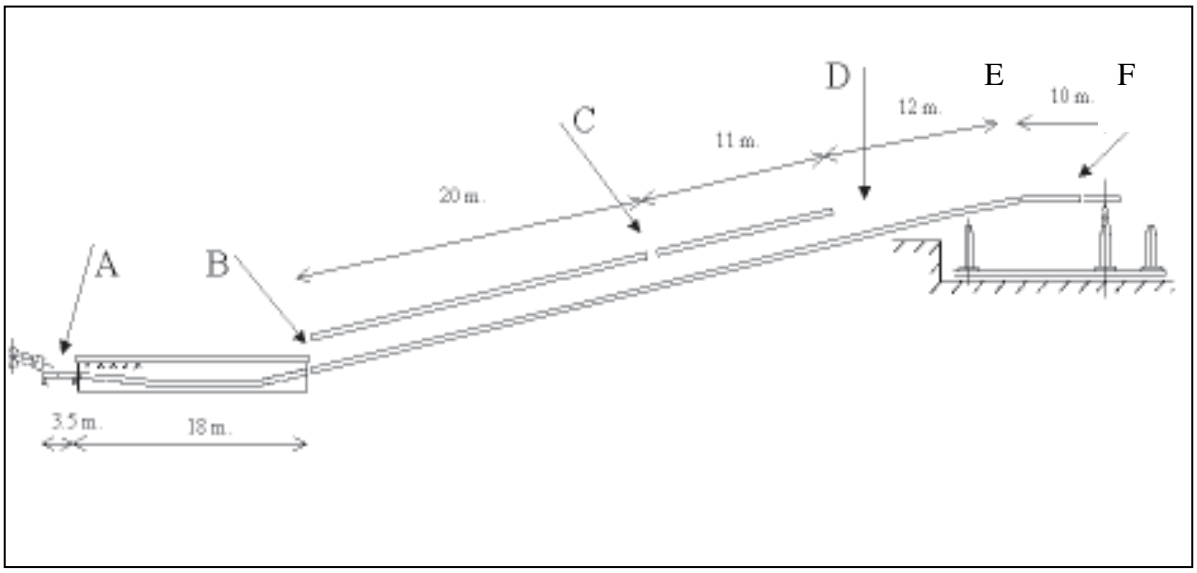


Fig.1.2 EDC conveyor of SIDENOR wire rod mill. Measurement points are marked.

Meas. No.	**Cooling Condition in EDC**		**Code**
	Conveyor Speed	Covers	
I	High	Yes	HY
II	Low	Yes	LY
III	High	No	HN

Table 1.8 Cooling conditions during the temperature measurements in the EDC of SIDENOR.

All the results are presented in Table 1.9 (38Cr4 steel) and Table 1.10 (54SiCrV6 steel).

	MEAS. No I			MEAS. No II			MEAS. No III		
	t [s]	Tc [ºC]	Te [ºC]	t [s]	Tc [ºC]	Te [ºC]	t [s]	Tc [ºC]	Te [ºC]
A	0	850	879	0	861	880	0	840	866
B	56	736	778	79	745	761	56	716	748
C	99	687	728	138	694	711	99	660	698
D	120	665	707	166	666	689	120	634	679
E	138	643	690	184	646	673	138	609	665
F	156	619	675	203	623	659	156	585	651

Table 1.9 Thermal evolution for 38Cr4 (∅13). Measurements No. I, II and III.

	MEAS. No I			MEAS. No II			MEAS. No III		
	t [s]	Tc [ºC]	Te [ºC]	t [s]	Tc [ºC]	Te [ºC]	t [s]	Tc [ºC]	Te [ºC]
A	872	890	872	0	866	892	0	859	875
B	743	788	743	79	769	790	56	748	768
C	691	734	691	138	719	748	99	681	722
D	669	712	669	166	685	724	120	648	699
E	647	694	647	184	656	704	138	621	679
F	623	678	623	203	630	686	156	595	659

Table 1.10 Thermal evolution for 54SiCrV6 (∅13). Measurements No II and III.

The influence of cooling conditions of table 1.8 on temperature evolution at the centre and edge of the loop in the EDC can be observed more clearly for 38Cr4 steel in the thermal profiles of Fig.1.3.

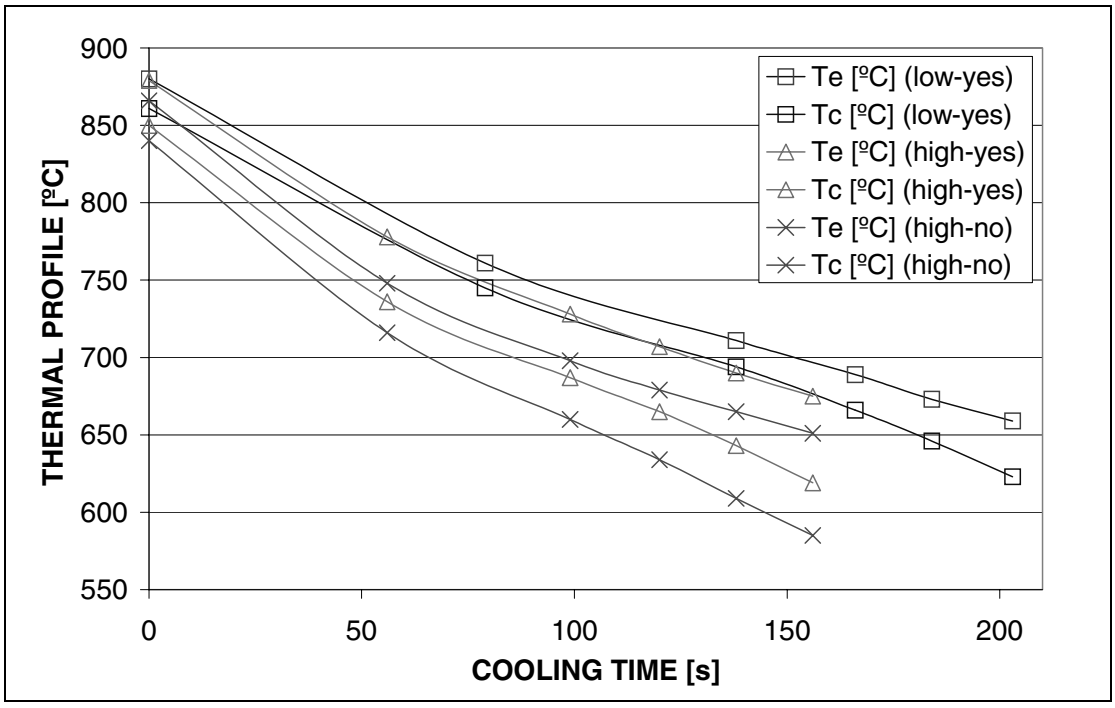


Fig.1.3 Temperature evolution in the EDC conveyor for 38Cr4 (∅13) under cooling conditions of table 1.9.

2. Description of the wire rod mills and collected data in Lucchini

In Table 1.11 a short description of the wire rod rolling mill in Piombino Steelworks is shown. All steel grades collected and supplied by Lucchini for the project have been produced in this plant. Nevertheless, to fulfil specific requirements and targets of the project (e.g. assessment of wire-rod cooling conditions), also the Caleotto wire rod rolling mill in Lecco was involved.

Piombino wire-rod mill

To collect process parameters and to sample material by the shears, the line No. 1 of the Piombino wire rod mill was selected and instrumented. The choice was induced by better accessibility of the operators for sampling activities. Process parameters refer to stock surface temperature evolution during rolling and, where necessary for a better understanding of initial rolling conditions, to furnace zone temperatures. Procedures were adjusted as follows:

- Thermal evolution

6 different positions were instrumented with pyrometers (called PIRO 1, 2, 3, 4, 5 and 6) to measure the following temperatures:

- PIRO 1: surface temperature of stock after reheating
- PIRO 2: temperature at shear D after stand No. 5 (after roughing block)
- PIRO 3: temperature at shear G after stand No. 11 (after first intermediate block)
- PIRO 4: temperature at shear H after stand No. 15 (before finishing block)
- PIRO 5: temperature at laying head (starting cooling temperature at Stelmor conveyor)
- PIRO 6: temperature at reforming pit.

The first site gives the surface temperature of the billet, as averaged both on different positions along each billet and on the overall number of billets for each steel grade rolled during the trials.

The second, third and fourth sites are controlled with manual pyrometers. Data collection is averaged for each billet on a wide number of points (frequency collection ~ 10 s^{-1}). The last two sites are equipped with fixed pyrometers with acquisition frequency of about 5 s^{-1}.

Zone	Reheating Furnace	Roughing Mill	Shear Stand N. 5	Intermediate Mill - I	Shear Stand N. 11
Description	- CC Billet : 140x140x16000 mm^3 - Walking beam - Productivity = 150 ton/h	- N. of stands = 6; - Stand N. 5: rolling speed min/max = 0.77/0.86 m/s - Exit diam.= 62.7 mm		- N. of stands = 6; - Stand N. 11: rolling speed min/max = 4.22/4.53 m/s - Exit diam. = 25.9 mm	
Process parameters and specimen sampling	T furnace zone N. 1, 2, 3, 4; T exit; Residence time $\%O_2$; $\%CO_2$	Mobile Pyrometer	X (after pass 5)	Mobile Pyrometer	X (after pass 11)

Zone	Intermediate Mill - II	Shear Stand 15	Finishing Stands	Water Box	Stelmor Conveyor
Description	- N. of lines = 2 - N. of stands = 4; - Stand N. 15: Speed Min/Max = 6.05/11.15 m/s - Exit diam. = 16.6 mm		- N. of lines = 2 - N. of stands=10; - Stand N. 25: rolling speed min/max = 74.2/100 m/s - Exit diam. = (*)		- N. of conveyors = 2 - Length=104000 mm; - N. of sections = 19; - N. of fans = 12; - N. of air locks = 12; - N. of copes = 40; Speed Min/Max = 0.05/1.3 m/s
Process parameters and specimen sampling	Mobile Pyrometer	X (after pass 15)	Mobile Pyrometers (T in finishing block)	Mobile Pyrometers (T exit finishing) Shape Meter (Orbis) Water Pressure Sensors	2 Fixed Pyrometers (T laying head) 2 Fixed Pyrometers (T reforming pit) 2 Mobile Pyrometers

dimensional range = 5.5 to 19 mm; progression = 0.5 mm; productivity = 600000 ton/y

Table 1.11 Characteristics of the facilities of the wire-rod mill in Piombino. Position of the pyrometers and shears.

- Sampling procedure

All shears (at stands No. 5, 11 and 15) can be involved in the sampling procedure in principle. Anyway, it has to be noticed that typical diameters at shear D (after pass 5) are quite large for an efficient direct quenching of material useful for further investigations in terms of austenite grain size (AGS) and microstructure evaluation. For this reason only samples at shear H (mainly) and G (where necessary) were collected. Sampling at shear H and G were preventively prepared in collaboration with plant personnel. At shear G, a large chest was placed immediately below the plant level and filled up with running water. At the rotative shear H, cut specimens were taken by a conveyor belt and immediately dipped into stirred water. All specimens sampled for the hot torsion tests to be carried out by TUBEF were collected at shear H and cooled in still air.

A schematic layout of the plant is shown in Fig.1.4, where the position of the different pyrometers and shears are indicated. From this picture it appears that manual pyrometers were located in coincidence of the shears to associate at any sampled material the temperature of the stock at that point.

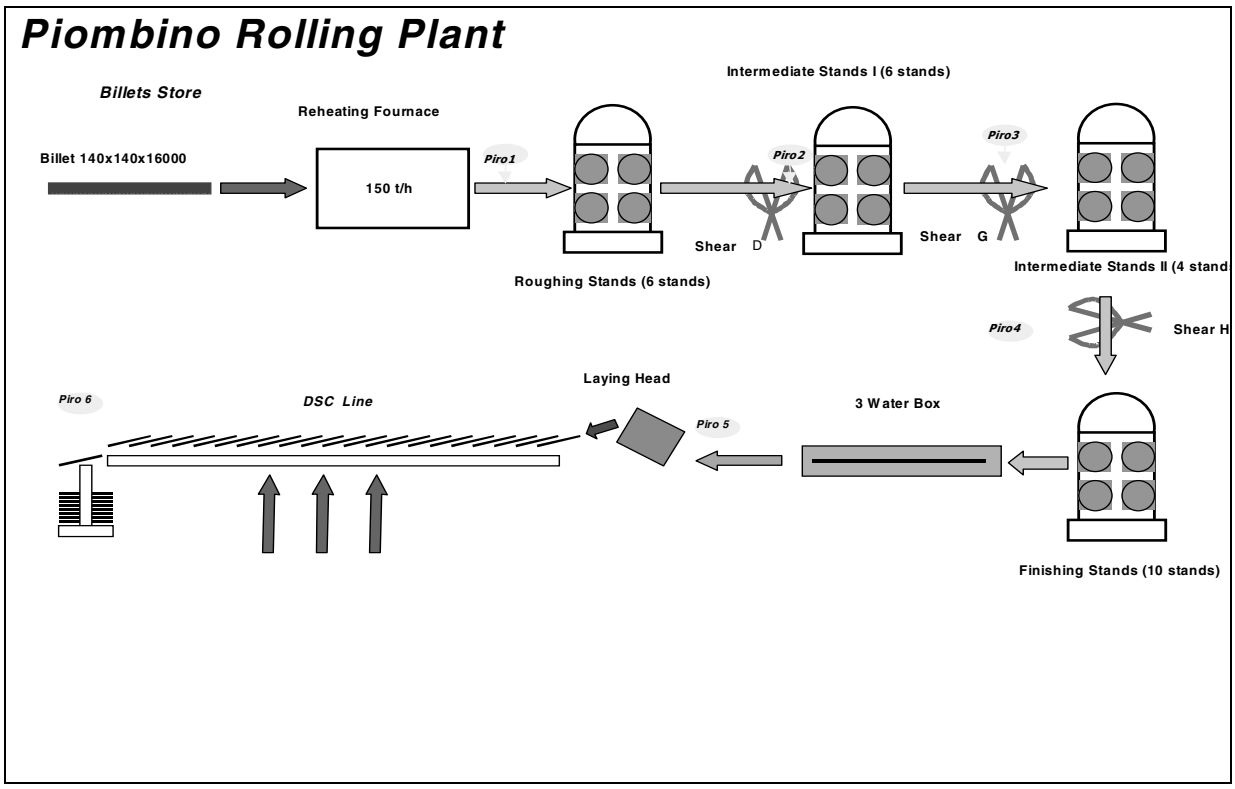


Fig.1.4 Piombino rolling mill.

All steel grades produced by Lucchini were rolled to 13 mm diameter and brightened in 6 m long straight bars of 12 mm diameter at VERTEK drawing facilities in Condove. Part of this material was delivered to TUBEF for the scheduled pilot hot rolling trials (WP3).

Some rolling parameters for the investigated steel grades, were collected and averaged on different campaigns (Table 1.12).

Steel	**Steel grade**	**Diameter** mm	**PIRO 1** °C	**PIRO 2** °C	**PIRO 3** °C	**PIRO 4** °C	**PIRO 5** °C
30MnB4	RM30B	13	1098	920	1010	1030	844
10MnSi7	ES16	13	1091	920	999	1029	845
10MnSi7	ES16	5.5	1097	900	970	990	811
54SiCrV6	54SiCrV6	13	1090	913	1022	1019	851

Table 1.12 Typical stock temperature evolution during hot rolling (Piombino steelworks).

For 10MnSi7 steel (high Si – high Mn steel for CO_2 gas shielding arc welding), wire rods with two diameters were processed. The reason was that 5.5 mm is the standard diameter for this product, while ∅ 13 mm was the size necessary to supply TUBEF with ∅ 12 mm straight bars. ∅ 13 mm is usual for steels 30MnB4 and 54SiCrV6.

The standard cooling procedures on the Stelmor conveyor are detailed in Table 1.13:

Steel grade	**Insulation procedure**	**Process Procedure**	PIRO 6 °C
RM30B	Covers	Low speed	<600 (@)
ES16	Covers	Low speed	<500 (@)
54SiCrV6 *	Covers	Low speed	<600 (@)

* cooling procedures for this steel grade are influenced (more than for the other cases) by customer specifications;
(@) averaged all over the loop

Table. 1.13 Typical process parameters of Stelmor conveyor.

At present all steel grades under investigation usually undergo thermal treatments (softening annealing) before being transformed in bright steels (cold drawn, peeled, shaved, etc.) due to the following reasons:

- undesired structures (martensite, bainite), when present, can reduce wire rods ductility in drawing or cold heading operations as well as increase as rolled ultimate tensile strength (and hardness) to values not compatible with a direct use of the material;

- slow cooling (necessary to reduce/eliminate undesired structures) can be difficult to be performed both for plant limits (Stelmor conveyor length, speed) and for production rate requirements (also when a slow cooling is technically possible, the drop in production rate necessary to obtain the correct transformation time could be extremely penalizing)
- different wire rod sections (ϕ) and loop densities (as a consequence of Stelmor conveyor speed) can lead to a wire rod temperature which is not sufficiently low or not well homogenized all over the loop at the reforming pit. This can also induce differences in the as-rolled microstructure.

Due to various combinations of all these conditions, undesired structures such as martensite islands or bainite can appear in a few percentage amount, sufficient to induce significant detrimental effects on the mechanical behaviour.

Desired as-rolled microstructure and mechanical characteristics are detailed in Table 1.14.

Steel grade	**Microstructure**	**UTS (max)**
RM30B	Ferrite + Pearlite	640 MPa
ES16	Ferrite + Pearlite	540 MPa
54SiCrV6	very fine Pearlite + Ferrite	1050 MPa *

* requirements on this steel grade, in terms of UTS, are mostly influenced by Customer specifications

Table. 1.14 Typical microstructures and strength values of steel wire-rod grades.

Caleotto wire-rod mill

This Lucchini mill is fully devoted to produce wire rods in the range 5.5÷19 mm for drawing and cold heading applications, mainly.

Typical steel grades are low-C steels (i.e. steel 10MnSi7), high-C steels (from C40 to C82), B-steels (from 0.1%C to 0.2%C Mn-B steels) for fasteners and cold heading applications.

Caleotto wire-rod mill has been selected for the unique feature of a Stelmor conveyor fully equipped with a pyrometer array and a thermographic scanner positioned at the laying head exit (Fig.1.5). The former measures the temperature evolution of steel rods from laying head down to final coil maker. In addition, standard process control gives the speed and the cooling time in the various conveyor sections, as well as the percentage of fan utilisation and cover position. Therefore, the temperature evolution along the line can be obtained (Fig.1.6).

An example of data collection, with length and time recorded from Caleotto wire-rod mill is shown in Table 1.15.

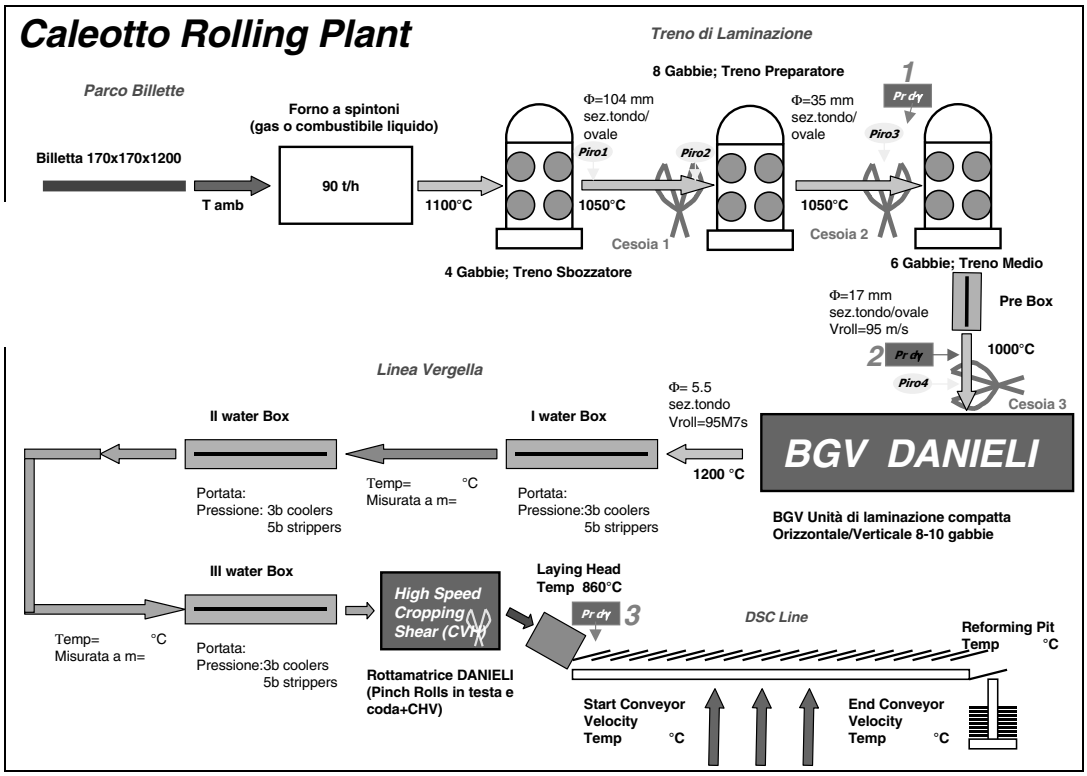


Fig.1.5 Layout of Caleotto plant with pyrometer positions.

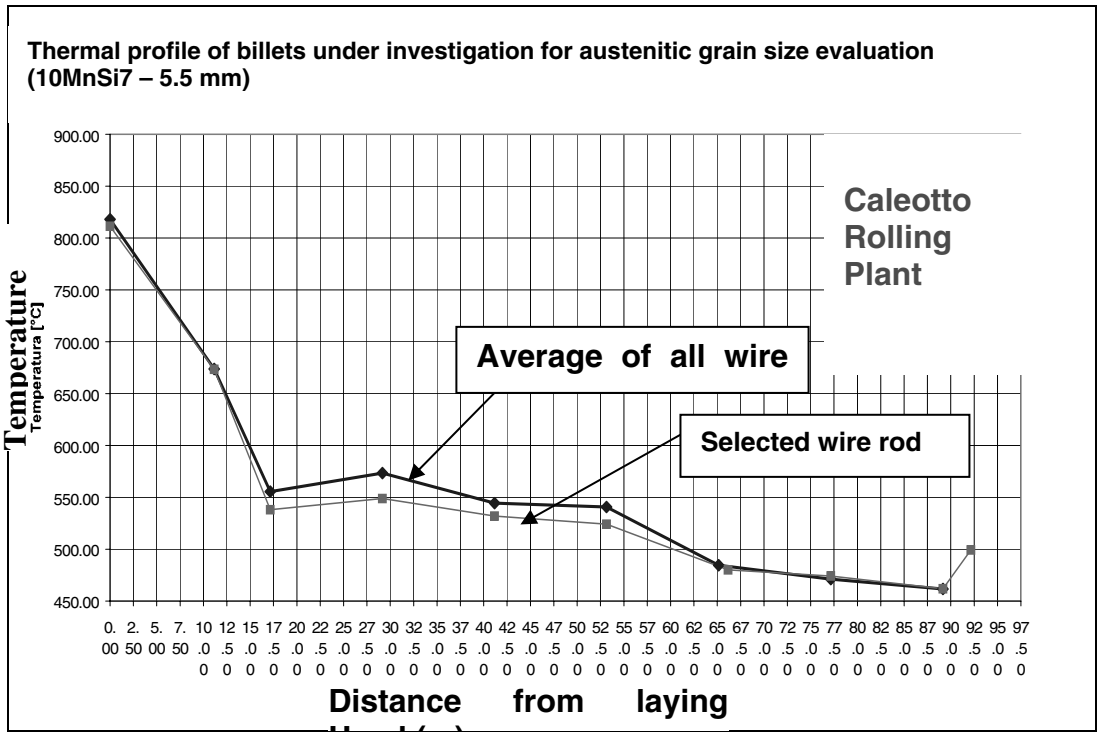


Fig.1.6 Caleotto Plant: Temperature evolution for 10MnSi7 Steel (ϕ 5.5 mm).

10MnSi7: Sampling from EDGE position.			
Temperature [°C]	**Length [m]**	**Time [s]**	**Position**
818.04	0.00	0.00	Scanner
673.83	11.15	50.91	P1
555.58	17.15	108.60	P2
573.46	29.15	221.81	P5
544.40	41.15	332.92	P7
540.63	53.15	442.01	P9
484.58	65.15	549.16	P2x
471.10	77.15	654.42	P4x
461.63	89.15	757.87	P6x
499.21	92.15	783.73	End

Table 1.15 Scheme of data collection.

Pyrometers are positioned in order to obtain thermal information from coil edges on one side of the cooling bed. Some of them, i.e. P1 to P9 are in a fixed configuration, while devices P2x, P4x and P6x are mobile pyrometers. All devices are supplied with water and forced air equipments to ensure correct cooling under operation and optical components (i.e. lens) cleanliness. "*End"* pyrometer is located at the reforming pit level to have thermal information at coil forming step.

Moreover, all pyrometers are electronically connected to a HW/SW process control system. This system systematically loads information about wire-rod thermal evolution during the cooling process, giving back the necessary support in terms of device synchronisation, rough data storage and first graphical analysis (i.e. temperature evolution) along the conveyor as a function of time or length.
This system is used both to verify the correct stock thermal evolution with respect to given cooling strategies and define new cooling patterns by matching thermal information and material characterisation.

ANNEX 2

Mathematical models to predict austenite evolution during hot rolling (WP2)

In order to study the austenite evolution during rolling, Sidenor has selected from the literature equations describing changes of austenite (e.g. recrystallisation fraction, AGS, accumulated strain). From the available mathematical models, one exhibiting good agreement with previous experimental characterisation of the austenite microstructure of the wire rod in the Vitoria's mill is that by Maccagno, Jonas and Hodgson [A2.1]. The main formulae of this model are summarised below.

Recrystallised grain size

Dynamic recrystallisation initiates at a critical strain, ε_c, that is calculated from the grain size at the start of the pass, d_0, the strain rate and the absolute temperature T (K):

$$\varepsilon_c = 5.6.10^{-4} d_0^{0.3} Z^{0.17} \qquad \text{/eq.1/}$$

where Z= ε'·exp (300000/8.31·T) is the Zener-Hollomon parameter

The grain size (μm) produced by complete recrystallisation, if the softening after hot working is by static recrystallisation ($\varepsilon < \varepsilon_c$) is described by:

$$d_{SRX} = 343\, \varepsilon^{-0.5}\, d_o^{\,0.4} \exp\left(\frac{-45000}{8.31\,T}\right) \qquad \text{/eq.2/}$$

Similarly, if it is described by metadynamic recrystallisation ($\varepsilon > \varepsilon_c$):

$$d_{MRX} = 2.6\ 10^4\ Z^{-0.23} \qquad \text{/eq.3/}$$

Austenitic grain growth

When recrystallisation is complete, further growth may take place even in the relative short times available between passes (t_{ip}) in hot rolling. The grain coarsening is well described with second order equations for times shorter than 1 s, and with seventh order equations starting from that time:

For SRX:

$$t < 1\,s \quad d^2 = d_{SRX}^2 + 4.0 \cdot 10^7\ (t_{ip} - 4.32\, t_{0.5}) \exp\left(\frac{-113000}{8.31\,T}\right) \qquad \text{/eq.4/}$$

$$t > 1\,s \quad d^7 = d_{SRX}^7 + 1.5 \cdot 10^{27}\ (t_{ip} - 4.32\, t_{0.5}) \exp\left(\frac{-400000}{8.31\,T}\right) \qquad \text{/eq.5/}$$

For MRX:

$$t<1\,s \quad d^2 = d_{MRX}^2 + 1.2\cdot 10^7\ (t_{ip} - 2.65\,t_{0.5}) \exp\left(\frac{-113000}{8.31\,T}\right) \qquad \text{/eq.6/}$$

$$t>1\,s \quad d^7 = d_{MRX}^7 + 8.2\cdot 10^{25}\ (t_{ip} - 2.65\,t_{0.5}) \exp\left(\frac{-400000}{8.31\,T}\right) \qquad \text{/eq.7/}$$

Recrystallisation kinetics

The 50% recrystallisation time, $t_{0.5}$, can be expressed as follows:

SRX :

$$t_{0,5} = 2.3\cdot 10^{-15}\ \varepsilon^{-2.5} \exp\left(\frac{230.000}{8.31\,T}\right) \qquad \text{/eq.8/}$$

MRX :

$$t_{0,5} = 1.1\,Z^{-0.8} \exp\left(\frac{230.000}{8.31\,T}\right) \qquad \text{/eq.9/}$$

These formulae describe the grain size evolution implying that the microstructural events occur under isothermal conditions. This assumption is inappropriate for subsequent water sprays cooling process into the water boxes placed between the finishing block and the laying head prior to the EDC conveyor or Stelmor conveyor.

To enable these isothermally determined equations to describe the behaviour under non-isothermal conditions, the use of 'temperature compensated time' (w) [A2.2] is needed. This is defined as:

$$w = \sum_i \delta t_i \exp\left(\frac{-Q}{RT_i}\right) \qquad \text{/eq.10/}$$

where δt_i is the finite difference time step at which the temperature was T_i.

Following equation will enable to assess the austenitic grain coarsening under non-isothermal conditions (inside water boxes):

$$d^m = d_{RX}^m + A\sum_i \delta ti \exp\left(\frac{-Q}{RT_i}\right) \qquad \text{/eq.11/}$$

These constitutive relations have been inserted in the CSM algorithms used to describe the changes of austenite (e.g. recrystallisation fraction, AGS, accumulated strain) during hot rolling.

The CSM models calculate the austenitic structure developed during each pass as a function of the local thermal and mechanical conditions (T, ε, ε'). The set of equations forecasting the austenite structure is applied sequentially according to procedures already presented [A2.3].
In the program, structures which differ significantly one from the other are treated as separate features.

For each homogeneous structure, a check is made to control whether the total strain, inclusive of the residual strain accumulated during previous passes (ε_r), is smaller (static recrystallisation) or greater than the critical strain required for the onset of dynamic recrystallisation (ε_{dyn}).

In order to have a flexible tool, in addition to the constitutive equations proposed by Sidenor, also those determined for steels similar to those under investigation have been inserted by CSM [A2.3].

Under isothermal conditions, the volume percent of material undergoing static recrystallisation (R%) within a certain time t from hot deformation is calculated with an Avrami-type equation:

$$R\% = 100 \{1 - \exp [0.693 (t/t_{50,\varepsilon'})^n]\} \qquad /eq.12/$$

The values of the exponent "n" and of the 50% recrystallisation time "$t_{50,\varepsilon'}$" have been expressed in terms of operating parameters T, ε, ε', and d_γ with the following equations:

$$t_{50} = d\, \varepsilon^e\, d_\gamma^f \cdot \exp (Q_1/T) \qquad /eq.13/$$

$$t_{50,\varepsilon'} = t_{50} (\varepsilon'_0 / \varepsilon')^r \qquad /eq.14/$$

$$n = g\, \varepsilon^h\, d_\gamma^m \cdot \exp (Q_2/T) \qquad /eq.15/$$

where the coefficients depend on steel composition and ε'_0 is the strain rate used in the hot deformation tests used to measure the recrystallisation kinetics. [A2.3]

In the case that $(\varepsilon + \varepsilon_r) > \varepsilon_{dyn}$, the (meta)-dynamic recrystallisation occurs and its kinetics is described using laws similar to those of static recrystallisation, but with different constants.

In the case of partial recrystallisation, the grains are smaller than those of a completely recrystallised material and are calculated by the following relationship:

$$d_\gamma'_{rex} = d_{\gamma\, rex} (R\%/ 100)^{1/3} \qquad /eq.16/$$

The fraction of material that does not recrystallise before the subsequent pass exhibits elongated grains whose effective size can be calculated by the relationship:

$$d_{\gamma\, ef} = 1.06\, d_\gamma (1 - R\%/100) \exp (-\varepsilon) \qquad /eq.17/$$

Once the recrystallisation is completed, further grain growth takes place as a function of time and temperature:

$d_{\gamma,t}{}^{y} = d_{\gamma rex}{}^{y} + G\ t \exp(-Q_{gg} / RT)$ /eq.18/

In the following Tables, the empirical coefficients for various steels are summarised.

Constitutive Equations For Recrystallisation

Steel	*Time for 50% Static Recrystallisation* t_{50} *(s)*	*Reference*
C-Mn	$t_{50} = \left(2.5 \cdot 10^{-19}\right) \cdot d_{\gamma_0}{}^{2} \cdot \varepsilon^{-4} \cdot e^{\left(\frac{300000}{RT}\right)}$	*Sellars* [A2.4]
	$t_{50} = \left(5.1 \cdot 10^{-21}\right) \cdot d_{\gamma_0}{}^{2} \cdot \varepsilon^{-4} \cdot e^{\left(\frac{330000}{RT}\right)}$	*Roberts et al.* [A2.5]
	$t_{50} = \left(8.31 \cdot 10^{-15}\right) \cdot \varepsilon^{-1.5} \cdot d_{\gamma_0}{}^{1.5} \cdot \dot{\varepsilon}^{-0.33} \cdot e^{\left(\frac{263000}{RT}\right)}$	*Muojekwu* [A2.6]
	$t_{50} = (1.4\ 10^{-14})\ d\gamma_0{}^{1.1}\ \varepsilon^{-1.9}\ e^{(274200/RT)}$	*Anelli* [A2.3]
	$t_{50} = \left(2.3 \cdot 10^{-15}\right) \cdot \varepsilon^{-2.5} \cdot d_{\gamma_0}{}^{-2} \cdot e^{\left(\frac{230000}{RT}\right)}$	*Hodgson et al.* [A2.7]
	$t_{50} = \left(3.6 \cdot 10^{-12}\right) \cdot \varepsilon^{-2.5} \cdot d_{\gamma_0}{}^{-1} \cdot e^{\left(\frac{230000}{RT}\right)}$	*Hodgson et al.* [A2.8]
Ti	$t_{50} = \left(2.52 \cdot 10^{-19}\right) \cdot d_{\gamma_0}{}^{2} \cdot \varepsilon^{-4} \cdot e^{\left(\frac{325000}{RT}\right)}$	*Sellars* [A2.4]
	$t_{50} = (3.2\ 10^{-9})\ d_{\gamma 0}{}^{0.9}\ \varepsilon^{-1.6}\ e^{(152900/RT)}$	*Anelli* [A2.3]

Steel	*Static Recrystallised Grain Size* d_{SRX} *(μm)*	*Reference*
C-Mn	$d_{SRX} = 0.5 \cdot d_{\gamma_0}^{0.67} \varepsilon^{-1}$ for $\varepsilon \le \varepsilon^*$ With: $d_{SRX} = 1.8 \cdot 10^3 \cdot Z^{-0.15}$ for $\varepsilon > \varepsilon^*$ $\varepsilon^* = 2.8 \cdot 10^{-4} \cdot d_{\gamma_0}^{0.67} \cdot Z^{-0.15}$ $Z = \dot{\varepsilon} \cdot e^{\left(\frac{312000}{RT}\right)}$	*Sellars* [A2.4]
	$d_{SRX} = 45 \cdot d_{\gamma_0}^{0.374} \cdot \varepsilon^{-0.6} \cdot \dot{\varepsilon}^{-0.1} \cdot e^{\left(\frac{-25000}{RT}\right)}$	*Choquet et al.* [A2.5]
	$d_{SRX} = 1150\, d_{\gamma 0}^{0.39}\, \varepsilon^{-0.3}\, \varepsilon'^{-0.1}\, e^{(-54000/RT)}$	*Anelli* [A2.3]
	$d_{SRX} = 343 \cdot d_{\gamma_0}^{0.4} \cdot \varepsilon^{-0.5} \cdot e^{\left(\frac{-45000}{RT}\right)}$	*Hodgson et al.* [A2.7]
	$d_{\gamma} = (88.96) \cdot \varepsilon^{-0.368} \cdot d_{\gamma_0}^{0.369} \cdot e^{\left(\frac{-28060}{RT}\right)}$	*Muojekwu et al.* [A2.6]
	$d_{SRX} = 4.3 + 195.7 \cdot d_{\gamma_0}^{0.15} \cdot \varepsilon^{-0.57} \cdot e^{\left(\frac{-38500}{RT}\right)}$	*Siwecki* [A2.9]
Ti	$d_{SRX} = 84\, d_{\gamma 0}^{0.33}\, \varepsilon^{-0.6}\, \varepsilon'^{-0.1}\, e^{(-29000/RT)}$	*Anelli* [A2.3]

Steel	*(Meta)-dynamic Recrystallisation* t_{50} *(μm)*	*Reference*
C-Mn	$t_{50} = 1.06 \cdot 10^{-5} \cdot Z^{0.6} \cdot e^{\left(\frac{300000}{RT}\right)}$ $Z = \dot{\varepsilon} \cdot e^{\left(\frac{312000}{RT_{def}}\right)}$	*Sellars et al.* [A2.4]
	$t_{50} = \left(2.13 \cdot 10^{-6}\right) \cdot \dot{\varepsilon}^{-0.68} \cdot e^{\left(\frac{132800}{RT}\right)}$	*Muojekwu et al.* [A2.6]
	$t_{50} = \left(1.14 \cdot 10^{-13}\right) \cdot \varepsilon^{-3.8} \cdot \dot{\varepsilon}^{-0.41} \cdot e^{\left(\frac{252000}{RT}\right)}$	*Laasraoui et al.* [A2.10]
	$t_{50} = \left(1.27 \cdot 10^{-18}\right) \cdot \varepsilon^{-3.81} \cdot \dot{\varepsilon}^{-0.36} \cdot e^{\left(\frac{404000}{RT}\right)}$	*Laasraoui et al.* [A2.10]

Steel	*(Meta)-dynamic Recrystallisation* ($\varepsilon \geq 0.85 \cdot \varepsilon_p$) ε_p, d_{dyn} *(μm)*	*Reference*
C-Mn	$\varepsilon_p = \left(1.318 \cdot 10^{-2}\right) \cdot d_{\gamma_0}^{\ 0.174} \cdot \varepsilon^{-0.165} \cdot e^{\left(\frac{24327}{RT}\right)}$	*Muojekwu et al.* [A2.6]
	$\varepsilon_p = (4.83\ 10^{-3})\ d_{\gamma 0}^{\ 0.09}\ Z^{-0.15}$	*Anelli* [A2.3]
	$d_{dyn} = \left(88.96\right) \cdot \varepsilon^{-0.368} \cdot d_{\gamma_0}^{\ 0.369} \cdot e^{\left(\frac{-28060}{RT}\right)}$	*Muojekwu et al.* [A2.6]
	$d_{dyn} = d_{0dyn} + 1.1 \cdot \left(5380 \cdot e^{\left(\frac{-6840}{T}\right)} - d_{0dyn} \right) \cdot \left(1 - e^{\left(-295 \cdot \dot{\varepsilon}^{0.1}\right)} \cdot e^{\left(\frac{-8000}{T}\right)} \cdot t \right)$ $d_{0dyn} = 22600 \cdot \dot{\varepsilon}^{-0.27} \cdot e^{\left(\frac{-8670}{T}\right)}$	*Yada, Senuma* [A2.11]
Ti	$\varepsilon_p = (2.73\ 10^{-3})\ d_{\gamma 0}^{\ 0.20}\ Z^{-0.15}$	*Anelli* [A2.3]

Steel	*Recrystallised Grain Growth* d_g (μm)	*Reference*
C-Mn	$d_{gg}^{10} = d_{rex}^{10} + 3.87 \cdot 10^{32} \cdot t \cdot e^{\left(\frac{-400000}{RT}\right)}$ T ≥ 1000°C $d_{gg}^{10} = d_{rex}^{10} + 5.02 \cdot 10^{53} \cdot t \cdot e^{\left(\frac{-914000}{RT}\right)}$ T<1000°C	*Sellars* [A2.12]
	$d_{gg} = \sqrt{d_{rex}^{2} + 1.44 \cdot 10^{12} \cdot t \cdot e^{\left(\frac{-266700}{RT}\right)}}$	*Yada et al.* [A2.13]
	$d_{gg} = \sqrt{d_{rex}^{2} + 1.1 \cdot 10^{7} \cdot t \cdot e^{\left(\frac{-113000}{RT}\right)}}$ t ≤ 1 s $d_{gg} = \sqrt[7]{d_{rex}^{7} + 1.5 \cdot 10^{27} \cdot t \cdot e^{\left(\frac{-400000}{RT}\right)}}$ t>1 s	*Hodgson et al.* [A2.14]

Ti	$d_{gg} = 95 \{1 - \exp [- 7.4\ 10^{-3} (T - 1173)]\}$	*Anelli* [A2.3]

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ANNEX 3.1

Hot torsion tests (WP3)

1. Theory

Yield stress is defined as the stress level, which initiates and maintains yielding in a material in its uni-axial-stress condition. It is governed by numerous factors of influence.
The exact knowledge of the yield stress as a function of the deformation conditions of a material is the main prerequisite for calculating the forces to be applied and the work to be done during each step of deformation and for defining the pertinent mechanical and control engineering measures that are apt to ensure satisfactory process efficiency and a high level of product quality. From the profile of the stress-strain curve, conclusions can be drawn about structure formation, softening and hardening processes and consequently product properties.

Awareness of the plasticity limit of a material, which is the deformation to rupture, enables us to determine the optimum ranges of temperature and strain rate necessary in a shaping process for full utilisation of the plasticity capabilities of the material and hence to resort to appropriate cold shaping to render heat treatment operations unnecessary. If shaping takes place under conditions where plasticity reaches its maximum values, the process output will be high and the surface finish and the accuracy to size of the product will be excellent.

The choice of the test method for determining the yield stress is of special importance.
Because it is difficult to investigate the influence of strain rate and temperature by means of tensile tests, the torsion test was preferred. The torsion test offers several advantages over other test methods, such as the upsetting test and the tensile test, because it allows the simultaneous determination of both yield stress and plasticity. It is frictionless and permits the simulation of thermo-mechanical conditions of practical interest. Main drawbacks are the uneven distribution of stresses and that shear deformations must be converted into a reference deformation for which there is more than one possibility in plasticity theory.

Table 3.1.1 shows a comparison of the most used test methods for the determination of yield stress and plasticity.

In the torsion test a round bar of uniform cross section area is gripped at one end loaded by a torque at the other end.

During the test
- torque,
- angle of rotation,
- revolutions and
- longitudinal force

can be measured until the specimen fractures. The reference yield stress, the reference strain and the reference strain rate can be calculated.

		Test method		
		Upsetting test	**Torsion test**	**Tensile test**
1.	**Test conditions**			
1.1	Temperature control	Only possible during intermittent deformation, otherwise specimen reheating	Same as for upsetting test	Same as for upsetting test
1.2	Max. strain rate	$\dot{\varphi} = 1000 s^{-1}$	$\dot{\varphi} = 50\ s^{-1}$	$\dot{\varphi} = 1000\ s^{-1}$
1.3	Max. deformation strain	$\varphi \leq 2$	Unlimited	$\varphi \leq 0.7$
1.4	Simulation of Thermomechanical conditions	Strongly limited	Possible	Strongly limited
2.	**Determination of**			
2.1	Yield stress	Yes	Yes	Limited
2.2	Plasticity	No	Yes	Yes
3.	**Specific Process Peculiarities**			
3.1	Stress condition	Compression stress, uni-axial only until beginning of bulging ($\varphi \leq 0.40$).	Shear stress with super-imposed longitudinal stress	Tensile stress, uni-axial only until beginning of area reduction ($\varphi \leq 0.20$)
3.2	Distribution of stress and strain variation in the specimen.	Homogeneous until beginning of bulging.	Inhomogeneous over the cross section: $\varphi = 0$ at specimen centre, φ = maximum at the surface. Inhomogeneous over specimen length if $L/D \geq 3.5$, Longitudinal stress results in cross section variations when strain $\varphi \leq 3.0$	Homogeneous until area reduction
3.3	Friction conditions	Friction on contact face between test piece-work piece, $\mu = f(\vartheta, \varphi, \dot{\varphi}$, material)	Frictionless	Frictionless

Table 3.1.1 Comparison of the most used test methods for the measurement of yield stress and plasticity.

If deformation is within the single phase range with constant strain and constant deformation rate, the temperature dependence of the yield stress will follow an exponential function.
Hot plasticity is influenced by the material (whether cast or wrought structure), by the temperature and by the deformation rate. The impact of each influencing parameter varies considerably from case to case.
Yield stress levels decrease steadily with rising temperature during hot deformation. The stress-strain curve variation with rising temperature can correctly expressed by yield stress calculations. Recovery dominates as the softening mechanism at high temperatures and the stress strain curve peak value disappears.
On the other side torsion test is one possible test method to determine flow curves. These flow curves are necessary for a precise calculation of rolling forces, torque, power consumption.
To characterise hardening and dynamic softening behaviour during hot deformation, hot flow curves can be used. Such flow curves can be measured by tensile tests, compression tests and/or torsion tests.
Flow curves show the relations between flow stress (σ_F) and strain (φ). Flow curves are the basic element for a calculation of stresses, forces, torque and power during deformation processes.
Dynamic softening can occur by recovery, polygonisation (curve type 1 in Fig.3.1.1) or additional by recrystallisation processes (curves of type 2 and 3).

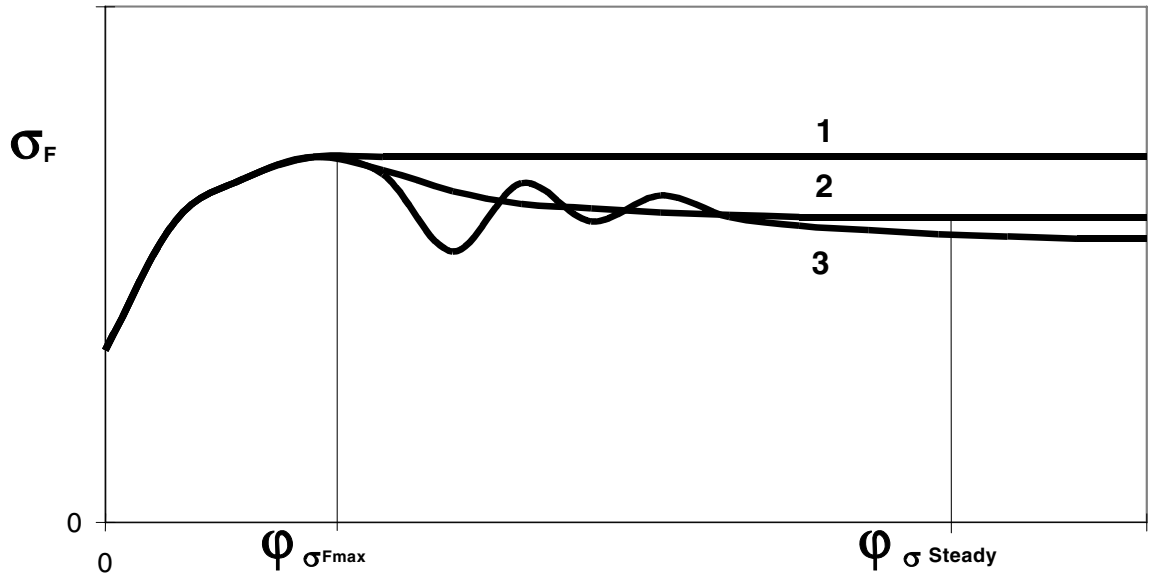


Fig. 3.1.1 Schematic flow curve at hot deformation

At strain values lower than $\varphi_{\sigma Fmax}$ in general predominates dynamic recovery. In materials with a low stacking fault energy dynamic recovery dominates too when $\varphi > \varphi_{\sigma Fmax}$.
Recovery and polygonisation include all reactions of dislocations, which take place before recrystallisation starts. They differ from each other, so at recovery only the lattice defect density will be reduced. Polygonisation on the other side creates new boundary surfaces, which have less differences in their orientations, and so the new substructure is created. In both cases the reduction of deformation resistance takes place by dislocation annihilation owing to interrelations of dislocations. This phenomenon takes place in regions with high concentrations

of point defects by cross-slipping of screw dislocations and climbing of edge dislocations and by a reorganisation of dislocations.

Dynamic polygonisation develops if dislocations relatively easily pull out of their glide plane and create sub-grain boundaries. This is the dominating recovery effect at deformations of $\varphi = 0.8\ \varphi_{\sigma Fmax}$ and takes place in such materials which have not a so strong work hardening at deformation. Results of recovery with polygonisation are:

- Redistribution, healing, migration and dissolution of point defects and dislocations,
- arrangement of dislocations at energetic favourable dislocation walls,
- creation of a substructure,
- weakening of the internal stress conditions.

Due to this facts plasticity, toughness and resistance against brittle fracture is improved.

Softening by recrystallisation is characterised by movement of large-angle grain boundaries. This includes the stages of nucleation and growth. The recrystallisation nuclei appear at grain boundaries, precipitates and glide bands, that means in regions with a high dislocation density, where dislocations pile up. The result is the creation of new grains and a simultaneous reduction of crystal defect density, decrease of deformation structures and disappearance of elongated grains. The majority of grain appears fine, too. Dynamic recrystallisation is the dominating mechanism at high temperatures in materials with low stacking fault energy. It occurs, if the local dislocation densities in the microstructure reach a critical level.

The softening rate gets equal or higher than the hardening rate, so a maximum appears in the flow curve. At deformations of $\varphi > \varphi_{\text{Steady}}$, that means in the steady region of hot deformation, the microstructure recrystallisation is finished.
Dynamic recrystallisation is characterised by the existence of new small grains at the boundaries of the initial grains.
The result of the dynamic recrystallisation is a remarkable reduction of dislocation density, a significant reduction of flow stress, a complete new arrangement of microstructure and appropriate favourable properties.

After deformation at $\varphi_{\sigma Fmax} > \varphi > \varphi_{\text{Steady}}$ and after a special incubation time, immediate meta-dynamic recrystallisation and static recrystallisation, respectively occurs. The kinetics of static recrystallisation increases with deformation temperature, strain, strain rate and prior austenite grain refinement.

2. Hot torsion tests in Freiberg

Two torsion plastometers are used for the determination of flow-stress curves by the torsion test at Institut für Metallformung in Freiberg. These equipments have the technical parameters shown in Table 3.1.2:

	Torsion plastometer No.	
	I	II
Specimen dimensions (mm)	∅ 6 * L 12 to 15	∅ 6 * L 12 to 15
Power [kW]	2.0	0.97
Nominal torque [Nm]	38.5	7.7
Speed range [min^{-1}]	10.to.2000	10.to.4000
Overload	6-fold	10-fold
Minimum torsion angle [°]	5	5
Deformation temperature (for steel) [°C]	0 to 1200	700 to 1200
Deformation rate [s^{-1}]	8	10

Table 3.1.2 Torsion plastometers at Institut für Metallformung, Freiberg

For hot torsion tests, the material delivered from Sidenor and Lucchini was rolled to a round rod with ∅ 16 mm. After the manufacturing of specimens in the workshop, followed the torsion test. Before testing all probe diameters were measured precisely. Fig.3.1.2 shows the dimensions of the specimen used for the tests. This is a standard specimen, which is internationally acknowledged.

The hot torsion tests were carried out for all materials received from Sidenor and Lucchini, Steel 54SiCrV6 was available from both companies:

Steel grades:	10MnSi7	(Lucchini)
	38Cr4	(Sidenor)
	30MnB4	(Lucchini)
	54SiCrV6	(Lucchini)
	54SiCrV6	(Sidenor).

Austenitisation temperature:	1150°C, (holding time 10 minutes), cooling to deformation temperature
Deformation temperature:	1050°C, 1000°C, 950°C, 900°C, 850°C, 800°C, 750°C
Strain rate $\dot{\varphi}$:	0.2, 2.0 5.0 s^{-1}.

Measurements of torque, number of revolutions, and angle of rotation were performed.

Fig.3.1.3 to Fig.3.1.40 show the flow-stress (flow curves) of all investigated steels at deformation temperatures between 750 °C and 1050 °C. Significant effects of temperature and deformation rate on deformation resistance as well as on deformation behaviour can be seen.

3. Mathematical modelling of flow stress

Flow-stress represents an essential factor of the hardening conditions and for the calculations of deformation factors such as force, work or power consumption. Therefore flow-stress is of importance for the development of new deformation facilities as well as for optimising the existing technologies.

On the other hand the flow-stress gives, in combination with an accurate evaluation of hardening and softening behaviour, information on the real material stress.
For a mathematical flow-stress prediction, based on torsion test results, specific mathematical equations were developed. For a phenomenological description, mathematical terms which combine thermodynamic multiplicative factors are proven. These thermodynamic factors can be used for calculations of rolling force, torque, power consumption, etc. The hardening and softening behaviour of each steel grade can also be evaluated.
Ten mathematical equations for the calculation of flow-stress σ_F were developed in Freiberg**.** Usually two mathematical statements are used today (Statements 1 and 8). These statements differ in consideration of the correlation between temperature (ϑ) and deformation degree (φ), and deformation rate ($\dot{\varphi}$):

Statement 1:

$$\sigma_F = A \cdot e^{m_1 \vartheta} \cdot \varphi^{m_2} \cdot e^{m_4/\varphi} \cdot \dot{\varphi}^{m_3}$$

Statement 8:

$$\sigma_F = A \cdot e^{m_1 \vartheta} \cdot \vartheta^{m_9} \cdot \varphi^{m_2} \cdot e^{(m_4/\varphi + m_7 \cdot \varphi)} \cdot (1+\varphi)^{m_5 \vartheta} \cdot \dot{\varphi}^{(m_3 + m_8 \vartheta)}$$

ϑ = Ambient temperature to 1050 °C
φ = 0.05 to 1.5

where:

A	Constant
$\mathbf{m_1 \ldots m_9}$	Exponent coefficients, which reflect the effects of deformation conditions on flow stress
ϑ	Deformation temperature in °C
φ	Deformation degree (strain)
$\dot{\varphi}$	Deformation rate in 1/s.

The statements for flow stress calculations give the most precise results either below A_{r1} or above A_{r3} For each region, the statements use other mathematical factors (Table 3.1.3).

Flow stress in the temperature region between A_{r1} and A_{r3} can be calculated by the following equation:

$$\sigma_x = \sigma_{(A_{r1})} + \Delta\vartheta_x * \frac{\Delta\sigma(A_{r3} - A_{r1})}{\Delta\vartheta(A_{r3} - A_{r1})}$$

Hot torsion tests permit also to assess the transformation points of each steel. In hot deformation processes the transformation temperatures (points) are effected by the deformation degree (φ) as well as by the deformation rate, $\dot{\varphi}$.The equations needed for the calculation of A_{r1} and A_{r3} are shown below:

$$A_{r1} = B_1 + C_1 * \varphi + D_1 * Ln\dot{\varphi}$$

$$A_{r3} = B_3 + C_3 * \varphi + D_3 * Ln\dot{\varphi}$$

The coefficients for the calculation of transformation points for all investigated steels are shown in Table 3.1.3. The found thermodynamic factors are listed in Table 3.1.4.

The best and most accurate equation is statement 8. Statement 1 should be used for rough calculations only.

Steel	B1	C1	D1	B3	C3	D3
10MnSi7	804.567	16.25	0.00	850.033	20.50	0.00
38Cr4	799.439	35.25	1.223	842.733	35.00	0.00
30MnB4	801.266	27.50	0.00	846.167	27.50	0.00
54SiCrV6 (LU)	802.800	32.00	0.00	838.933	47.75	0.00
54SiCrV6 (SI)	804.733	25.75	0.00	840.933	35.90	0.00

Table 3.1.3 Coefficients for calculation of transformation points.

The flow curves presented in the Figures from 3.1.3 to 3.1.40 are actual measured values, which means flow stresses without any temperature corrections.

Especially at low deformation temperatures and for steels with a high deformation resistance, a heating-up effect appears during torsion testing as in any other deformation process. The real deformation temperature differs from the desired test temperature. Therefore, temperature corrections have to be done. These corrections are based on previous experiences, especially on former investigations with thermocouples, using different torsion conditions and steel grades. Such factors as density and specific heat of the investigated steels were calculated and introduced into the corrections.

An example for steel 10MnSi7 is shown in Fig.3.1.41, whilst Fig.3.1.42 reports a comparison of calculations according statement 8 with and without temperature correction. Especially at high deformation values, differences between both calculation methods can be seen. This is the result of a heating-up effect during deformation.
Fig.3.1.43 shows flow stress calculations for the same steel compared with measured data. Already without any temperature corrections we have a good agreement between experiment and calculation.

Statement 1 below Ar1:

Steel	A	m1	m2	m3	m4	Corr..
10MnSi7	1684.61	-0.00280008	0.146098	0.0815625	-0.000212576	0.9395
38Cr4	230.745	-0.000191658	0.117799	0.0755139	-0.00176244	0.9171
30MnB4	865.854	-0.00179017	0.159172	0.0867776	-0.0011783	0.9717
54SiCrV6 (LU)	6922.54	-0.0041914	0.185133	0.100915	-0.000600042	0.9530
54SiCrV6 (SI)	1200.58	-0.00203843	0.101537	0.0804712	-0.00217236	0.9357

Statement 1 above Ar3:

Steel	A	m1	m2	m3	m4	Corr..
10MnSi7	6320.64	-0.0039868	0.1659	0.126347	-0.00148481	0.9779
38Cr4	3940.62	-0.0036135	0.148969	0.1108	-0.00149707	0.9700
30MnB4	6969.32	-0.00412664	0.186763	0.161433	-0.00131308	0.9657
54SiCrV6 (LU)	9595.55	-0.00428253	0.239824	0.158794	0.000571626	0.9803
54SiCrV6 (SI)	10369.9	-0.00438903	0.150502	0.110337	-0.00303078	0.9750

Statement 8 below Ar1:

Steel	A	m1	m2	m4	m5
10MnSi7	2573.71	-0.00287839	0.296604	0.00302558	-0.00051194
38Cr4	368.248	-0.000281388	0.301753	0.00288609	-0.00028791
30MnB4	908.768	-0.00171875	0.21393	9.65913e-005	0.00103697
54SiCrV6 (LU)	6741.33	-0.00424086	0.194694	0.000643869	0.00262508
54SiCrV6 (SI)	1449.58	-0.00208776	0.195719	0.00124156	0.00168938

Steel	m7	m8	Correlation
10MnSi7	-0.114726	0.000102649	0.9768
38Cr4	-0.262696	9.92169e-005	0.9676
30MnB4	-0.682823	0.000108887	0.9875
54SiCrV6 (LU)	-1.3886	0.000131522	0.9783
54SiCrV6 (SI)	-1.11554	0.00010202869	0.9832

Statement 8 above Ar3:

Steel	A	m1	m2	m4	m5
10MnSi7	8434.71	-0.00385297	0.345634	0.00260012	-0.000269912
38Cr4	4509.12	-0.00309881	0.396474	0.00306038	-0.00171069
30MnB4	8273.25	-0.00383554	0.390981	0.00280372	-0.000932992
54SiCrV6 (LU)	11403.0	-0.00435093	0.284152	0.00200734	-7.80437e-006
54SiCrV6 (SI)	10601.7	-0.00387638	0.374836	0.00321833	-0.000567575

Steel	m7	m8	correlation
10MnSi7	-0.244021	0.000127871	0.9921
38Cr4	0.459906	0.000119208	0.9800
30MnB4	0.175134	0.000162266	0.9785
54SiCrV6 (LU)	-0.100054	0.000171618	0.9823
54SiCrV6 (SI)	-0.163577	0.000123951	0.9865

Table 3.1.4 Thermodynamic factors of the investigated steels.

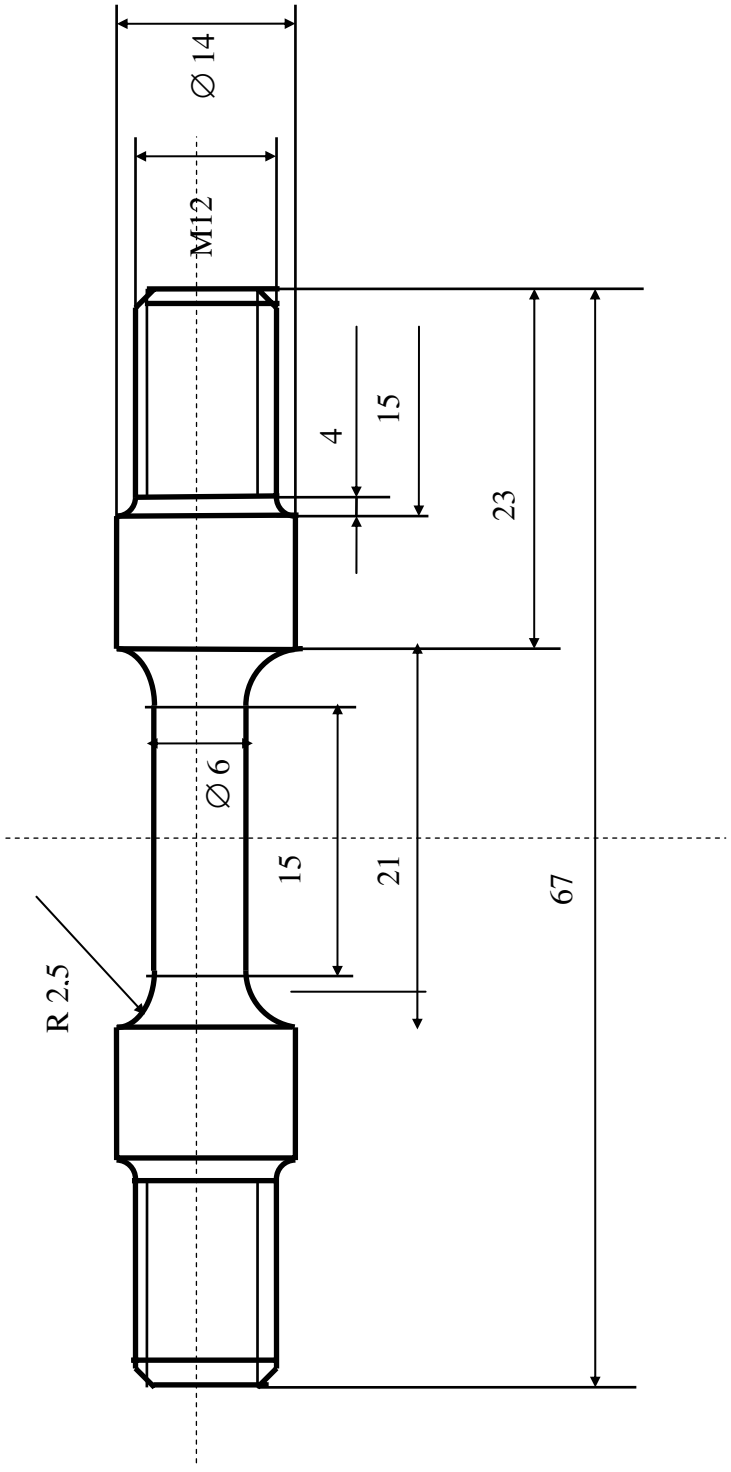


Fig. 3.1.2 Specimen for hot torsion tests

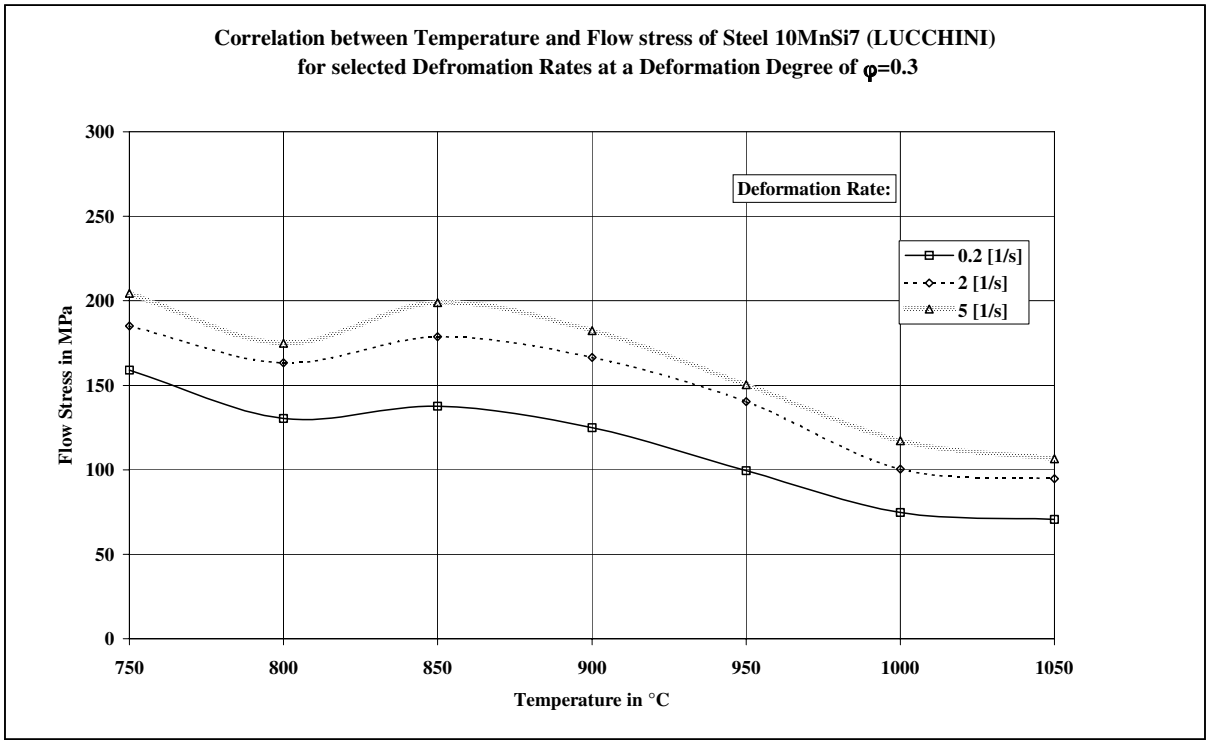


Fig. 3.1.3 Correlation between Temperature and flow stress, steel 10MnSi7 (LUCCHINI).

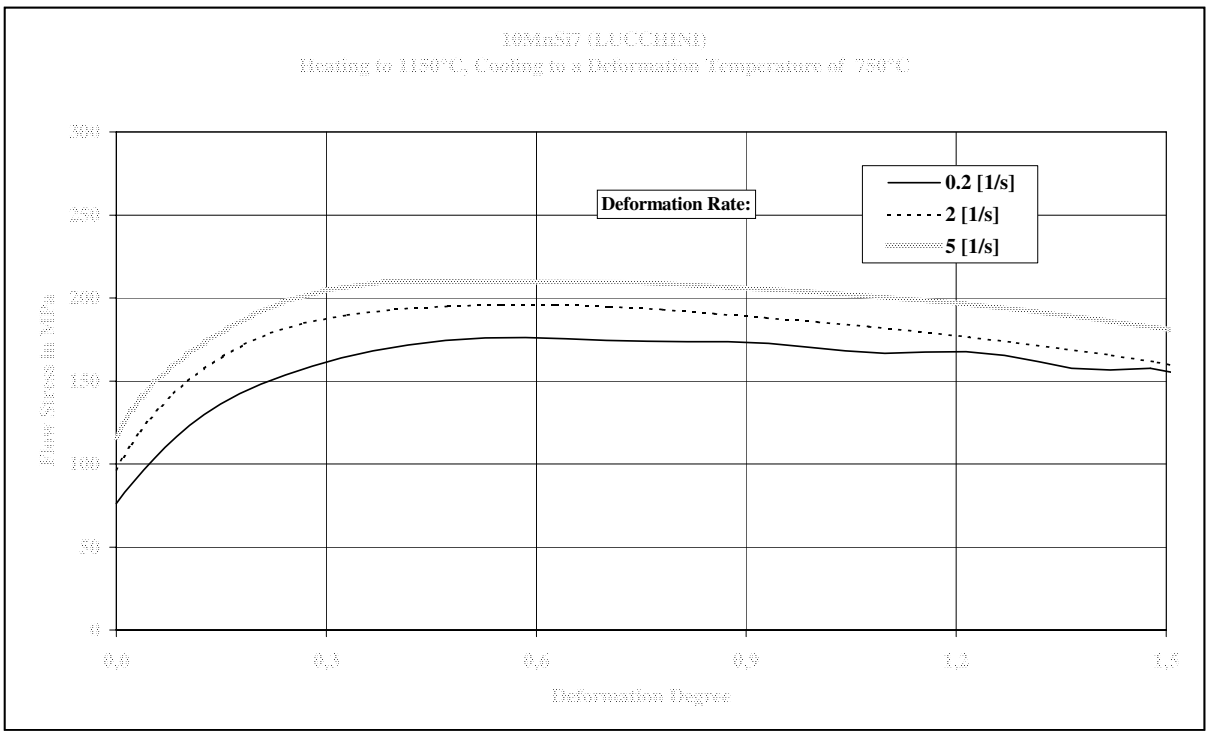


Fig. 3.1.4 Flow stress (flow curves) of steel 10MnSi7 (LUCCHINI) at 750°C.

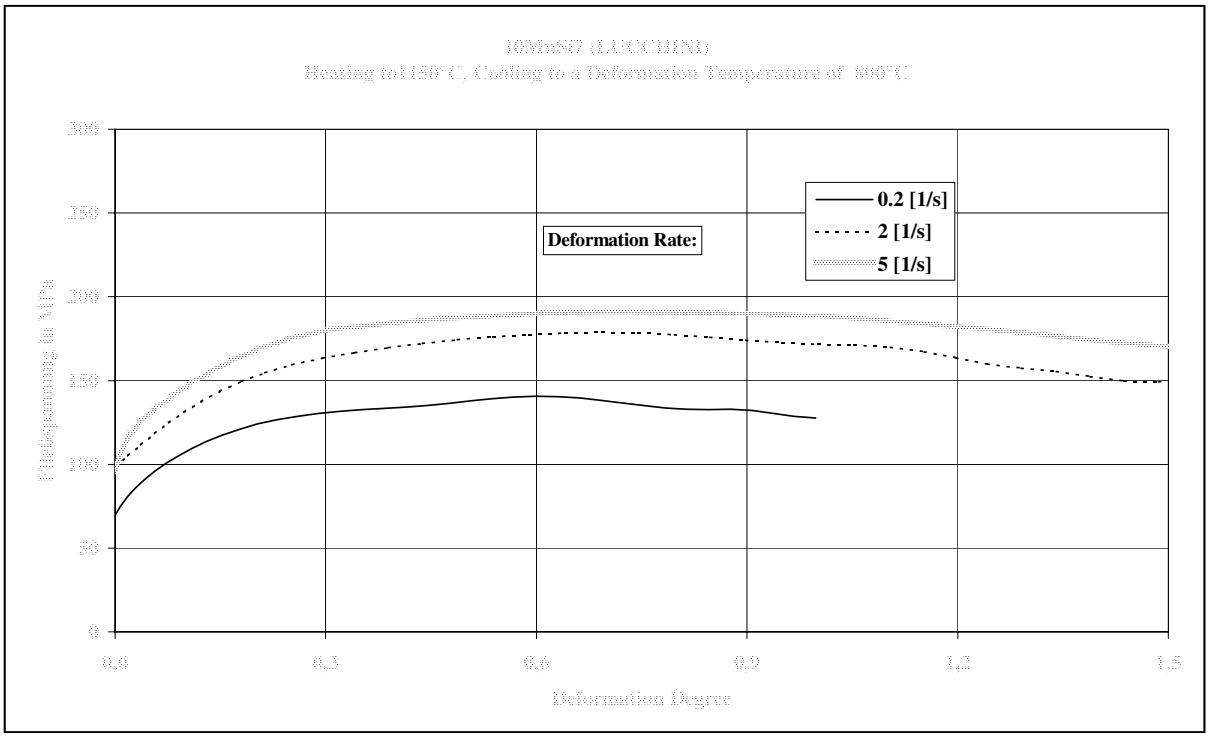


Fig. 3.1.5 Flow stress (flow curves) of steel 10MnSi7 (LUCCHINI) at 800°C.

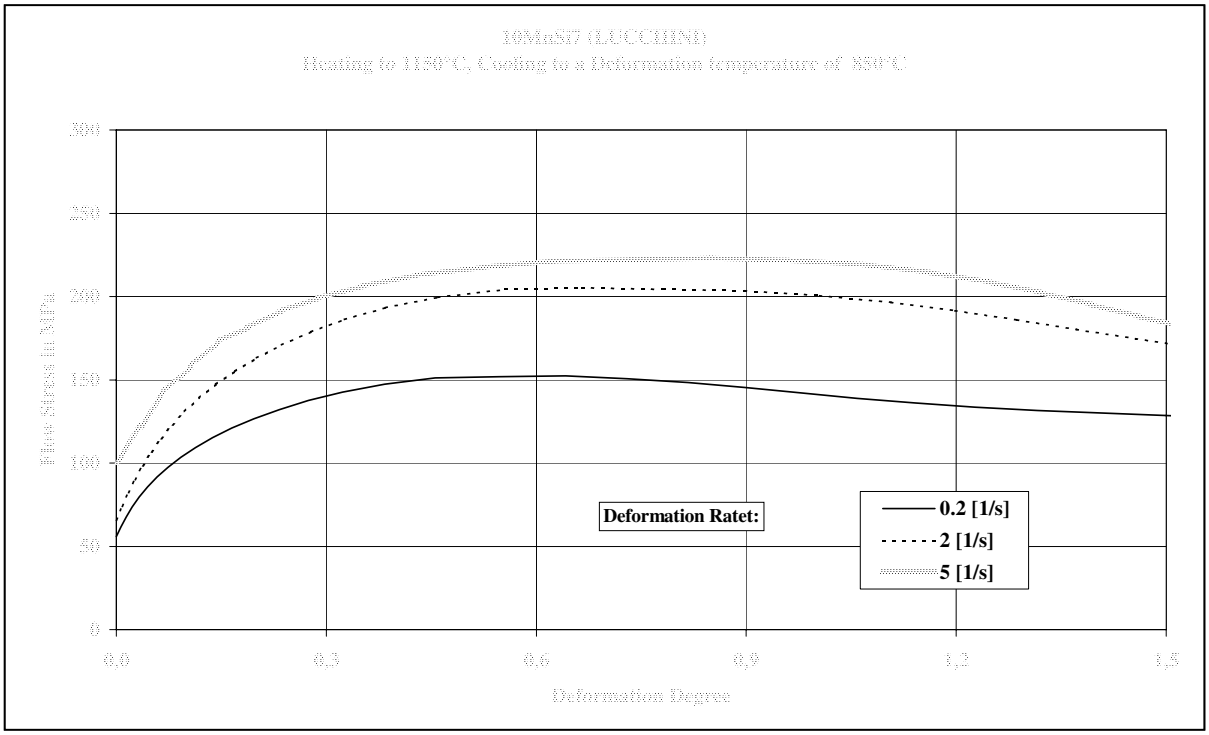


Fig. 3.1.6 Flow stress (flow curves) of steel 10MnSi7 (LUCCHINI) at 850°C.

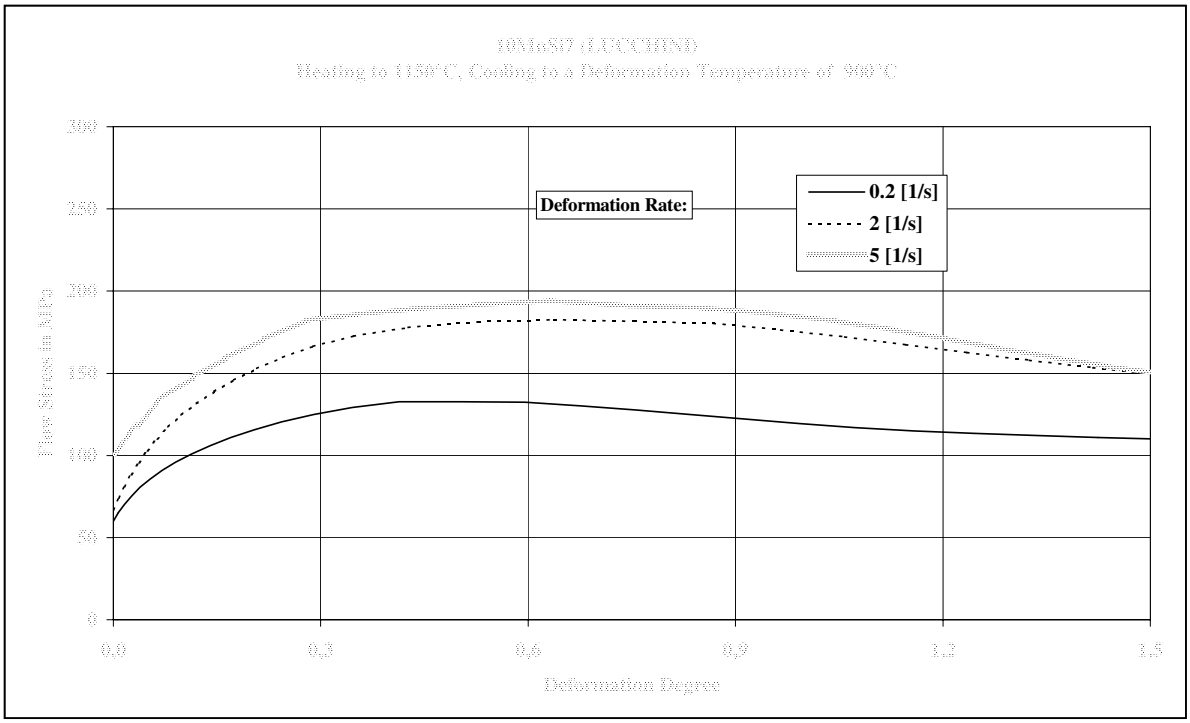


Fig. 3.1.7 Flow stress (flow curves) of steel 10MnSi7 (LUCCHINI) at 900°C.

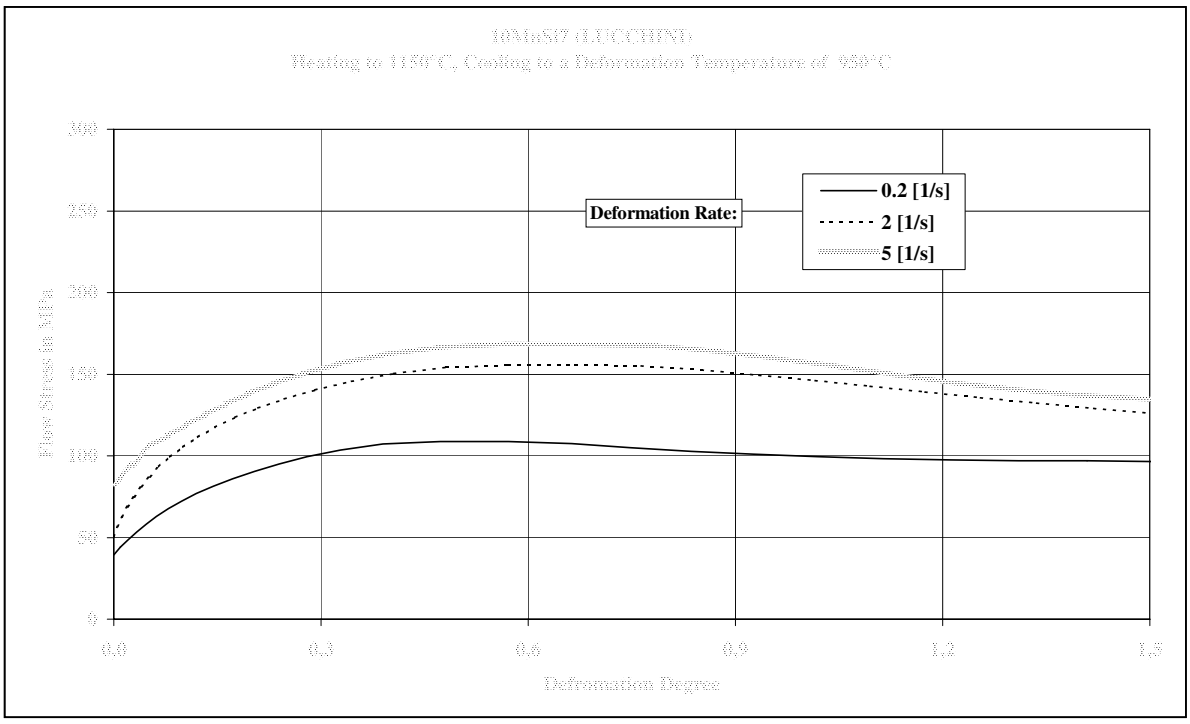


Fig. 3.1.8 Flow stress (flow curves) of steel 10MnSi7 (LUCCHINI) at 950°C.

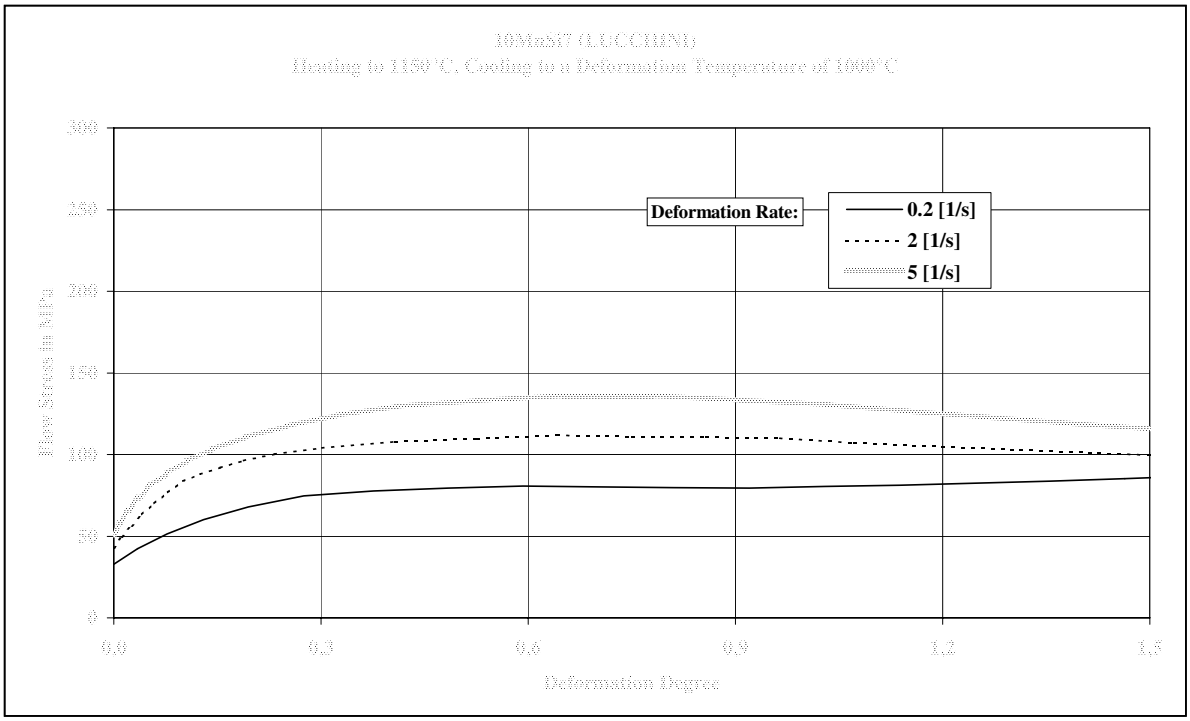


Fig. 3.1.7 Flow stress (flow curves) of steel 10MnSi7 (LUCCHINI) at 1000°C.

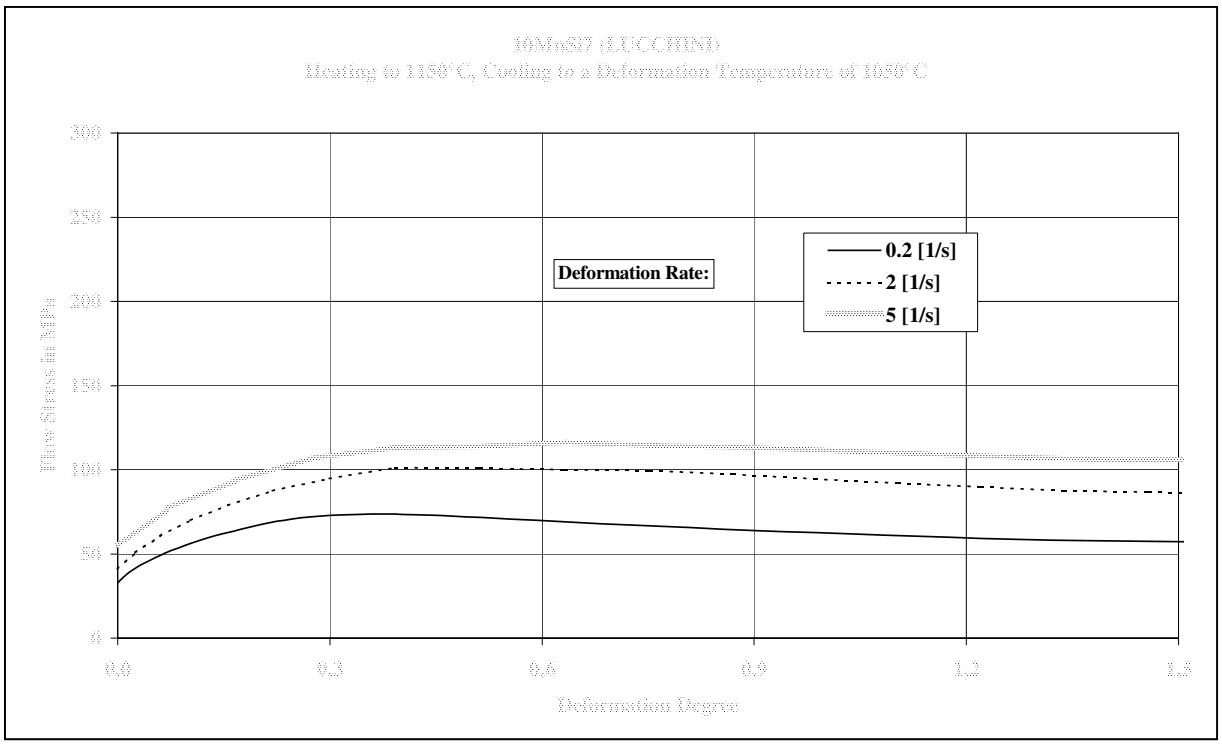


Fig. 3.1.8 Flow stress (flow curves) of steel 10MnSi7 (LUCCHINI)) at 1050°C.

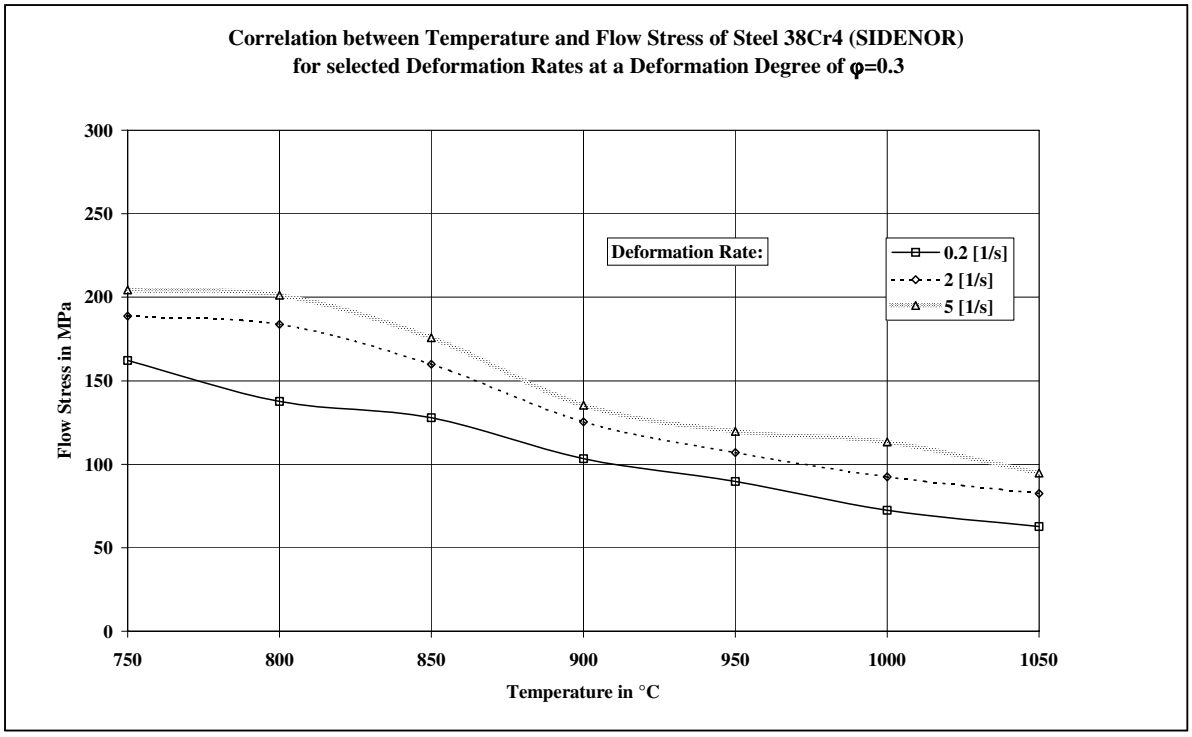


Fig. 3.1.9 Correlation between Temperature and flow stress, steel 38Cr4 (SIDENOR).

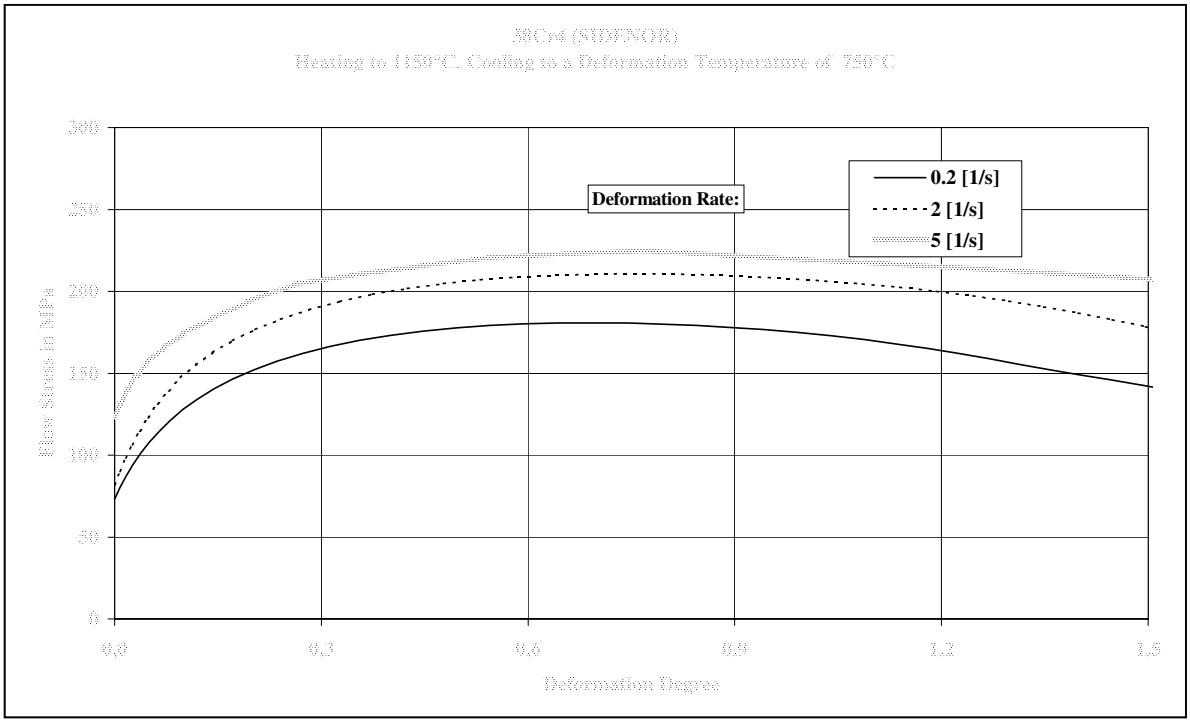


Fig. 3.1.10 Flow stress (flow curves) of steel 38Cr4 (SIDENOR) at 750°C.

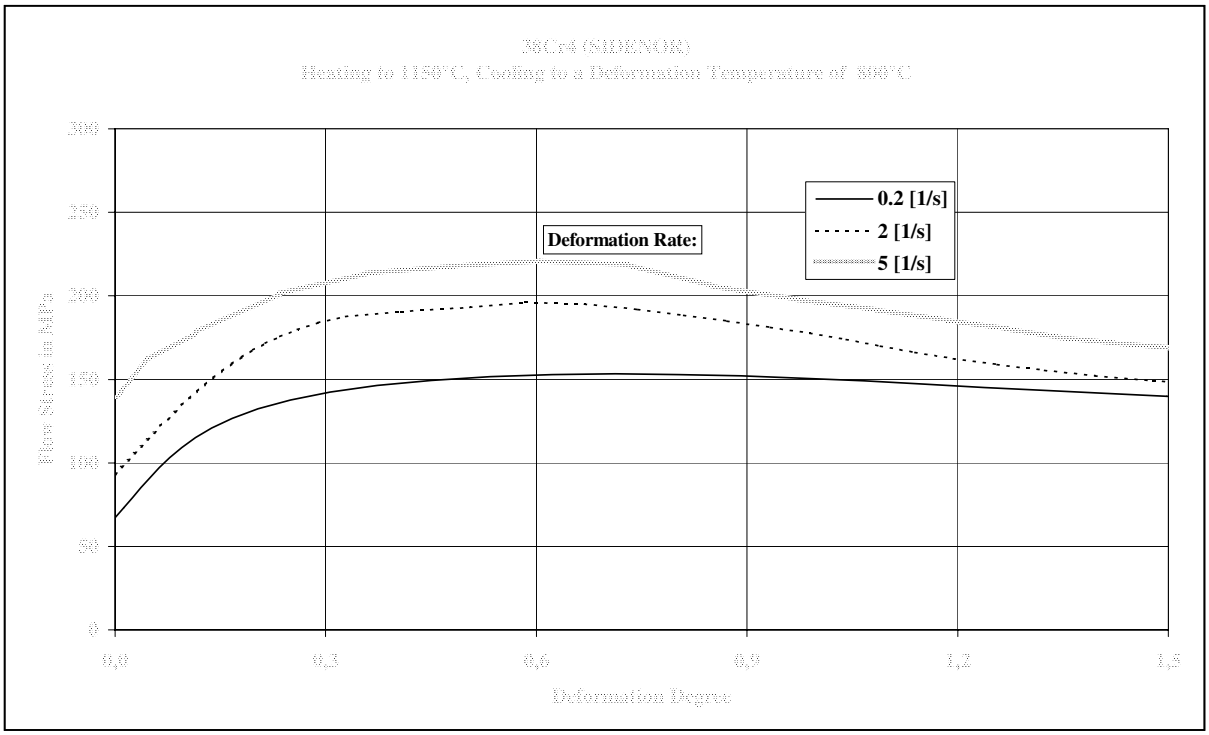


Fig. 3.1.11 Flow stress (flow curves) of steel 38Cr4 (SIDENOR) at 800°C.

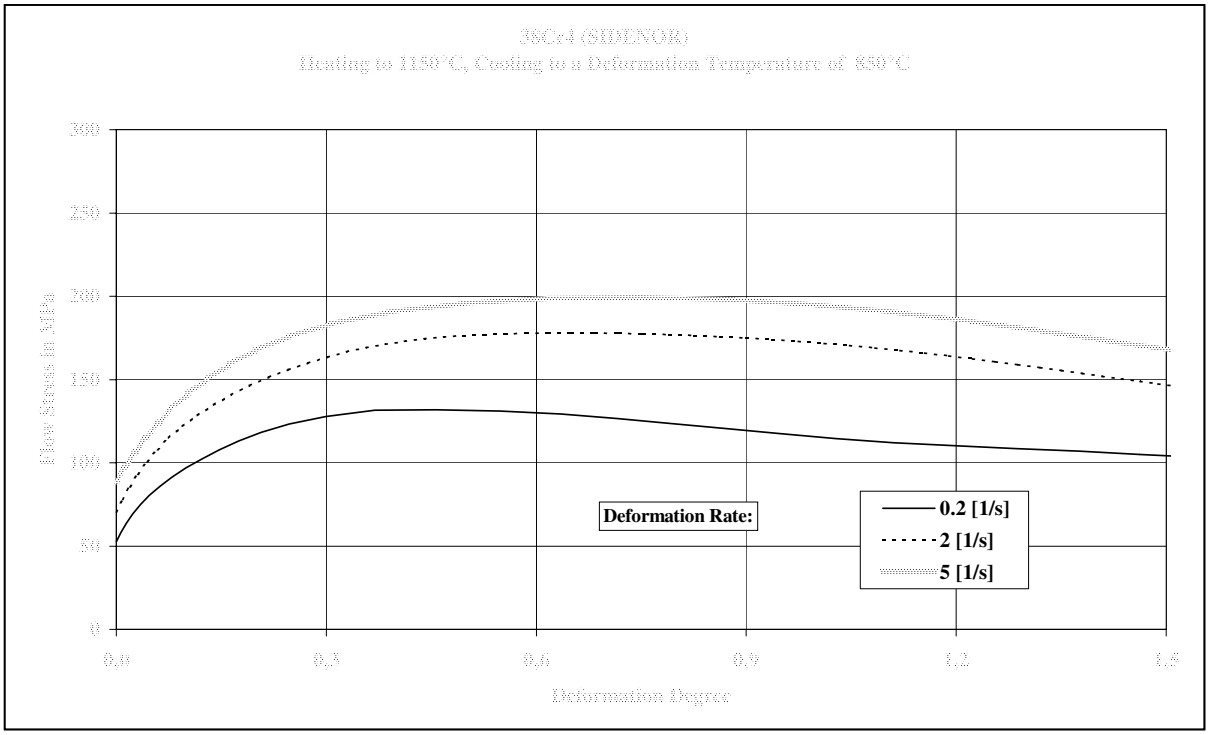


Fig. 3.1.12 Flow stress (flow curves) of steel 38Cr4 (SIDENOR) at 850°C.

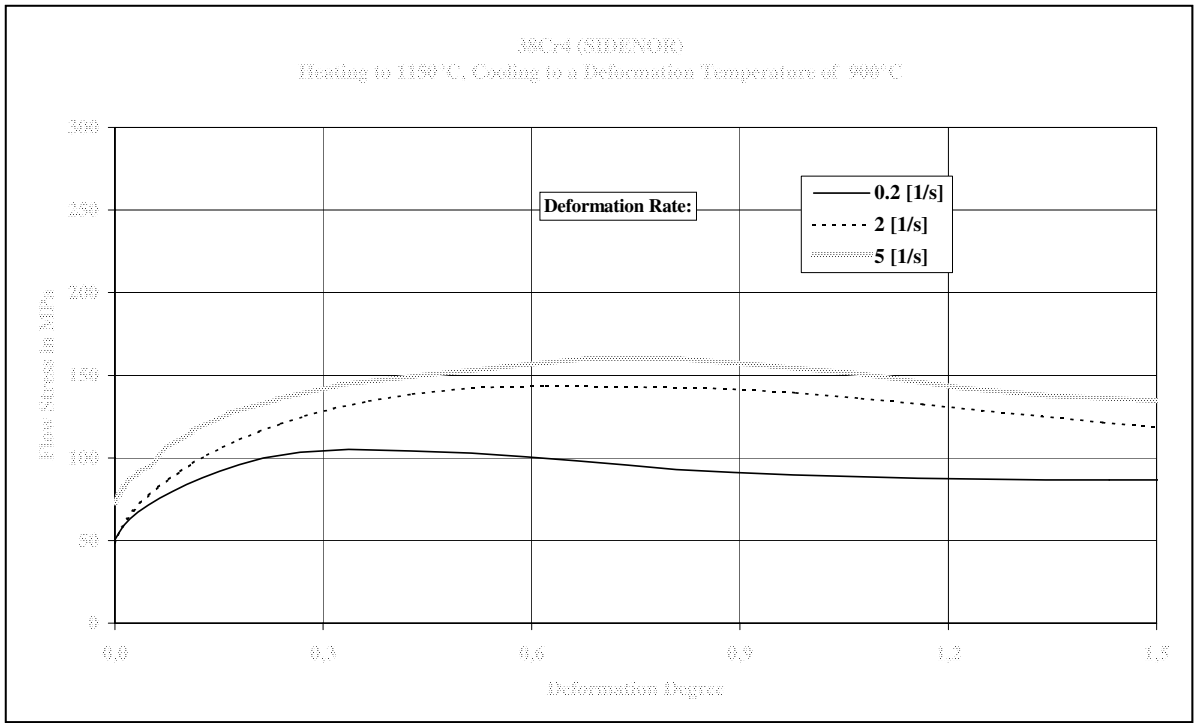


Fig. 3.1.13 Flow stress (flow curves) of steel 38Cr4 (SIDENOR) at 900°C.

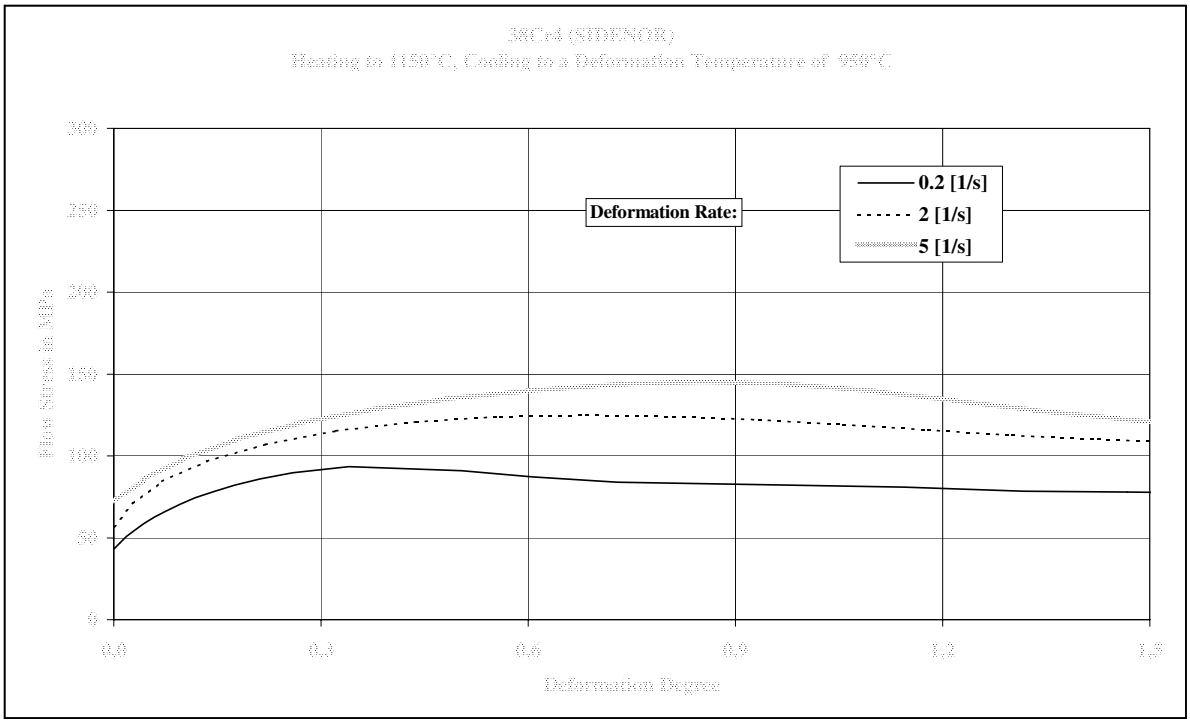


Fig. 3.1.14 Flow stress (flow curves) of steel 38Cr4 (SIDENOR) at 950°C.

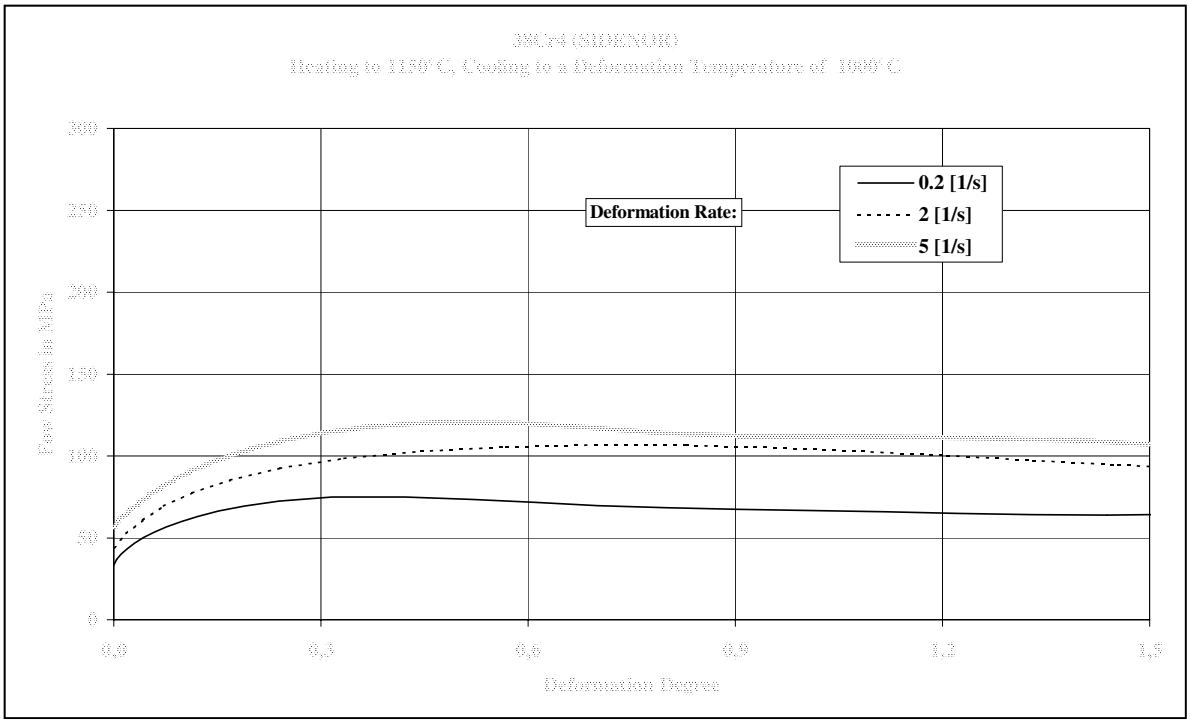


Fig. 3.1.15 Flow stress (flow curves) of steel 38Cr4 (SIDENOR) at 1000°C.

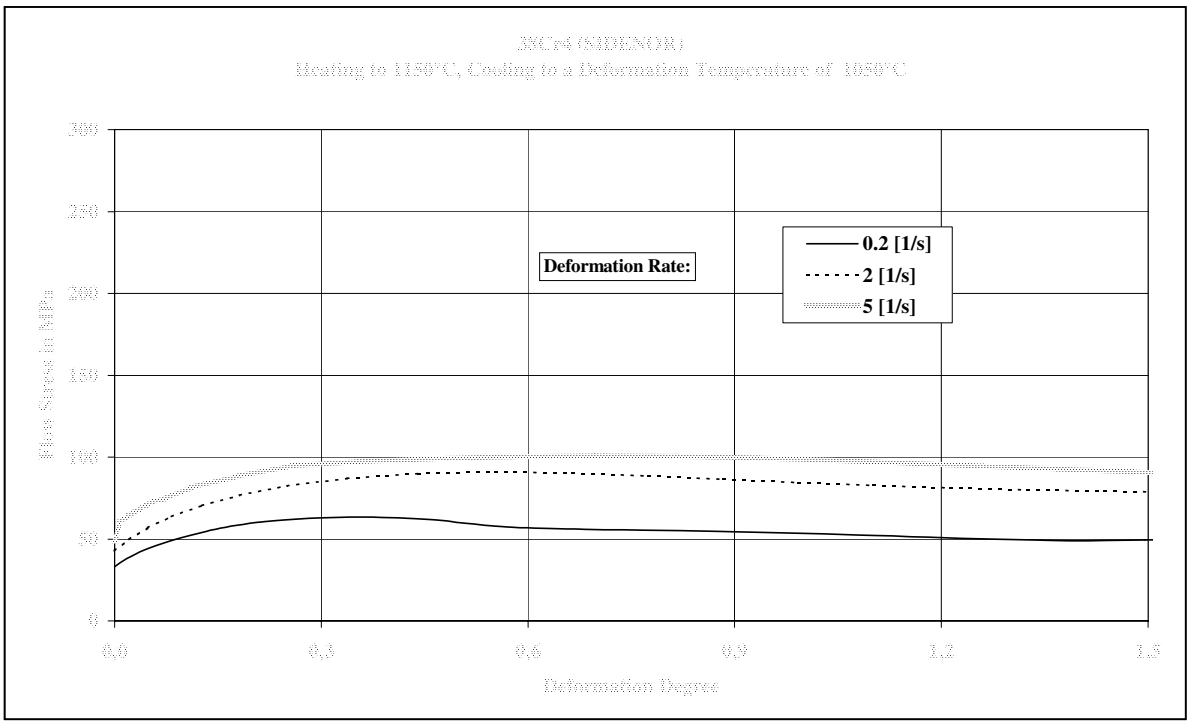


Fig. 3.1.16 Flow stress (flow curves) of steel 38Cr4 (SIDENOR) at 1050°C.

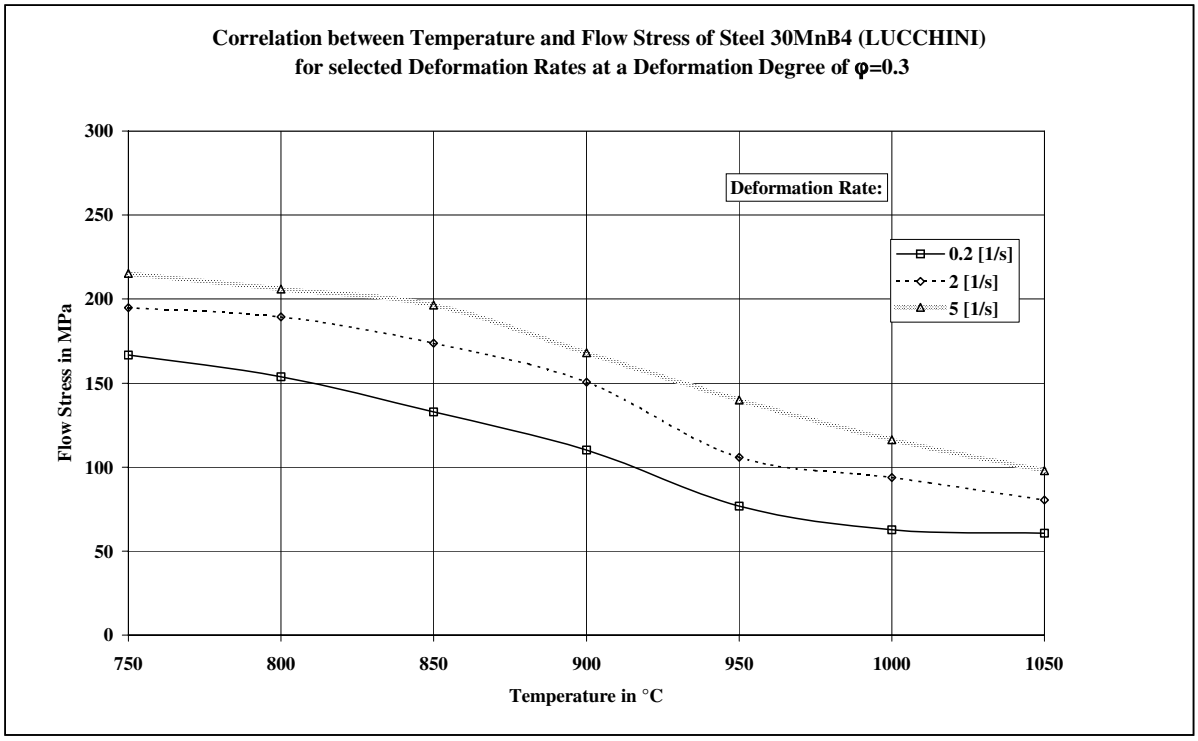


Fig. 3.1.17 Correlation between Temperature and flow stress, steel 30MnB4 (LUCCHINI).

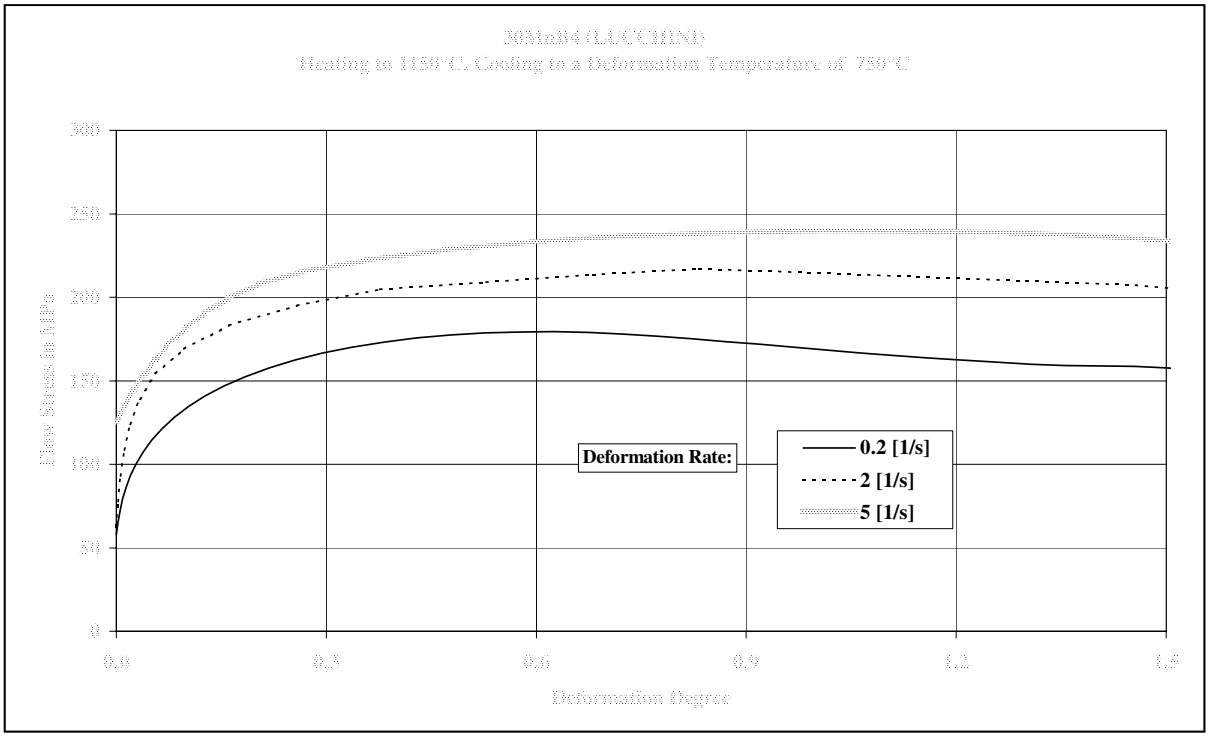


Fig. 3.1.18 Flow stress (flow curves) of steel 38MnB4 (LUCCHINI) at 750°C.

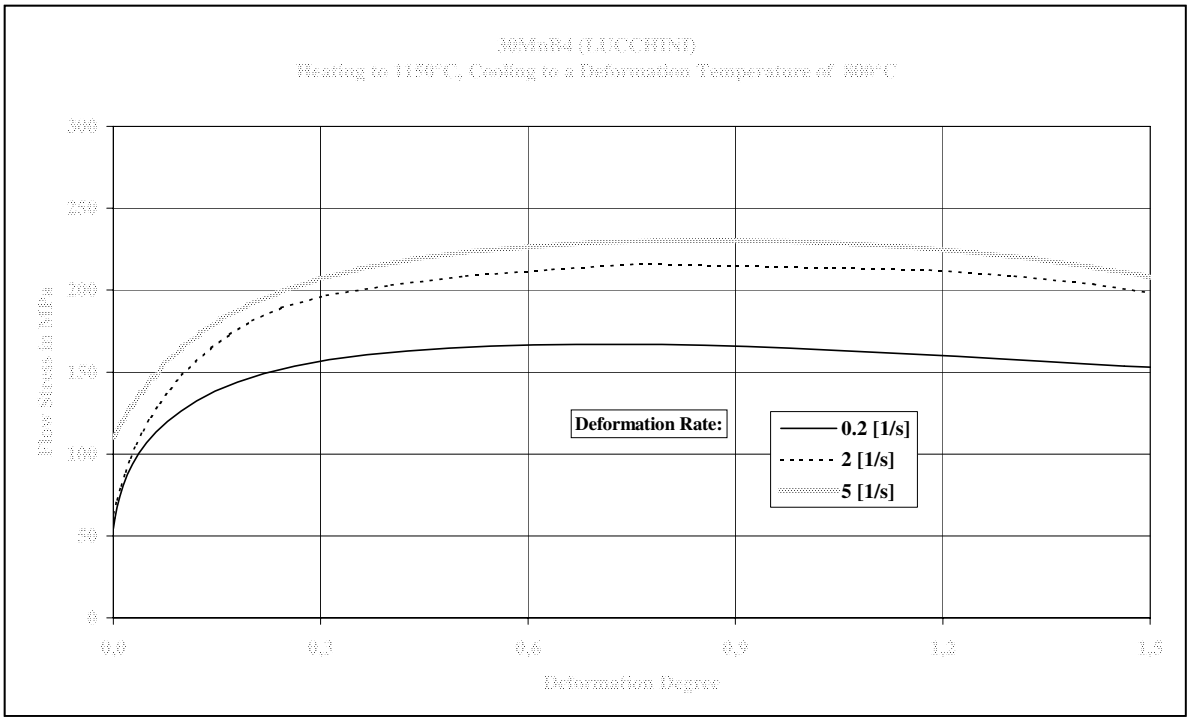


Fig. 3.1.19 Flow stress (flow curves) of steel 38MnB4 (LUCCHINI) at 800°C.

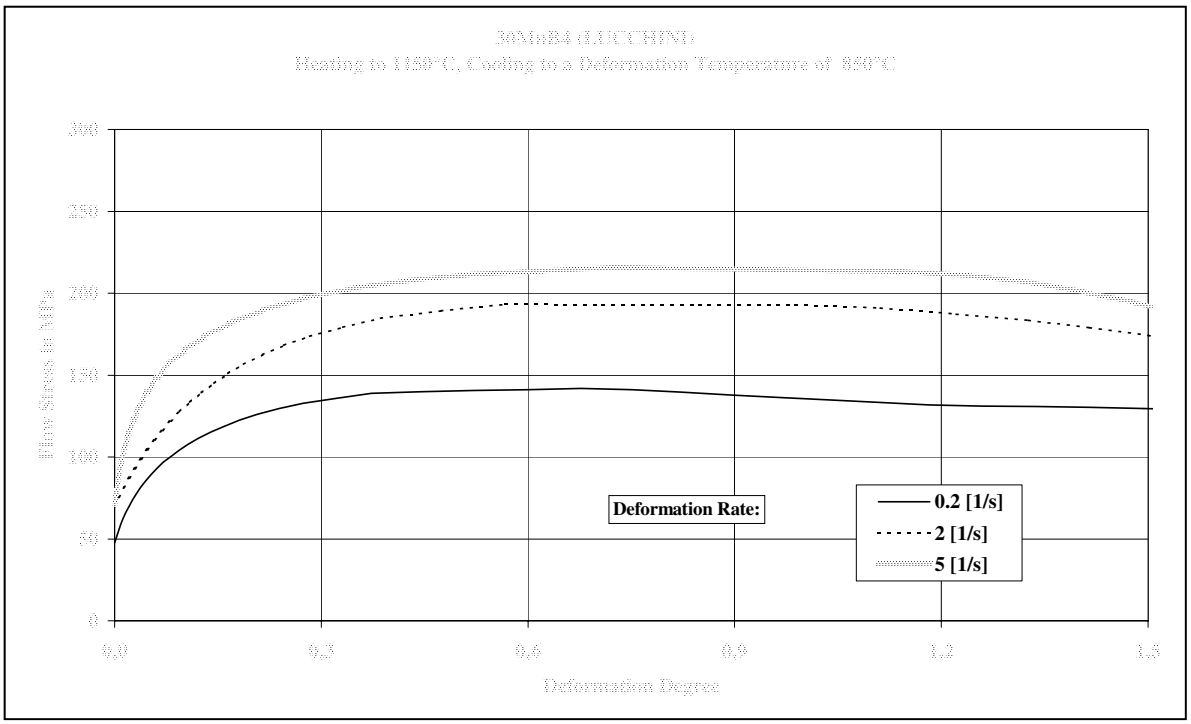


Fig. 3.1.20 Flow stress (flow curves) of steel 38MnB4 (LUCCHINI) at 850°C

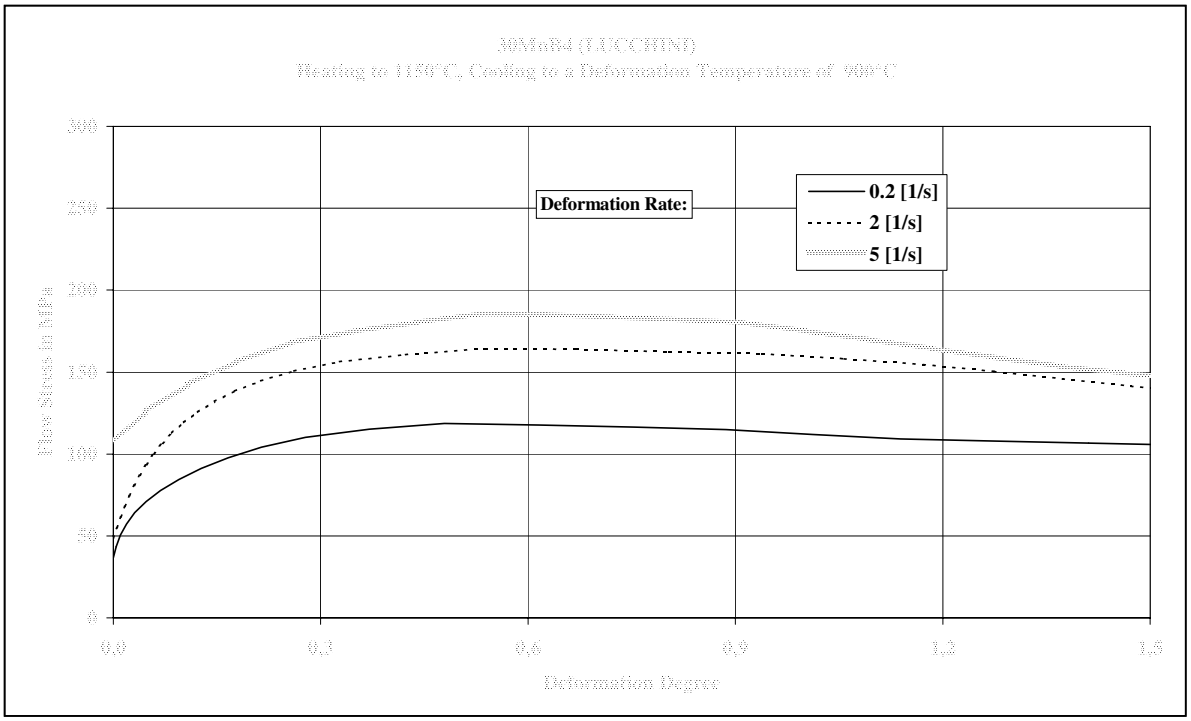


Fig. 3.1.21 Flow stress (flow curves) of steel 38MnB4 (LUCCHINI) at 900°C.

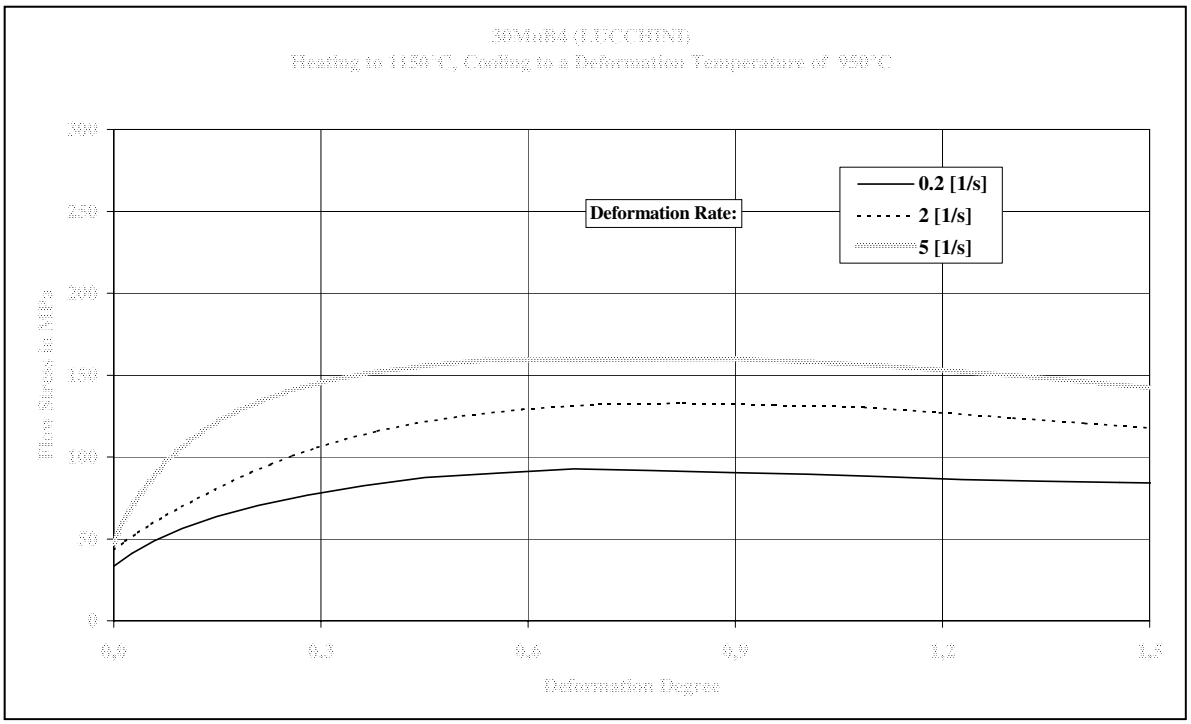


Fig. 3.1.22 Flow stress (flow curves) of steel 38MnB4 (LUCCHINI) at 950°C.

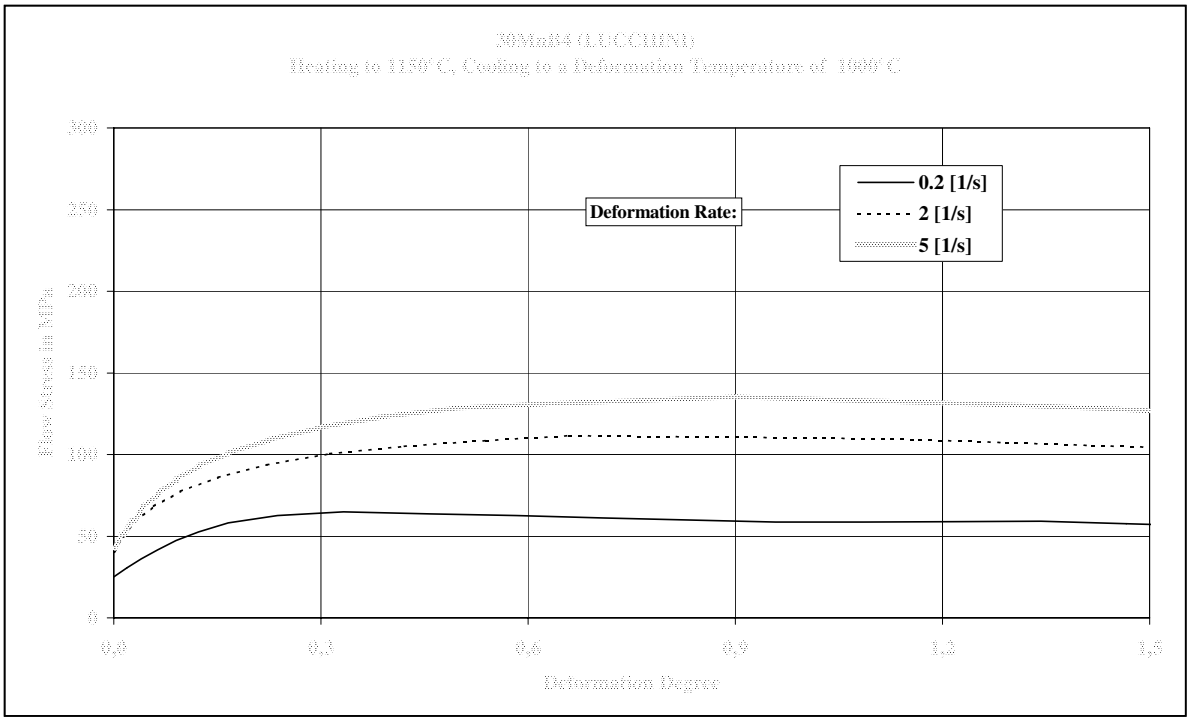


Fig. 3.1.23 Flow stress (flow curves) of steel 38MnB4 (LUCCHINI) at 1000°C.

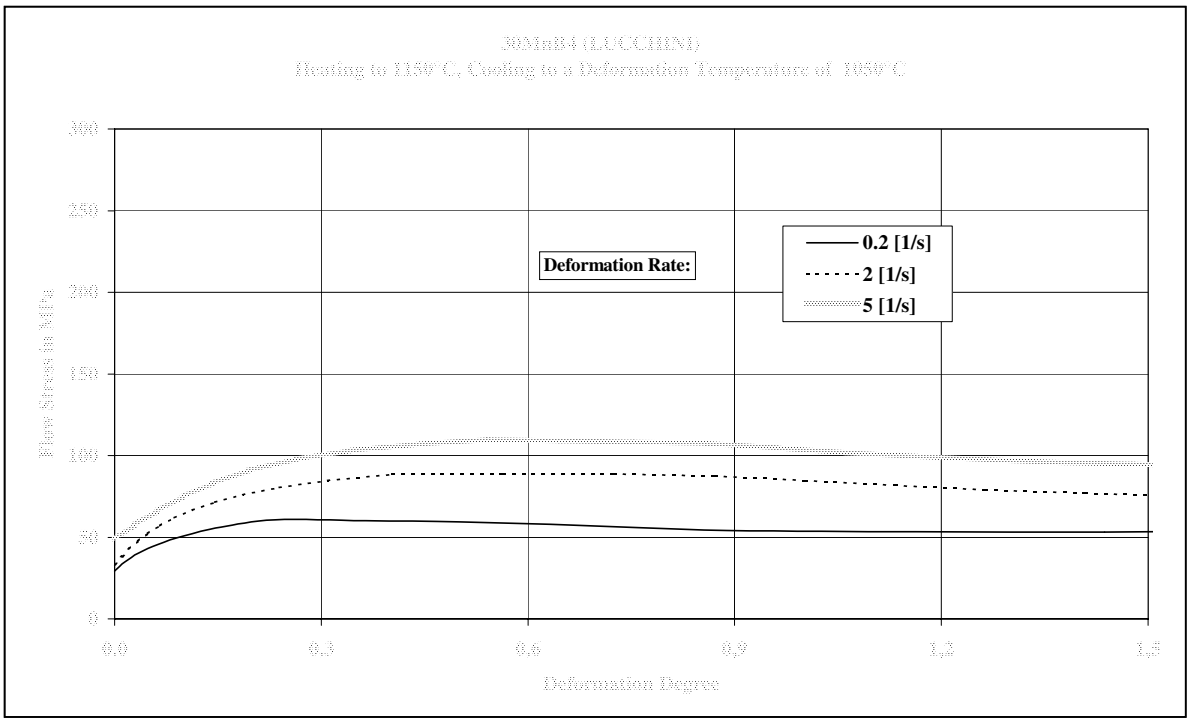


Fig. 3.1.24 Flow stress (flow curves) of steel 38MnB4 (LUCCHINI) at 1050°C.

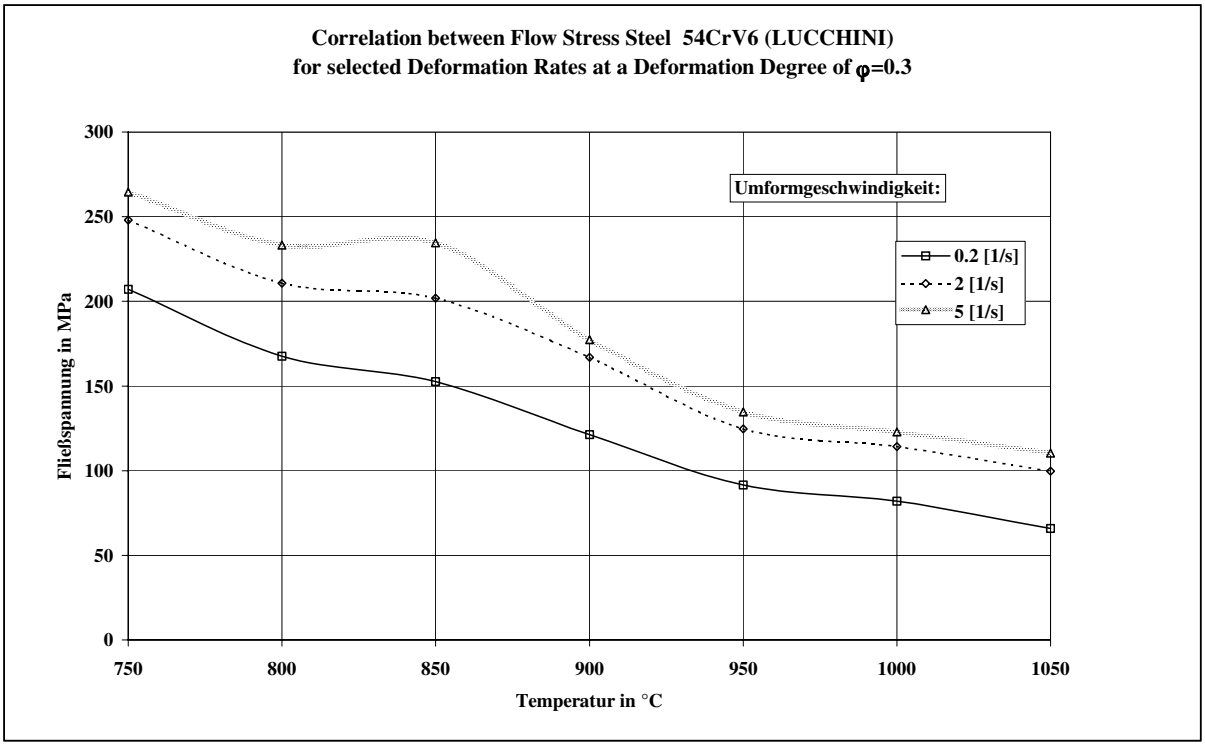


Fig. 3.1.25 Correlation between Temperature and flow stress, steel 54CrV6 (LUCCHINI).

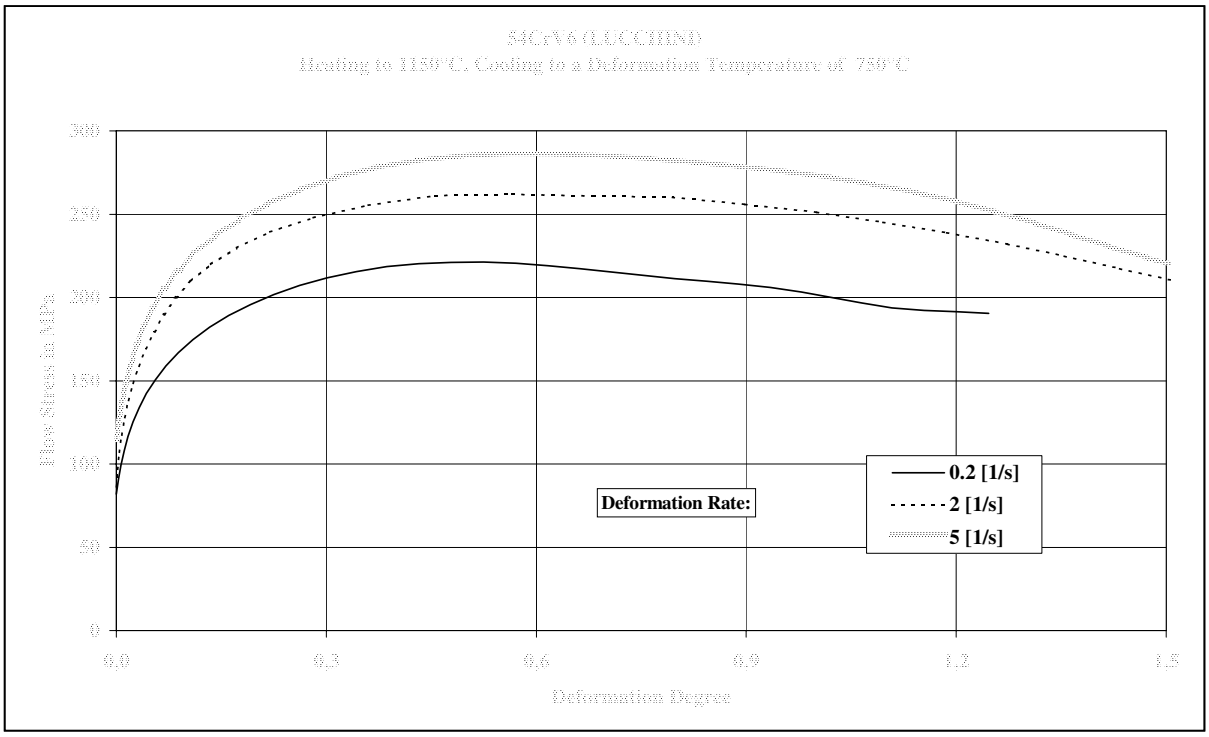


Fig. 3.1.26 Flow stress (flow curves) of steel 54CrV6 (LUCCHINI) at 750°C.

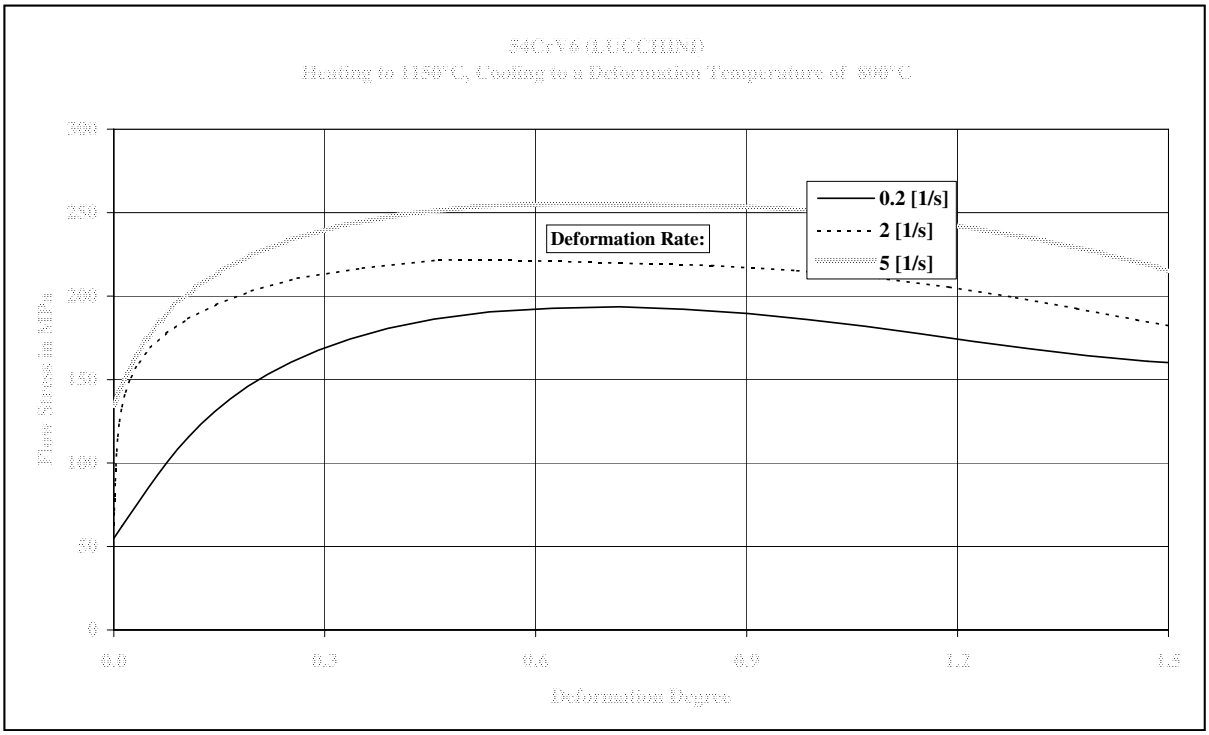


Fig. 3.1.27 Flow stress (flow curves) of steel 54CrV6 (LUCCHINI) at 800°C.

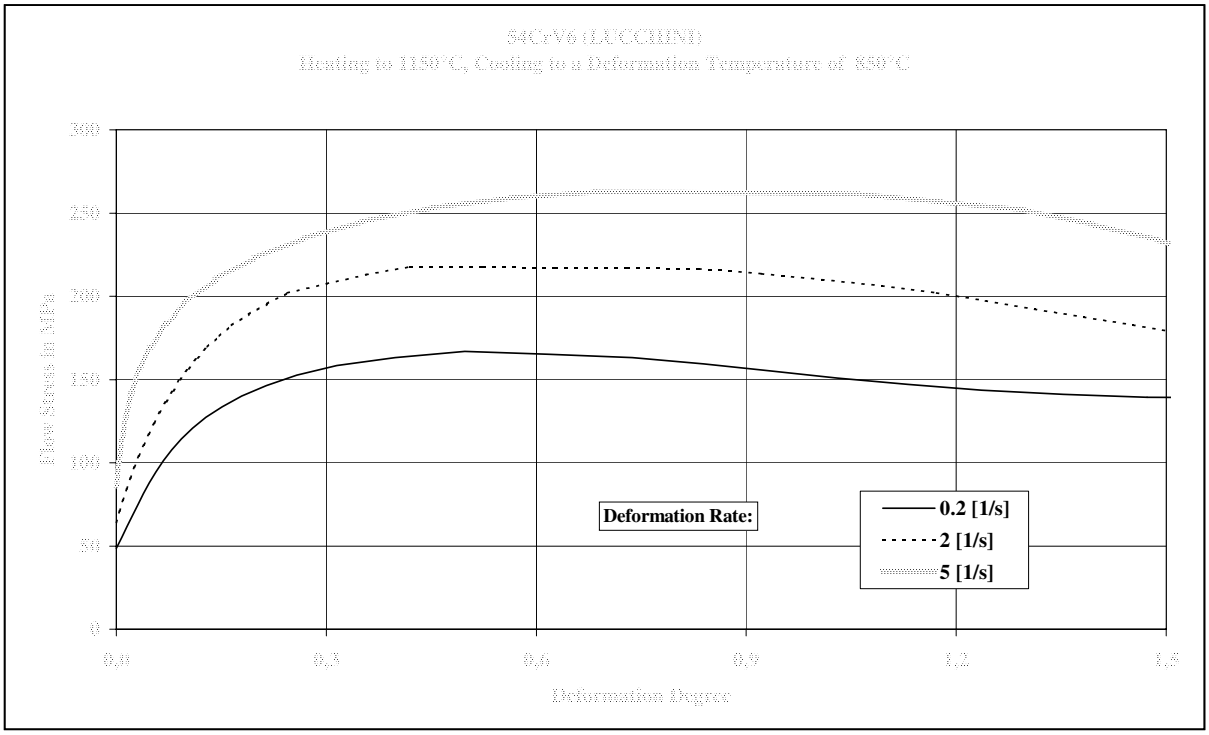


Fig. 3.1.28 Flow stress (flow curves) of steel 54CrV6 (LUCCHINI) at 850°C.

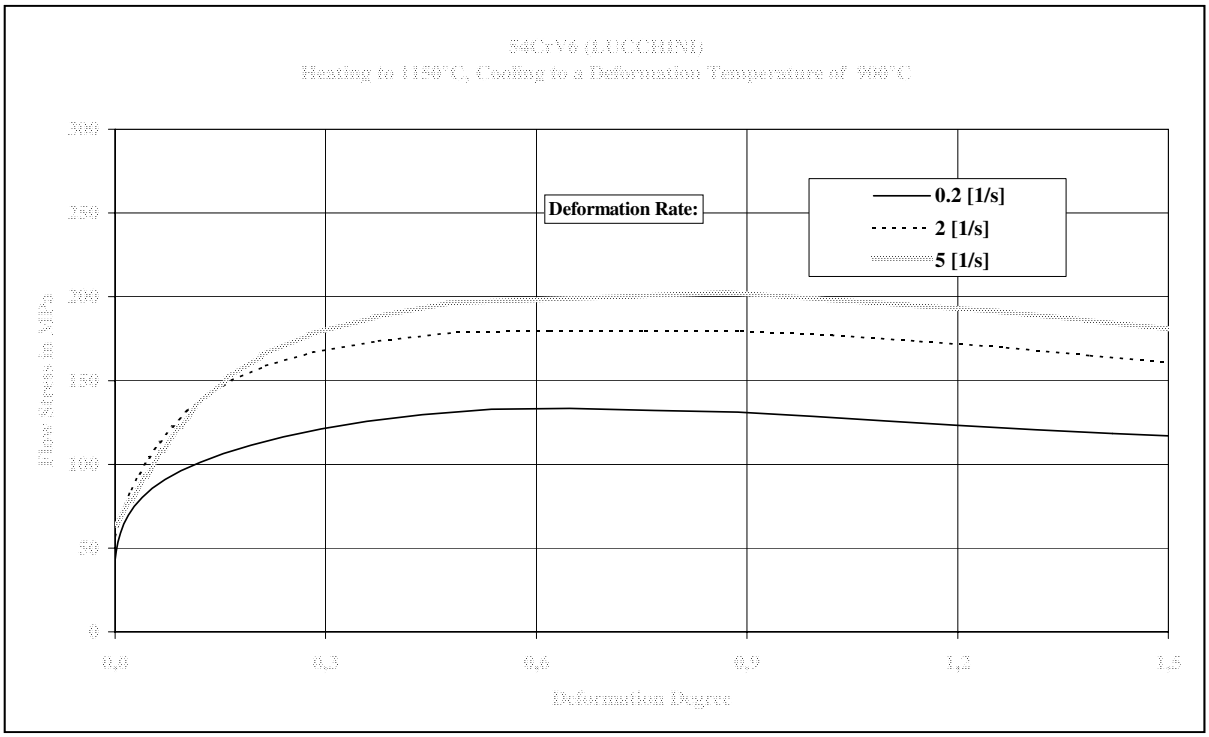


Fig. 3.1.29 Flow stress (flow curves) of steel 54CrV6 (LUCCHINI) at 900°C.

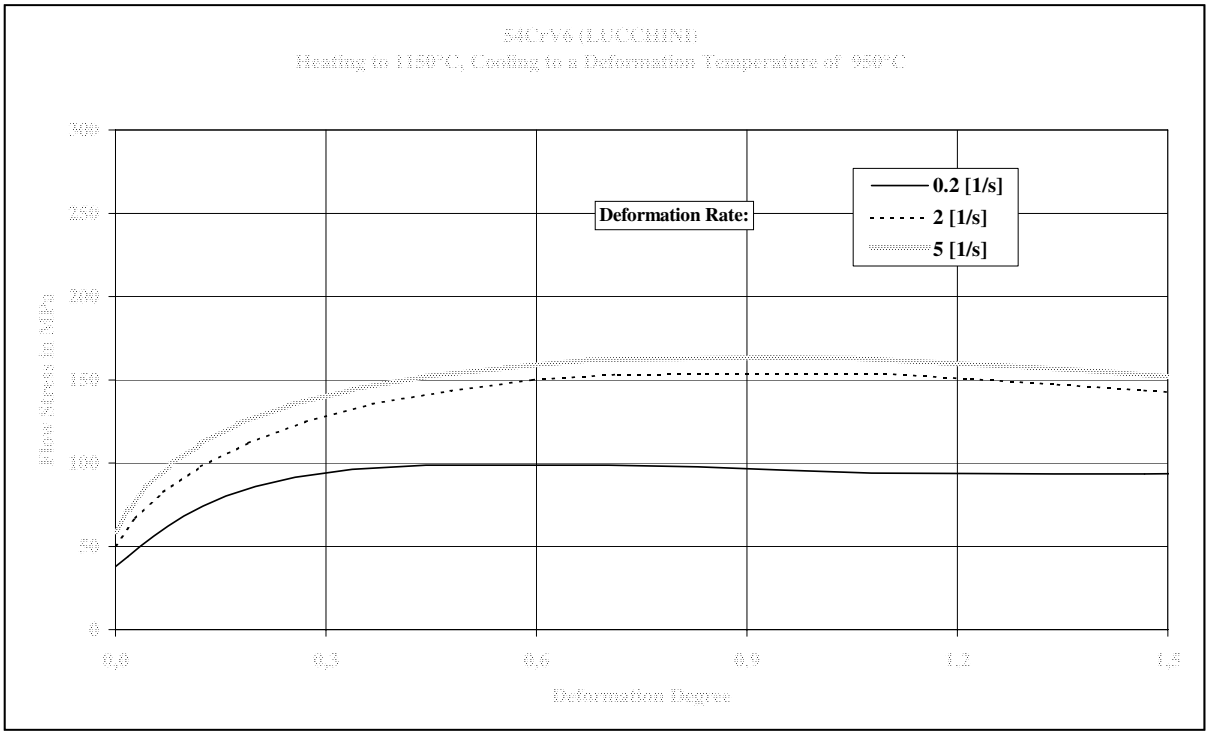


Fig. 3.1.30 Flow stress (flow curves) of steel 54CrV6 (LUCCHINI) at 950°C.

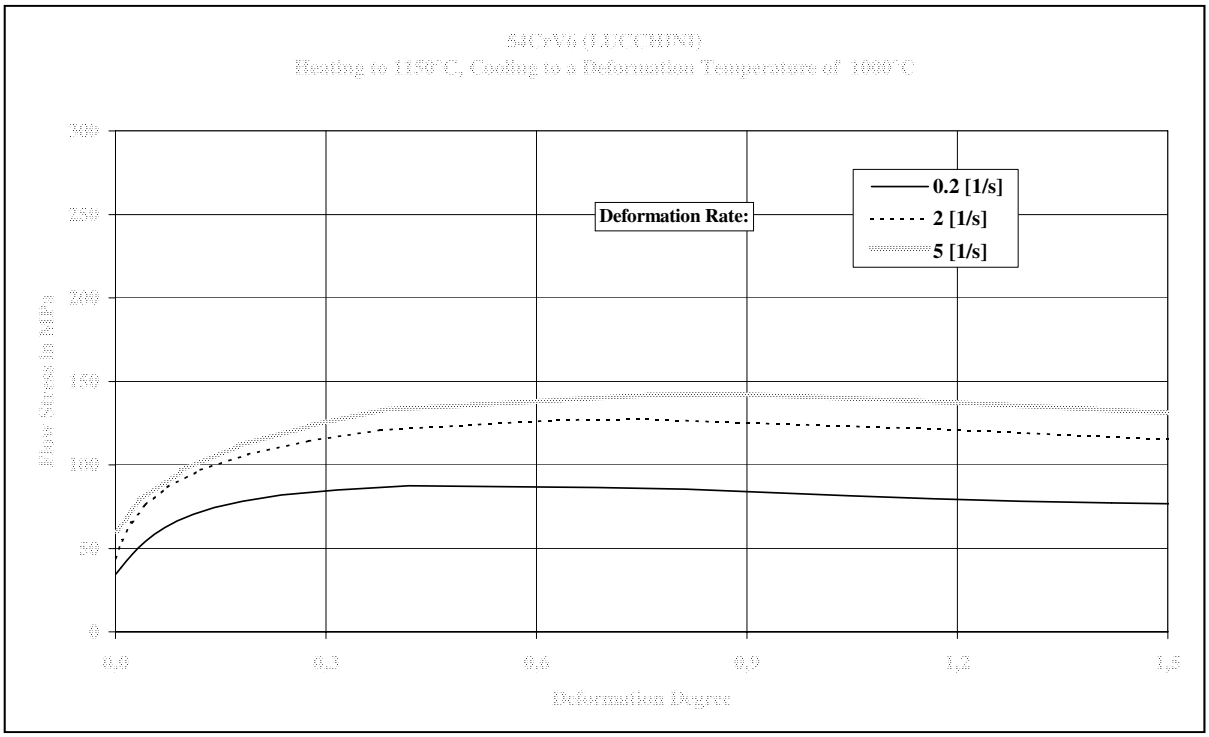


Fig. 3.1.31 Flow stress (flow curves) of steel 54CrV6 (LUCCHINI) at 1000°C.

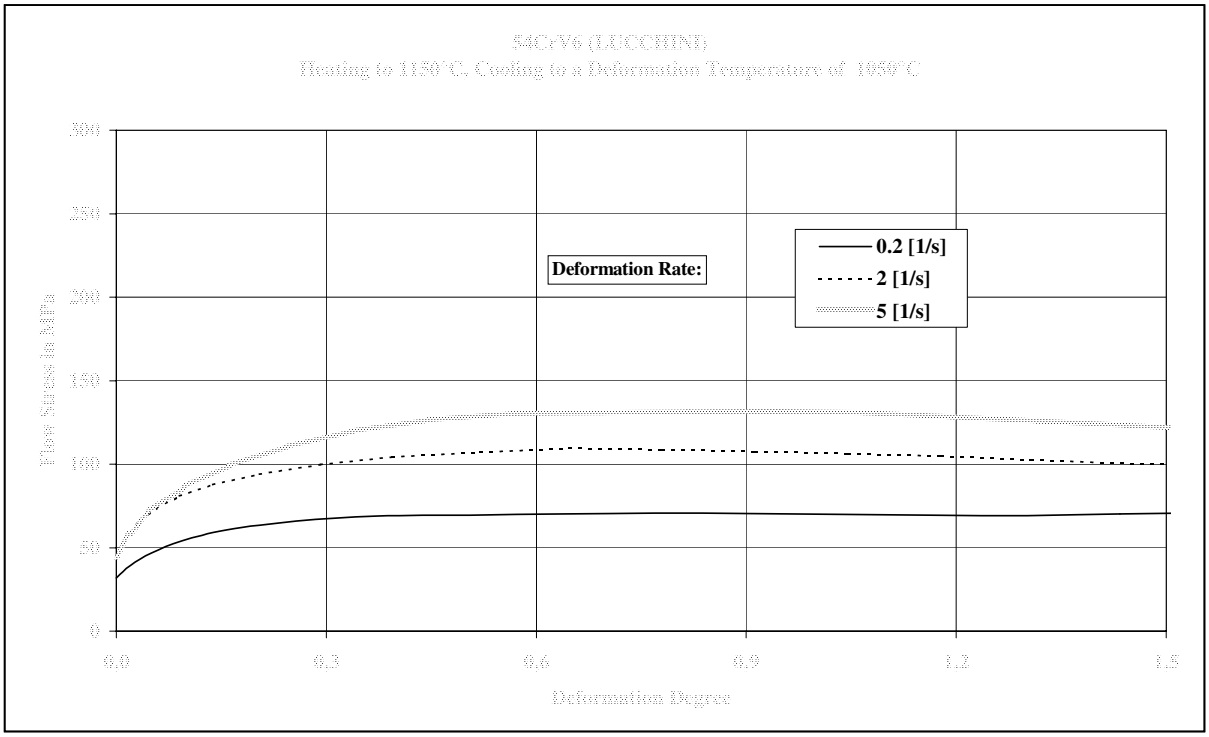


Fig. 3.1.32 Flow stress (flow curves) of steel 54CrV6 (LUCCHINI) at 1050°C.

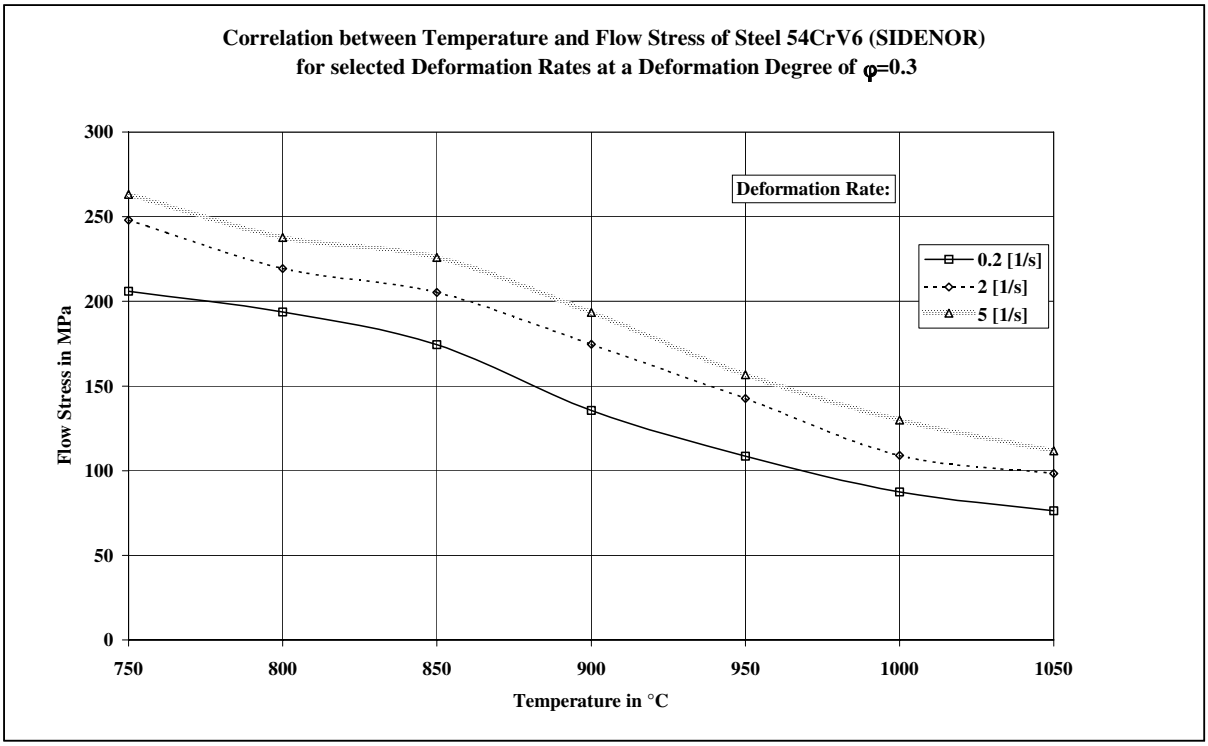


Fig. 3.1.33 Correlation between Temperature and flow stress, steel 54CrV6 (SIDENOR).

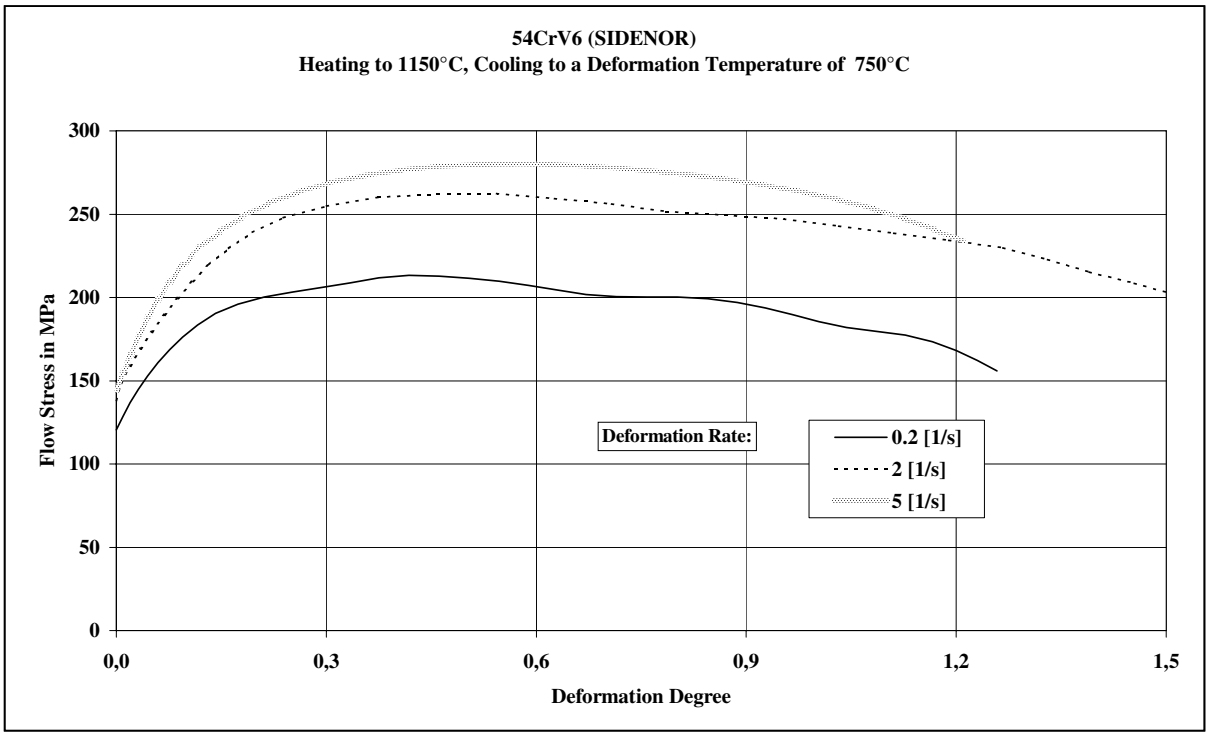


Fig. 3.1.34 Flow stress (flow curves) of steel 54CrV6 (SIDENOR) at 750°C.

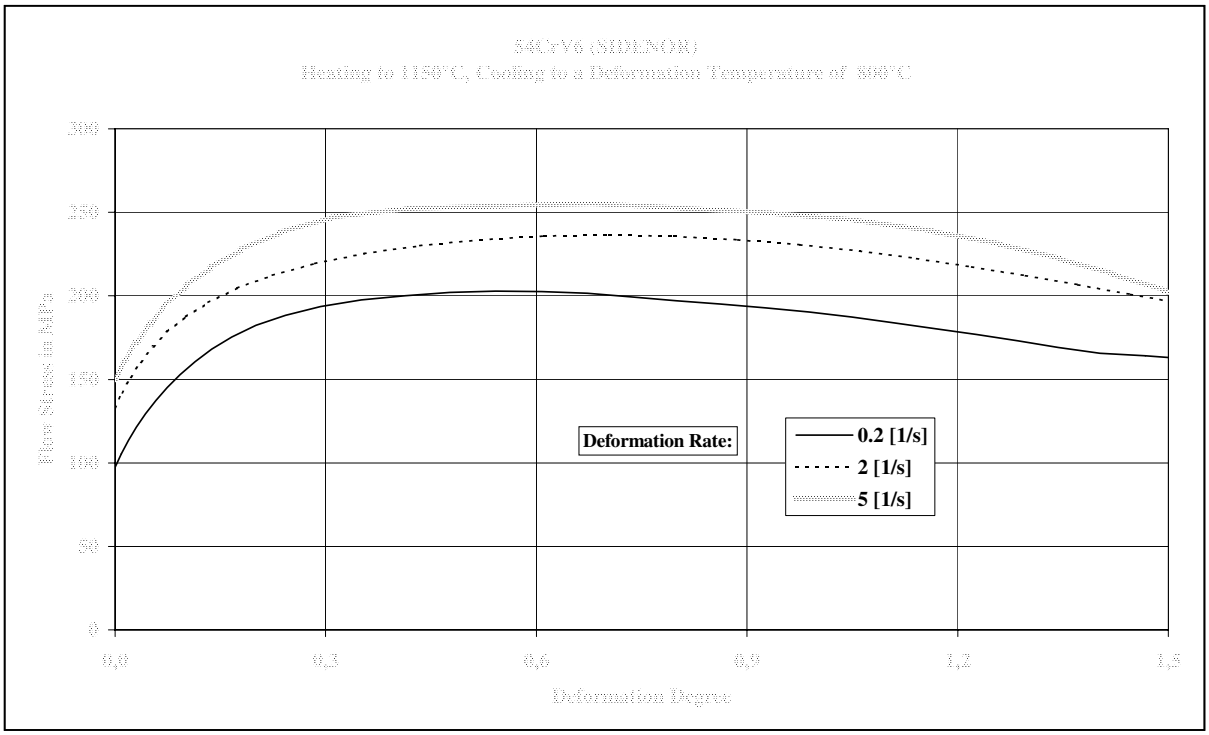


Fig. 3.1.35 Flow stress (flow curves) of steel 54CrV6 (SIDENOR) at 800°C.

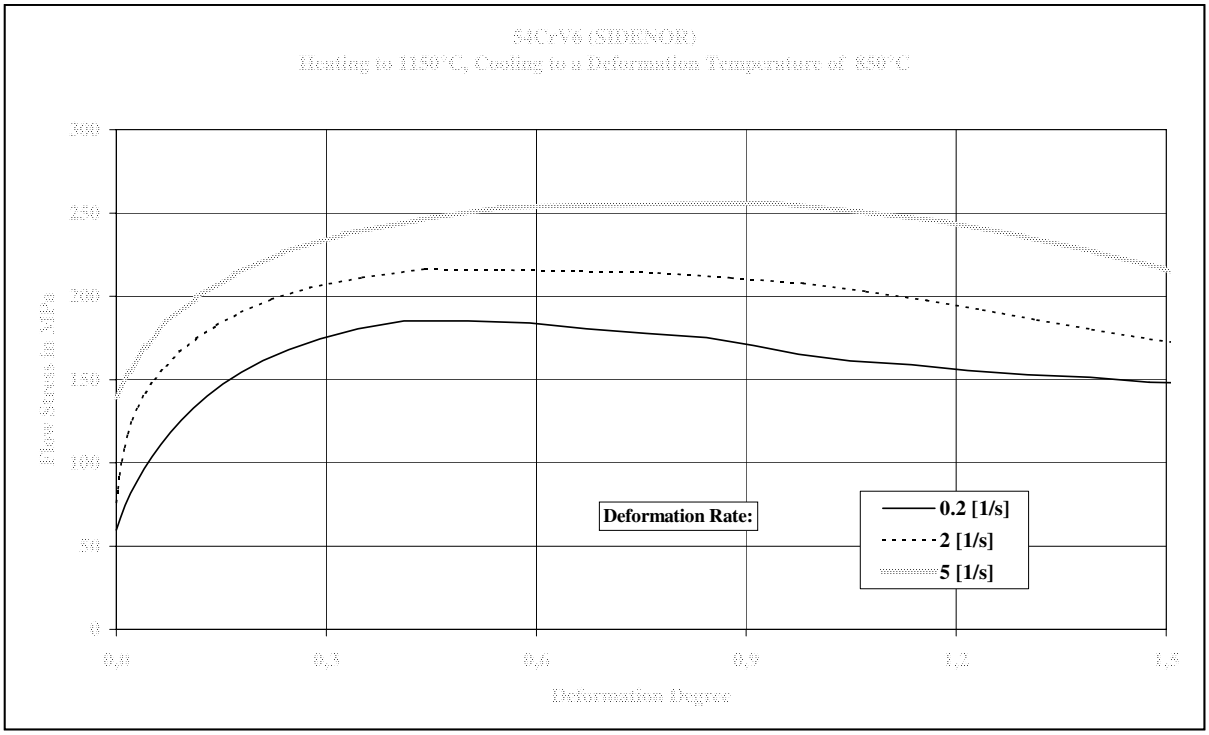


Fig. 3.1.36 Flow stress (flow curves) of steel 54CrV6 (SIDENOR) at 850°C.

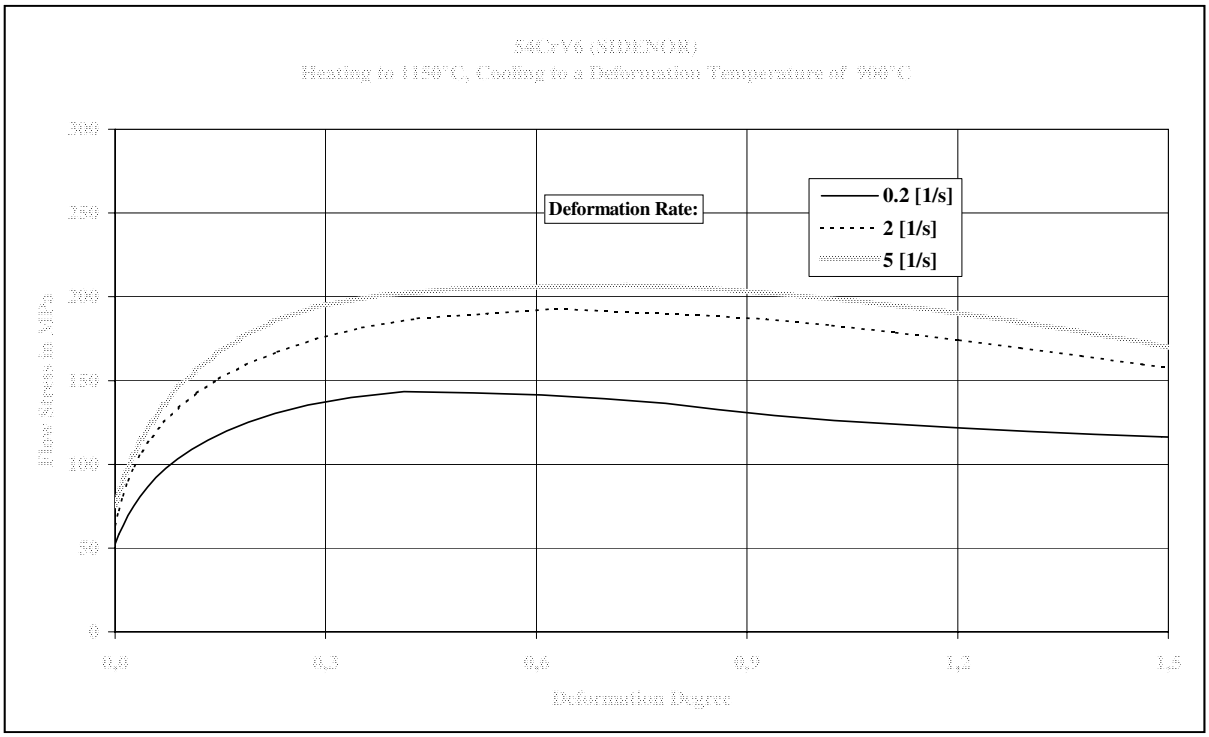


Fig. 3.1.37 Flow stress (flow curves) of steel 54CrV6 (SIDENOR) at 900°C.

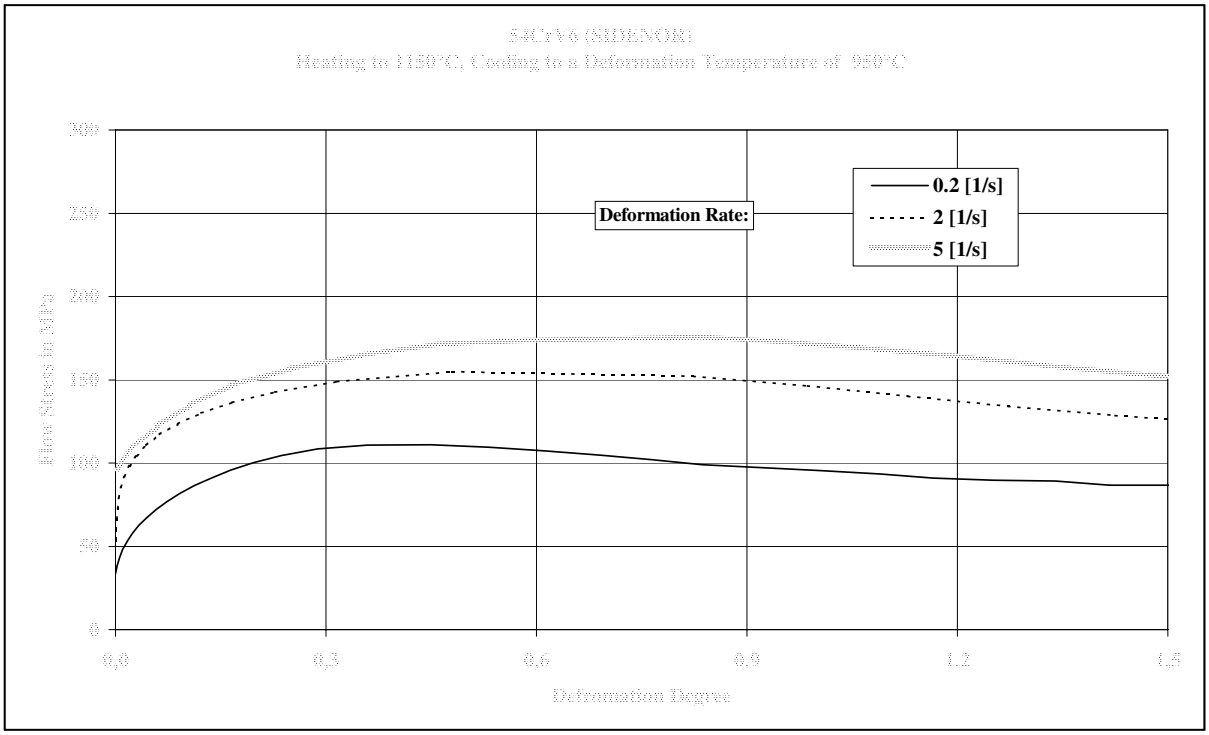


Fig. 3.1.38 Flow stress (flow curves) of steel 54CrV6 (SIDENOR) at 950°C.

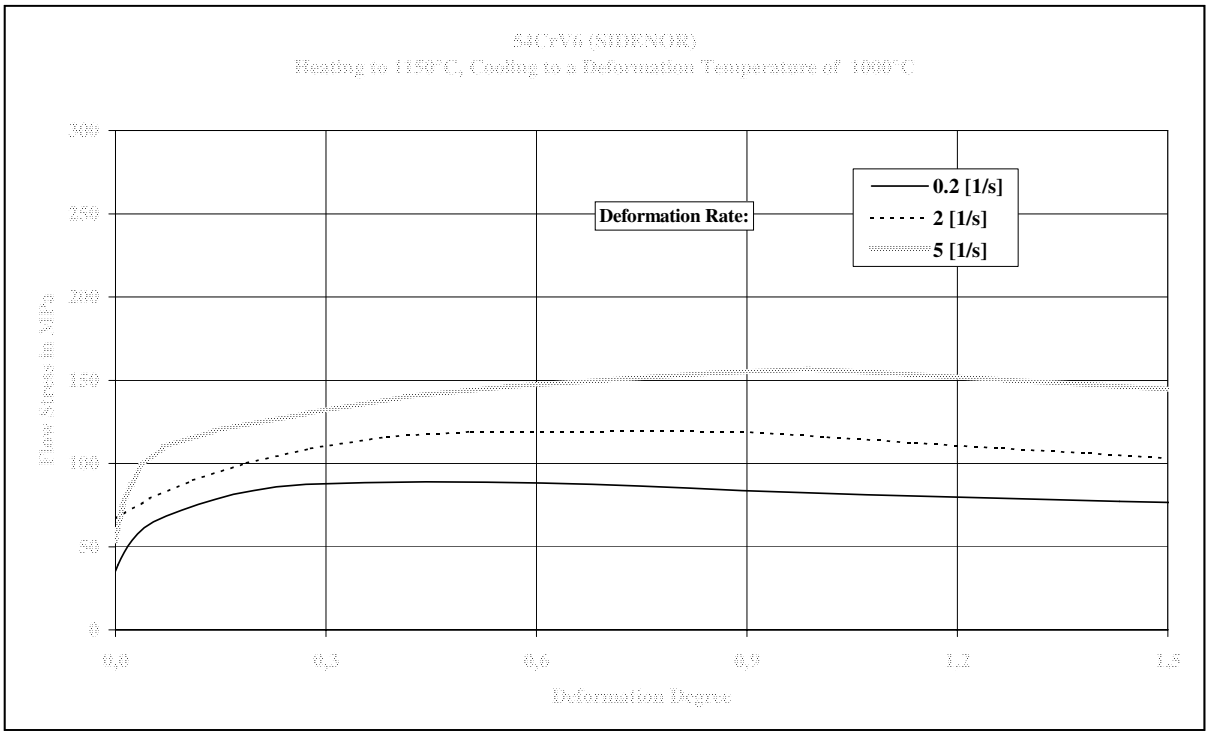


Fig. 3.1.39 Flow stress (flow curves) of steel 54CrV6 (SIDENOR) at 1000°C.

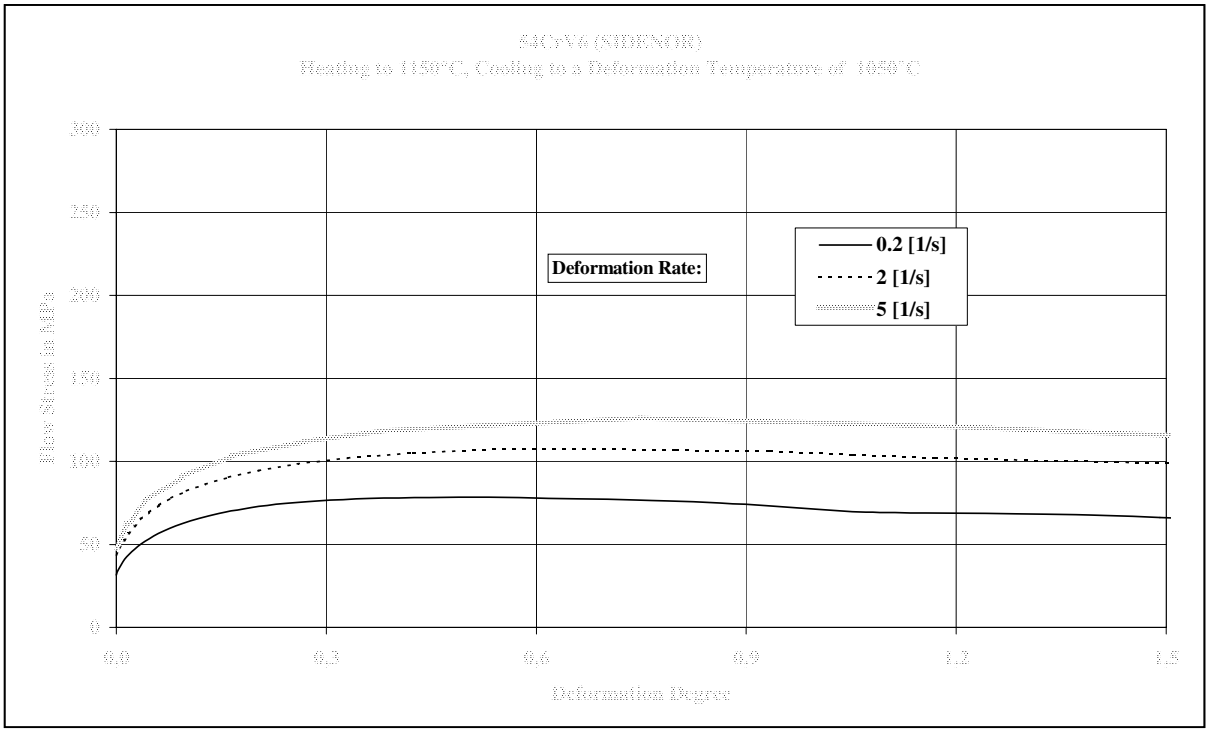


Fig. 3.1.40 Flow stress (flow curves) of steel 54CrV6 (SIDENOR) at 1050°C.

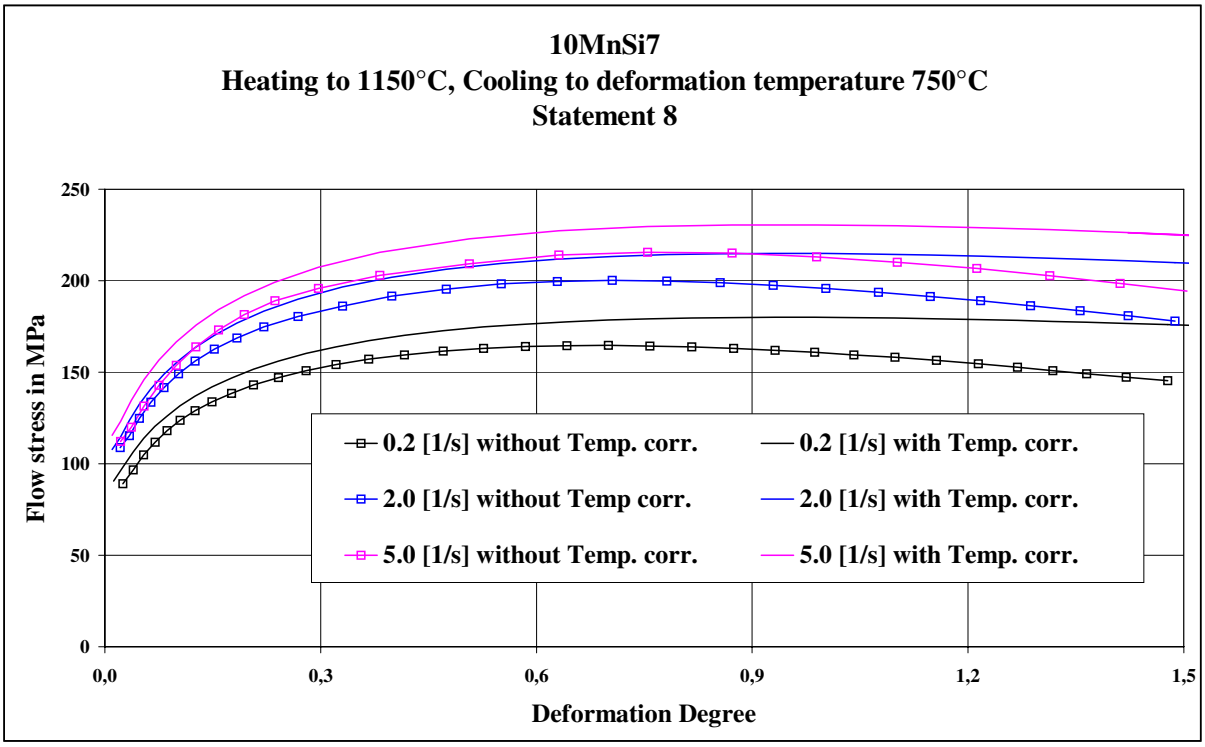


Fig. 3.1.41 Flow stress calculations steel 10MnSi7, 750°C, comparison of calculations by statement 8 with and without temperature correction.

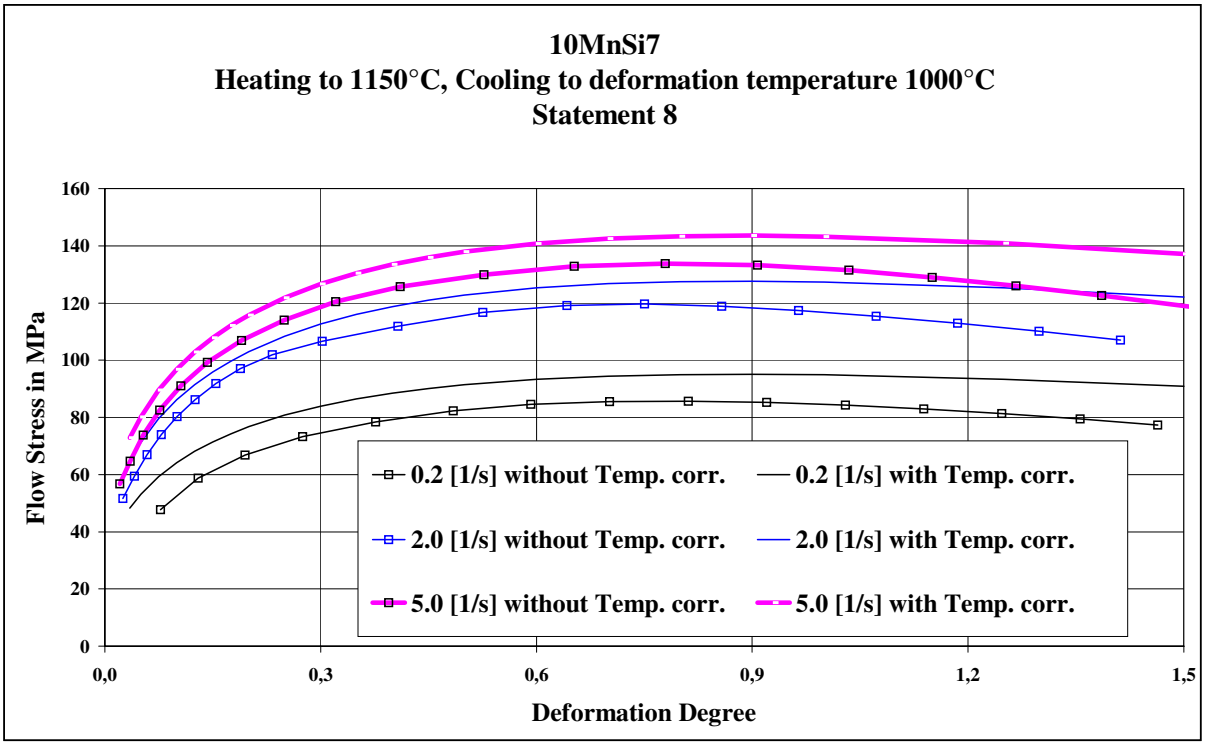


Fig. 3.1.42 Flow stress calculations steel 10MnSi7, 1000°C, comparison of calculations by statement 8 with and without temperature correction.

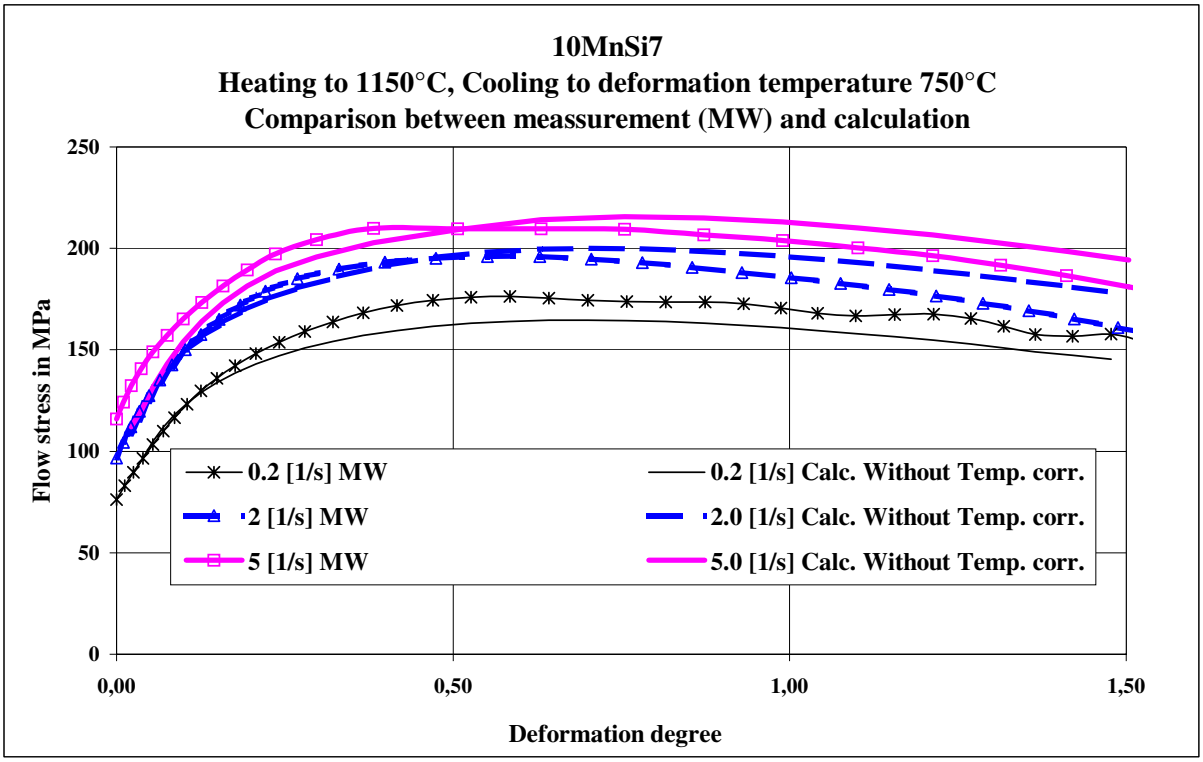


Fig. 3.1.43 Flow stress calculations steel 10MnSi7, 750°C, comparison between measurement (MW) and calculation by statement 8.

ANNEX 3.2

Grain growth after recrystallization (WP3)

Hot compression trials have been performed by the Gleeble 1500/20 of SIDENOR in order to study the austenitic grain growth after recrystallisation.
The testing schedule is presented in Fig.3.2.1.

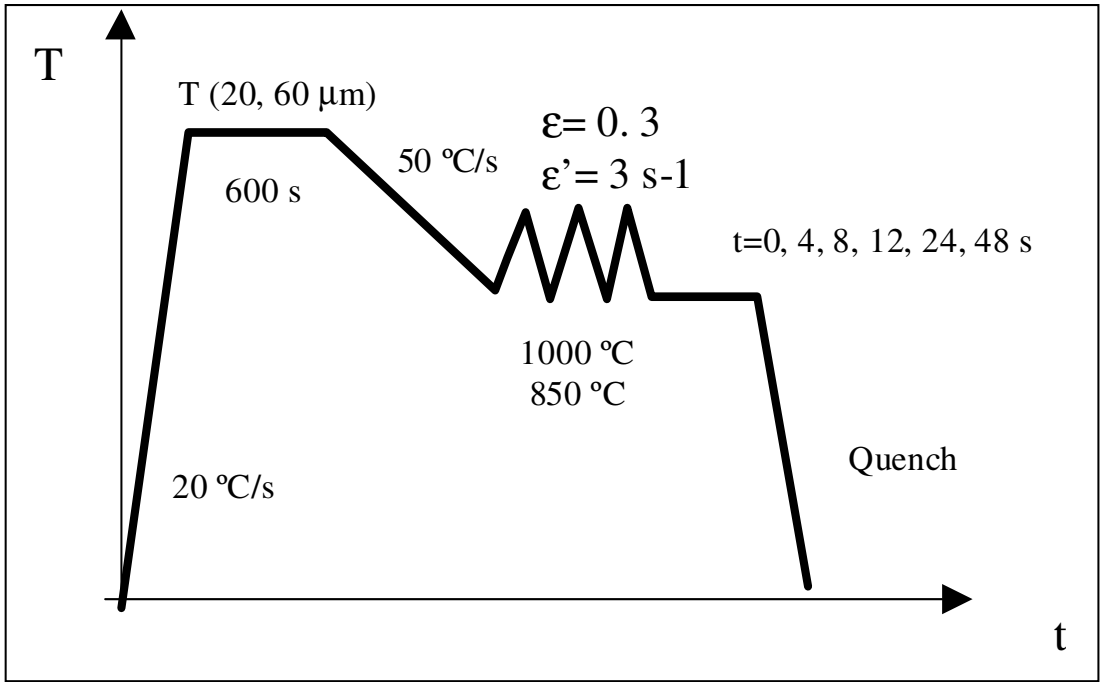


Fig.3.2.1 Hot compression schedule.

Friction and thermal gradient arising during uniaxial compression tests imply deformation heterogeneity and as a result important differences in recrystallisation and grain growth processes take place.
In order to quantify these differences FEM simulation of this test has been performed (Fig.3.2.2).

At the centre of the specimen a local strain of 0.5 is evaluated; near the surface the local strain is similar to the average one (0.3). The measurements of average austenite grain size (AGS) have been carried out at this point by linear intercept method.

The average AGS of 38Cr4 steel grade, measured after different holding times from hot deformation, is reported in Fig.3.2.4.

The typical morphology of austenite grains for the various specimens is shown in Fig.3.2.5 to Fig. 3.2.7.

It is worth to be noted that a mean value of AGS has been measured by the linear intercept method. In the trial with a deformation temperature of 1000ºC holding time of 48 s, a heterogeneous grain growth has taken place, appearing grain with sizes up to 400 μm.

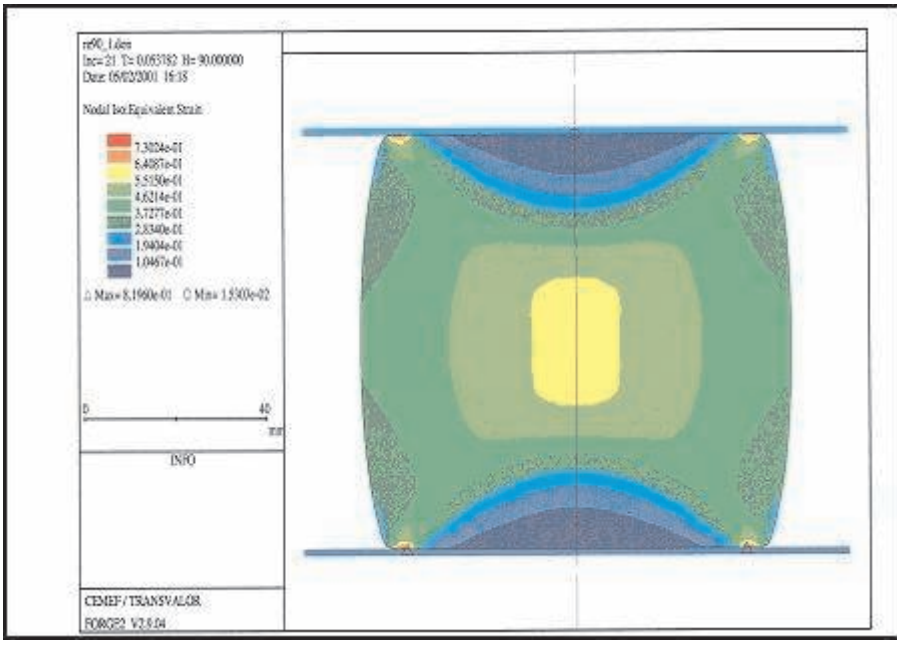


Fig.3.2.2 Uniaxial compression testing FEM simulation. Mean strain = 0.3.

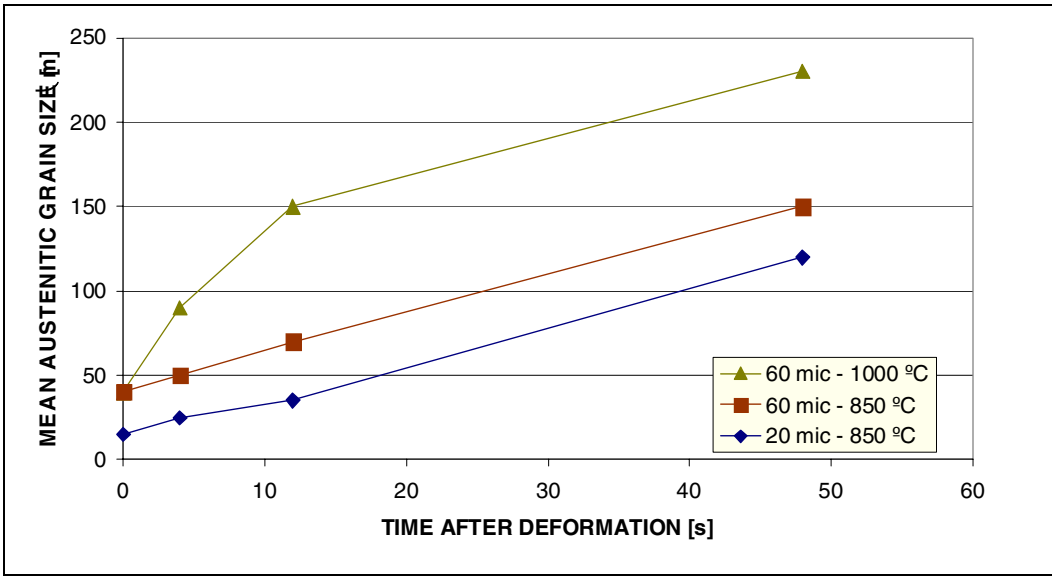
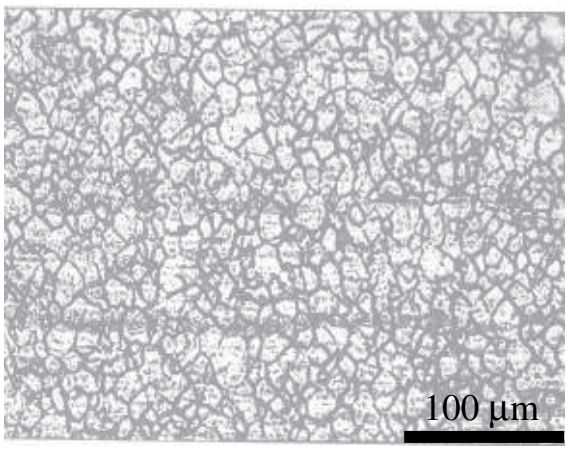


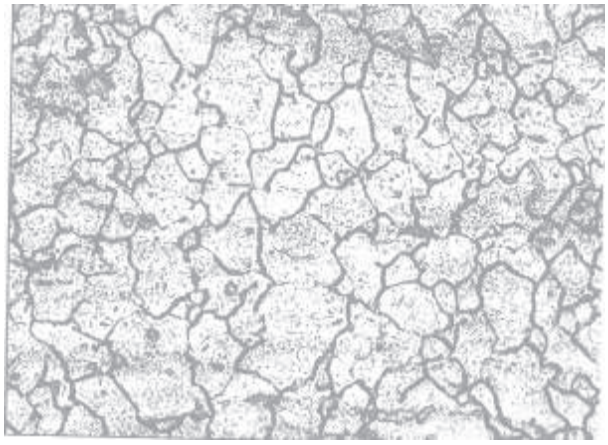
Fig.3.2.3 Austenitic grain growth of 38Cr4 steel grade of SIDENOR.

AUSTENITIC GRAIN GROWTH OF 38Cr4 STEEL GRADE

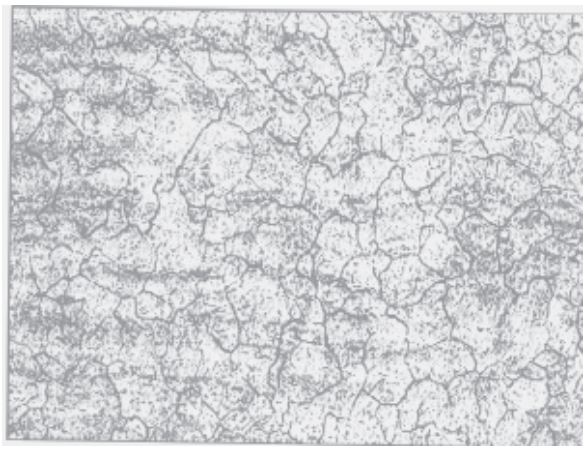
Deformation temperature: 850 ºC. Initial AGS: 20 μm.



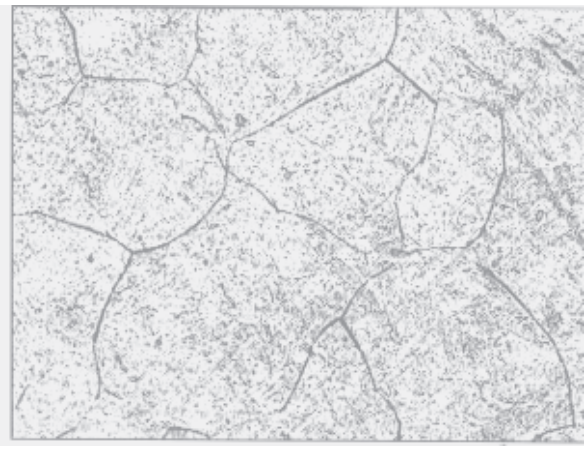
Holding time: 0 s (200x).



Holding time: 4 s (200x).



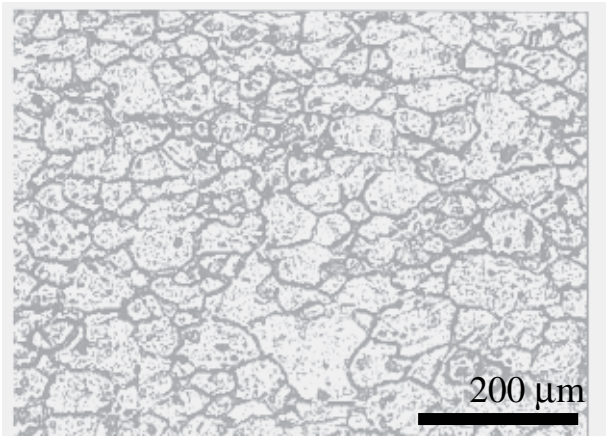
Holding time: 12 s (200x).



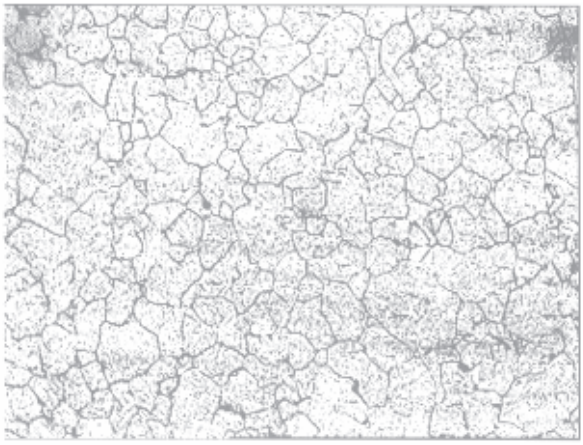
Holding time: 48 s (200x).

Fig.3.2.4 Austenitic grain growth of 38Cr4 steel grade of SIDENOR. Deformation temperature: 850 ºC. Initial AGS: 20 μm.

Deformation temperature: 850 ºC. Initial AGS: 60 μm.



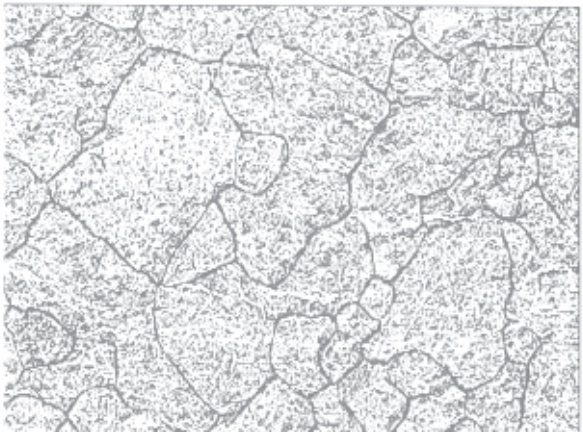
Holding time: 0 s (100x).



Holding time: 4 s (100x).



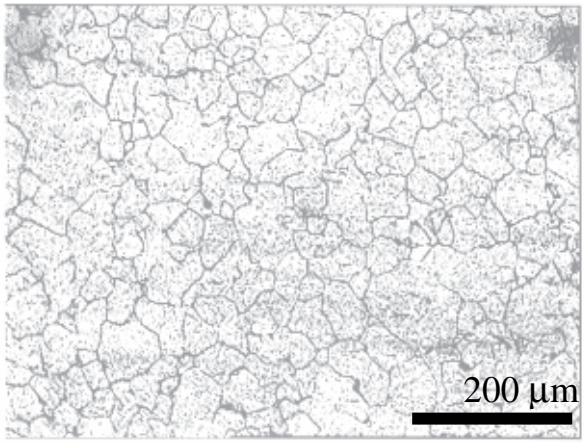
Holding time: 12 s (100x).



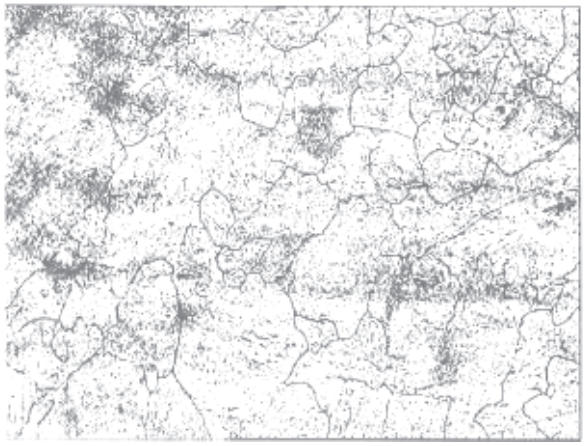
Holding time: 48 s (100x).

Fig.3.2.5 Austenitic grain growth of 38Cr4 steel grade of SIDENOR. Deformation temperature: 850 ºC. Initial AGS: 60 μm.

Deformation temperature: 1000 ºC. Initial AGS: 60 μm.



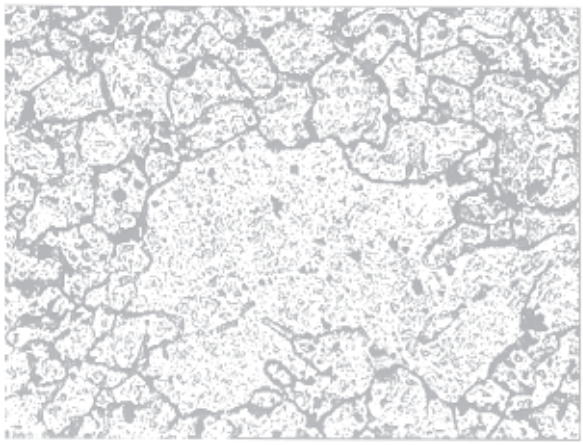
Holding time: 0 s (100x).



Holding time: 4 s (100x).



Holding time: 12 s (100x).



Holding time: 48 s (100x).

Fig.3.2.6 Austenitic grain growth of 38Cr4 steel grade of SIDENOR. Deformation temperature: 1000 ºC. Initial AGS: 60 μm.

ANNEX 3.3

Phase Transformation (WP3)

Dilatometry Experiments

Isothermal holding and intermediate quenching have been used to study grain growth during austenitizing stage.
TTT and CCT diagrams were constructed for various prior austenite grain sizes (AGS).
Sidenor investigated its steels 2 and 4, whilst CSM and Lucchini focused on steels 1, 3 and 4 produced by Lucchini.

Austenitising Tests

In order to define initial AGS in the planned dilatometry tests, different austenitising temperatures were tested as shown in Fig.3.3.1.

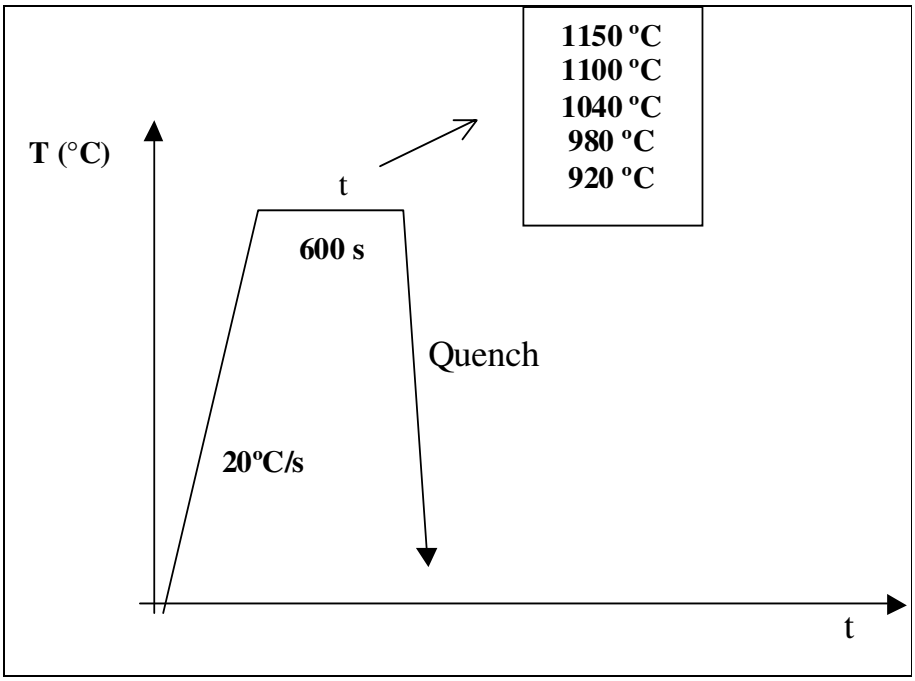


Fig.3.3.1 Testing procedure to define initial AGS.

These tests were performed in the same equipments to be used for the dilatometry and hot compression tests (Theta Dilatronic III, Adamel L'Homargy DT-1000, and Gleeble 1500/20, respectively).

In Fig.3.3.2, a schematic description of the specimens is shown for Gleeble and Adamel DT-1000. The specimens of the Theta dilatometer are similar, but the diameter is 3.25 mm and the height is 6 mm.

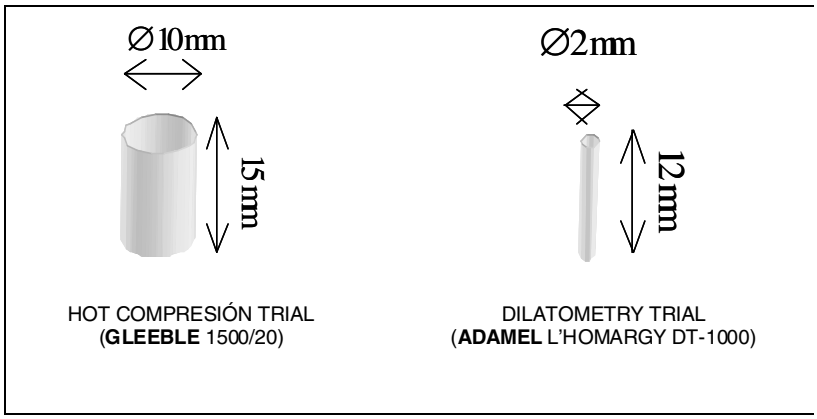


Fig.3.3.2 Description of the specimens.

The results in terms of AGS on Sidenor steels 38Cr4 and 54SiCrV6 are shown in Fig.3.3.3.

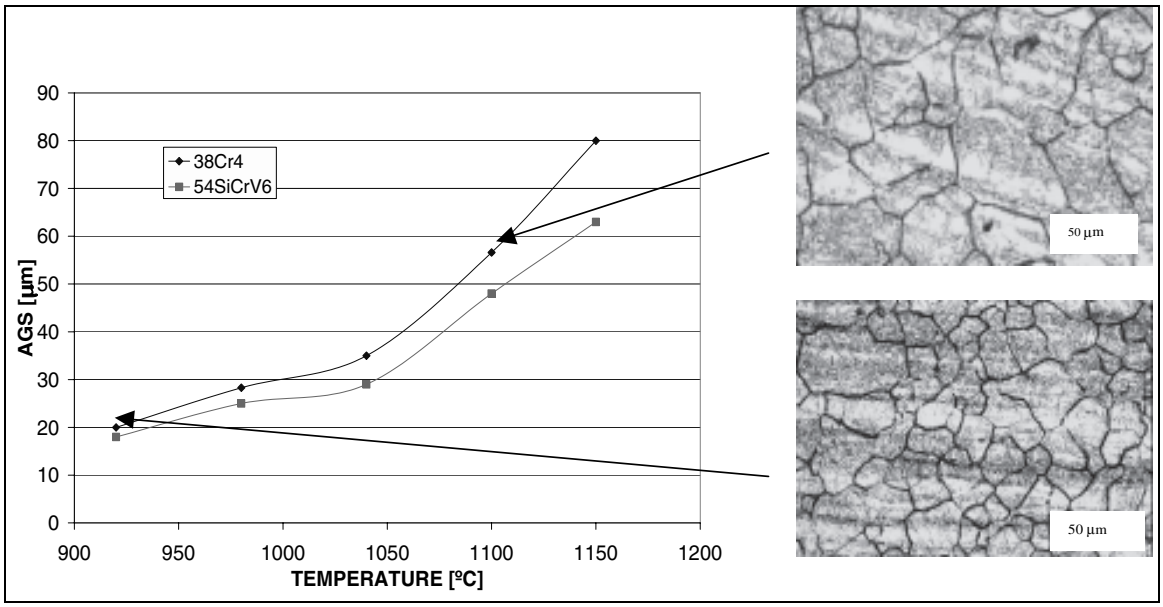


Fig.3.3.3 AGS at different austenitising temperatures for Sidenor steels 38Cr4 and 54SiCrV6, and examples of AGS in 38Cr4 steel.

As far as the target austenitic grain sizes of 20 and 60 µm, they can be achieved using the selected austenitising conditions in Table 3.3.1:

	Conditions to obtain AGS of	
Steel	**20 µm**	**60 µm**
38Cr4	920 ºC-10 min	1100 ºC-10 min
54SiCrV6	920 ºC-10 min	1150 ºC-10 min

Table 3.3.1 Austenitising conditions to obtain a given average prior AGS.

Similar experiments were carried out by CSM on the steels manufactured by Lucchini. The results are shown in Fig.3.3.4.
Other heating cycles were simulated by Lucchini in order to reproduce those used for the torsion tests. The dilatometric samples after heating at 1150 °C were cooled from the various temperatures used for the hot torsion (i.e. 750, 800, 850, 900, 950, and 1000 °C) using liquid nitrogen as cooling agent to have a very effective quenching. The AGS in ASTM No. was measured for some steels (Table 3.3.2).
The only grade which gave always some problems in the determination of AGS was steel 10MnSi7, for which, furthermore, the sample quenched from 750°C, exhibited also the presence of ferrite that starts to form around 800°C.
No problems were met with the other two steel grades: 30MnB4 and 54SiCrV6.

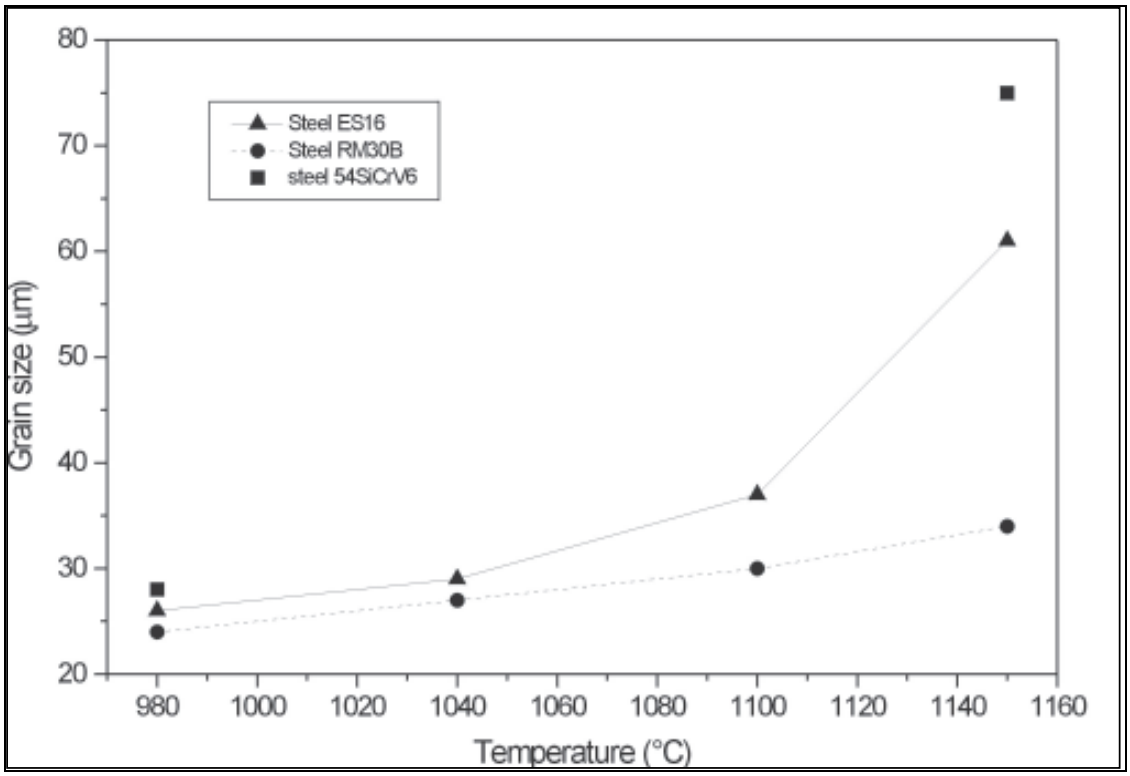


Fig.3.3.4 Evolution of average prior austenitic grain size as a function of temperature (holding time 600 s) for the steels produced by Lucchini.

	AGS (ASTM No.)					
Steel	750 °C	800 ° C	850 °C	900 ° C	950 °C	1000 °C
10MnSi7	5/6	5/6	5/6	6/5	6/5	6/5
30MnB4				4/5		4/5
54SiCrV6	4/5	5/4	5/4	5/4	5	6/5

Table 3.3.2 AGS (ASTM No.) after simulation of thermal cycles used in the hot torsion tests.

Critical Temperatures

The critical temperatures Ac_1, Ac_3 and Ms were determined by CSM on all the considered steels (Table 3.3.3).

Steel Code	Steel Type	Steel Producer	Ac_1	Ac_3	Ms
0 & 1	54SiCrV6	Lucchini	767	834	315
			765	837	
		Sidenor	762	830	-
			762	825	
2	30MnB4	Lucchini	729	831	420
			732	827	
3	38Cr4	Sidenor	741	807	-
			743	803	
4	10MnSi7	Lucchini	715	936	517
			720	924	

Table 3.3.3 Critical temperatures determined by dilatometry.

Continuous Cooling Transformation Diagrams

CCT diagrams were determined for various AGS values (Table 3.3.4):

Steels	No. of CCT	AGS (No. ASTM)	AGS (μm)	Notes
54SiCrV6	3	≈ 5 8 9/10	60 20 15/10	-
30MnB4	1	9	15	Coarse AGS is not possible in Ti steel due to TiN that inhibit grain growth
38Cr4	2	5 8	56 20	-
10MnSi7	3	5 7-8 8	56 25 20	-

Table 3.3.4 Constructed CCT diagrams for various AGS values.

Some CCT diagrams from Lucchini and CSM are shown in Figures from 3.3.5 to 3.3.9. Other data on continuous cooling transformation from Sidenor are reported in Tables 3.3.5 to 3.3.8.

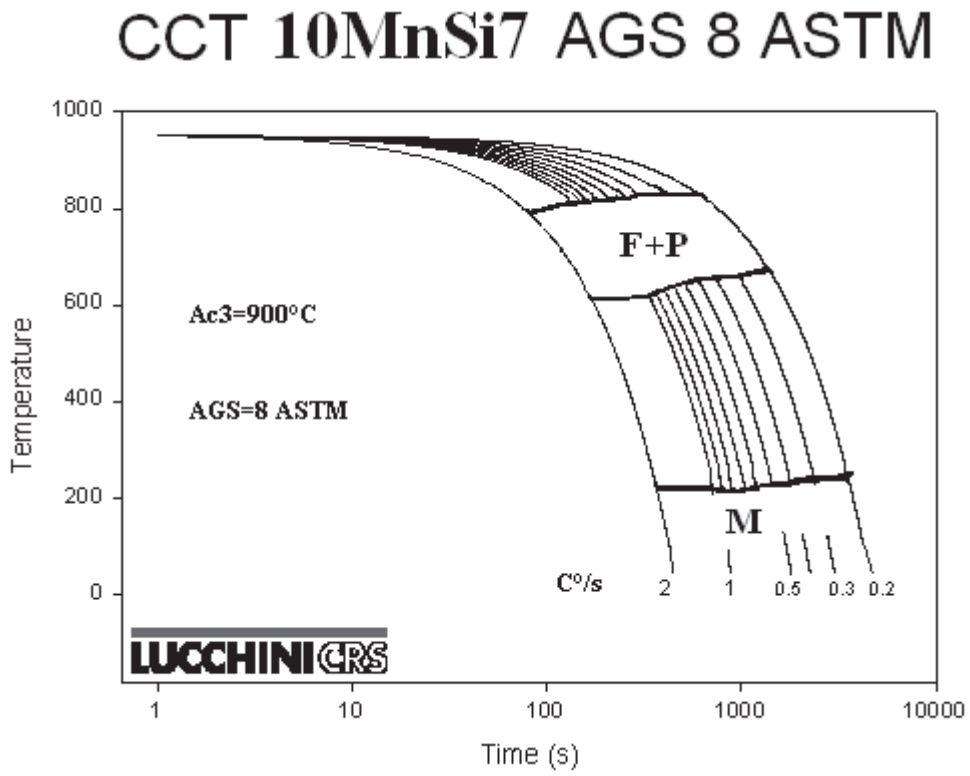


Fig. 3.3.5 CCT diagram for 10MnSi7 steel (AGS = ASTM No. 8).

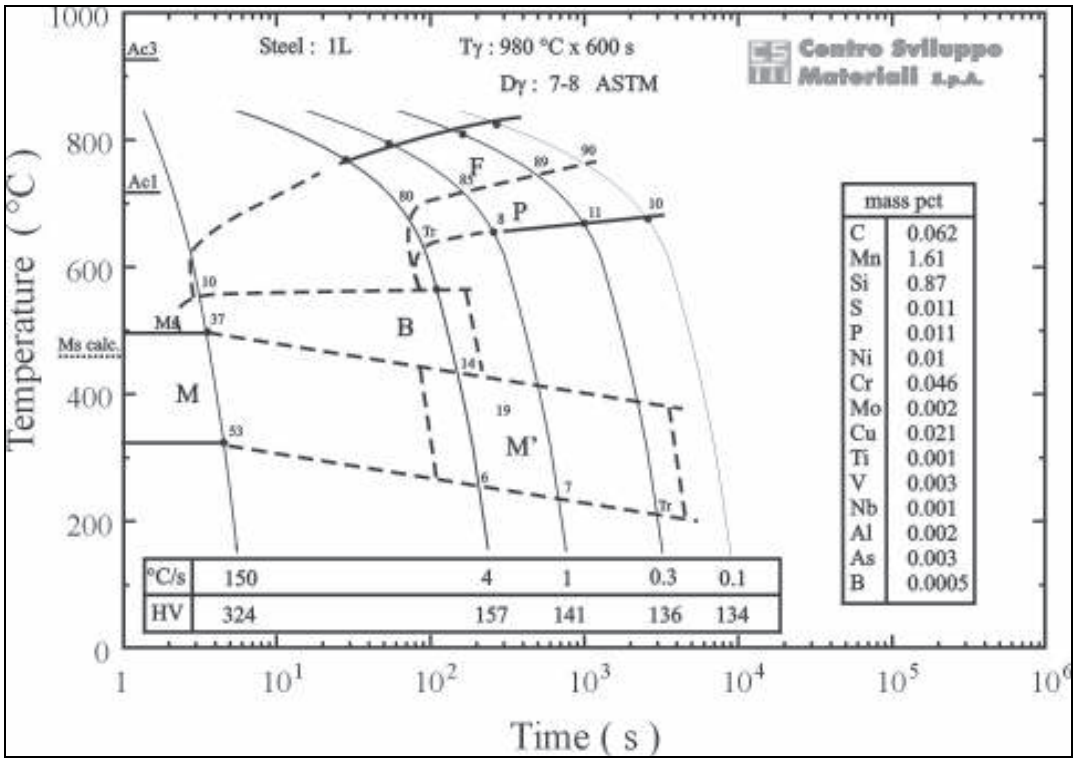


Fig. 3.3.6 CCT diagram for 10MnSi7 steel (AGS = ASTM No. 7-8).

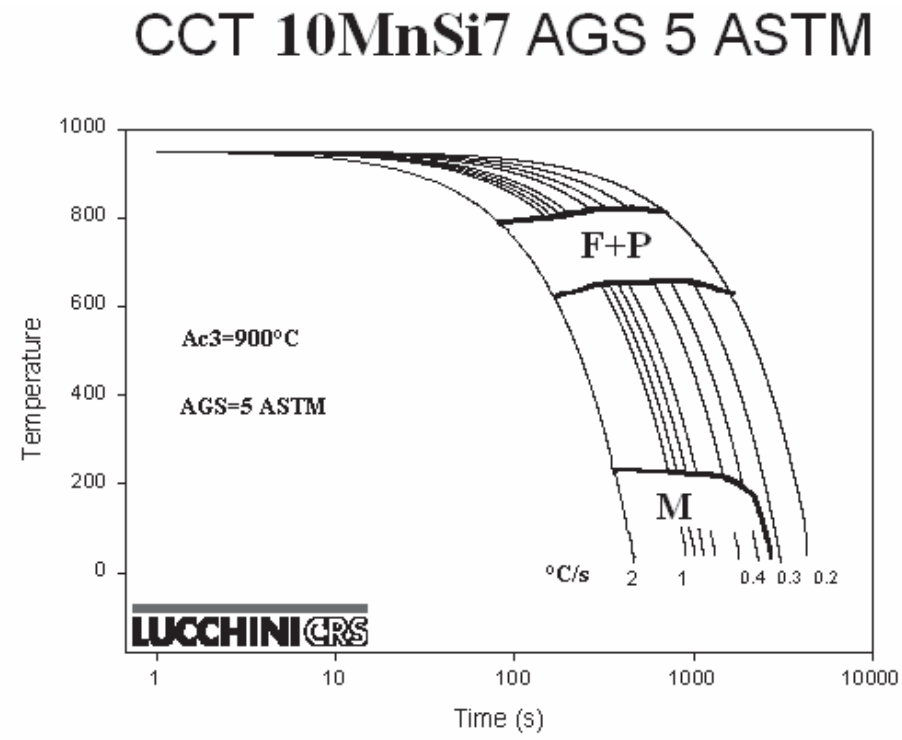


Fig. 3.3.7 CCT diagram for 10MnSi7 steel (AGS = ASTM No. 5).

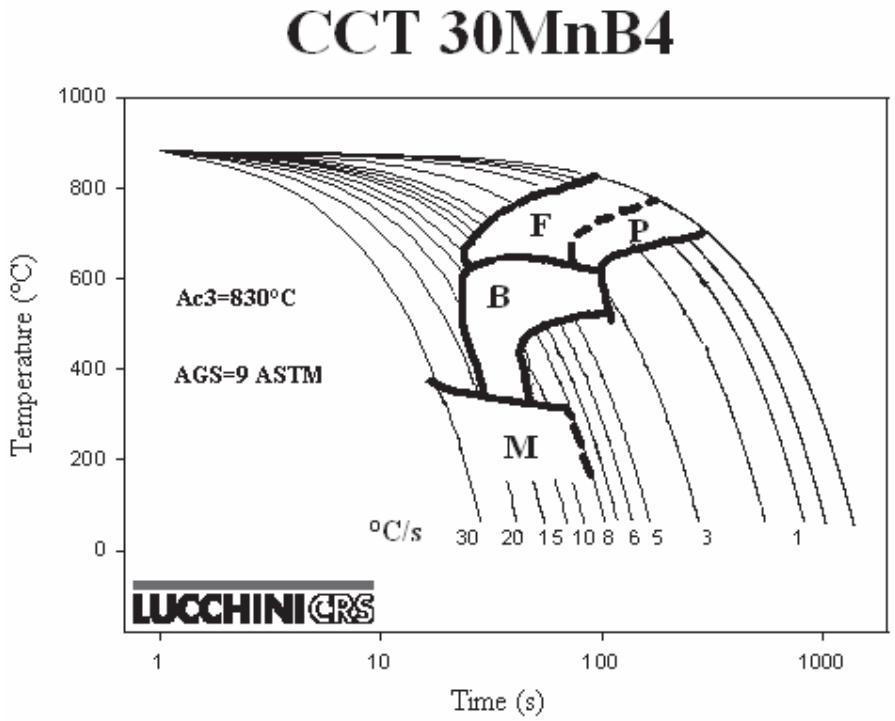


Fig. 3.3.8 CCT diagram for 30MnB4 steel (AGS = ASTM No. 9).

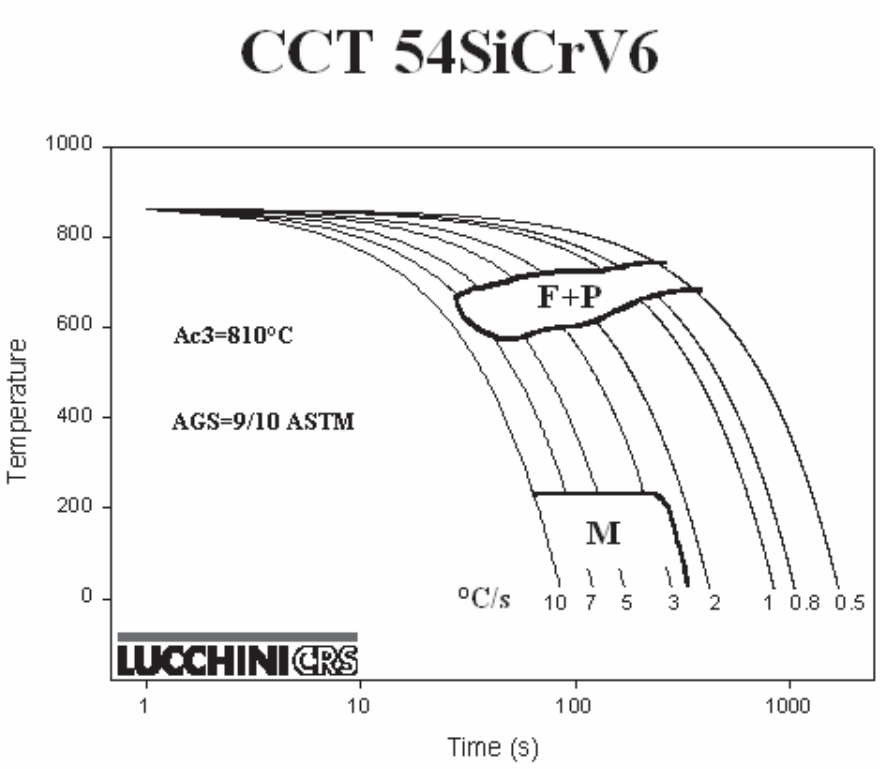


Fig. 3.3.9 CCT diagram for 54SiCrV6 steel (AGS = ASTM No. 9-10).

CR	TRANSFORMATION TEMPERATURE [°C]				HARDNESS
[°C/s]	F	P	B	M	HV30
0.5	756-710	710-669			283
1	754-650	650-662			289
4	726-600	600-542	542-350	283	472
10	712-540			314	593

Table 3.3.5 54SiCrV6 steel grade. Initial AGS: 20 µm. Transformation temperature and hardness at different cooling rates.

CR	TRANSFORMATION TEMPERATURE [°C]				HARDNESS
[°C/s]	F	P	B	M	HV30
0.5	738-710	710-654			325
1	725-704	704-624			342
4	716-545		545-306	306	535
10			543-346	346	691

Table 3.3.6 54SiCrV6 steel grade. Initial AGS: 60 µm. Transformation temperature and hardness at different cooling rates.

CR	TRANSFORMATION TEMPERATURE [ºC]				HARDNESS
[ºC/s]	F	P	B	M	HV30
0.5	753-680	680-626			192
1	750-659	659-578			205
4	735-590	590-470	470-344	344	301
10	568-510		510-362	362	390

Table 3.3.7 38Cr4 steel grade. Initial AGS: 20 µm. Transformation temperature and hardness at different cooling rates.

CR	TRANSFORMATION TEMPERATURE [ºC]				HARDNESS
[ºC/s]	F	P	B	M	HV30
0.5	753-727	727-602			261
1	739-655	655-575	575-400		296
4	610-484		484-350	350	420
10	597-446		446-385	385	430

Table 3.3.8 38Cr4 steel grade. Initial AGS: 60 µm. Transformation temperature and hardness at different cooling rates.

Isothermal Transformation Diagrams

The TTT diagrams of the various steels are shown in Figures 3.3.10 to 3.3.16

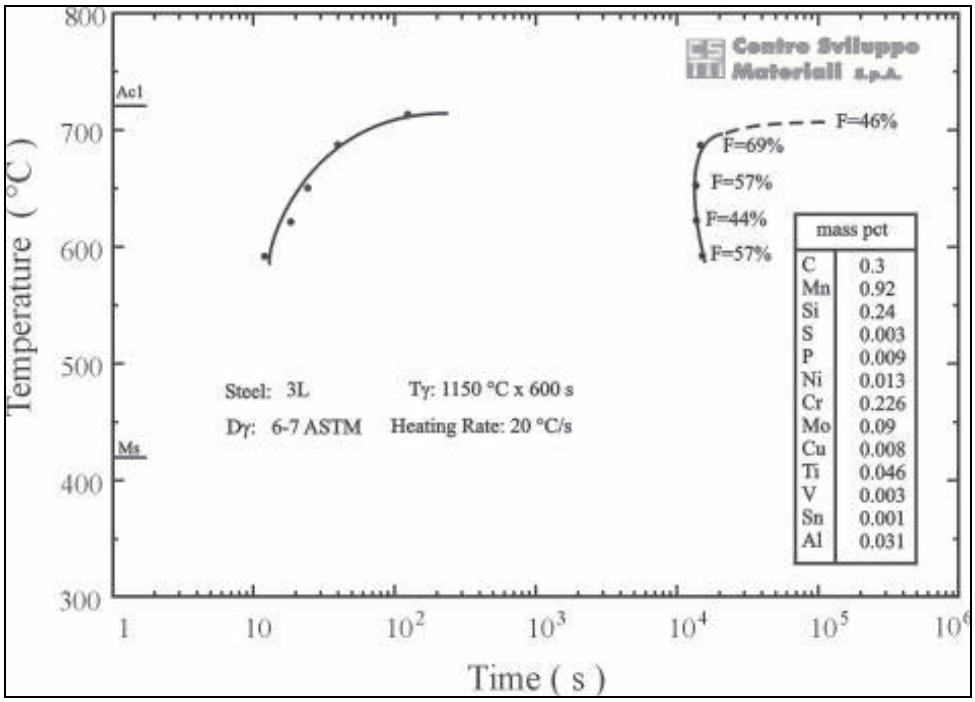


Fig. 3.3.10 TTT diagram of 30MnB4 steel (AGS = ASTM No. 6-7).

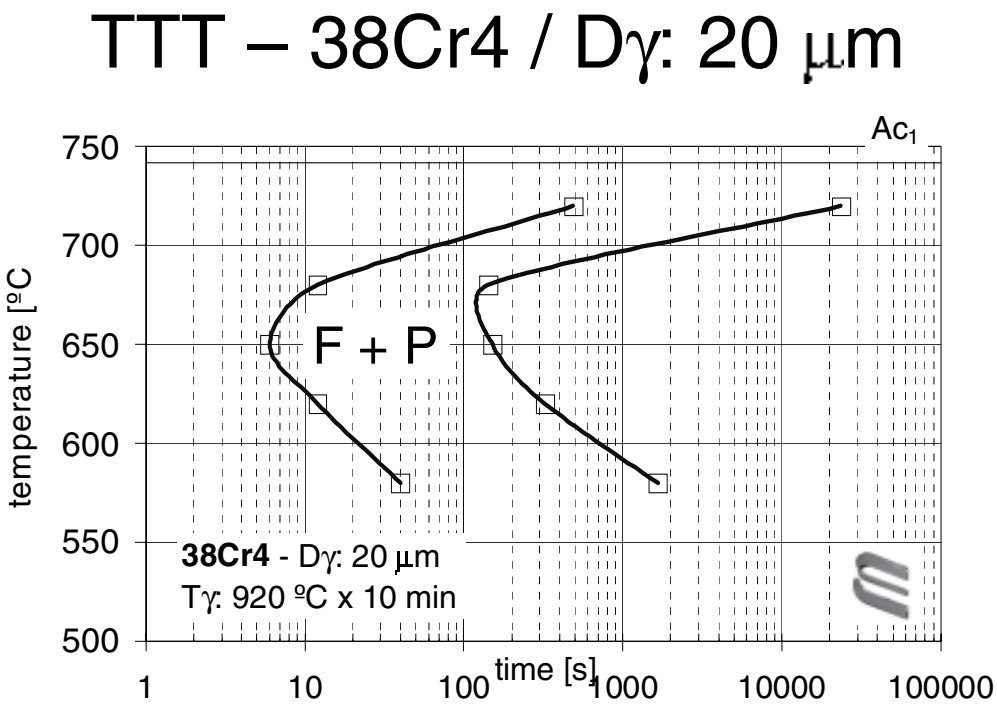


Fig.3.3.11 TTT diagram of 30Cr4 steel (AGS = ASTM No. 8).

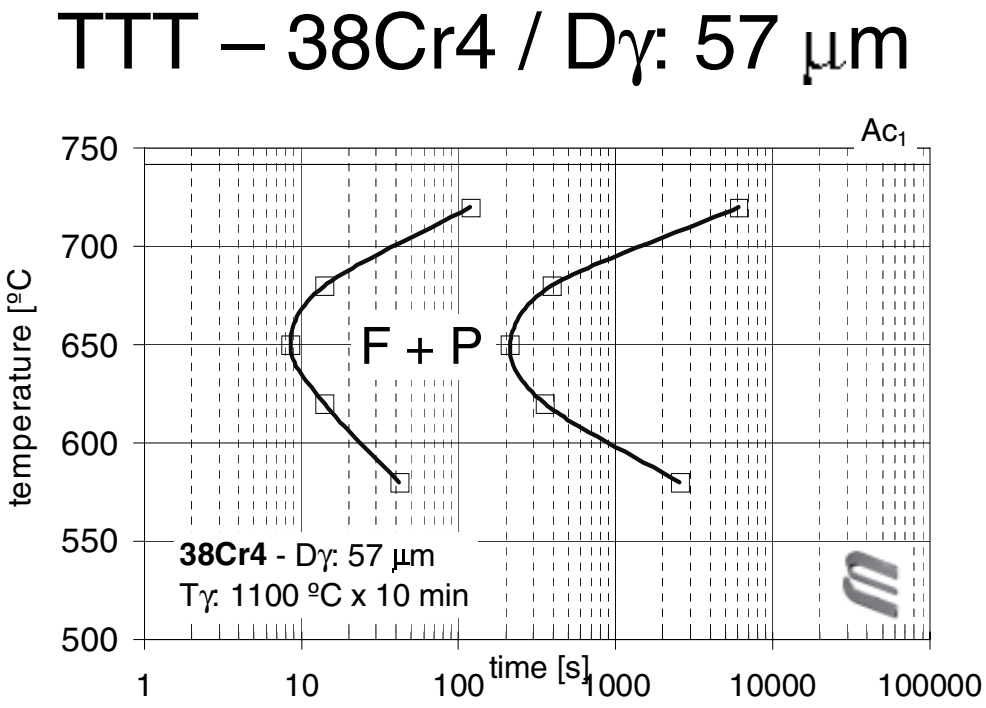


Fig.3.3.12 TTT diagram of 30Cr4 steel (AGS = ASTM No. 5).

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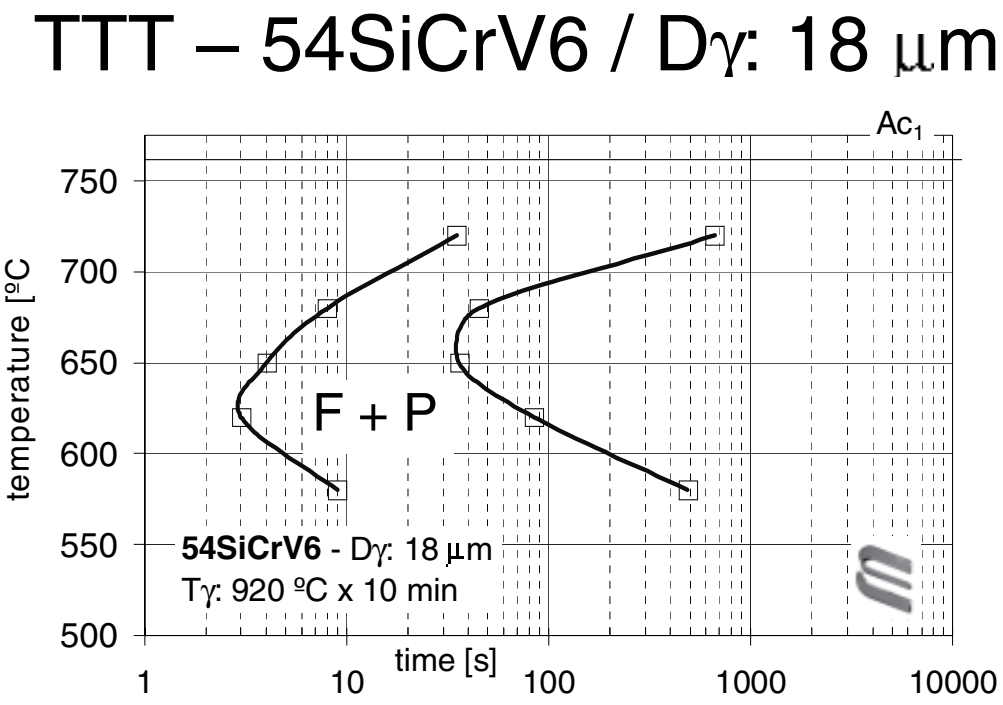


Fig.3.3.13 TTT diagram of 54SiCrV6 steel (AGS = ASTM No. 8).

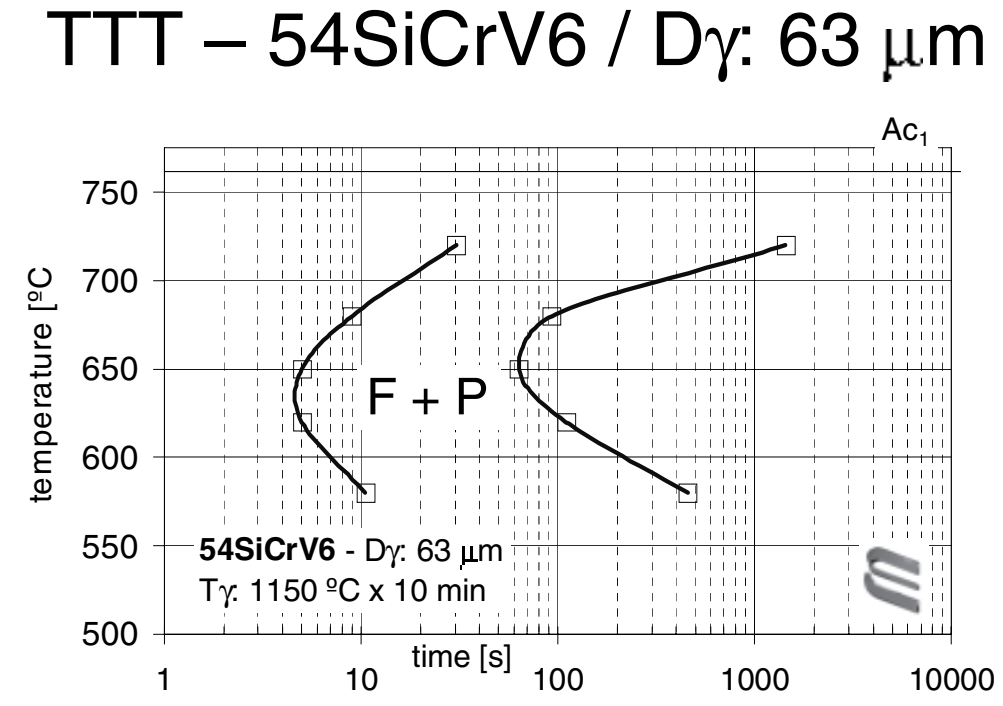


Fig.3.3.14 TTT diagram of 54SiCrV6 steel (AGS = ASTM No. 5).

ANNEX 4

Validation and Application of Mathematical models (WP4)

Austenite Evolution during Hot Rolling

With reference to the average thermo-mechanical evolution of 13 mm diameter wire rods processed at Vitoria's mill, given by Sidenor, the mean austenite grain sizes at the entry of each stand were calculated by various equations and approaches by Sidenor and CSM (Table 4.1). These calculated data were compared with the values measured on 38Cr4 steel specimens taken at various locations along the mill and rapidly quenched (< 3 s) to freeze the austenitic structure (Table 4.2).
The initial AGS after billet heating was estimated 200 μm in the CSM simulations.
The following aspects can be outlined from the data shown in Tables 4.1 and 4.2:

The AGS calculated during hot rolling by CSM approach is closer to the experimental value.

Sidenor approach appears to underestimate the real AGS, that is 36 – 46 μm after passes No. 13, 18 and 21. The experimental AGS after the laying head (18 μm) was better predicted by Sidenor. However, quenching at the mill requires 2-3 s during which grain growth can appear. This, together with the real recrystallisation mechanism operating, can explain some of the differences in AGS between calculated and experimental values.

The final AGS of 38Cr4 steel and that of 10MnSi7 steel were predicted very similar, whilst the 30MnB4 steel is finer because of the presence of Ti (C, N) particles that inhibits the grain growth.

Also the data from the hot torsion tests and the hot compression tests (WP3) were used to assess the equations used to predict the critical strain for (meta)-dynamic recrystallisation and grain growth kinetics of recrystallised austenitic grains.

Concerning the peak strain, an experimental data analysis was performed on Lucchini steel 10MnSi7 in order to verify the effect of strain rates and temperatures (Fig.4.1).

For a given strain rate a lower peak strain is obtained increasing the temperature. Increasing the strain rate from 0.2 s^{-1} up to 5 s^{-1} a 30-40% increase of peak strain has been noted as shown in Fig.4.1.

		Sidenor Approach		CSM Approach		
		38Cr4		38Cr4	10MnSi7	30MnB4
Stand	**T (°C)**	**Rex AGS**	**AGS at the entry of following pass**	**AGS at the entry of following pass**	**AGS at the entry of following pass**	**AGS at the entry of following pass**
1 D	1150	47.8	55.4	138	123	80
2 D	1140	44.9	53.1	108	102	67
3 D	1129	41.6	50.6	97	92	56
4 D	1115	38.3	47.8	88	85	50
5 D	1099	39.0	46.0	98	76	76
6 D	1079	31.3	41.4	72	71	44
7 D	1060	30.1	42.8	77	67	45
8C	1038	36.5	39.8	46	67	35
9C	1028	33.7	36.7	39	59	36
10C	1022	29.4	35.7	35	53	31
11C	1010	26.9	31.8	36	54	33
12C	1007	22.7	29.3	31	47	27
13C	**1008**	**23.1**	**30.7**	**47**	**39**	**32**
14C	1005	20.8	26.3	52	43	32
15C	1008	21.4	26.1	45	38	29
16C	1011	19.8	24.4	38	36	25
17C	1016	19.2	25.0	40	33	23
18C	**1023**	**17.8**	**22.2**	**44**	**34**	**23**
19C	1032	17.7	27.1	45	32	20
20C	1040	17.4	22.2	53	37	29
21C	**1045**	**15.5**	**29.2**	**47**	**38**	**21**
22 B	1036	14.8	15.3	58	37	34
31 B	1033	13.4				
after 0.5 s	1033		18.8	40	37	27
after 1 s	1033		23.1	42	42	27
after 4 s	1033		31.7	49	51	27
after 12 s	1033		37.1	59	60	27

Table 4.1 Calculated evolution of average austenite grain size (μm) for 13 mm diameter wire-rod.

		AGS calculated at the entry of following pass (μm)		Experimental AGS (μm) *	
Stand	**T (°C)**	Sidenor Approach	CSM Approach	Centre	Surface
13C	1008	30.7	47	42	39
18C	1023	22.2	44	46	39
21C	1045	29.2	47	40	36
* including 2-3 s for quenching					

Table 4.2 Comparison of calculated and experimental average austenite grain size (in μm) for 13 mm diameter wire-rod in 38Cr4 steel.

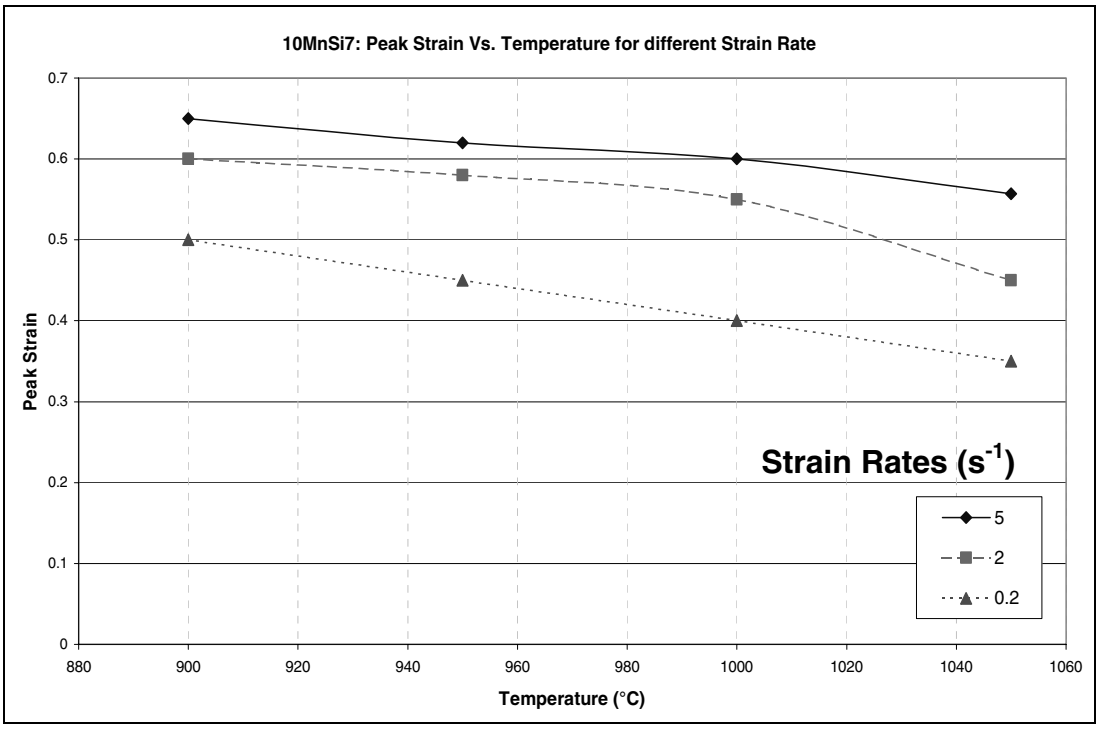


Fig.4.1 Experimental Peak strain Vs. temperature for different strain rate for 10Mnsi7 steel.

In order to check the prediction capability of the model, the comparison between the experimental values of peak strain (ε_{peak}) and those calculated by empirical equations was carried out.

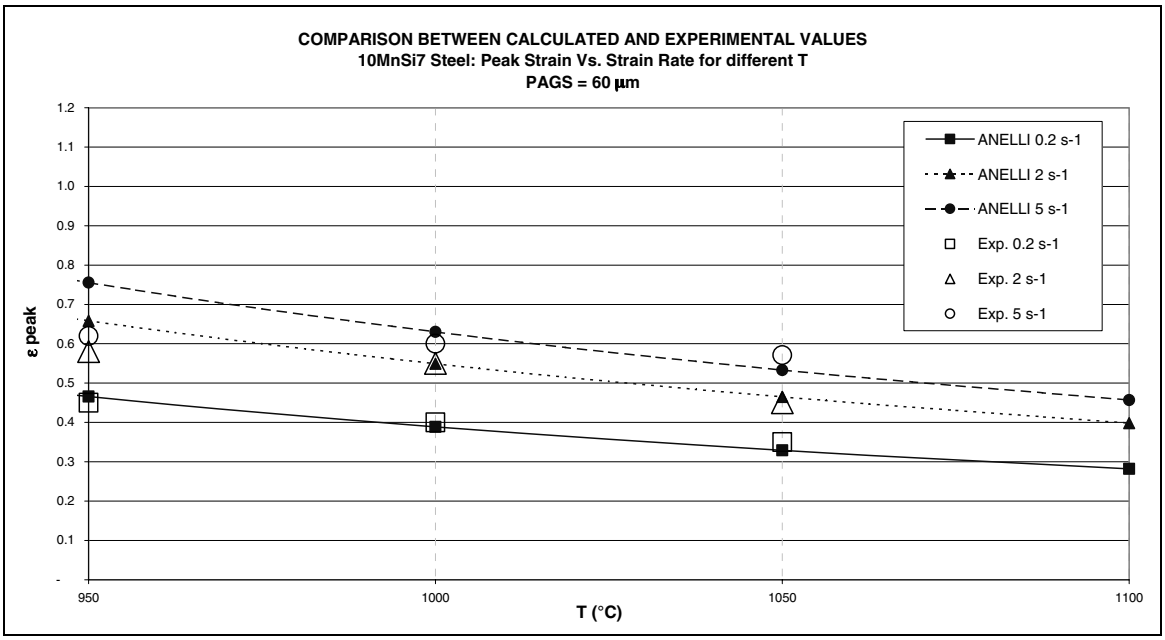


Fig.4.2 Comparison between calculated and experimental values of peak strain for 10MnSi7 steel.

The experimental values of peak strain for 10MnSi7 steel in the selected temperature range are in good agreement (Fig.4.2) with those calculated by the CSM model.
A good agreement between calculated and experimental peak strain was also found for steel 30MnB4 (Fig.4.3 and Fig.4.4).

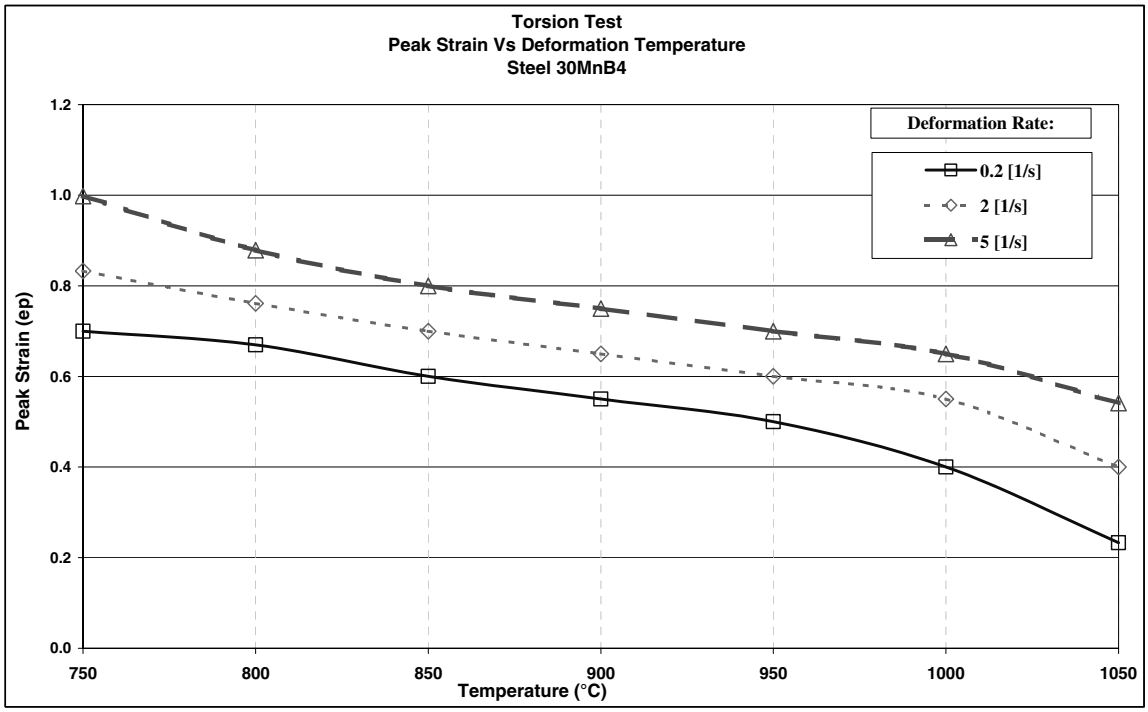


Fig.4.3 Torsion tests: peak strain deformation temperature of 30MnB4 steel.

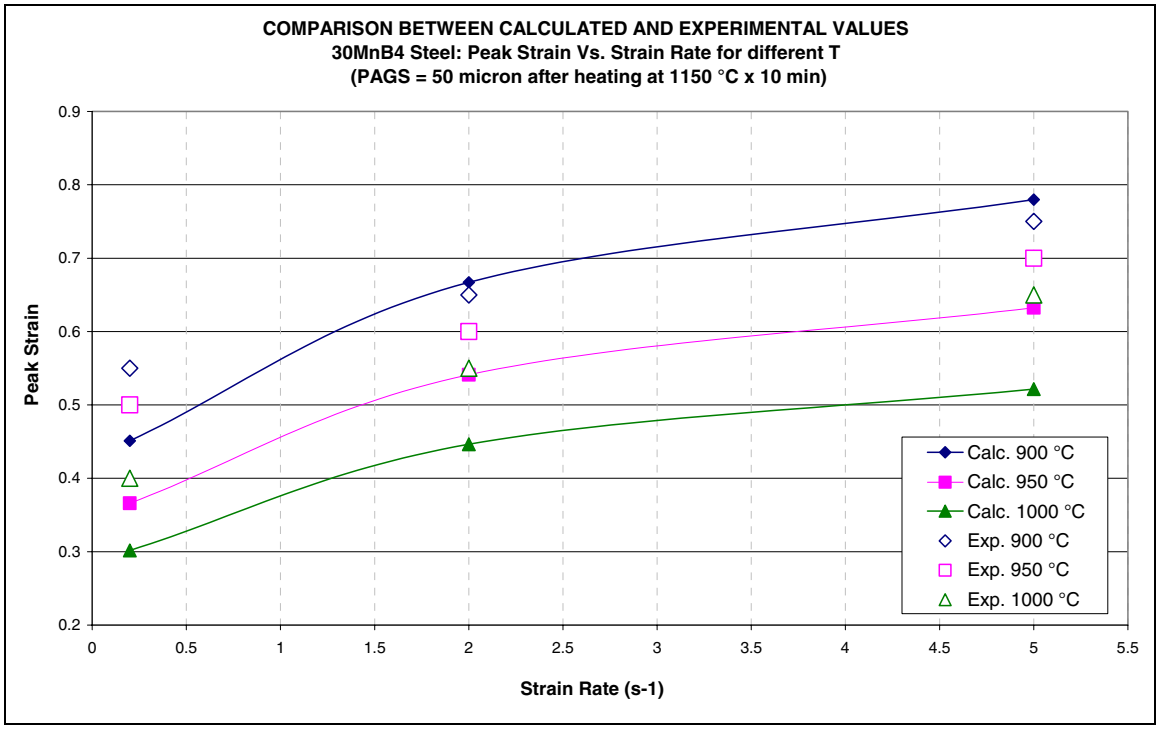


Fig.4.4 Comparison between calculated and experimental values of peak strain for 30MnB4 steel.

Phase Transformation Model

A computing algorithm having high flexibility for simulating different cooling conditions (e.g. Stelmor conveyor, EDC) has been developed to forecast the thermal and microstructural evolution after hot rolling.
The general Fourier heat equation, which takes into account the heat generated by the phase transformation, is integrated by finite differences, describing concurrently the evolution of austenite during cooling.

The austenite transformation is calculated by considering the real cooling curve as a series of micro-isotherms and applying the laws governing the isothermal austenite transformation, as derived from TTT diagrams, in accordance with the procedures proposed by various authors.

Different boundary conditions can be selected for various cooling zones, each characterised by its length, speed and heat transfer coefficient. This parameter was established for the specific plant set-up on the basis of the thermal evolution, measured by either pyrometers or equivalent devices, during cooling on the conveyor (WP1).

An example of cooling pattern simulation carried out on 30MnB4 steel, for which a TTT diagram was constructed by CSM, is given in Fig.4.5. Various values for the heat transfer coefficient were used to reproduce different average cooling rate,. The typical heat transfer coefficients which gave a calculated cooling pattern close to the experimental one were determined for the various cases experienced in the pilot trials in Freiberg and in the industrial mills of Sidenor and Lucchini.

Also phase transformation can be calculated in order to predict the constituents volume fraction along the conveyor (Fig.4.6).

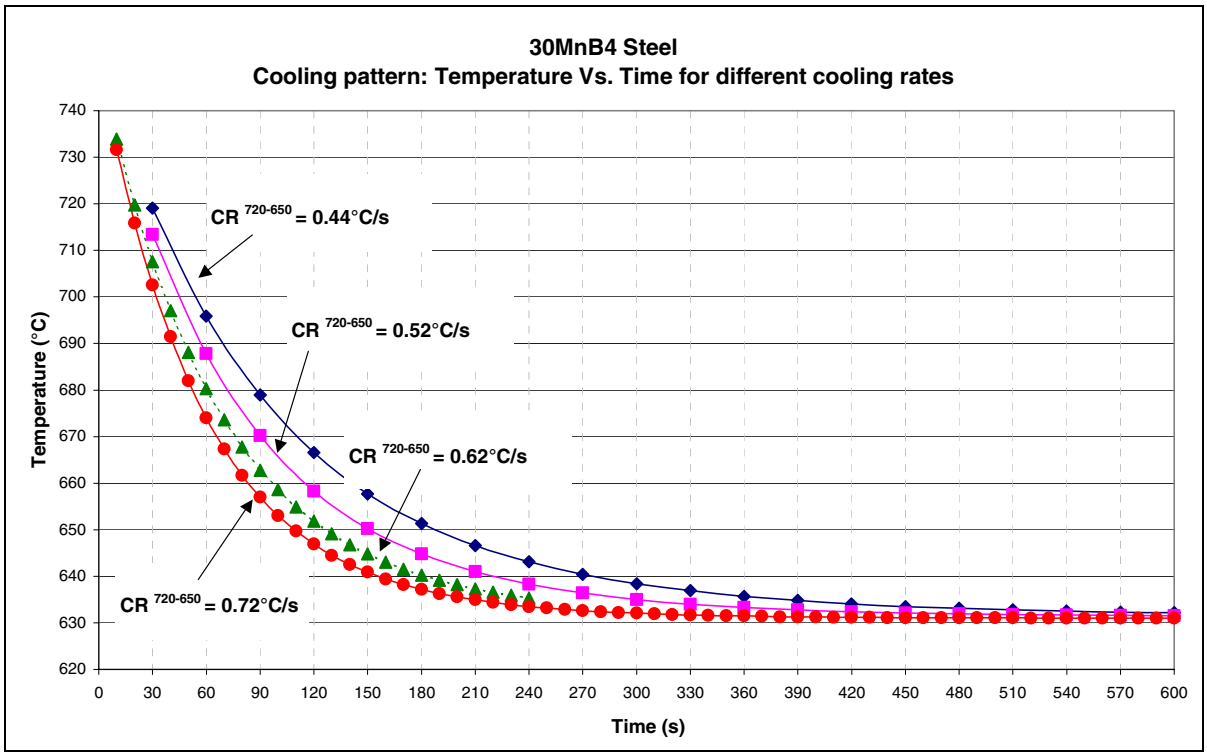


Fig.4.5 Cooling pattern of 30MnB4 for different cooling rates.

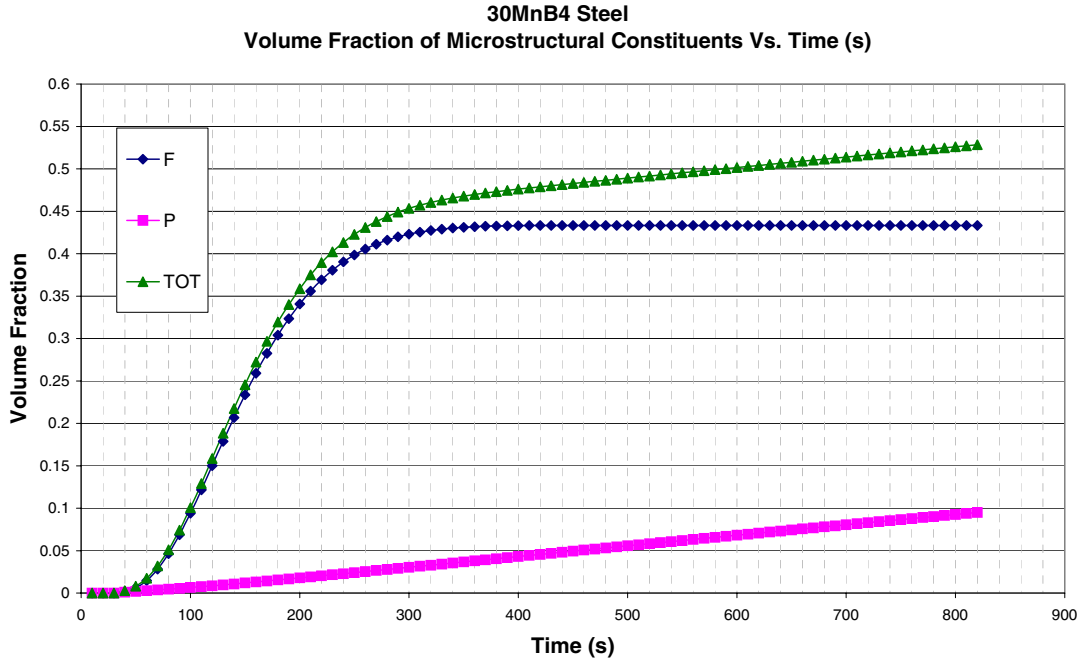


Fig.4.6 Evolution of phase transformation of 30MnB4 for a given cooling pattern.

ANNEX 5.1

Hot rolling trials at Freiberg
Pilot rolling mill (WP5)

Distances at the Freiberg four-stand continuous pilot rolling mill

Shear → G1	G1 → G2	G2 → Int. cool	Interst. Cooling	Inter. Cool → G3	G3 → G4	G4 → Pipe 1	Pipe 1	Pipe 2	Pipe 3	Pipe 4	Pipe 5	Pipe 6	Pipe 6 → Lay.head
4.000	**1.500**	**650**	**2.100**	**650**	**1.100**	**6.300**	**1.600**	**1.600**	**1.600**	**1.600**	**1.600**	**1.600**	**6.000**
5.500													
6.150													
8.250													
8.900													
10.000													
16.300													
17.900													
19.500													
21.100													
22.700													
24.300													
25.900													
32.000													

Rolling speed at single parts of the pilot mill ($\lambda_1 = 1.30$, $\lambda_2 = 1.15$, $\lambda_3 = 1.26$, $\lambda_4 = 1.20$)

Finish speed	G1	G2	Interst. Cooling	G3	G4	Pipe 1	Pipe 2	Pipe 4	Pipe 5	Pipe 6	Laying head (*)
16.0m/s	9.20m/s	10.58m/s	10.58m/s	13.33m/s	16.0m/s	16.0m/s	16.0m/s	16.0m/s	16.0m/s	16.0m/s	20.0m/s
17.0m/s	9.78m/s	11.24m/s	11.24m/s	14.17m/s	17.0m/s	17.0m/s	17.0m/s	17.0m/s	17.0m/s	17.0m/s	20.0m/s
18.0m/s	10.35m/s	11.90m/s	11.90m/s	15.00m/s	18.0m/s	18.0m/s	18.0m/s	18.0m/s	18.0m/s	18.0m/s	20.0m/s
19.0m/s	10.92m/s	12.57m/s	12.57m/s	15.83m/s	19.0m/s	19.0m/s	19.0m/s	19.0m/s	19.0m/s	19.0m/s	20.0m/s
20.0m/s	11.50m/s	13.23m/s	13.23m/s	16.67m/s	20.0m/s	20.0m/s	20.0m/s	20.0m/s	20.0m/s	20.0m/s	20.0m/s
21.0m/s	12.08m/s	13.89m/s	13.89m/s	17.50m/s	21.0m/s	21.0m/s	21.0m/s	21.0m/s	21.0m/s	21.0m/s	21.0m/s
22.0m/s	12.65m/s	14.55m/s	14.55m/s	18.33m/s	22.0m/s	22.0m/s	22.0m/s	22.0m/s	22.0m/s	22.0m/s	22.0m/s
23.0m/s	13.23m/s	15.21m/s	15.21m/s	19.17m/s	23.0m/s	23.0m/s	23.0m/s	23.0m/s	23.0m/s	23.0m/s	23.0m/s
24.0m/s	13.80m/s	15.87m/s	15.87m/s	20.00m/s	24.0m/s	24.0m/s	24.0m/s	24.0m/s	24.0m/s	24.0m/s	24.0m/s

(*) in most cases the minimum laying head speed was 20m/s

Deformation (Phi) and Deformation rate (Phi point) in each stand, depending on the rolling speed

	G1	G2	G3	G4
	Phi			
	0.456	0.442	0.452	0.493
	Phi point			
16.0m/s	336.2	291.0	363.4	461.3
17.0m/s	357.2	309.2	386.1	490.1
18.0m/s	378.3	327.4	408.8	518.9
19.0m/s	399.3	345.6	431.6	547.8
20.0m/s	420.3	363.8	454.3	576.6
21.0m/s	441.3	382.0	477.0	605.4
22.0m/s	462.3	400.2	499.7	634.3
23.0m/s	483.3	418.4	522.4	663.1
24.0m/s	504.3	436.0	545.1	691.9

Steel: 54SiCrV6 (SIDENOR), Heating to 1050°C, no holding, (Trial (I))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
001	1000	1010	1070	840	75	37	75	44	1.76	1.03	1.64	0.97	20	No	4	Quenching
002	900	920	980	670	96	54	130	62	2.15	1.42	1.61	2.06	20	No	4	Quenching
003	900	920	980	890	95	52	75	60	2.12	1.39	0.49	1.39	20	No	3	Quenching
004	890	920	990	820	94	52	110	57	2.06	1.39	1.27	1.48	22	No	4	Quenching
005	810	850	930	750	117	61	113	73	2.33	1.45	2.18	1.45	22	No	4	Quenching
006	810	850	920	760	115	60	112	73	2.30	1.42	2.15	1.45	24	No	4	Quenching
007	800	845	920	820	118	62	114	72	2.33	1.48	2.09	1.61	22	No	3	Quenching
008	1000	1010	1060	800	75	37	114	72	1.79	0.96	2.09	1.61	22	No	5	Quenching
009	1000	1020	1075	740	75	37	79	48	1.79	0.97	1.64	1.09	20	No	5	Quenching
010	1000	1010	1070	760	76	35	79	46	1.82	1.00	1.69	1.12	18	No	5	Quenching
011	1000	1020	1000	830	76	38	76	44	1.79	0.97	1.57	0.97	20	Yes	5	Quenching
012	1000	1010	980	730	76	37	78	46	1.70	1.03	1.57	1.06	18	Yes	5	Quenching
013	900	920	890	780	99	57	102	62	2.21	1.18	1.94	1.30	20	Yes	4	Quenching
014	895	920	890	760	98	50	102	62	2.06	1.33	1.85	1.30	19	Yes	4	Quenching
015	850	860	840	810	108	56	114	72	2.24	1.78	2.09	1.61	19	Yes	3	Quenching
016	850	870	840	760	108	56	112	70	2.21	1.33	2.09	1.45	19	Yes	4	Quenching

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, behind interstand cooling
ϑ 3: Finish rolling temperature after 4 passes,
ϑ 4: Temperature behind cooling line at the laying head

Steel: 54SiCrV6 (SIDENOR), Heating to 1050°C, no holding, (Trial (I))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
021	1000	1010	990	750	78	43	88	35	1.79	0.97	1.79	0.85	20	Yes	5	600°C / 600s
022	1000	1010	990	850	76	40	81	35	1.76	1.03	1.79	0.85	20	Yes	4	600°C / 600s
023	850	870	850	660	102	56	132	62	2.33	1.49	2.12	1.61	20	Yes	4	600°C / 600s
024	870	900	860	770	102	60	132	76	2.18	1.36	2.09	1.55	20	Yes	3	600°C / 600s
025	870	910	870	800	100	54	122	60	2.12	1.36	2.12	1.30	22	Yes	3	600°C / 600s
026	900	910	900	810	94	50	112	54	2.06	1.33	2.09	1.24	22	Yes	3	600°C / 600s
027	900	910	900	800	94	50	114	56	2.09	1.30	1.97	1.21	20	Yes	3	Air cooling
028	900	920	920	760	94	50	114	58	2.03	1.52	1.97	1.36	24	Yes	4	600°C / 600s

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, behind interstand cooling
ϑ 3: Finish rolling temperature after 4 passes,
ϑ 4: Temperature behind cooling line at the laying head

Steel: 54SiCrV6 (SIDENOR), Heating to 1050°C, no holding, (Trial (II))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
040	980	860	930	730									18	Yes	3	Air cooling
041	1000	860	930	860									18	Yes	2	Air cooling
042	960	850	910	820									20	Yes	2	Air cooling
043	1060	970	990	690									20	Yes	4	Air cooling
044	1060	960	980	850									20	Yes	3	Air cooling
045	1060	960	970	800									18	Yes	3	Air cooling
046	1060	980	980	760									22	Yes	4	Air cooling
047	910	780	860	900									20	Yes	1	Air cooling
048	900	800	850	700									20	Yes	3	Air cooling
049	900	780	860	800									20	Yes	2	Air cooling
050	910	790	860	760									20	Yes	2	Air cooling

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, behind interstand cooling
ϑ 3: Finish rolling temperature after 4 passes,
ϑ 4: Temperature behind cooling line at the laying head

Steel: 54SiCrV6 (SIDENOR), Heating to 1150°C, holding 150s, (Trial (III))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
060	1060	1100	1005	770									20	Yes	5	200°C / 600s
061	1070	1100	1000	750									22	Yes	5	200°C / 600s
062	1070	1110	1005	830									24	Yes	5	200°C / 600s
063	980	1015	930	800									20	Yes	3	200°C / 600s
064	1000	1050	930	750									18	Yes	3	200°C / 600s
065	950	1000	905	840									18	Yes	2	200°C / 600s
No more rods available																

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, in front of interstand cooling
ϑ 3: Finish rolling temperature after 4 passes,
ϑ 4: Temperature behind cooling line at the laying head

Steel: 54SiCrV6 (LUCCHINI), Heating to 1050°C, no holding, (Trial (I))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
101	995	1010	1010	860	81	44	90	54	1.88	1.06	1.52	1.09	20	Yes	4	Quenching
102	1000	1010	990	780	80	42	89	48	1.94	1.03	1.70	1.09	20	Yes	5	Quenching
103	1010	1020	990	730	75	36	82	44	1.67	0.97	1.42	1.03	18	Yes	5	Quenching
104	900	920	900	770	99	52	108	58	2.27	1.18	1.94	1.36	20	Yes	4	Quenching
105	900	920	870	740	100	49	114	74	2.18	1.12	2.00	1.36	22	Yes	4	Quenching
106	900	920	890	840	100	49	118	84	2.21	1.18	2.00	1.67	22	Yes	3	Quenching
107	895	920	900	780	99	50	108	62	2.15	1.12	1.97	1.52	22	Yes	4	Quenching
108	810	830	830	790	114	62	128	78	2.36	1.51	2.06	1.64	20	Yes	3	Quenching

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, behind interstand cooling
ϑ 3: Finish rolling temperature after 4 passes,
ϑ 4: Temperature behind cooling line at the laying head

Steel: 54SiCrV6 (LUCCHINI), Heating to 1050°C, no holding, (Trial (I))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
121	850	890	860	850	105	56	128	64	2.27	1.45	2.03	1.36	20	Yes	2	600°C / 600s
122	840	880	860	820	104	54	124	62	2.09	1.33	2.12	1.30	18	Yes	2	600°C / 600s
123	830	860	850	820	108	56	126	66	2.24	1.42	2.18	1.45	22	Yes	3	600°C / 600s
124	850	870	850	750	104	54	124	66	2.18	1.33	2.15	1.45	22	Yes	4	600°C / 600s
125	890	925	900	760	96	52	110	58	2.09	1.24	1.97	1.30	22	Yes	4	600°C / 600s
126	900	930	900	850	92	48	108	60	2.03	1.12	1.97	1.18	22	Yes	3	600°C / 600s
127	920	925	910	850	92	50	109	60	2.09	1.30	1.88	1.33	20	Yes	3	600°C / 600s
128	910	930	910	820	94	54	102	62	2.24	1.21	1.88	1.24	18	Yes	3	600°C / 600s
129	1000	1000	1000	810	78	44	94	50	2.00	1.12	1.48	1.12	20	Yes	4	600°C / 600s
130	990	1010	990	860	80	42	92	50	1.73	0.82	1.45	1.03	22	Yes	4	600°C / 600s
131	990	1010	990	760	80	44	88	48	1.76	0.97	1.48	1.00	22	Yes	5	600°C / 600s

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, behind interstand cooling
ϑ 3: Finish rolling temperature after 4 passes,
ϑ 4: Temperature behind cooling line at the laying head

Steel: 54SiCrV6 (LUCCHINI), Heating to 1050°C, no holding, (Trial (II))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
140	900	800	870	800									20	Yes	2	Air cooling
141	900	790	860	770									18	Yes	2	Air cooling
142	970	850	930	700									18	Yes	3	Air cooling
143	980	860	920	820									18	Yes	2	Air cooling
144	990	890	930	810									22	Yes	3	Air cooling
145	970	850	920	700									18	Yes	3	Air cooling
146	970	850	910	750									20	Yes	3	Air cooling
147	1070	970	990	830									20	Yes	3	Air cooling
148	1070	940	950	920									20	Yes	2	Air cooling
149	1060	965	1000	910									24	Yes	4	Air cooling
150	1060	970	990	790									22	Yes	4	Air cooling
151	1060	960	980	735									20	Yes	4	Air cooling

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, behind interstand cooling
ϑ 3: Finish rolling temperature after 4 passes,
ϑ 4: Temperature behind cooling line at the laying head

Steel: 54SiCrV6 (LUCCHINI), Heating to 1150°C, holding 150s, (Trial (III)),

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
160	1000	1010	915	790									18	Yes	3	Air cooling
161	950	990	890	770									18	Yes	3	Air cooling
162	960	1000	900	750									18	Yes	3	500°C / 600s
163	990	1000	910	750									18	Yes	3	Air cooling
164	970	995	890	790									20	Yes	3	500°C / 600s
165	1070	1080	1000	900									20	Yes	3	Air cooling
166	1050	1080	990	860									20	Yes	4	Air cooling
167	1110	1120	1020	900									20	Yes	4	**500°C / 200s**
168	1045	1090	1010	860									20	Yes	4	500°C / 600s
169	1010	1040	980	770									20	Yes	5	500°C / 600s
170	1030	1050	980	730									18	Yes	5	Air cooling
171	905	-	860	870									22	Yes	2	Air cooling
172	925	-	900	870									20	Yes	3	**500°C / 300s**
173	950	990	890	790									20	Yes	3	Air cooling
174	915	970	850	750									20	Yes	3	Air cooling
175	960	995	925	800									20	Yes	3	500°C / 600s
176	950	985	870	770									20	Yes	3	500°C / 600s
177	870	910	820	760									18	Yes	3	500°C / 600s
178	1050	1100	990	900									18	Yes	3	**200°C / 200s**
179	980	1000	900	770									20	Yes	3	Air cooling

Successful rods:

ϑ 1: Start temperature

ϑ 2: Temperature after pass 2, in front of interstand cooling

ϑ 3: Finish rolling temperature after 4 passes

ϑ 4: Temperature behind cooling line at the laying head

Steel: 54SiCrV6 (LUCCHINI), Heating to 1150°C, holding 150s, (Trial (III))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
180	980	1030	940	790									18	Yes	3	Air
181	990	1010	900	760									18	Yes	3	Air
182	960	1000	840	750									18	Yes	3	200°C / 600s
183	980	1000	860	790									20	Yes	3	200°C / 600s
184	1015	1040	940	800									18	Yes	3	200°C / 600s
185	1010	1030	930	770									18	Yes	3	200°C / 600s
186	1050	1085	970	900									20	Yes	4	**200°C / 200s**
187	1070	1105	1000	850									20	Yes	5	200°C / 600s
188	1050	1110	1010	770									18	Yes	5	200°C / 600s
189	1025	1060	930	820									18	Yes	3	350°C / 600s
190	1025	1060	940	750									18	Yes	4	350°C / 600s
191	950	985	1000	770									18	No	4	350°C / 600s
192	960	990	1020	850									18	No	3	350°C / 600s
193	1000	1100	940	830									20	Yes	3	350°C / 600s
194	970	1000	870	750									18	Yes	3	350°C / 600s
195	960	1005	890	880									18	Yes	2	350°C / 600s
196	930	970	860	820									18	Yes	2	350°C / 600s

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, in front of interstand cooling
ϑ 3: Finish rolling temperature after 4 passes,
ϑ 4: Temperature behind cooling line at the laying head

Steel: 54SiCrV6 (LUCCHINI), Heating to 1150°C, holding 150s, (Trial (III))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
197	1040	1080	940	820									18	Yes	4	Quenching
198	1000	1050	930	760									18	Yes	4	Quenching
199	950	1020	1020	750									18	No	4	Quenching
100A	980	1000	1010	850									18	No	3	Quenching
101A	980	1000	910	770									18	Yes	3	Quenching
102A	930	960	865	820									18	Yes	2	Quenching
103A	950	980	860	770									16	Yes	2	Quenching
1003	1010	1060	**General damage, but quenched by interstand cooling after pass 2!**													

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, in front of interstand cooling
ϑ 3: Finish rolling temperature after 4 passes,
ϑ 4: Temperature behind cooling line at the laying head

Steel: 30MnB4 (LUCCHINI), Heating to 1050°C, no holding, (Trial (I))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
201	970	1010	980	850	74	38	80	56	1.76	1.00	1.52	1.39	20	Yes	5	Quenching
202	1000	1040	1020	900	68	35	72	44	1.73	0.85	1.55	1.12	20	Yes	5	Quenching
203	950	1000	980	760	75	38	82	53	1.76	1.03	1.55	1.27	18	Yes	5	Quenching
204	950	1000	980	850	76	39	79	40	1.76	1.12	1.49	1.45	20	Yes	5	Quenching
205	970	1000	990	840	74	38	77	39	1.76	1.06	1.49	1.45	20	Yes	5	Quenching
206	870	930	920	820	87	48	98	60	2.00	1.12	1.91	1.61	20	Yes	4	Quenching
207	910	980	930	840	82	44	84	45	1.97	1.03	1.79	1.58	18	Yes	4	Quenching
208	900	960	930	730	82	44	82	44	1.94	1.03	1.61	1.73	18	Yes	5	Quenching
209	850	890	870	770	92	52	104	58	2.12	1.09	1.91	1.79	20	Yes	4	Quenching
210	810	880	860	820	94	50	96	46	1.97	1.27	1.73	2.06	18	Yes	3	Quenching

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, behind interstand cooling
ϑ 3: Finish rolling temperature after 4 passes,
ϑ 4: Temperature behind cooling line at the laying head

Steel: 30MnB4 (LUCCHINI), Heating to 1050°C, no holding, (Trial (I))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
221	960	1010	970	700	74	39	96	40	1.82	1.00	1.58	1.00	20	Yes	5	600°C / 600s
222	940	1000	950	750	72	38	94	36	1.82	1.06	1.67	0.97	20	Yes	4	600°C / 600s
223	1000	1020	990	830	66	34	88	34	1.73	1.06	1.58	0.88	20	Yes	4	600°C / 600s
224	1000	1020	990	730	71	38	90	36	1.82	1.07	1.70	0.91	20	Yes	5	600°C / 600s
225	900	950	920	730	82	45	104	44	2.03	1.18	1.85	1.09	20	Yes	4	600°C / 600s
226	990	1030	1020	900	67	35	84	34	1.67	0.94	1.58	0.94	22	Yes	4	600°C / 600s
227	910	970	930	780	80	44	100	42	1.94	1.09	1.79	1.03	22	Yes	4	600°C / 600s
228	950	990	980	840	80	44	100	40	1.85	1.09	1.70	1.03	24	Yes	4	600°C / 600s
229	860	910	900	800	84	47	108	46	2.00	1.12	2.03	1.12	20	Yes	4	600°C / 600s
230	860	920	870	750	86	46	112	50	1.97	1.27	1.94	1.21	18	Yes	3	600°C / 600s
231	850	900	870	850	86	48	113	54	2.00	1.24	1.94	1.21	18	Yes	3	600°C / 600s
232	860	910	880	850	86	48	110	52	1.91	1.27	1.91	1.15	18	Yes	1	600°C / 600s
233	820	870	850	780	90	52	120	58	2.03	1.36	2.09	1.24	18	Yes	2	600°C / 600s

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, behind interstand cooling
ϑ 3: Finish rolling temperature after 4 passes,
ϑ 4: Temperature behind cooling line at the laying head

Steel: 30MnB4 (LUCCHINI), Heating to 1050°C, no holding, (Trial (II))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
240	1010	-	940	800									20	Yes	4	Air cooling
241	990	910	950	760									20	Yes	4	Air cooling
242	970	910	930	740									20	Yes	4	Air cooling
243	970	910	950	810									20	Yes	3	Air cooling
244	980	880	940	900									20	Yes	2	Air cooling
245	960	900	940	850									18	Yes	2	Air cooling
246	990	1010	1010	850									20	No	4	Air cooling
247	970	1010	1010	750									20	No	5	Air cooling
248	915	840	900	770									20	Yes	3	Air cooling
249	900	820	870	750									20	Yes	3	Air cooling
250	890	810	870	850									20	Yes	2	Air cooling
251	880	780	860	800									18	Yes	2	Air cooling

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, behind interstand cooling
ϑ 3: Finish rolling temperature after 4 passes,
ϑ 4: Temperature behind cooling line at the laying head

Steel: 30MnB4 (LUCCHINI), Heating to 1150°C, holding 150s, (Trial (III))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
260	970	1035	980	750									20	Yes	4	Air cooling
261	970	1060	980	760									20	Yes	4	Quenching
262	980	1070	970	880									20	Yes	3	Quenching
263	980	1040	980	820									18	Yes	3	Quenching
264	980	1050	980	820									18	Yes	3	Air cooling
265	930	980	900	800									20	Yes	3	Air cooling
266	910	980	920	810									20	Yes	3	Quenching
267	940	1010	920	730									20	Yes	3	Quenching
268	930	980	930	735									20	Yes	3	Air cooling
269	850	920	850	880									20	Yes	1	Air cooling
270	860	905	845	810									20	Yes	1	Air cooling
271	860	900	860	810									20	Yes	1	Quenching

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, in front of interstand cooling
ϑ 3: Finish rolling temperature after 4 passes
ϑ 4: Temperature behind cooling line at the laying head

Steel: 38Cr4 (SIDENOR), Heating to 1050°C, no holding, (Trial (I))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
301	970	975	960	900	76	40	98	48	1.73	0.97	1.61	1.21	20	Yes	4	Quenching
302	970	990	980	850	72	38	92	42	1.76	0.91	1.57	1.15	20	Yes	5	Quenching
303	970	980	960	850	70	36	95	46	1.67	0.85	1.61	1.09	18	Yes	5	Quenching
304	990	1000	980	750	66	36	89	45	1.67	0.79	1.55	1.09	18	Yes	6	Quenching
305	890	920	890	750	82	44	110	56	2.00	1.00	1.79	1.39	18	Yes	4	Quenching
306	910	940	910	800	80	41	104	54	1.88	0.97	1.76	1.33	20	Yes	4	Quenching
307	800	830	800	750	102	56	130	68	2.18	1.21	2.09	1.58	20	Yes	3	Quenching
308	930	860	850	820	94	52	121	62	2.18	1.09	2.06	1.52	22	Yes	3	Quenching
309	800	825	900	840	104	60	110	54	2.27	1.09	2.06	1.49	20	No	3	Quenching
310	800	840	925	850	100	56	108	54	2.15	1.33	1.94	1.45	18	No	3	Quenching
311	800	830	920	730	102	58	112	56	2.21	1.30	1.97	1.45	18	No	4	Quenching
312	970	1000	975	830	73	41	86	48	1.67	0.85	1.49	1.12	22	Yes	4	Quenching
313	1000	1020	1005	740	68	34	79	40	1.58	0.76	1.45	1.03	20	Yes	5	Quenching
314	820	850	820	700	100	54	122	62	2.27	1.09	2.06	1.58	20	Yes	4	Quenching
315	830	865	840	760	96	52	116	56	2.18	1.12	2.12	1.52	22	Yes	3	Quenching

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, behind interstand cooling
ϑ 3: Finish rolling temperature after 4 passes,
ϑ 4: Temperature behind cooling line at the laying head

Steel: 38Cr4 (SIDENOR), Heating to 1050°C, no holding, (Trial (I))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
321	980	1000	960	700	73	42	94	46	1.70	0.85	1.48	1.15	20	Yes	5	600°C / 600s
322	1030	1050	1020	790	60	30	74	32	1.49	0.67	1.33	0.85	20	Yes	5	600°C / 600s
323	1015	1040	1010	840	60	32	80	38	1.49	0.82	1.27	0.85	22	Yes	5	600°C / 600s
324	1000	1020	980	680	66	38	85	46	1.49	0.73	1.39	0.97	18	Yes	5	600°C / 600s
325	1020	1040	980	770	83	46	104	54	1.97	0.97	1.70	1.15	18	Yes	4	600°C / 600s
326	910	930	880	670	82	44	105	54	1.88	0.97	1.79	1.27	20	Yes	4	600°C / 600s
327	900	930	900	760	82	44	105	54	1.85	0.94	1.64	1.21	20	Yes	3	600°C / 600s
328	930	940	930	800	77	42	100	50	2.03	0.94	1.91	1.30	22	Yes	3	600°C / 600s
329	870	920	870	830	86	47	108	55	2.06	0.82	1.91	1.33	22	Yes	2	600°C / 600s
330	855	895	870	840	87	48	113	58	2.06	0.83	1.94	1.33	22	Yes	2	600°C / 600s
331	840	870	850	800	92	54	120	62	2.30	0.97	2.00	1.36	20	Yes	2	600°C / 600s
332	870	900	850	760	88	52	110	56	2.21	0.91	1.94	1.39	22	Yes	3	600°C / 600s

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, behind interstand cooling
ϑ 3: Finish rolling temperature after 4 passes,
ϑ 4: Temperature behind cooling line at the laying head

Steel: 38Cr4 (SIDENOR), Heating to 1050°C, no holding, (Trial (II))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
340	-	-	-	700									22	Yes	5	Air cooling
341	1025	970	1010	670									22	Yes	5	Air cooling
342	1000	940	950	920									22	Yes	2	Air cooling
343	1020	940	985	900									22	Yes	3	Air cooling
344	1010	950	980	830									22	Yes	4	Air cooling
345	1030	960	1000	750									22	Yes	5	Air cooling
346	920	830	880	820									22	Yes	3	Air cooling
347	950	860	910	800									20	Yes	2	Air cooling
348	960	880	920	760									18	Yes	2	Air cooling
349	870	800	850	700									18	Yes	2	Air cooling
350	830	770	800	730									18	Yes	1	Air cooling
351	830	770	790	780									18	Yes	0	Air cooling

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, behind interstand cooling
ϑ 3: Finish rolling temperature after 4 passes,
ϑ 4: Temperature behind cooling line at the laying head

Steel: 38Cr4 (SIDENOR), Heating to 1150°C, holding 150s, (Trial (III))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
360	1015	1050	970	890									22	Yes	4	Quenching
361	990	1060	1000	790									22	Yes	5	Quenching
362	970	1060	980	740									20	Yes	4	Quenching
363	1000	1070	980	730									20	Yes	4	Air cooling
364	1030	1080	980	865									20	Yes	5	Air cooling
365	1020	1080	990	870									20	Yes	5	Quenching
366	950	1020	910	900									20	Yes	2	Air cooling
367	970	1010	920	820									20	Yes	3	Air cooling
368	960	1020	925	820									20	Yes	3	Quenching
369	940	1010	900	770									18	Yes	3	Air cooling
370	940	1015	890	790									18	Yes	3	Quenching
371	970	1010	880	755									17	Yes	3	Quenching
372	890	860	840	900									18	Yes	1	Quenching
373	890	860	840	820									18	Yes	2	Quenching
374	870	840	830	800									18	Yes	2	Air cooling
375	930	1000	870	770									18	Yes	3	Air cooling
376	930	990	870	760									18	Yes	3	Quenching

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, in front of interstand cooling
ϑ 3: Finish rolling temperature after 4 passes
ϑ 4: Temperature behind cooling line at the laying head

Steel: 10MnSi7 (LUCCHINI), Heating to 1050°C, no holding, (Trial (I))

No.	Temperature				Rolling force				Rolling torque				Fin. Speed	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	[°C]				[kN]				[kNm]				[m/s]			
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
401	1000	1010	980	800	62	38	85	52	1.36	0.97	1.36	1.06	20	Yes	4	Quenching
402	1000	1010	1070	850	Scrap								20	No	4	Quenching
403	1000	1000	880	680	60	34	78	41	1.33	0.98	1.52	0.94	20	Yes	5	Quenching
404	1000	1010	1070	700	Scrap								18	No	5	Quenching
405	1000	1020	1000	770	58	31	87	40	1.21	1.03	1.33	0.91	18	Yes	4	Quenching
406	1000	1005	990	840	58	31	86	40	1.24	1.06	1.21	0.97	18	Yes	3	Quenching
407	880	890	980	760	72	44	105	56	1.52	1.24	1.45	1.21	20	Yes	4	Quenching
408	900	920	910	790	72	42	87	52	1.45	1.06	1.39	1.27	22	Yes	3	Quenching
409	900	910	900	830	70	40	102	48	1.45	1.09	1.39	1.27	22	Yes	2	Quenching
410	910	920	900	750	74	44	112	52	1.61	1.12	1.52	1.30	22	Yes	4	Quenching
411	860	870	860	820	72	40	108	52	1.55	1.27	1.52	1.27	22	Yes	1	Quenching
412	850	870	870	890	74	42	110	54	1.58	1.18	1.55	1.27	22	Yes	0	Quenching
413	850	870	840	870	74	42	113	60	1.52	1.30	1.61	1.21	18	Yes	0	Quenching

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, behind interstand cooling
ϑ 3: Finish rolling temperature after 4 passes,
ϑ 4: Temperature behind cooling line at the laying head

Steel: 10MnSi7 (LUCCHINI), Heating to 1050°C, no holding, (Trial (I))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
421	1020	1020	990	800	57	30	81	38	1.18	0.97	1.27	0.85	20	Yes	4	600°C / 600s
422	1005	1010	990	800	56	31	80	38	1.27	1.00	1.36	0.91	22	Yes	4	600°C / 600s
423	1010	1020	990	710	58	32	81	39	1.30	1.03	1.21	0.88	22	Yes	5	600°C / 600s
424	1010	1030	995	690	56	30	79	40	1.18	1.00	1.21	0.91	22	Yes	5	600°C / 600s
425	1020	1025	990	760	58	32	83	42	1.24	1.00	1.21	0.91	18	Yes	4	600°C / 600s
426	1010	1030	990	840	58	32	83	44	1.21	0.94	1.21	0.94	18	Yes	2	600°C / 600s
427	900	920	900	750	74	46	106	58	1.49	1.18	1.45	1.24	20	Yes	3	600°C / 600s
428	900	920	870	790	73	46	102	53	1.61	1.12	1.51	1.33	22	Yes	3	600°C / 600s
429	960	970	920	840	65	39	91	46	1.45	1.03	1.27	0.97	22	Yes	3	600°C / 600s
430	860	870	840	760	75	46	111	57	1.61	0.97	1.55	1.27	22	Yes	2	600°C / 600s
431	850	870	820	800	74	45	110	57	1.64	0.88	1.76	1.30	22	Yes	1	600°C / 600s
432	865	890	850	790	74	45	108	56	1.70	1.58	1.70	1.30	22	Yes	2	600°C / 600s

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, behind interstand cooling
ϑ 3: Finish rolling temperature after 4 passes,
ϑ 4: Temperature behind cooling line at the laying head

Steel: 10MnSi7 (LUCCHINI), Heating to 1050°C, no holding, (Trial (II))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
440	1000	900	920	840									20	Yes	3	Air cooling
441	1010	885	920	780									18	Yes	3	Air cooling
442	1010	940	950	790									24	Yes	4	Air cooling
443	1010	920	930	750									22	Yes	4	Air cooling
444	900	790	840	760									22	Yes	3	Air cooling
445	910	810	860	800									24	Yes	3	Air cooling
446	1000	1020	1000	880									24	No	4	Air cooling
447	1010	1000	1020	850									22	No	4	Air cooling
448	1010	1000	1020	800									20	No	4	Air cooling
449	1000	1010	990	770									18	No	4	Air cooling

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, behind interstand cooling
ϑ 3: Finish rolling temperature after 4 passes,
ϑ 4: Temperature behind cooling line at the laying head

Steel: 10MnSi7 (LUCCHINI), Heating to 1150°C, holding 150s, (Trial (III))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. Pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
460	980	1020	920	820									20	Yes	3	Quenching
461	1010	1050	940	880									22	Yes	3	Quenching
462	1010	1050	950	920									20	Yes	2	Quenching
463	1010	1050	920	810									20	Yes	4	Quenching
464	1025	1050	920	740									20	Yes	5	Quenching
465	940	1000	900	800									22	Yes	3	Quenching
466	940	980	900	750									24	Yes	4	Quenching
467	920	970	960	790									24	Yes	4	Quenching
468	990	1040	1050	900									24	No	4	Quenching
469	960	1000	1005	750									24	No	5	Quenching
470	980	1020	1015	860									24	No	4	Quenching
471	820	860	900	890									20	No	1	Quenching
472	890	930	830	900									20	No	0	Quenching
473	910	950	810	820									20	No	1	Quenching
474	910	960	850	850									20	Yes	2	Quenching
475	920	950	850	760									20	Yes	3	Quenching
476	910	940	850	790									22	Yes	3	Quenching

Successful rods:

ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, in front of interstand cooling
ϑ 3: Finish rolling temperature after 4 passes
ϑ 4: Temperature behind cooling line at the laying head

Steel: 10MnSi7 (LUCCHINI), Heating to 1150°C, holding 150s, (Trial (III))

No.	Temperature [°C]				Rolling force [kN]				Rolling torque [kNm]				Fin. Speed [m/s]	Interst. cooling 2 - 3	Finish cool. Pipes	Treatment
	ϑ 1	ϑ 2	ϑ 3	ϑ 4	Pass 1	Pass 2	Pass 3	Pass 4	Pass 1	Pass 2	Pass 3	Pass 4				
477	970	1020	940	820									20	Yes	3	600°C / 600s
478	1010	1040	920	770									20	Yes	4	600°C / 600s
479	960	1010	930	830									22	Yes	3	**600°C / 60s**
480	910	940	850	770									22	Yes	3	600°C / 600s
481	920	960	850	800									24	Yes	3	600°C / 600s
482	980	1020	920	810									24	Yes	4	600°C / 600s
483	980	1010	1020	890									24	No	4	**600°C / 60s**
484	1000	1040	1020	830									24	No	5	600°C / 600s
485	950	1020	1000	720									20	No	5	600°C / 600s

Successful rods:

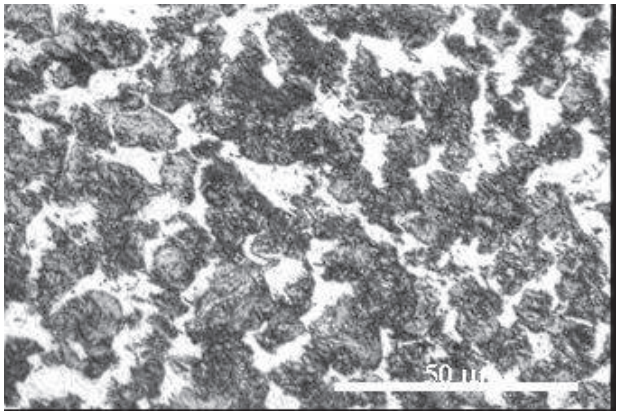
ϑ 1: Start temperature
ϑ 2: Temperature after pass 2, in front of interstand cooling
ϑ 3: Finish rolling temperature after 4 passes
ϑ 4: Temperature behind cooling line at the laying head

ANNEX 5.2

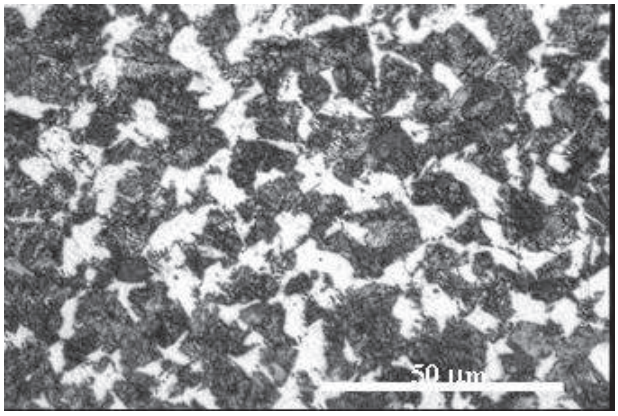
Examples of microstructures of ms-rolled and quenched wire-rods manufactured by Freiberg pilot mill (WP5)

AS-ROLLED MICROSTRUCTURE OF WIRE ROD PROCESSED BY FREIBERG PILOT MILL

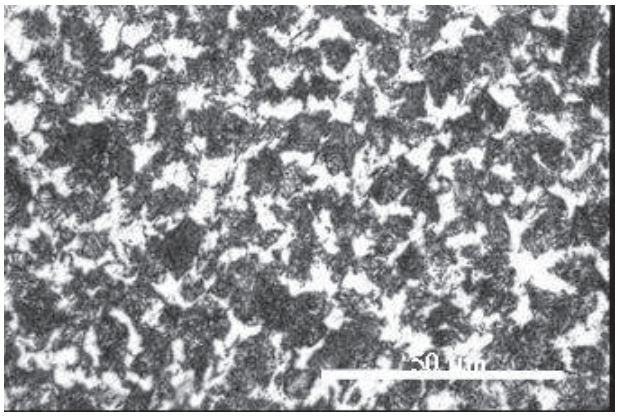
38Cr4 – FIRST TRIAL



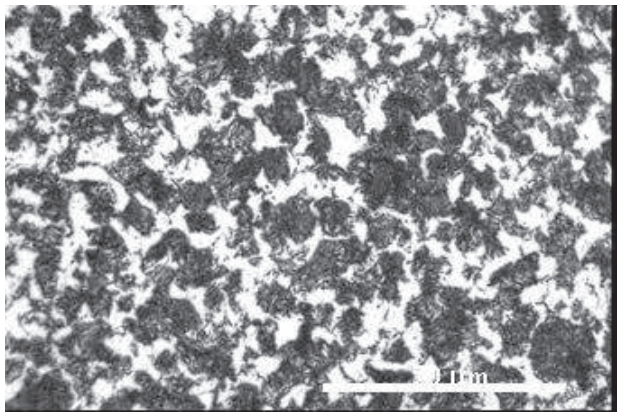
Sample No 323. (500x)



Sample No 325. (500x)



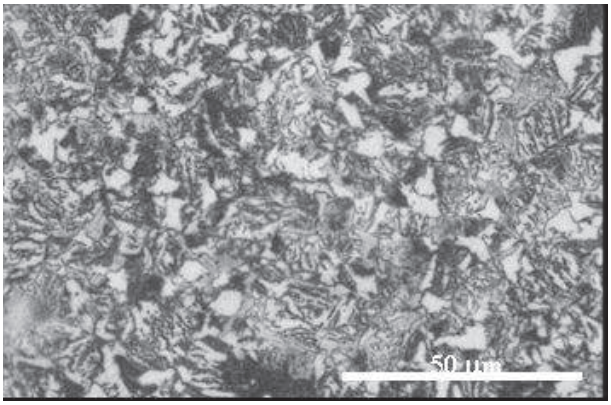
Sample No 331. (500x)



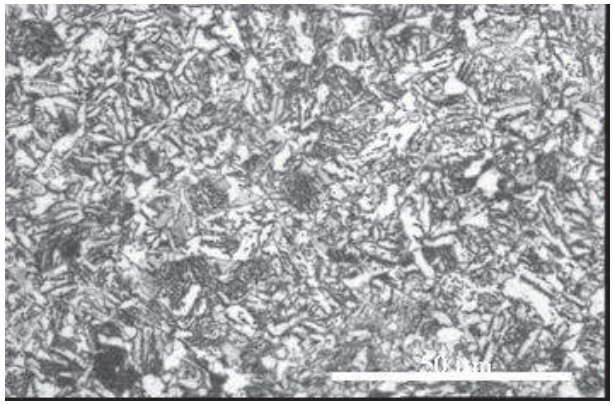
Sample No 332. (500x)

AS-ROLLED MICROSTRUCTURE OF WIRE ROD PROCESSED BY FREIBERG PILOT MILL

38Cr4 – SECOND TRIAL



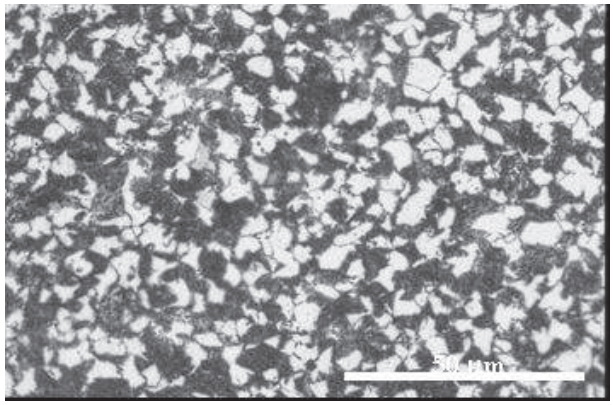
Sample No 344. (500x)



Sample No 347. (500x)



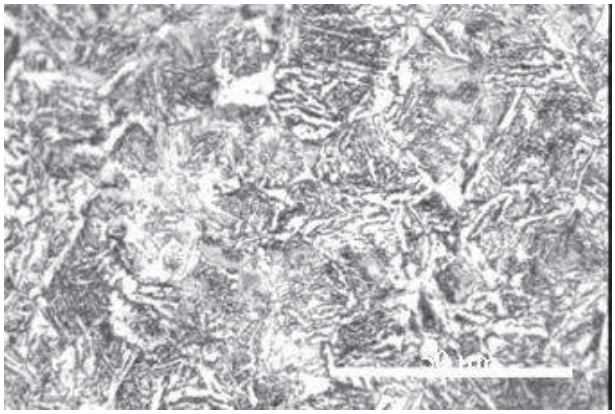
Sample No 348. (500x)



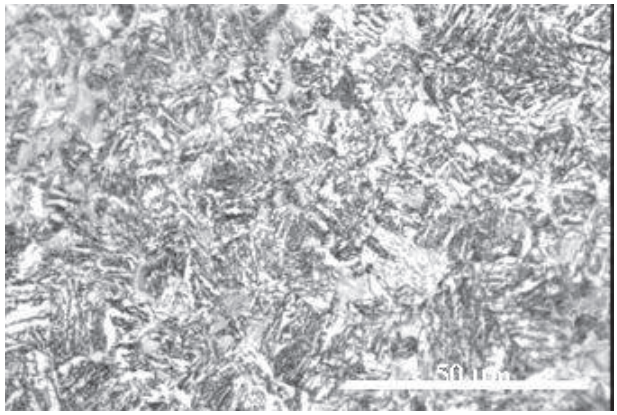
Sample No 351. (500x)

AS-ROLLED MICROSTRUCTURE OF WIRE ROD PROCESSED BY FREIBERG PILOT MILL

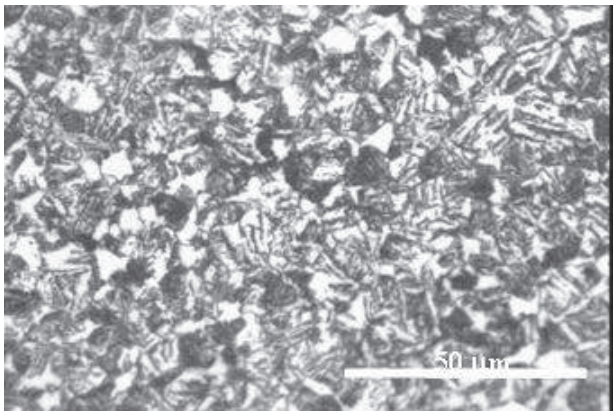
38Cr4 – THIRD TRIAL



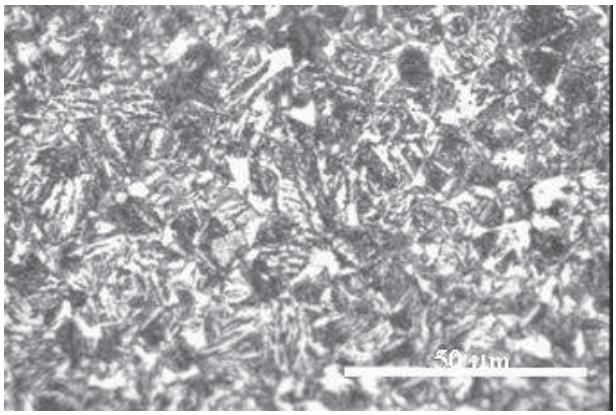
Sample No 363. (500x)



Sample No 364. (500x)



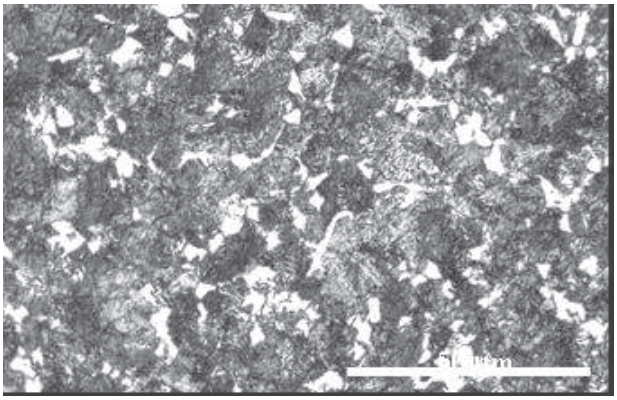
Sample No 374. (500x)



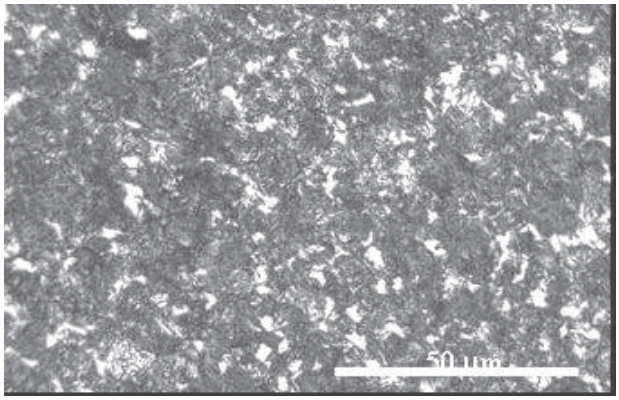
Sample No 375. (500x)

AS-ROLLED MICROSTRUCTURE OF WIRE ROD PROCESSED BY FREIBERG PILOT MILL

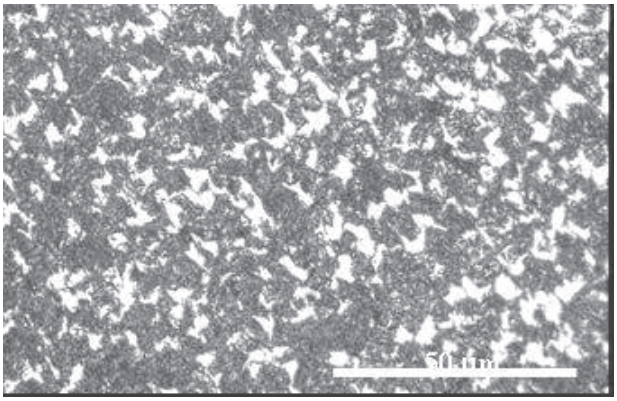
54SiCrV6 – FIRST TRIAL



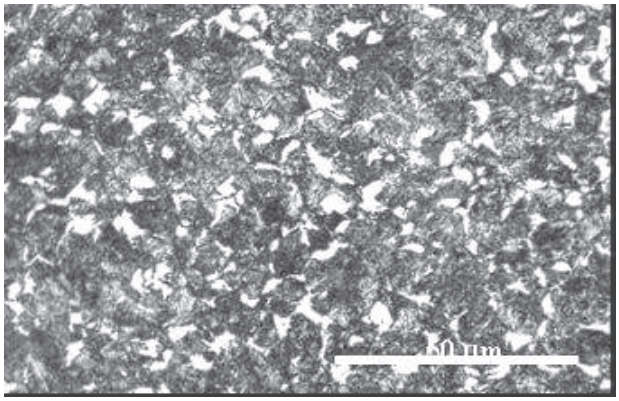
Sample No 021. (500x)



Sample No 022. (500x)



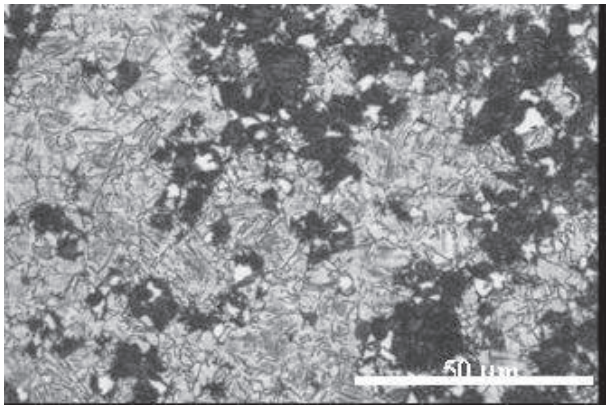
Sample No 024. (500x)



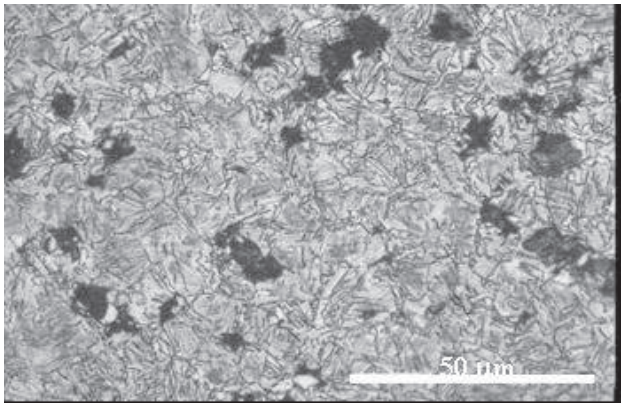
Sample No 025. (500x)

AS-ROLLED MICROSTRUCTURE OF WIRE ROD PROCESSED BY FREIBERG PILOT MILL

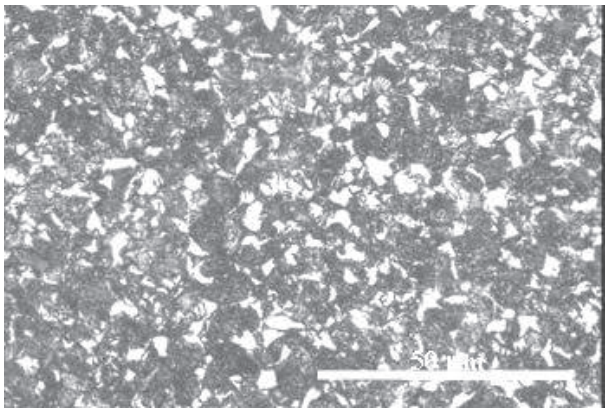
54SiCrV6 – SECOND TRIAL



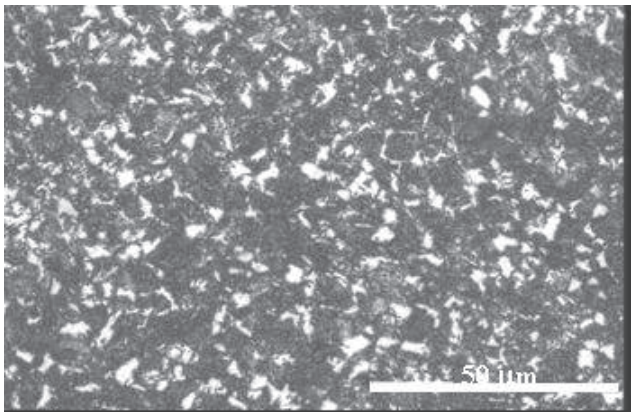
Sample No 042. (500x)



Sample No 044. (500x)



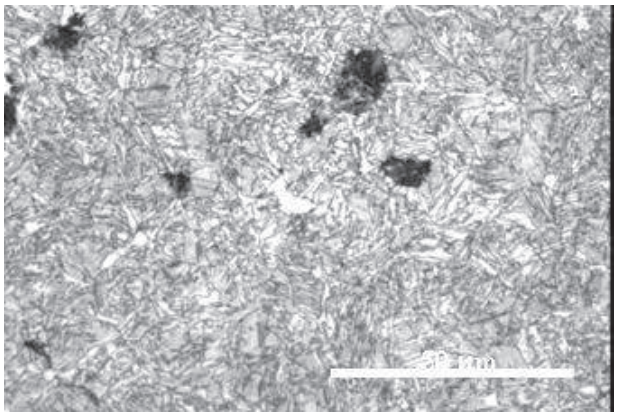
Sample No 049. (500x)



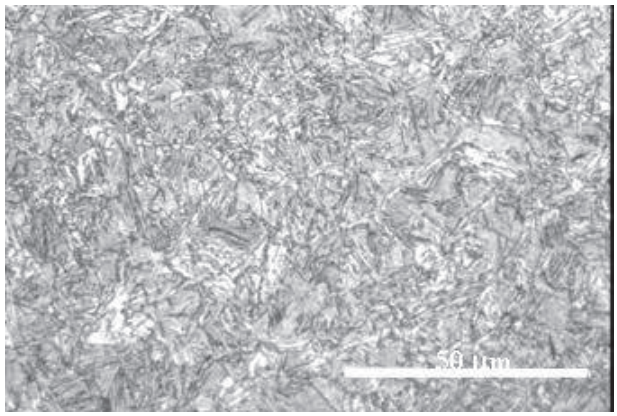
Sample No 050. (500x)

AS-ROLLED MICROSTRUCTURE OF WIRE ROD PROCESSED BY FREIBERG PILOT MILL

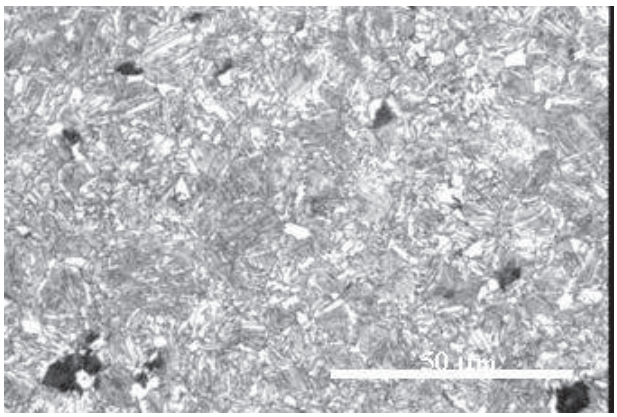
54SiCrV6 – THIRD TRIAL



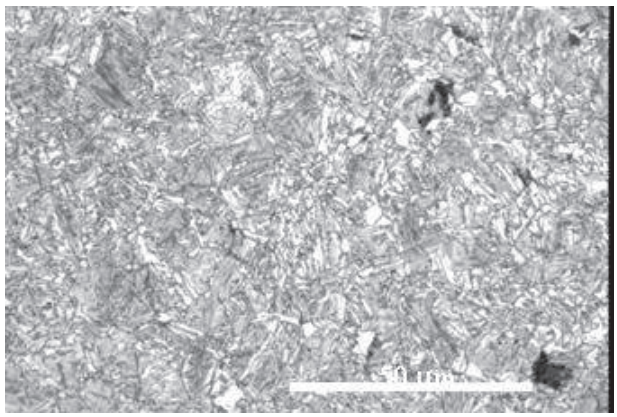
Sample No 061. (500x)



Sample No 062. (500x)



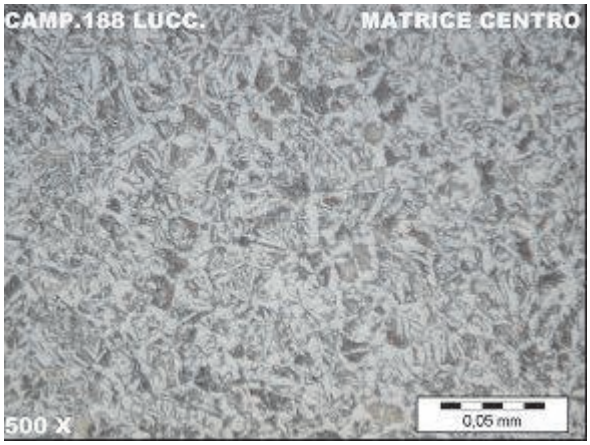
Sample No 063. (500x)



Sample No 064. (500x)

AS-ROLLED MICROSTRUCTURE OF WIRE ROD PROCESSED BY FREIBERG PILOT MILL

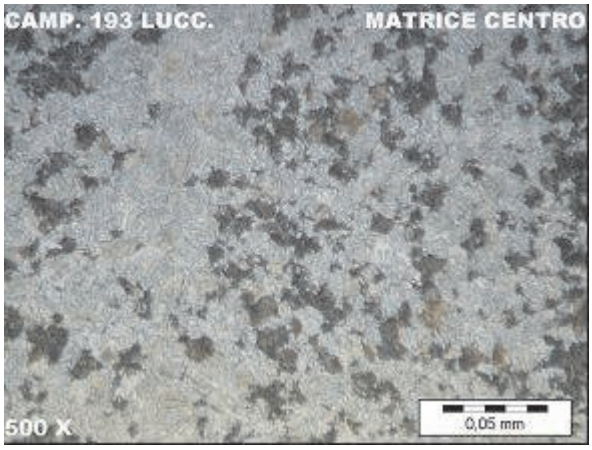
54SiCrV6 – THIRD TRIAL



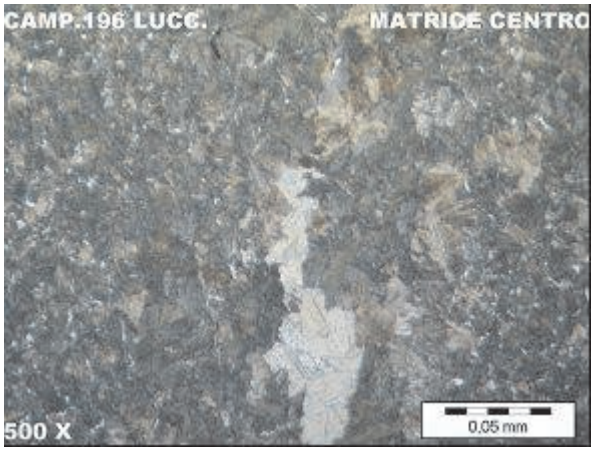
Sample No 172. (500x)



Sample No 188. (500x)



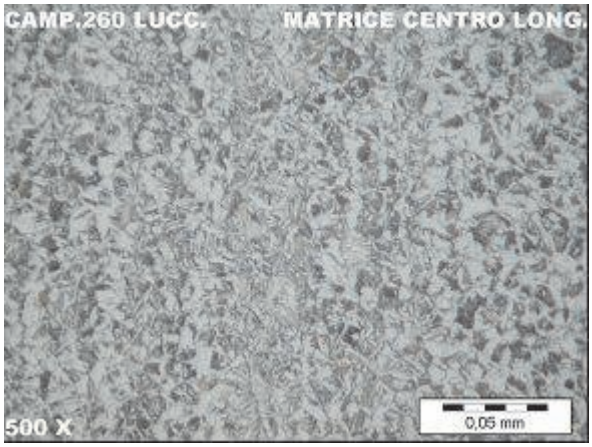
Sample No 193. (500x)



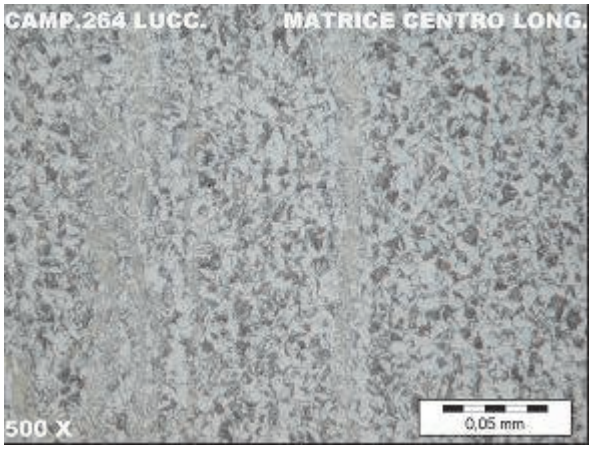
Sample No 196. (500x)

AS-ROLLED MICROSTRUCTURE OF WIRE ROD PROCESSED BY FREIBERG PILOT MILL

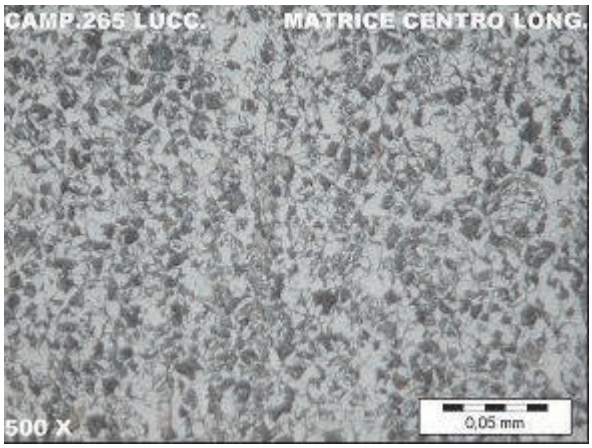
30MnB4 – THIRD TRIAL



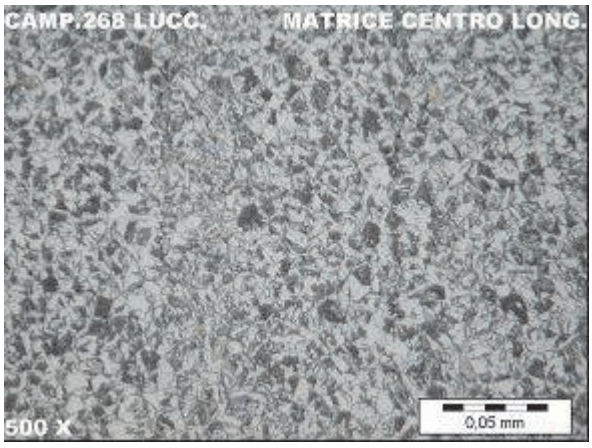
Sample No 260. (500x)



Sample No 264. (500x)



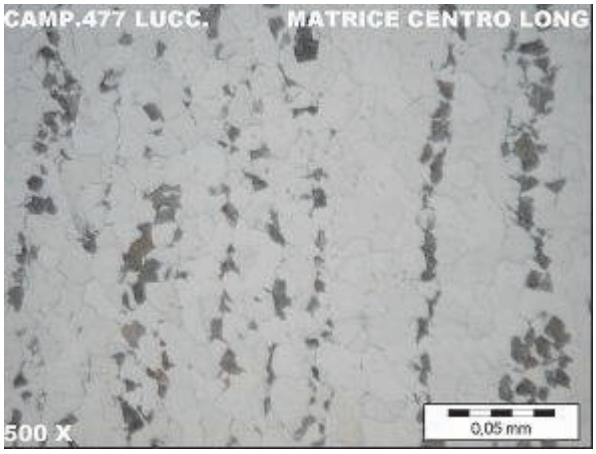
Sample No 265. (500x)



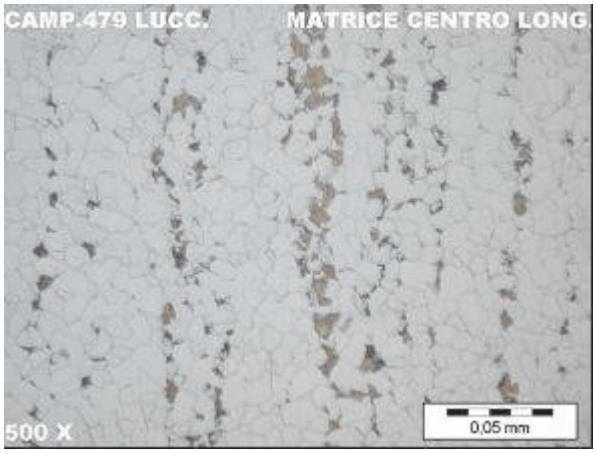
Sample No 268. (500x)

AS-ROLLED MICROSTRUCTURE OF WIRE ROD PROCESSED BY FREIBERG PILOT MILL

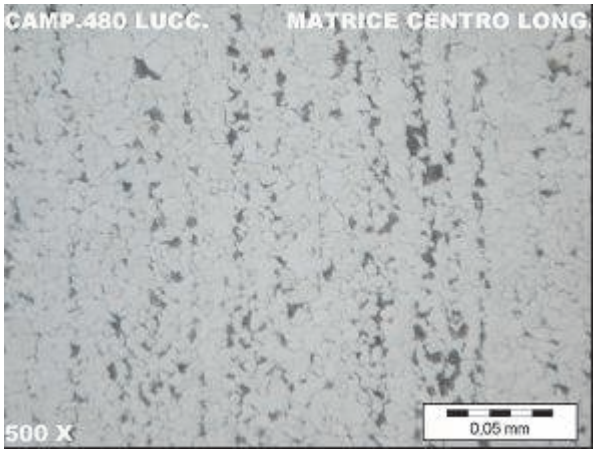
10MnSi7 – THIRD TRIAL



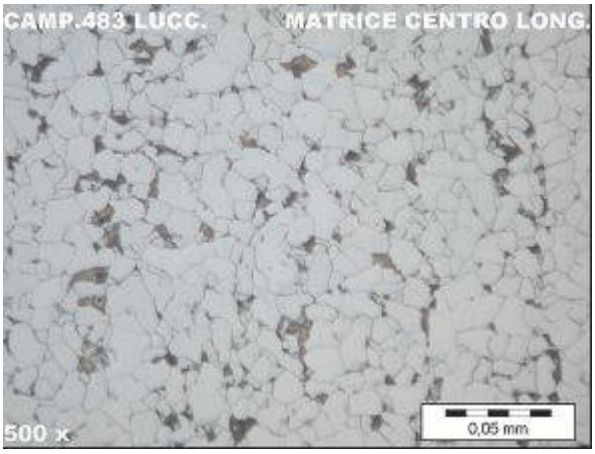
Sample No 477. (500x)



Sample No 479. (500x)



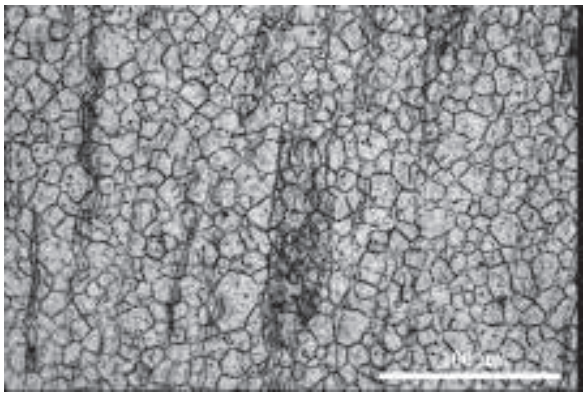
Sample No 480. (500x)



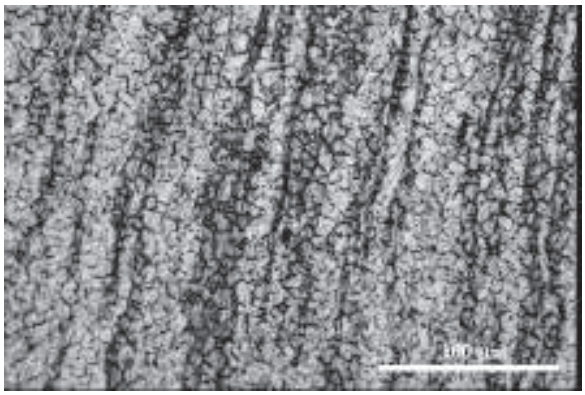
Sample No 483. (500x)

AGS OF WIRE ROD PROCESSED IN FREIBERG PILOT MILL

30MnB4

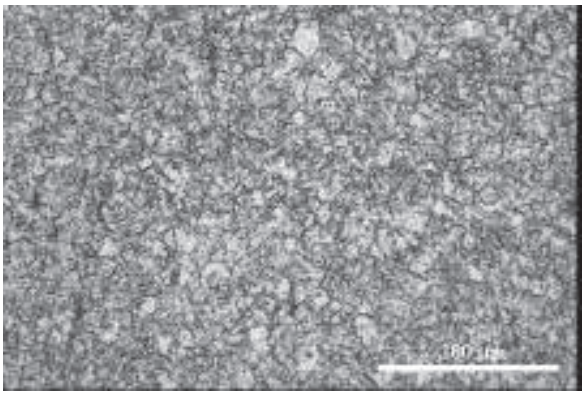


Sample No 205. (200x)

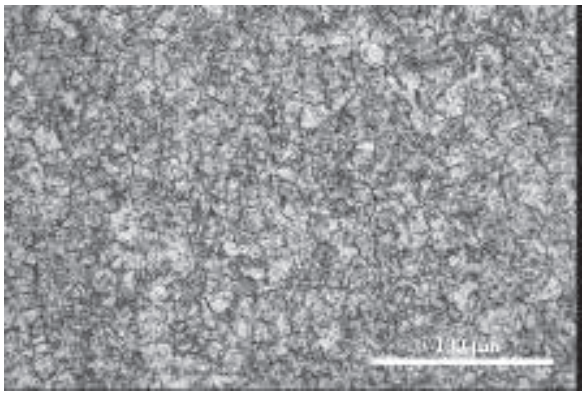


Sample No 209. (200x)

10MnSi7



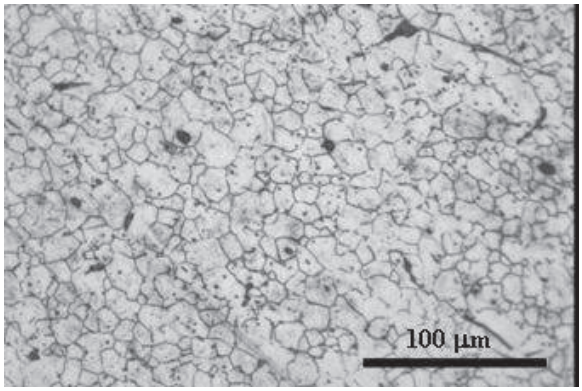
Sample No 401 (200x)



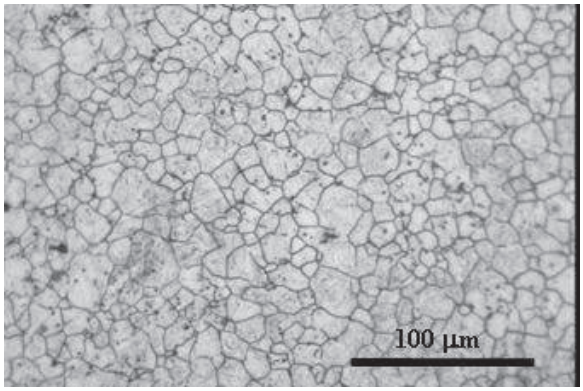
Sample No 410. (200x)

AGS OF WIRE ROD PROCESSED IN FREIBERG PILOT MILL

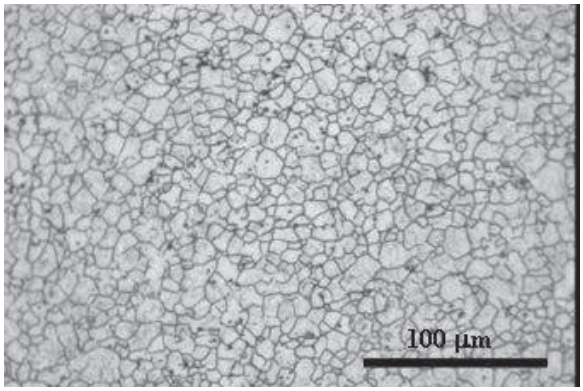
Steel 38Cr4



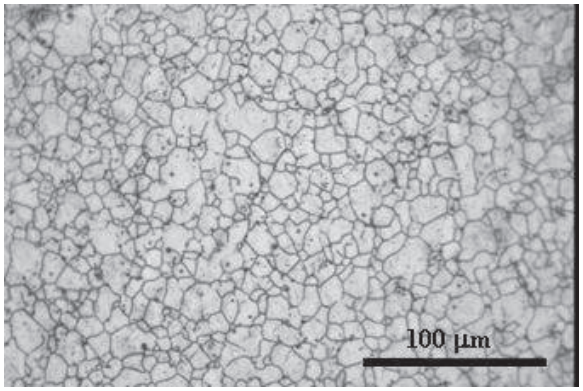
Sample No 362. (200x)



Sample No 365. (200x)



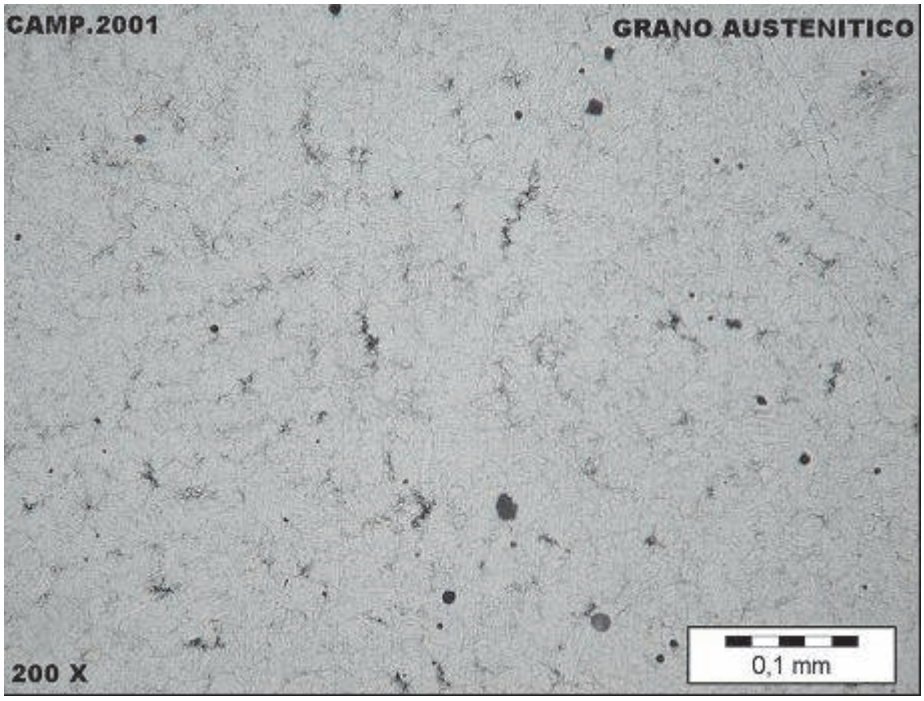
Sample No 373 (200x)



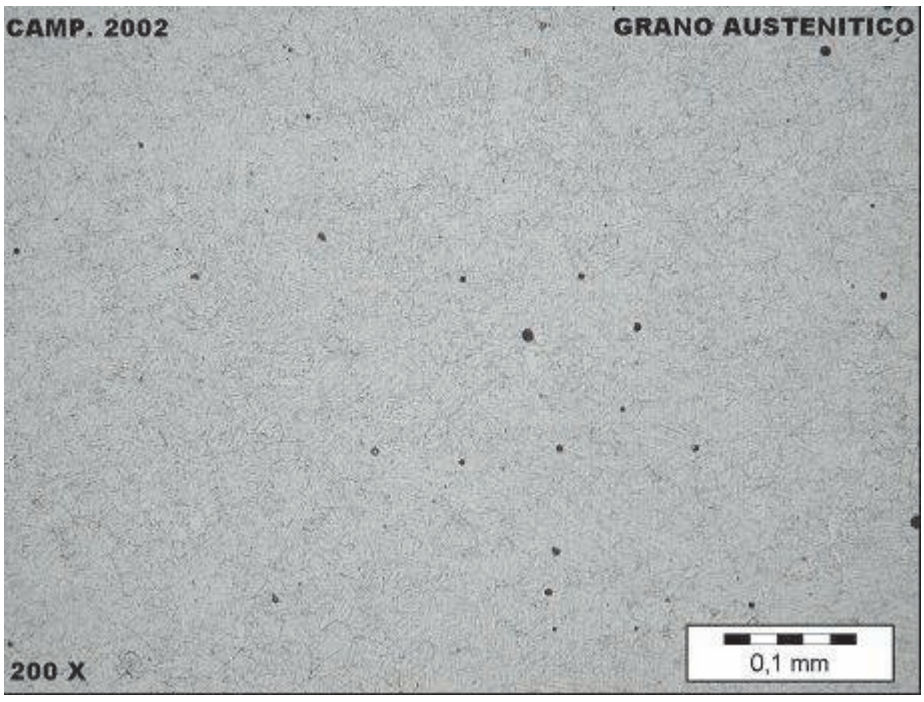
Sample No 376. (200x)

AGS AFTER HEATING TRIALS BEFORE FREIBERG PILOT MILL ROLLING

Steel 30MnB4



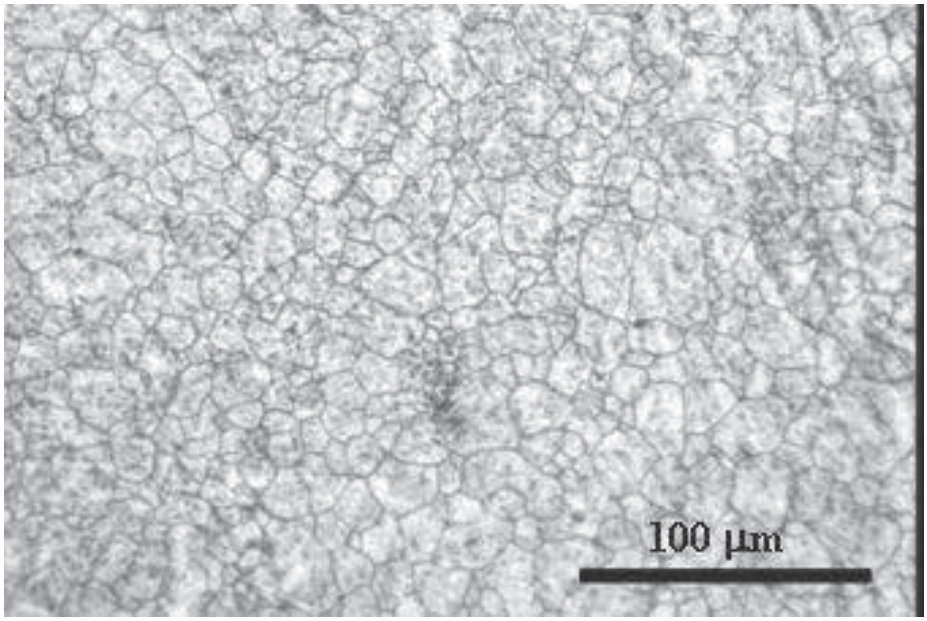
Sample No 2001 (no holding time). (200x)



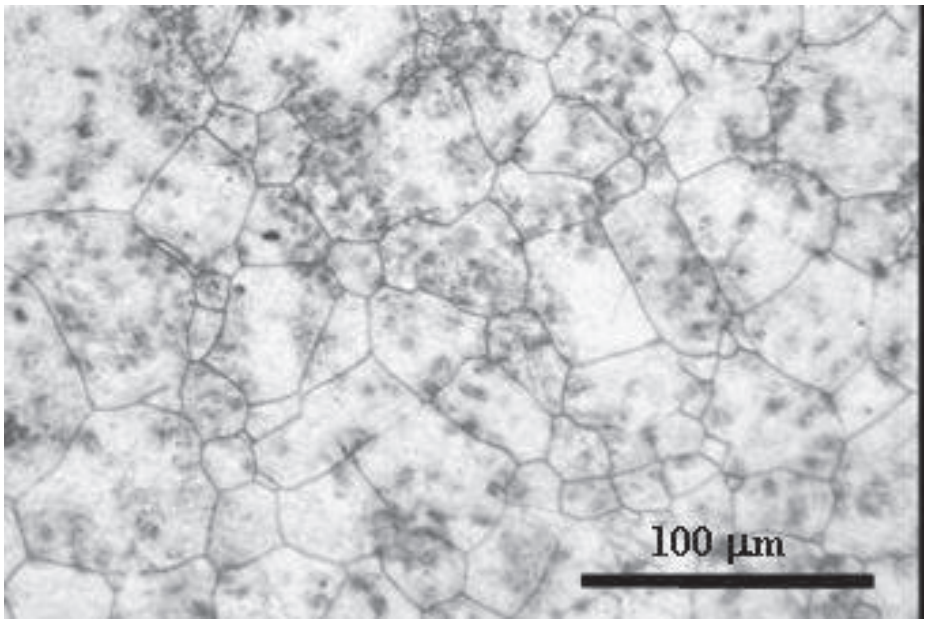
Sample No 2002 (holding time 150 s). (200x)

AGS AFTER HEATING TRIALS BEFORE FREIBERG PILOT MILL ROLLING

Steel 38Cr4



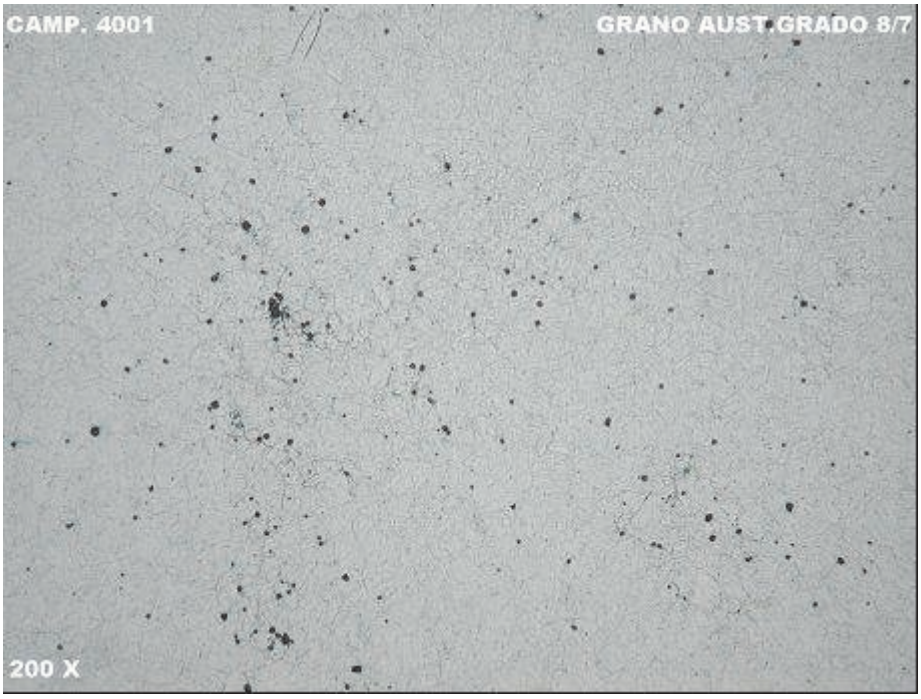
Sample No 3001 (no holding time). (200x)



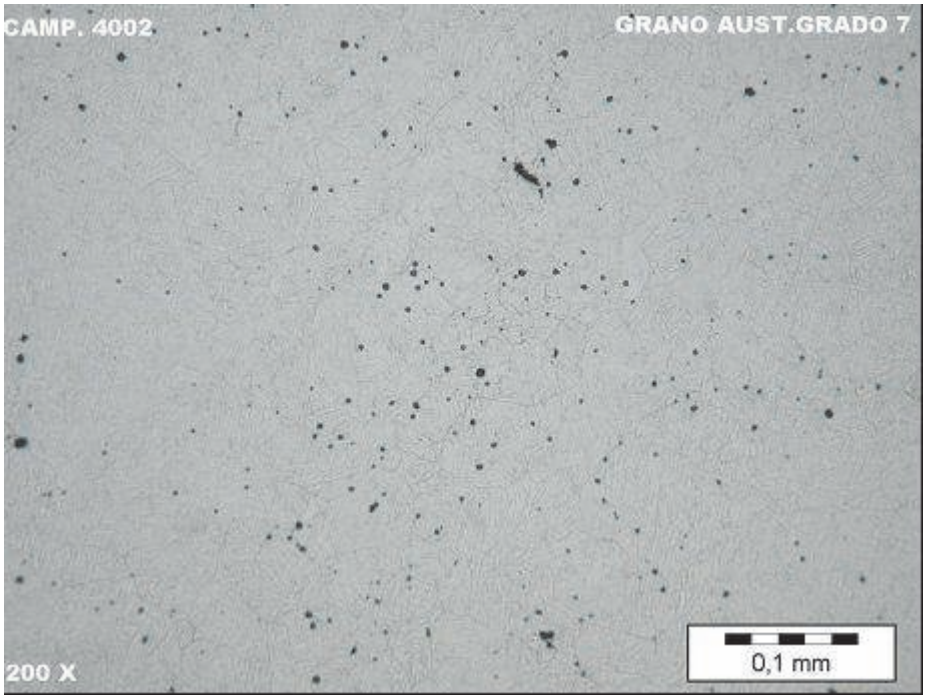
Sample No 3002 (holding time 150 s). (200x)

AGS AFTER HEATING TRIALS BEFORE FREIBERG PILOT MILL ROLLING

Steel 30MnB4



Sample No 4001 (no holding time). (200x)



Sample No 4002 (holding time 150 s). (200x)

ANNEX 5.3

Summary of mechanical properties and final microstructures of selected prototype wire-rods (WP5)

Tensile tests and metallographic observations were performed on the materials processed by pilot rolling mill in Freiberg.

The results obtained for these materials are presented in the following tables, where rolling temperatures, ultimate tensile strength (UTS), reduction of area (ROA) and percentage of the various microstructural constituent in the final product are shown.

The microstructures of selected specimens are reported in Annex 5.2.

STEEL 38Cr4: FIRST TRIAL (SLOW AIR COOLING)							
Nº	T1 [ºC]	T2 [ºC]	T3 [ºC]	T4 [ºC]	UTS [MPa]	ROA [%]	Bain+Mart
323	1015	1040	1010	840	740	70	0%
325	1020	1040	980	770	720	69	0%
331	840	870	850	800	750	70	0%
332	870	900	850	760	738	70	0%

Table 5.1 Process data and mechanical properties of steel 38Cr4 (trial I).

STEEL 38Cr4: SECOND TRIAL (NATURAL AIR COOLING)							
Nº	T1 [ºC]	T2 [ºC]	T3 [ºC]	T4 [ºC]	UTS [MPa]	ROA [%]	Bain+Mart
344	1010	950	980	830	991	48	75%
347	950	860	910	800	970	47	70%
348	960	880	920	760	954	49	85%
351	830	770	790	780	857	69	0%

Table 5.2 Process data and mechanical properties of steel 38Cr4 (trial II).

STEEL 38Cr4: THIRD TRIAL (NATURAL AIR COOLING – HOLDING 150 s)							
Nº	T1 [ºC]	T2 [ºC]	T3 [ºC]	T4 [ºC]	UTS [MPa]	ROA [%]	Bain+Mart
363	1000	1070	980	730	1006	34	75%
364	1030	1080	980	865	996	36	80%
374	870	840	830	800	985	42	55%
375	930	1000	870	770	990	63	65%

Table 5.3 Process data and mechanical properties of steel 38Cr4 (trial III).

STEEL 54SiCrV6: FIRST TRIAL (SLOW AIR COOLING)							
Nº	T1 [ºC]	T2 [ºC]	T3 [ºC]	T4 [ºC]	UTS [MPa]	ROA [%]	Bain+Mart
021	1000	1010	990	750	1157	60	0%
022	1000	1010	990	850	1088	60	0%
024	870	900	860	770	1123	54	0%
025	870	910	870	800	1125	61	0%

Table 5.4 Process data and mechanical properties of steel 54SiCrV6 (trial I).

STEEL 54SiCrV6: SECOND TRIAL (NATURAL AIR COOLING)							
Nº	T1 [ºC]	T2 [ºC]	T3 [ºC]	T4 [ºC]	UTS [MPa]	ROA [%]	Bain+Mart
042	960	850	910	820	1713	5	65%
044	1060	960	980	850	1551	5	88%
049	900	780	860	800	1295	58	0%
050	910	790	860	760	1311	57	0%

Table 5.5 Process data and mechanical properties of steel 54SiCrV6 (trial II).

STEEL 54SiCrV6: THIRD TRIAL (SLOW AIR COOLING – HOLDING 150 s)							
Nº	T1 [ºC]	T2 [ºC]	T3 [ºC]	T4 [ºC]	UTS [MPa]	ROA [%]	Bain+Mart
061	1070	1100	1000	750	1243	19	70%
062	1070	1110	1005	830	1331	17	90%
063	980	1015	930	800	1482	14	90%
064	1000	1050	930	750	1666	14	85%

Table 5.6 Process data and mechanical properties of steel 54SiCrV6 (trial III).

STEEL 10MnSi7: FIRST TRIAL (SLOW AIR COOLING)								
Nº	T1 [ºC]	T2 [ºC]	T3 [ºC]	T4 [ºC]	UTS [MPa]	ROA [%]	HV30	F-P-B-M [%]
421	1020	1020	990	800	502	78	165	83-16-0-1
422	1005	1010	990	800	507	72	165	n.d.
423	1010	1020	990	710	489	78	166	n.d.
424	1010	1030	995	690	508	79	166	n.d.
425	1020	1025	990	760	497	83	165	85-15-0-0
426	1010	1030	990	840	497	78	164	n.d.
427	900	920	900	750	511	82	165	n.d.
428	900	920	870	790	502	80	163	85-15-0-0
429	960	970	920	840	503	76	161	n.d.
430	860	870	840	760	510	78	165	85-15-0-0
431	850	870	820	800	508	79	162	n.d.
432	865	890	850	790	502	78	164	n.d.
Aver.	951	965	929	778	503	78	164	85-15-0-0

Table 5.7 Process data and mechanical properties of steel 10MnSi7 (trial I). Cooling rate, CR = 0.36 to 0.8 °C/s.

STEEL 10MnSi7: SECOND TRIAL (NATURAL AIR COOLING)								
Nº	T1 [ºC]	T2 [ºC]	T3 [ºC]	T4 [ºC]	UTS [MPa]	ROA [%]	HV30	F-P-B-M [%]
440	1000	900	920	840	477	76	155	87-8-0-5
441	1010	885	920	780	473	77	157	80-10-0-10
442	1010	940	950	790	469	76	159	80-10-0-10
443	1010	920	930	750	492	77	165	87-8-0-5
444	900	790	840	760	495	76	165	87-8-0-5
445	910	810	860	800	481	77	168	80-10-0-10
446	1000	1020	1000	880	461	76	164	80-10-0-10
447	1010	1000	1020	850	456	79	169	87-8-0-5
448	1010	1000	1020	800	467	76	166	87-8-0-5
449	1000	1010	990	770	474	76	169	87-8-0-5
Aver.	986	928	945	802	475	77	164	84-9-0-7

Table 5.8 Process data and mechanical properties of steel 10MnSi7 (trial II). Cooling rate, CR = 1.4 °C/s.

STEEL 10MnSi7: THIRD TRIAL (MIXED COOLING CONDITIONS)								
Nº	T1 [ºC]	T2 [ºC]	T3 [ºC]	T4 [ºC]	UTS [MPa]	ROA [%]	HV30	F-P-B-M [%]
477	970	1020	940	820	482	82	162	94-6-0-0
478	1010	1040	920	770	517	82	166	90-10-0-0
480	910	940	850	770	505	82	163	96-4-0-0
481	920	960	850	800	507	81	165	94-6-0-0
482	980	1020	920	810	451	85	170	96-4-0-0
484	1000	1040	1020	830	475	83	167	95-5-0-0
485	950	1020	1000	720	519	82	171	94-6-0-0
Aver. 1	963	1006	929	789	494	83	166	n.d.
479	960	1010	930	830	500	75	164	90-7-0-3
483	980	1010	1020	890	504	73	170	93-4-0-3
Aver. 2	970	1010	975	860	502	74	167	n.d.

Table 5.9 Process data and mechanical properties of steel 10MnSi7 (trial III). Cooling rate, CR = 0.2 to 0.4 °C/s (except specimens 479 and 483 with CR > 0.9 °C/s).

STEEL 30MnB4: FIRST TRIAL (SLOW AIR COOLING)								
Nº	T1 [ºC]	T2 [ºC]	T3 [ºC]	T4 [ºC]	UTS [MPa]	ROA [%]	HV30	F-P-B-M [%]
221	960	1010	970	700	653	67	208	45-35-14-1
222	940	1000	950	750	614	64	194	n.d.
223	1000	1020	990	830	604	65	191	n.d.
224	1000	1020	990	730	617	64	194	n.d.
225	900	950	920	730	663	67	207	n.d.
226	990	1030	1020	900	593	66	186	n.d.
227	910	970	930	780	599	61	178	n.d.
228	950	990	980	840	597	65	188	62-35-0-3
229	860	910	900	800	632	66	196	n.d.
230	860	920	870	750	641	66	195	n.d.
231	850	900	870	850	619	64	188	n.d.
232	860	910	880	850	628	66	193	60-40-0-0
233	820	870	850	780	627	66	195	60-40-0-0
Aver.	915	962	932	792	622	65	193	56-38-4-2

Table 5.10 Process data and mechanical properties of steel 30MnB4 (trial I). Cooling rate, CR = 0.39 to 0.95 °C/s.

STEEL 30MnB4: SECOND TRIAL (NATURAL AIR COOLING)								
Nº	T1 [ºC]	T2 [ºC]	T3 [ºC]	T4 [ºC]	UTS [MPa]	ROA [%]	HV30	F-P-B-M [%]
240	1010	1000	940	800	613	68	209	45-10-35-10
241	990	910	950	760	648	68	221	45-10-35-10
242	970	910	930	740	658	68	225	50-5-40-5
243	970	910	950	810	629	69	213	45-10-35-10
244	980	880	940	900	632	65	210	50-5-40-5
245	960	900	940	850	646	66	217	50-5-40-5
246	990	1010	1010	850	640	69	227	40-5-40-15
247	970	1010	1010	750	639	68	222	40-5-40-15
248	915	840	900	770	663	66	228	45-10-35-10
249	900	820	870	750	653	63	224	45-10-35-10
250	890	810	870	850	641	64	224	45-10-35-10
251	880	780	860	800	627	64	218	45-10-35-10
Aver.	952	898	931	803	641	67	220	46-9-36-9

Table 5.11 Process data and mechanical properties of steel 30MnB4 (trial II). Cooling rate, CR = 1.4 °C/s.

STEEL 30MnB4: THIRD TRIAL (NATURAL AIR COOLING)								
Nº	T1 [ºC]	T2 [ºC]	T3 [ºC]	T4 [ºC]	UTS [MPa]	ROA [%]	HV30	F-P-B [%]
260	970	1035	980	750	728	69	232	35-10-55*
264	980	1050	980	820	729	67	234	45-10-45*
265	930	980	900	800	721	69	230	55-10-35*
268	930	980	930	735	729	67	236	45-10-45*
269	850	920	850	880	734	65	238	65-15-20*
270	860	905	845	810	730	70	236	70-10-20*
Aver.	920	978	914	799	728	70	234	n.d.

* granular low C bainite

Table 5.12 Process data and mechanical properties of steel 30MnB4 (trial III).

STEEL 54SiCrV6: FIRST TRIAL (SLOW AIR COOLING)								
Nº	T1 [ºC]	T2 [ºC]	T3 [ºC]	T4 [ºC]	UTS [MPa]	ROA [%]	HV30	F-P-B-M [%]
121	850	890	860	850	1207	58	344	n.d.
122	840	880	860	820	1128	53	317	25-75-0-0
123	830	860	850	820	1113	53	311	n.d.
124	850	870	850	750	1230	56	343	25-75-0-0
125	890	925	900	760	1169	56	335	n.d.
126	900	930	900	850	1104	57	313	n.d.
127	920	925	910	850	1129	53	316	n.d.
128	910	930	910	820	1114	56	314	n.d.
129	1000	1000	1000	810	1112	53	311	25-75-0-0
130	990	1010	990	860	1143	52	328	25-75-0-0
131	990	1010	990	760	1182	56	339	n.d.
Aver.	906	930	911	814	1148	55	325	25-75-0-0

Table 5.13 Process data and mechanical properties of steel 54SiCrV6 (trial I). Cooling rate, CR = 0.50 to 0.85 °C/s.

STEEL 54SiCrV6: SECOND TRIAL (NATURAL AIR COOLING)								
Nº	T1 [ºC]	T2 [ºC]	T3 [ºC]	T4 [ºC]	UTS [MPa]	ROA [%]	HV30	F-P-B-M [%]
140	900	800	870	800	1150	59	352	20-50-30-0
141	900	790	860	770	1162	59	354	15-40-45-0
142	970	850	930	700	1245	59	388	10-44-45-1
143	980	860	920	820	1191	57	358	20-60-20-0
144	990	890	930	810	1229	58	378	20-60-20-0
145	970	850	920	700	1221	59	378	15-54-30-1
146	970	850	910	750	1199	59	363	15-55-30-0
147	1070	970	990	830	1246	58	399	15-62-22-1
148	1070	940	950	920	1161	60	371	10-45-45-0
149	1060	965	1000	910	1235	58	389	10-50-40-0
Aver.	988	877	928	801	1204	59	373	15-52-33-0

Table 5.14 Process data and mechanical properties of steel 54SiCrV6 (trial II). Cooling rate, CR = 1.4 °C/s.

STEEL 54SiCrV6: THIRD TRIAL (MIXED COOLING CONDITIONS)								
Nº	T1 [ºC]	T2 [ºC]	T3 [ºC]	T4 [ºC]	UTS [MPa]	ROA [%]	HV30	F-P-B+M [%]
172	925	-	900	870	1256	59	394	5-50-45
173	950	990	890	790	1248	58	391	5-45-50
175	960	995	925	800	1205	61	379	5-60-35
176	950	985	870	770	1271	60	398	5-40-55
177	870	910	820	760	1180	63	372	5-70-25
180	980	1030	940	790	1274	52	393	5-35-60
189	1025	1060	930	820	1197	61	377	5-55-40
190	1025	1060	940	750	1232	56	388	5-65-30
191	950	985	1000	770	1215	57	382	5-30-65
192	960	990	1020	850	1198	58	380	5-70-25
194	970	1000	870	750	1222	60	384	2-58-40
195	960	1005	890	880	1213	57	386	4-80-16
196	930	970	860	820	1254	59	394	4-70-26
Aver. 1	958	998	912	801	1228	58	386	n.d.
188	1050	1110	1010	770	721	68	236	10-10-80*
193	1000	1100	940	830	1210	33	383	0-20-80
Aver. 2	1025	1105	975	800	n.d.	n.d.	n.d.	n.d.

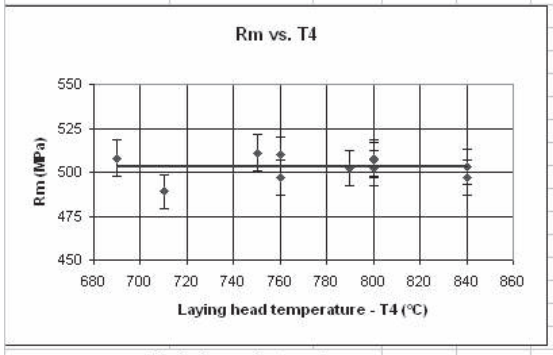
* granular low C bainite

Table 5.15 Process data and mechanical properties of steel 54SiCrV6 (trial III). Cooling rate, CR = 0.40 to 1.2 °C/s.

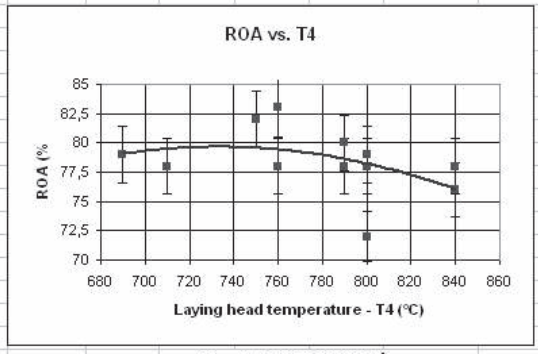
Trends of main mechanical parameters (UTS, ROA, Hardness) as a function of processing temperature were investigated by Lucchini on the materials produced from Lucchini steels 10MnSi7, 30MnB4 and 54SiCrV6 in the trials I and II. Results are given in the following sheets.

RELATIONSHIP BETWEEN MECHANICAL PROPERTIES AND PROCESS CONDITIONS

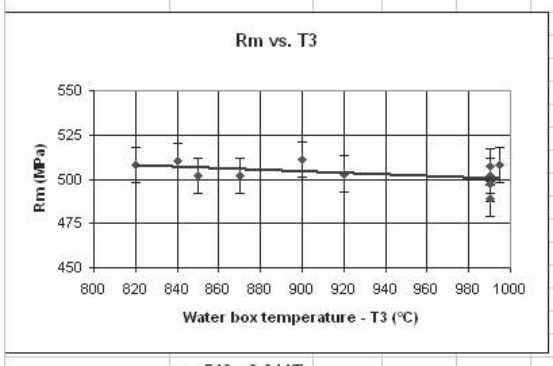
Steel grade: 10MnSi7; Supplier: Lucchini Piombino SpA; TUBEF trial: I
Tensile properties



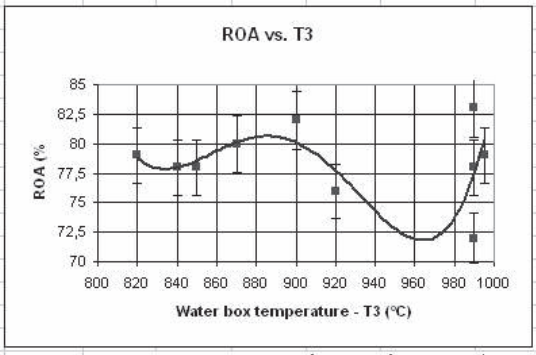
R_m independent on T_4



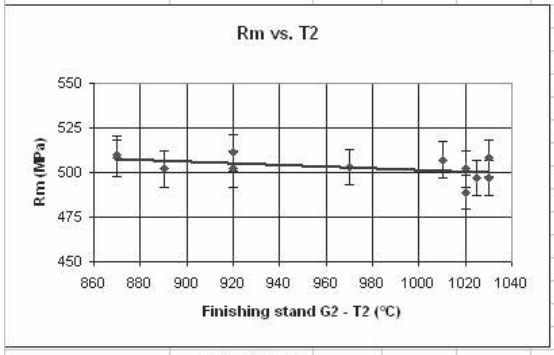
$y = -90 + 0{,}463T - 0{,}0003T_4^2$



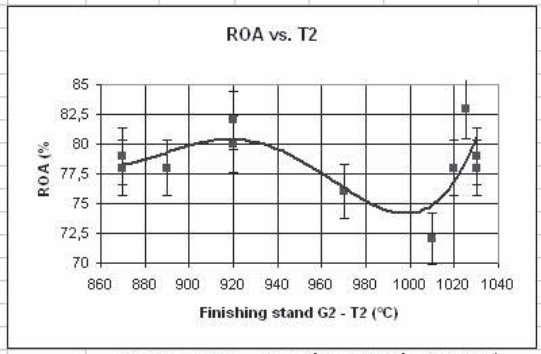
$y = 543 - 0{,}044T_3$



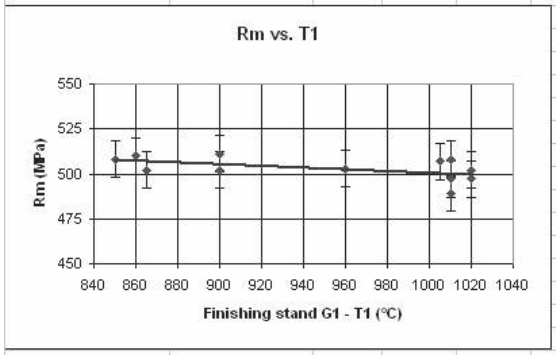
$y = 190000 - 853T_3 + 1{,}44T_3^2 - 0{,}001T_3^3 + 3E\text{-}07T_3^4$



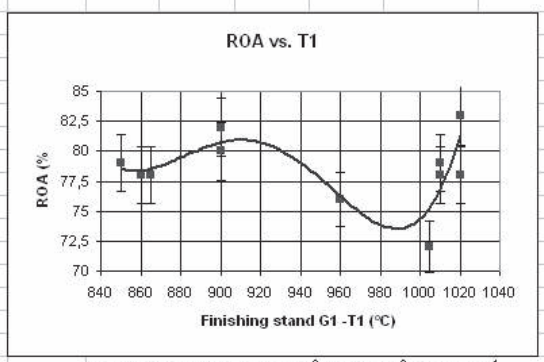
$y = 552 - 0{,}051T_2$



$y = 145000 - 628T_2 + 1{,}02T_2^2 - 0{,}0007T_2^3 + 2E\text{-}07T_2^4$

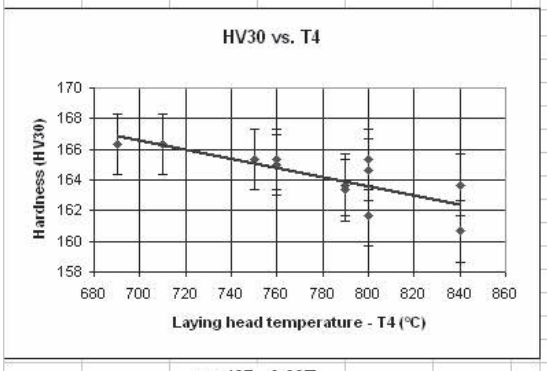


$y = 548 - 0{,}048T_1$

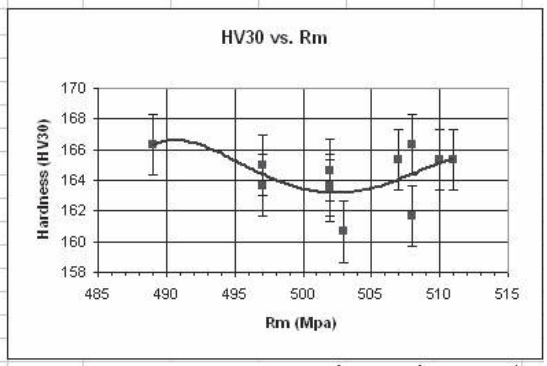


$y = 180000 - 784T_1 + 1{,}28T_1^2 - 0{,}0009T_1^3 + 3E\text{-}07T_1^4$

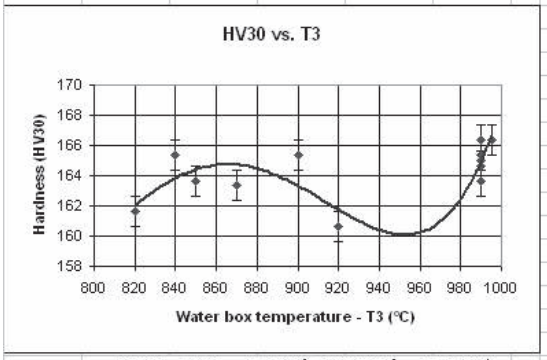
Steel grade: 10MnSi7; Supplier: Lucchini Piombino SpA; TUBEF trial: I
Hardness



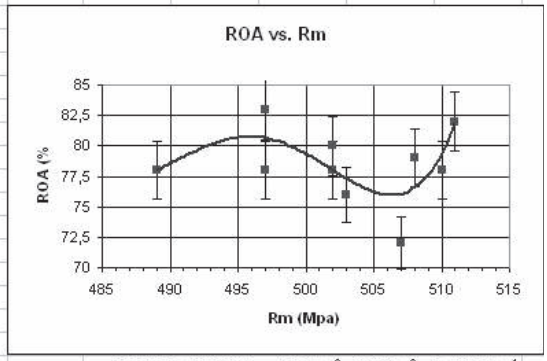
$y = 187 - 0{,}03T_4$



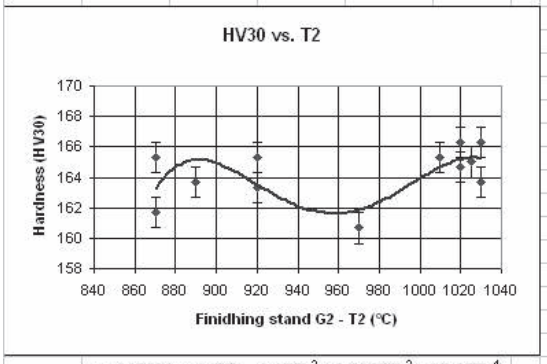
$y = -1E{+}07 + 10000R_m - 300R_m^2 + 0{,}40R_m^3 - 0{,}0002R_m^4$



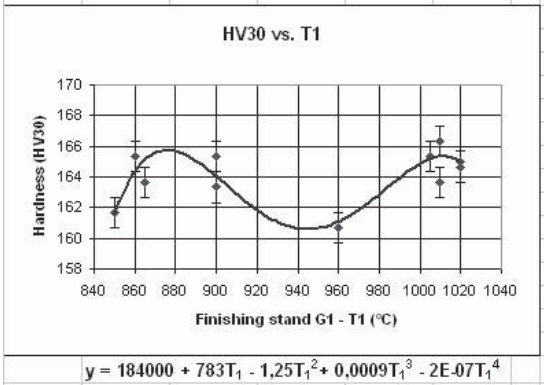
$y = 36000 - 170T_3 + 0{,}30T_3^2 - 0{,}0002T_3^3 + 7E{-}08T_3^4$



$y = 3E{+}07 - 200000R_m + 622R_m^2 - 0{,}84R_m^3 + 0{,}0004R_m^4$

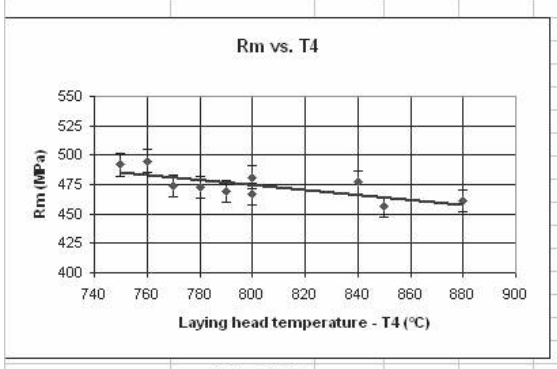


$y = 147000 + 617T_2 - 0{,}97T_2^2 + 0{,}0007T_2^3 - 2E{-}07T_2^4$

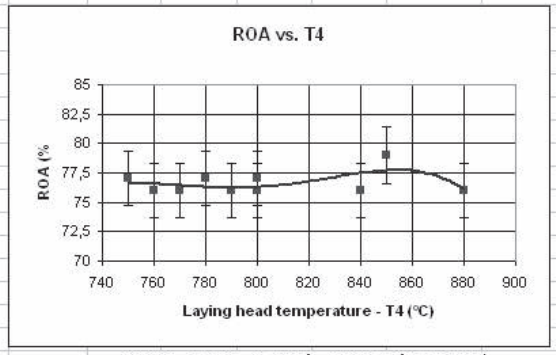


$y = 184000 + 783T_1 - 1{,}25T_1^2 + 0{,}0009T_1^3 - 2E{-}07T_1^4$

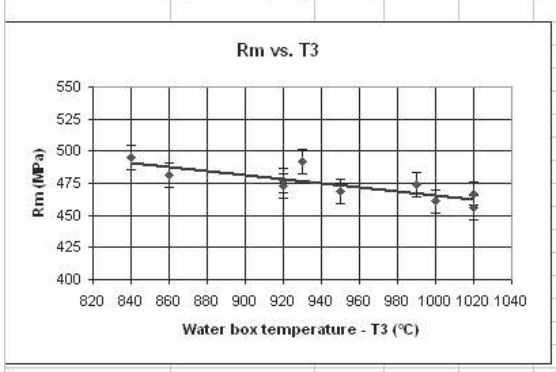
Steel grade: 10MnSi7; Supplier: Lucchini Piombino SpA; TUBEF trial: II
Tensile properties



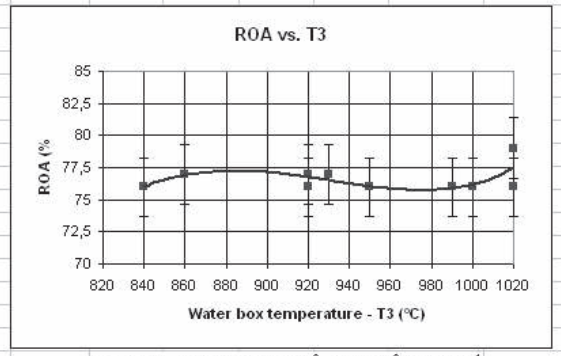
$y = 650 - 0{,}22T_4$



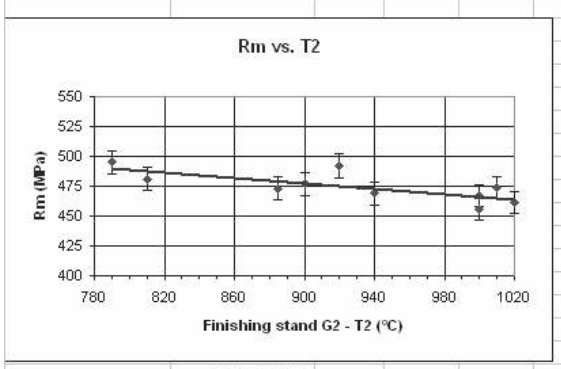
$y = 50000 + 250T_4 - 0{,}47T_4^2 + 0{,}0004T_4^3 - 1E\text{-}07T_4^4$



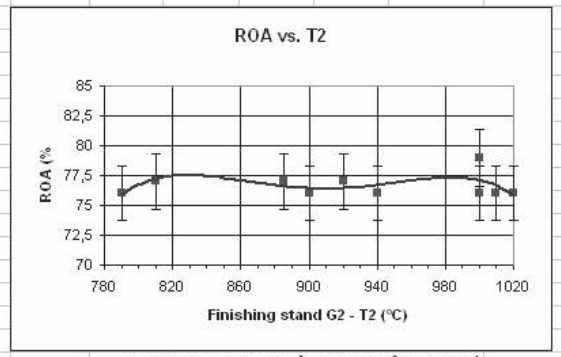
$y = 625 - 0{,}16T_3$



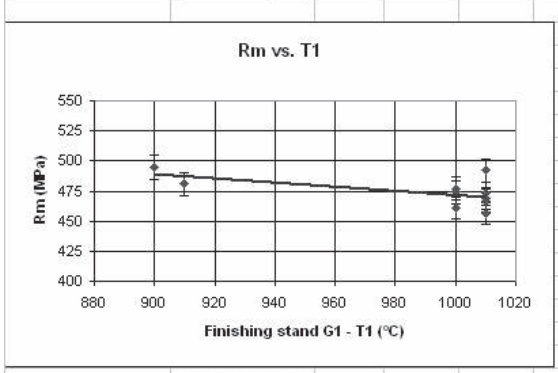
$y = 2500 - 13{,}15T_3 + 0{,}03T_3^2 - 2E\text{-}05T_3^3 - 7E\text{-}09T_3^4$



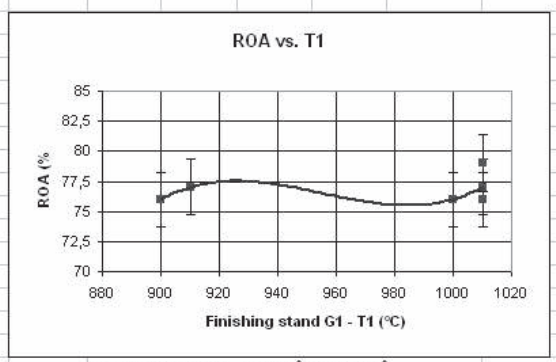
$y = 580 - 0{,}11T_2$



$y = 20000 + 86T_2 - 0{,}14T_2^2 + 0{,}0001T_2^3 - 3E\text{-}08T_2^4$

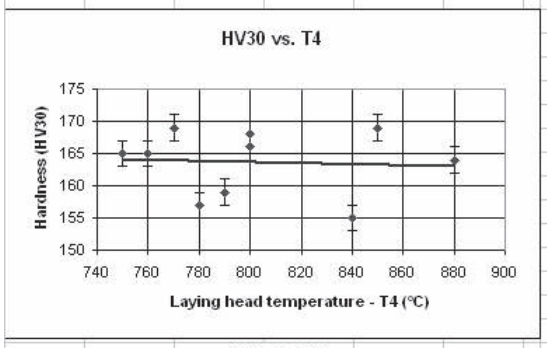


$y = 640 - 0{,}17T_1$

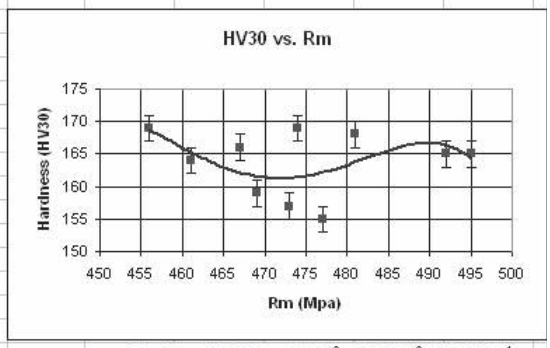


$y = 18000 + 55T_1 - 0{,}06T_1^2 + 2E\text{-}05T_1^3$

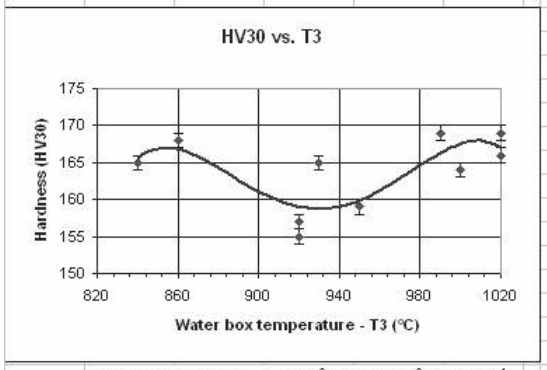
Steel grade: 10MnSi7; Supplier: Lucchini Piombino SpA; TUBEF trial: II
Hardness



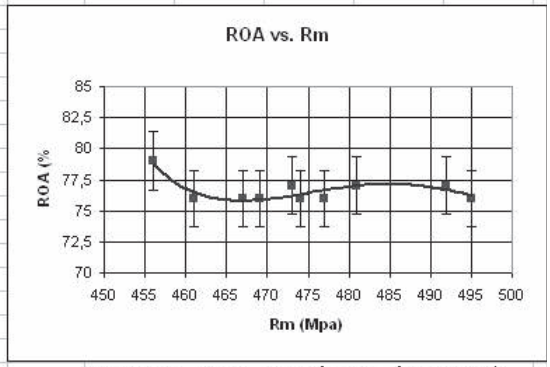
$y = 171 - 0{,}01T_4$



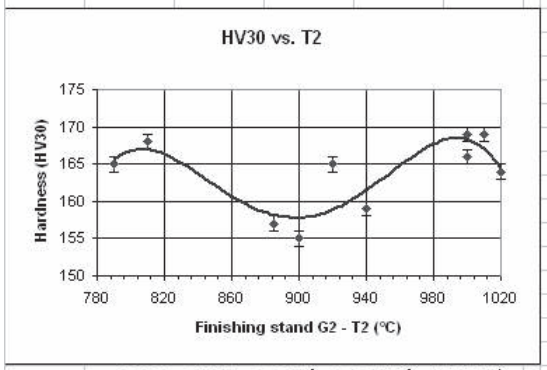
$y = -2E{+}06 + 19000R_m - 60R_m^2 + 0{,}09R_m^3 - 5E{-}05R_m^4$



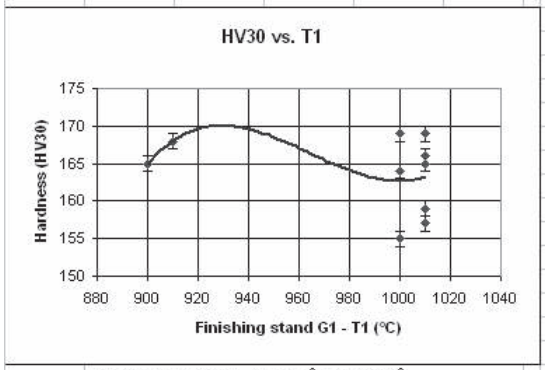
$y = 180000 + 780T_3 - 1{,}26T_3^2 + 0{,}0009T_3^3 - 2E{-}07T_3^4$



$y = 570000 - 4670R_m + 14R_m^2 - 0{,}02R_m^3 + 1E{-}05R_m^4$

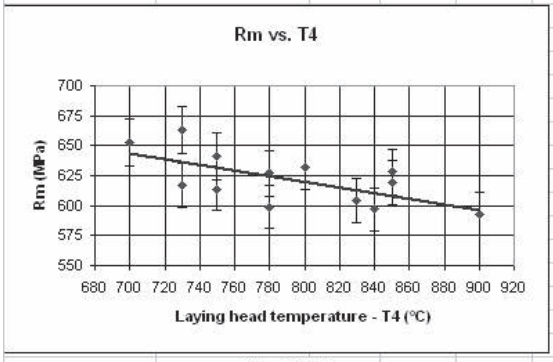


$y = 80000 + 370T_2 - 0{,}62T_2^2 + 0{,}0005T_2^3 - 1E{-}07T_2^4$

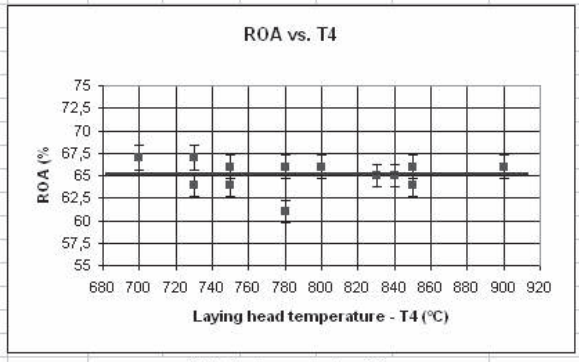


$y = 40000 + 120T_1 - 0{,}12T_1^2 + 4E{-}05T_1^3$

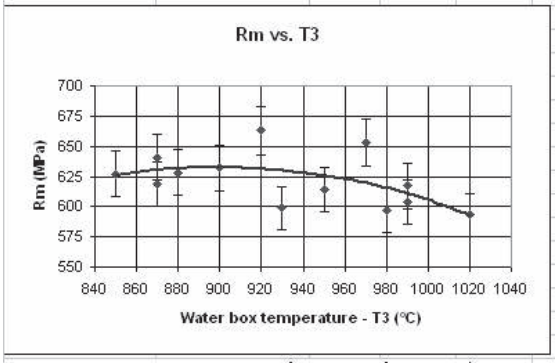
Steel grade: 30MnB4; Supplier: Lucchini Piombino SpA; TUBEF trial: I
Tensile properties



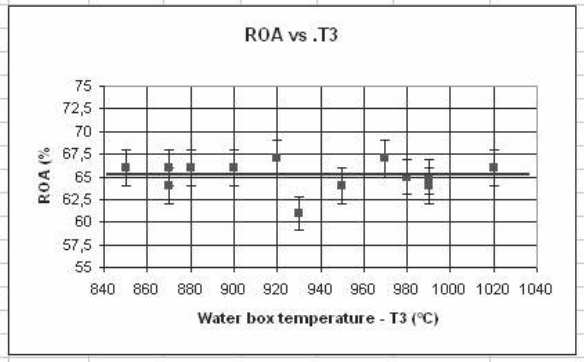
$y = 811 - 0{,}24T_4$



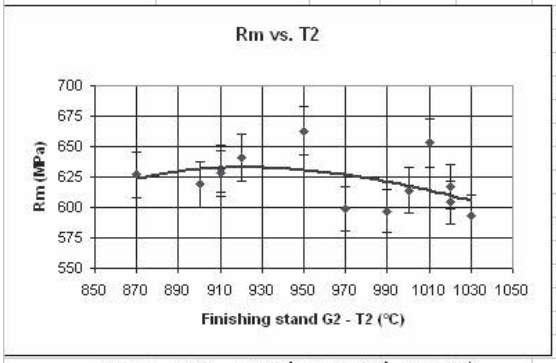
ROA independent on T_4



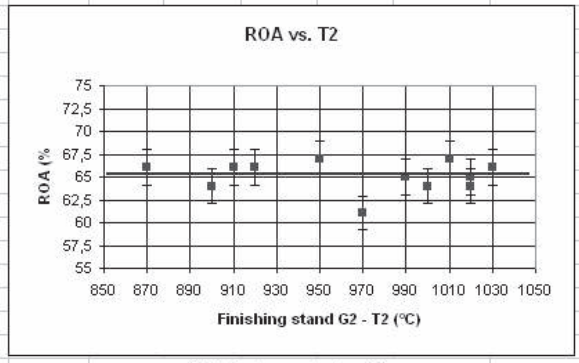
$y = -45000 + 200T_3 - 0{,}32T_3^2 + 0{,}0002T_3^3 - 6E\text{-}08T_3^4$



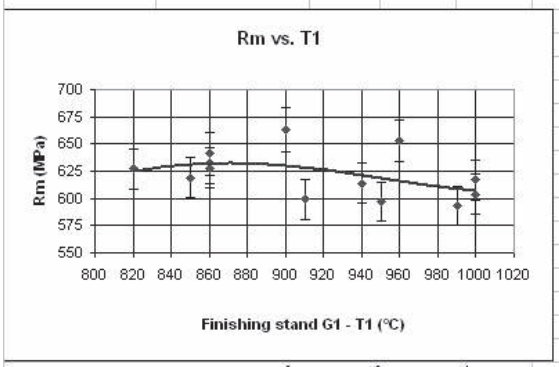
ROA independent on T_3



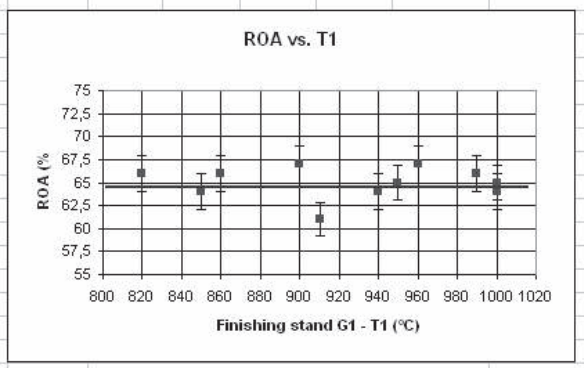
$y = -36000 + 145T_2 - 0{,}22T_2^2 + 0{,}0001T_2^3 - 4E\text{-}08T_2^4$



ROA independent on T_2

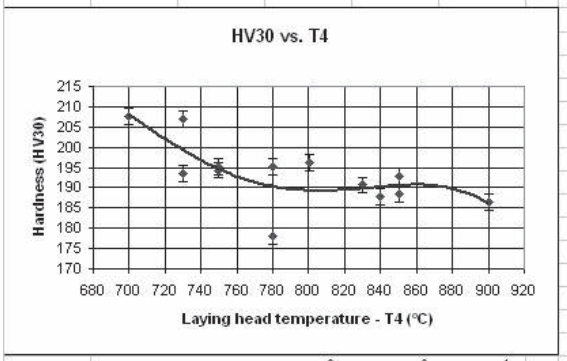


$y = 25000 - 115T_1 + 0{,}21T_1^2 - 0{,}0002T_1^3 + 5E\text{-}08T_1^4$

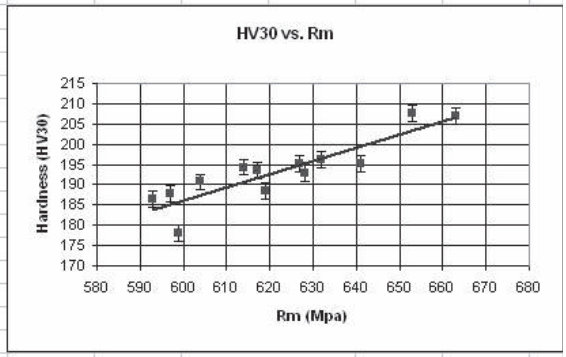


ROA independent on T_1

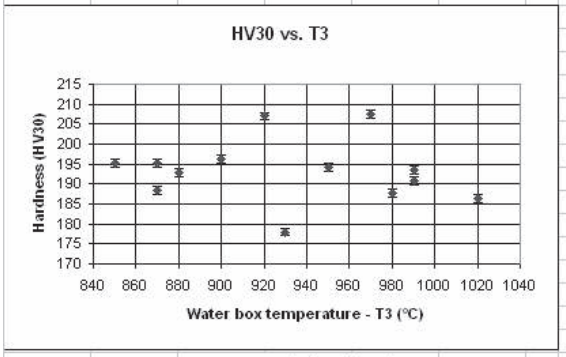
Steel grade: 30MnB4; Supplier: Lucchini Piombino SpA; TUBEF trial: I
Hardness



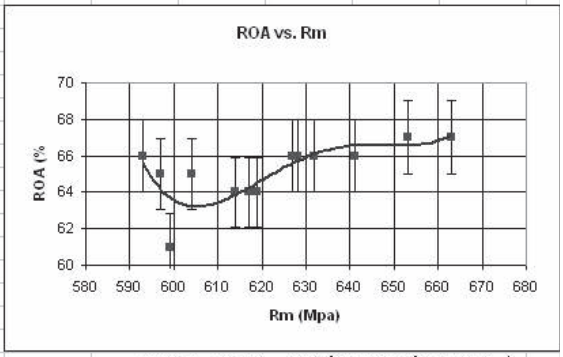
$y = -23000 + 125T_4 - 0{,}25T_4^2 + 0{,}0002T_4^3 - 7E\text{-}08T_4^4$



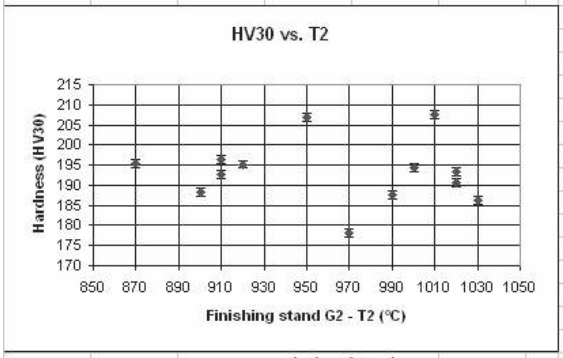
$y = -10 + 0{,}33R_m$



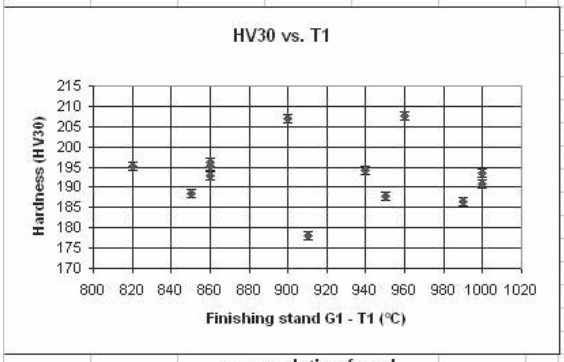
no correlation found



$y = 480000 - 3000R_m + 7R_m^2 - 0{,}008R_m^3 + 3E\text{-}06R_m^4$

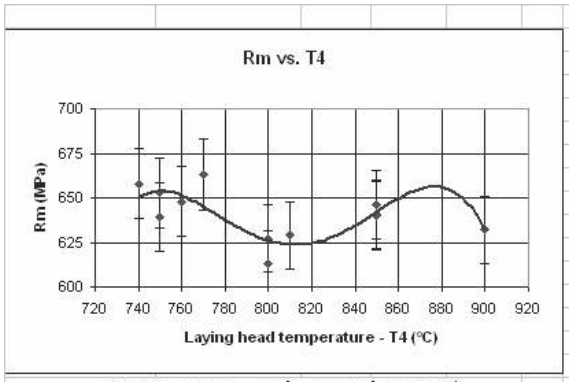


no correlation found

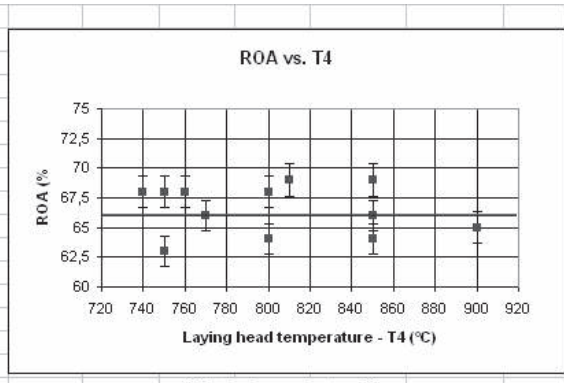


no correlation found

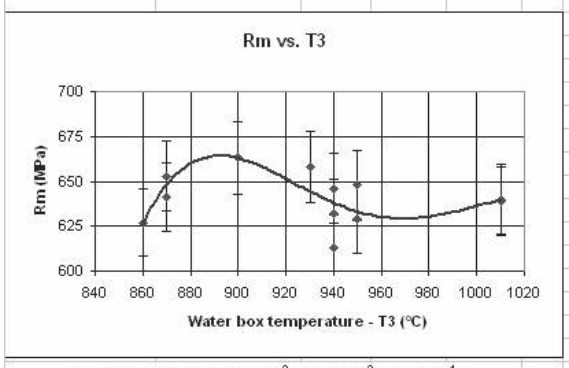
Steel grade: 30MnB4; Supplier: Lucchini Piombino SpA; TUBEF trial: II
Tensile properties



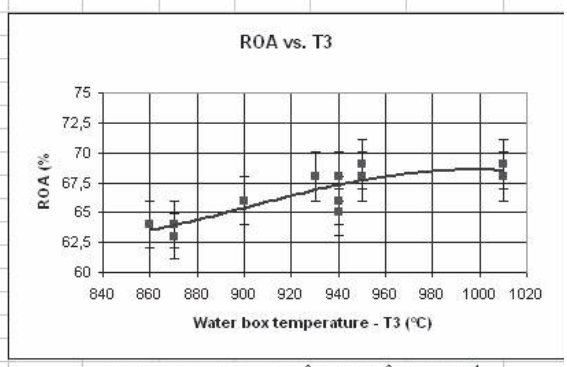
$y = -850000 + 4200T_4 - 8T_4^2 + 0{,}006T_4^3 - 2E\text{-}06T_4^4$



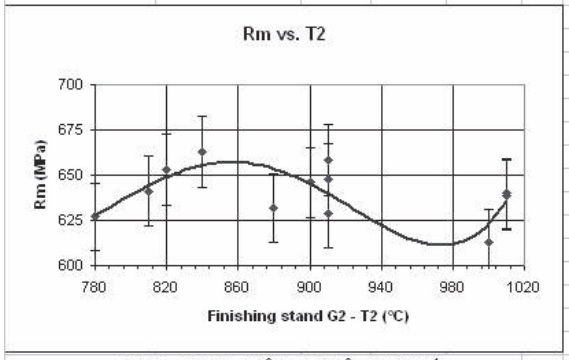
ROA independent on T_4



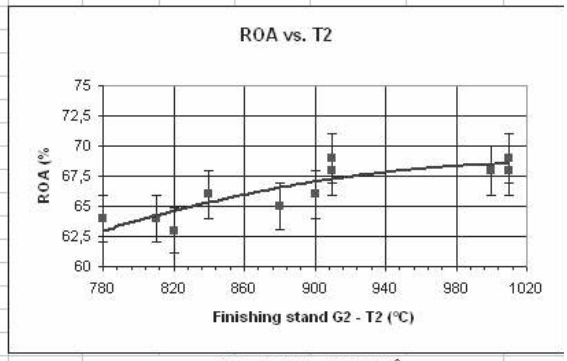
$y = -1E+06 + 4400T_3 - 7T_3^2 + 0{,}005T_3^3 - 1E\text{-}06T_3^4$



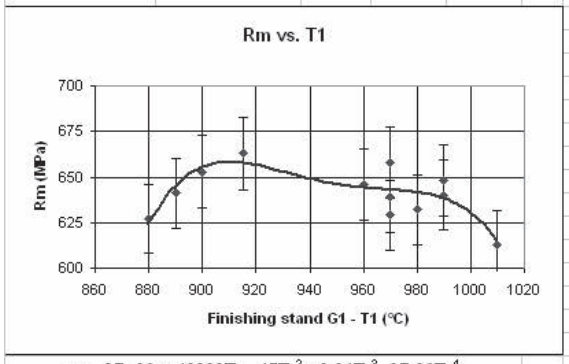
$y = 7400 - 30T_3 + 0{,}05T_3^2 - 3E\text{-}05T_3^3 + 8E\text{-}09T_3^4$



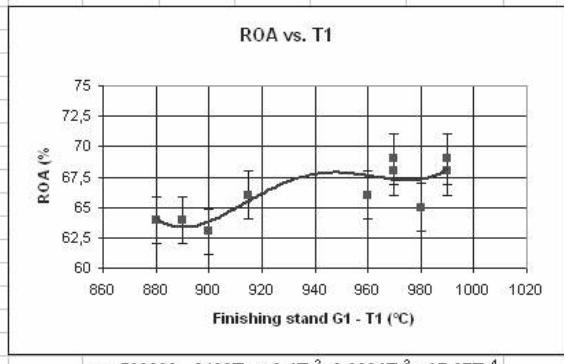
$y = 13000 - 620T_2 + T_2^2 - 0{,}001T_2^3 + 3E\text{-}07T_2^4$



$y = -27 + 0{,}19T_2 - 9E\text{-}05T_2^2$

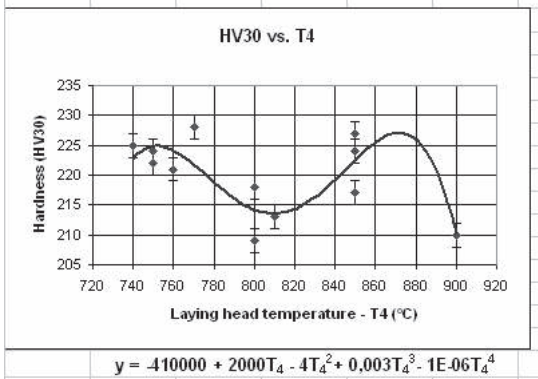


$y = -2E+06 + 10000T_1 - 15T_1^2 + 0{,}01T_1^3 - 3E\text{-}06T_1^4$

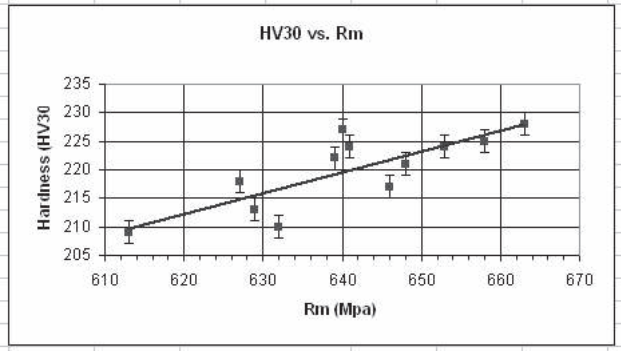


$y = 500000 - 2100T_1 + 3{,}4T_1^2 - 0{,}0024T_1^3 + 6E\text{-}07T_1^4$

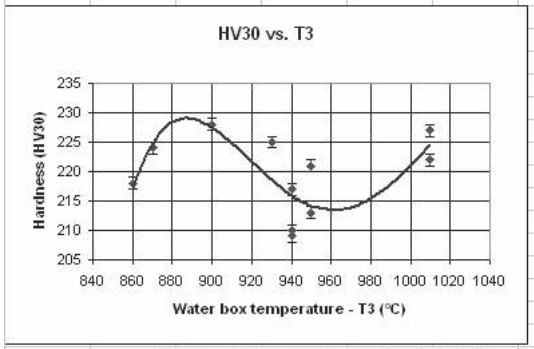
Steel grade: 30MnB4; Supplier: Lucchini Piombino SpA; TUBEF trial: II
Hardness



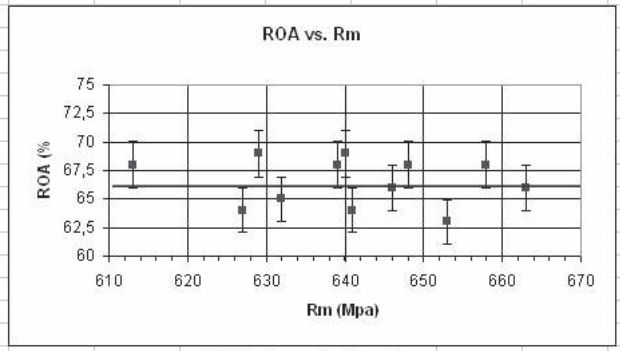
$y = -410000 + 2000T_4 - 4T_4^2 + 0,003T_4^3 - 1E\text{-}06T_4^4$



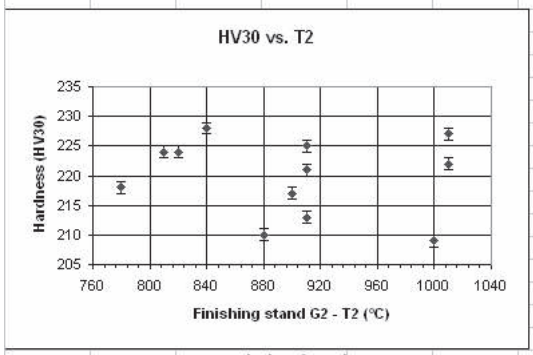
$y = -13 + 0,36R_m$



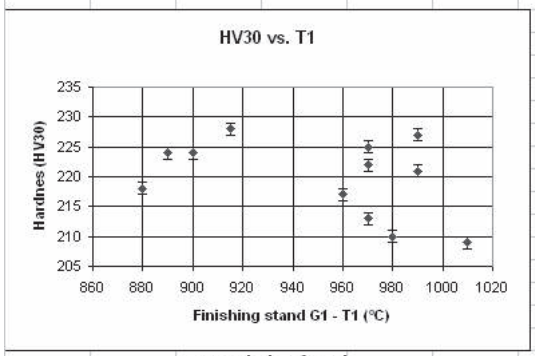
$y = -430000 + 1800T_4 - 3T_4^2 + 0,002T_4^3 - 5E\text{-}07T_4^4$



ROA independent on Rm

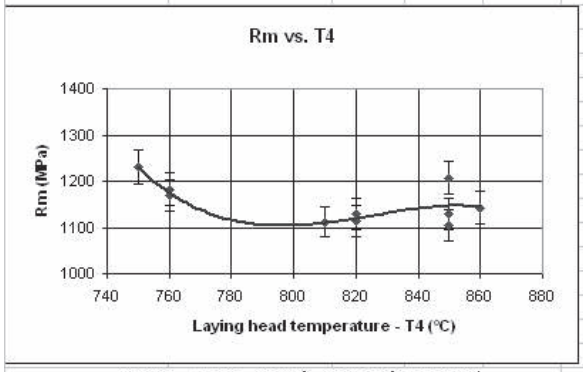


no correlation found

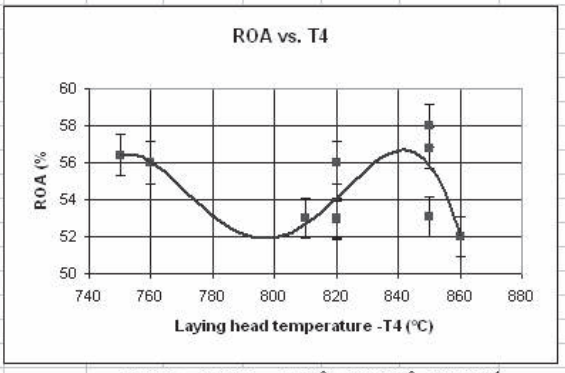


no correlation found

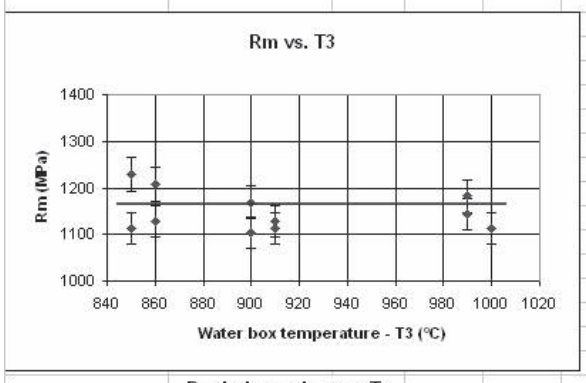
Steel grade: 54SiCrV6; Supplier: Lucchini Piombino SpA; TUBEF trial: I
Tensile properties



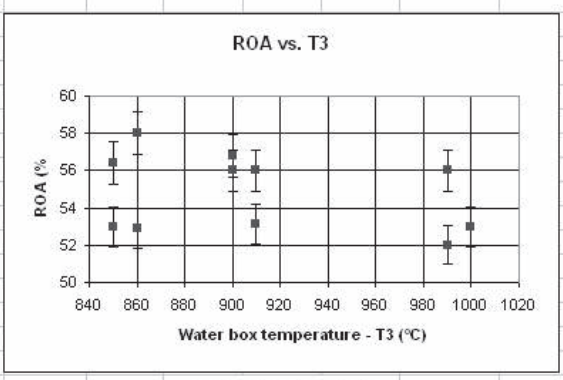
$y = 38000 + 155T_4 - 0{,}9T_4^2 + 0{,}001T_4^3 - 5E\text{-}07T_4^4$



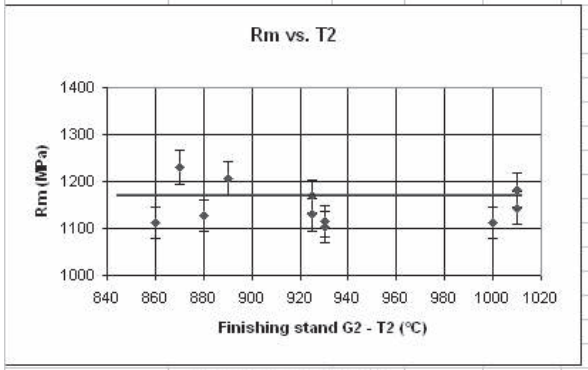
$y = 480000 + 2400T_4 - 4{,}5T_4^2 + 0{,}004T_4^3 - 1E\text{-}06T_4^4$



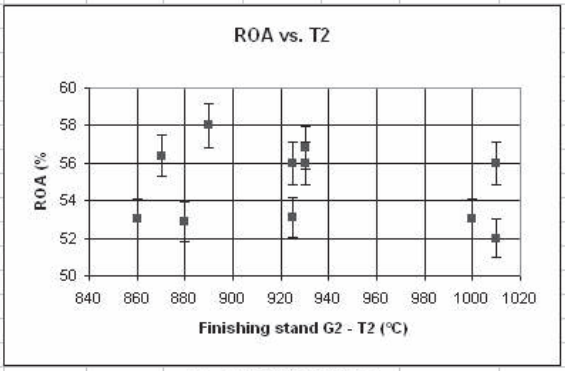
Rm independent on T_3



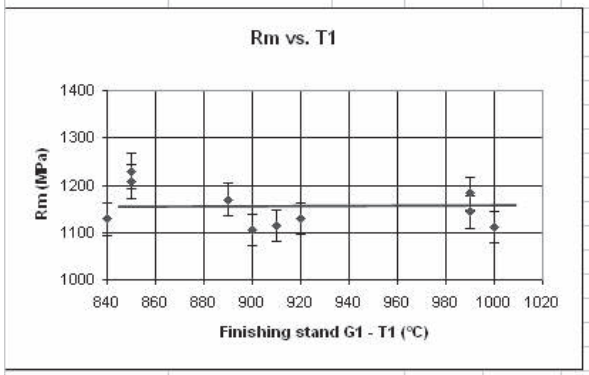
no correlation found



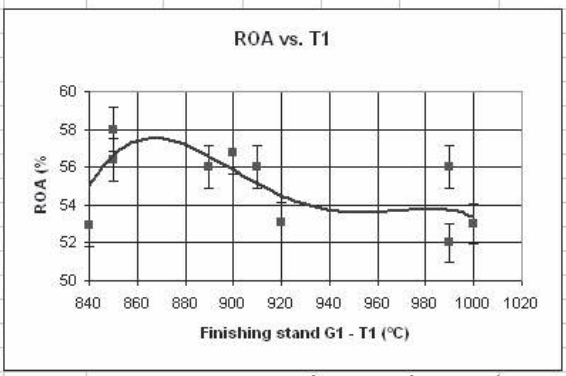
Rm independent on T_2



no correlation found

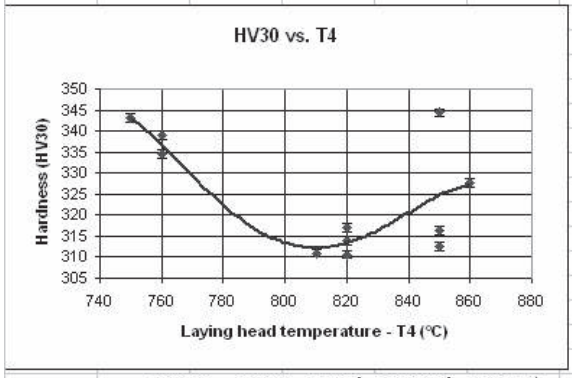


Rm independent on T_1

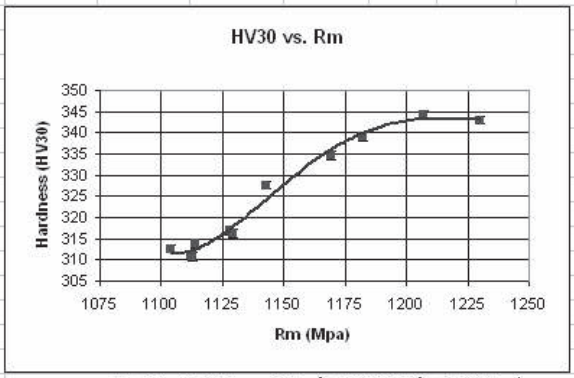


$y = 100000 + 410T_1 - 0{,}7T_1^2 + 0{,}0005T_1^3 - 1E\text{-}07T_1^4$

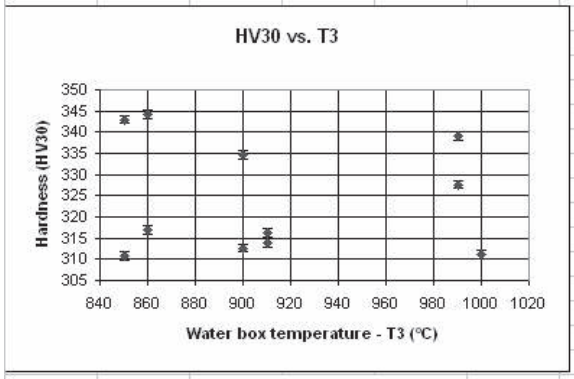
Steel grade: 54SiCrV6; Supplier: Lucchini Piombino SpA; TUBEF trial: I
Hardness



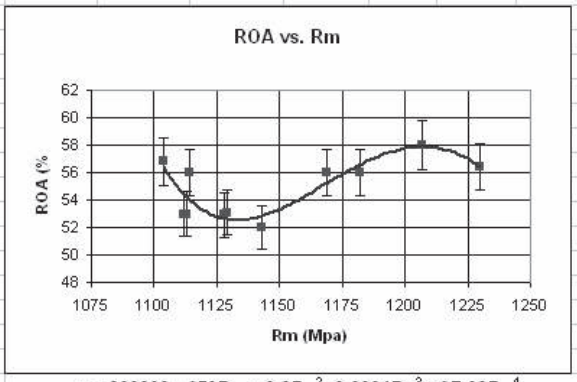
$y = 600000 + 3000T_4 - 5{,}6T_4^2 + 0{,}005T_4^3 - 1E\text{-}06T_4^4$



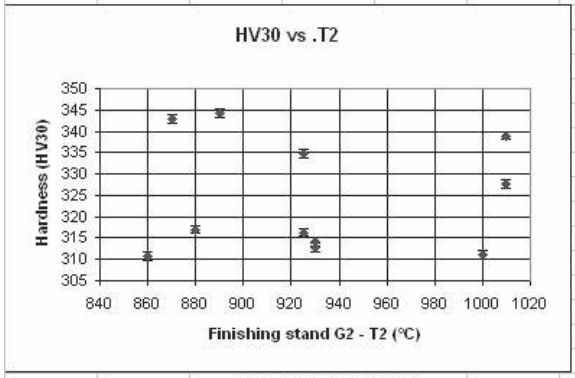
$y = 1E{+}06 - 3500R_m + 4{,}5R_m^2 - 0{,}0025R_m^3 + 5E\text{-}07R_m^4$



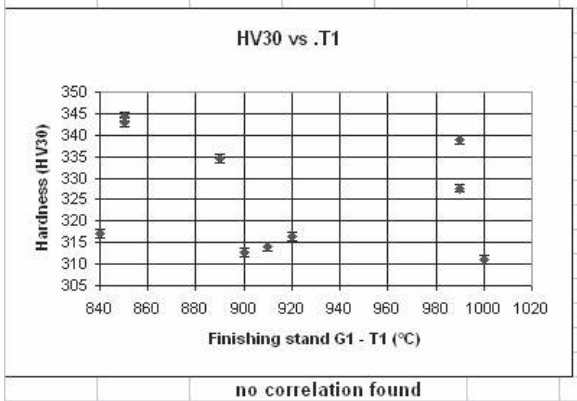
no correlation found



$y = 200000 - 650R_m + 0{,}8R_m^2 - 0{,}0004R_m^3 + 9E\text{-}08R_m^4$

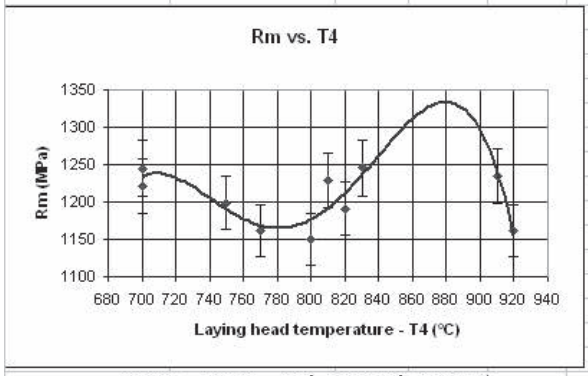


no correlation found

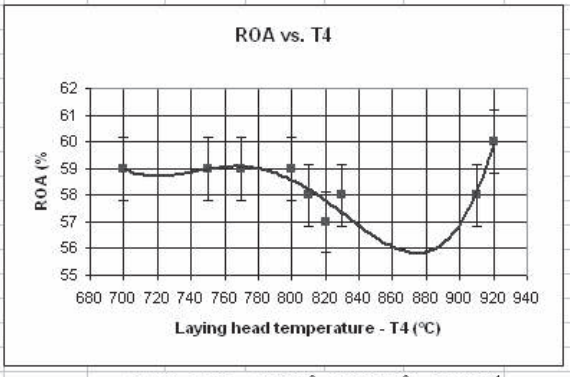


no correlation found

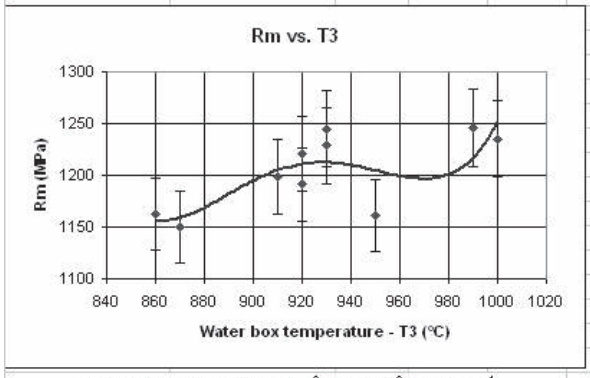
Steel grade: 54SiCrV6; Supplier: Lucchini Piombino SpA; TUBEF trial: II
Tensile properties



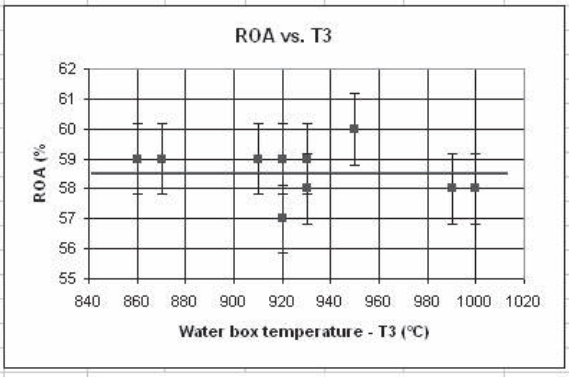
$y = -800000 + 4200T_4 - 8T_4^2 + 0{,}007T_4^3 - 2E\text{-}06T_4^4$



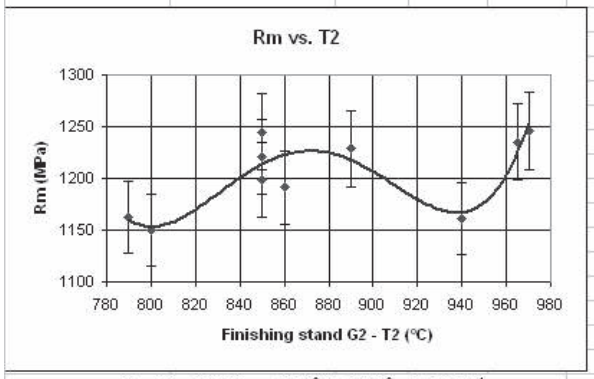
$y = 14500 - 74T_4 + 0{,}14T_4^2 - 0{,}0001T_4^3 + 4E\text{-}08T_4^4$



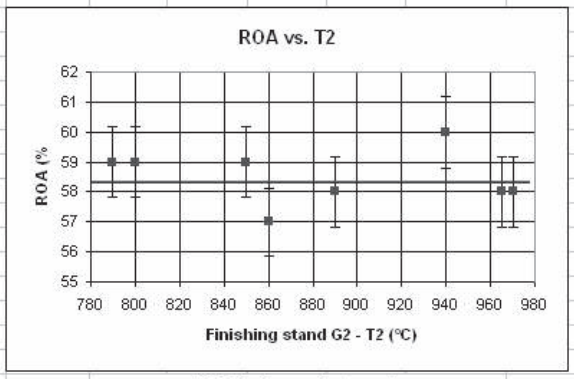
$y = 3E+06 - 12000T_3 + 20T_3^2 - 0{,}014T_3^3 + 4E\text{-}06T_3^4$



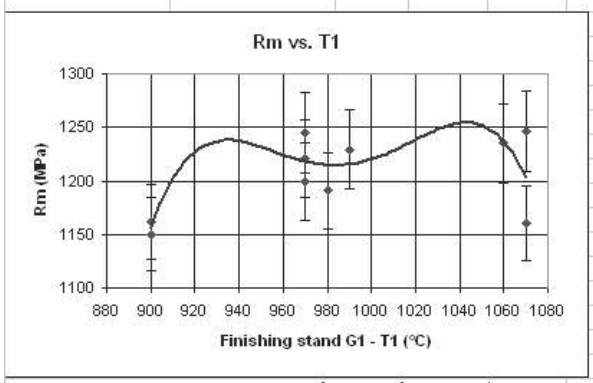
ROA independent on T_3



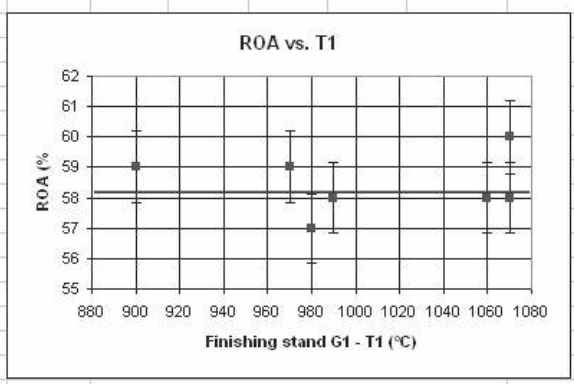
$y = 2E+06 - 8000T_2 + 13T_2^2 - 0{,}01T_2^3 + 3E\text{-}06T_2^4$



ROA independent on T_2

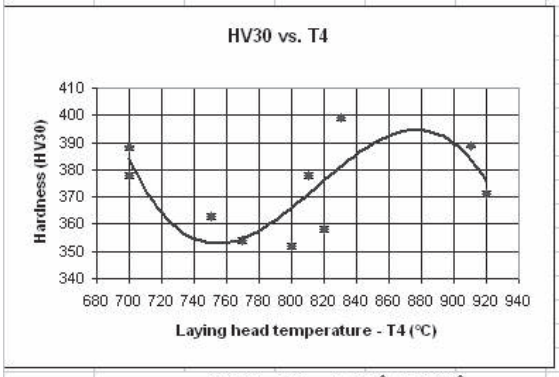


$y = -3E+06 + 14000T_1 - 21T_1^2 + 0{,}01T_1^3 - 4E\text{-}06T_1^4$

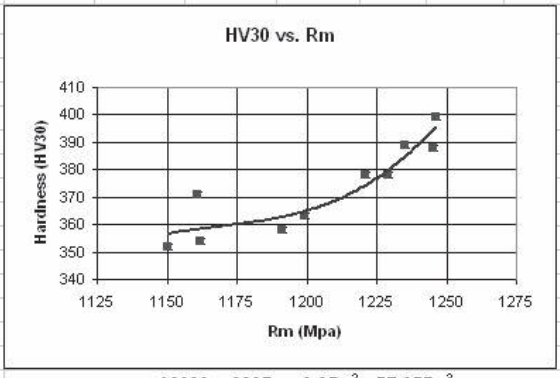


ROA independent on T_1

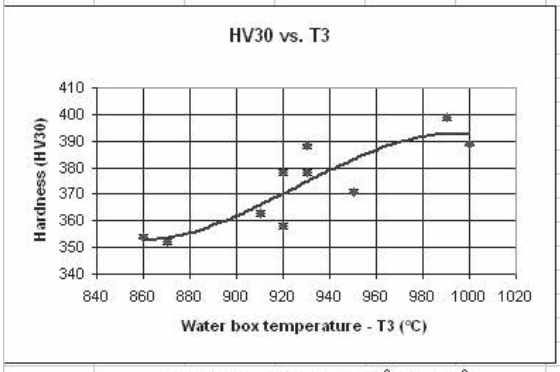
Steel grade: 54SiCrV6; Supplier: Lucchini Piombino SpA; TUBEF trial: II
Hardness



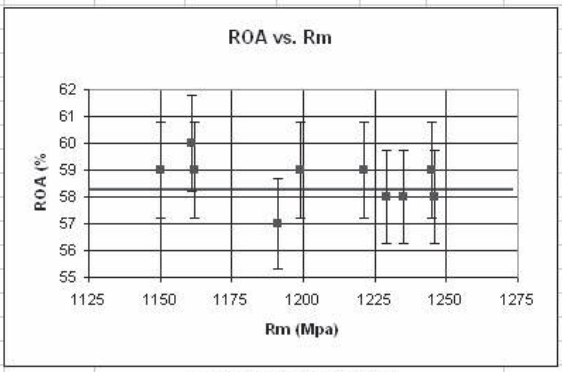
$y = 25000 - 90T_4 + 0,1T_4^2 - 4E\text{-}05T_4^3$



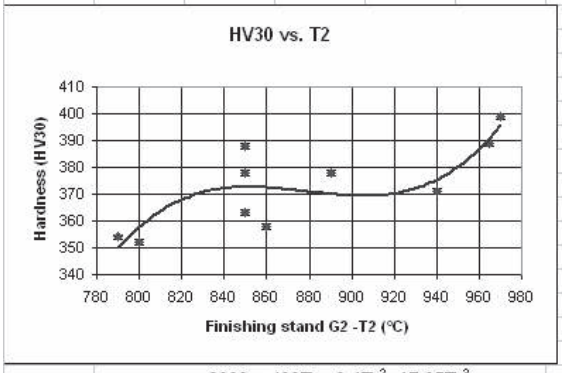
$y = 90000 + 220R_m - 0,2R_m^2 + 5E\text{-}05R_m^3$



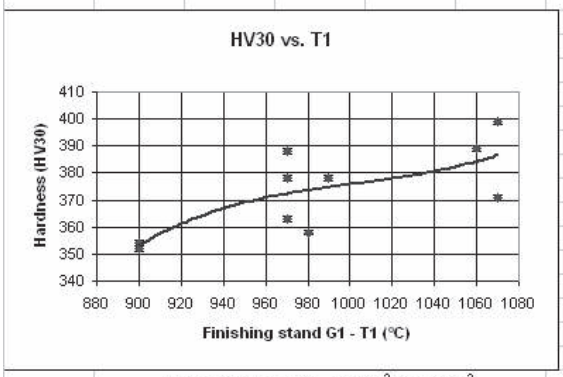
$y = 25000 - 80T_3 + 0,09T_3^2 - 3E\text{-}05T_3^3$



ROA independent on R_m



$y = -3000 + 100T_2 - 0,1T_2^2 + 4E\text{-}05T_2^3$



$y = -11500 + 35T_1 - 0,04T_1^2 + 1E\text{-}05T_1^3$

ANNEX 5.4

Cold compression tests of selected
prototype wire-rods (WP5)

Testing Procedure

Wire rods of trial I and trial II were submitted to cold compression testes. The objective was, in addition to the determination of true stress-strain curves, to verify if surface cracks appears during compression.

For this purpose, specimens from each steel rolled by Freiberg pilot mill were selected.
The compression tests were performed on cylindrical specimens (as-rolled and machined surface condition) at room temperature (RT) and using oil as lubricant (Table 5.4.1).

.	**Start Height**	**Start Diameter**	**Finish Height**	**Red.**	**Strain**	**Strain Rate**
	mm					**s^{-1}**
As-rolled condition	14.4	8	5.25	0.444	0.588	10
Machined (turned)	12.4	7	4.40	0.436	0.572	10

Table 5.4.1 Dimension of specimen and deformation conditions for compression tests.

Results

Rolling conditions, mechanical properties of as-rolled wire-rods, specimen dimensions, as well as the results of the compression tests are summarised in Tables 5.4.2 and 5.4.3. After compression test, an inspection of the specimen was performed in terms of shape and surface defects (i.e. barelling, askew, small cracks, fracture, etc.).

For technical reason, specimen No. 021 had different dimensions before compression test (h_0=9.91 mm and ∅ = 5.5 mm).

The stress-strain curves measured during the cold compression tests are reported in Figs.5.4.1 to 5.4.5.

Wire rods of Trials I and II														
No.	**Rolling**					**Tensile**		**Hardness**	**Compression Test**					
									as-rolled			machined		
	T1	T2	T3	T4	Heat treatment	UTS	ROA	HV30	h0	h1	Re.	h0	h1	Re.
	[°C]	[°C]	[°C]	[°C]	[s/°C]	[MPa]	[%]	-	[mm]	[mm]	-	[mm]	[mm]	-
54SiCrV6 (LUCCHINI)														
121	850	890	860	850	660/660	1207	58	344	14.39	5.48	-	12.62	4.70	-
124	850	870	850	750	660/660	1230	56	343	14.39	5.47	C	12.64	4.69	C
142	970	850	930	700	Air	1245	59	388	14.32	5.41	F	12.66	4.74	F
144	990	890	930	810	Air	1229	58	378	14.41	5.48	C	12.66	-	F
145	970	850	920	700	Air	1221	59	378	14.42	5.47	C	12.67	4.72	C
146	970	850	910	750	Air	1199	59	363	14.24	5.44	C	12.66	4.71	C
147	107	970	990	830	Air	1246	58	399	14.44	5.48	F	12.65	4.72	C
149	106	965	100	910	Air	1235	58	389	14.39	5.46	C	12.62	4.71	C
30MnB4 (LUCCHINI)														
221	960	101	970	700	660/660	653	67	208	14.38	5.22	C	12.66	4.50	-
225	900	950	920	730	660/660	663	67	207	14.41	5.27	C	12.63	4.55	-
229	860	910	900	800	660/660	632	66	196	14.45	5.27	C	12.64	4.57	-
230	860	920	870	750	660/660	641	66	195	14.41	5.20	A	12.65	4.51	-
241	990	910	950	760	Air	648	68	221	14.43	5.26	-	12.60	4.59	-
242	970	910	930	740	Air	658	68	225	14.38	5.29	C	12.63	4.57	-
244	980	880	940	900	Air	632	65	210	14.42	5.27	-	12.60	4.53	-
245	960	900	940	850	Air	646	66	217	14.40	5.30	-	12.60	4.52	-
246	990	101	101	850	Air	640	69	227	14.45	5.28	A	12.61	4.50	-
247	970	101	101	750	Air	639	68	222	14.42	5.23	-	12.63	4.54	-
248	915	840	900	770	Air	663	66	228	14.50	5.26	-	12.60	4.54	-
249	900	820	870	750	Air	653	63	224	14.47	5.29	A	12.63	4.52	-
250	890	810	870	850	Air	641	64	224	14.38	5.25	-	12.60	4.51	-
10MnSi7 (LUCCHINI)														
421	102	102	990	800	660/660	502	78	165	14.40	5.24	-	12.61	4.47	-
422	100	101	990	800	660/660	507	72	165	14.36	5.19	-	12.63	4.47	-
423	101	102	990	710	660/660	489	78	166	14.38	5.20	C	12.64	4.43	-
424	101	103	995	690	660/660	508	79	166	14.39	5.20	C	12.62	4.46	C
425	102	102	990	760	660/660	497	83	165	14.36	5.16	C	12.60	4.45	-
426	101	103	990	840	660/660	497	78	164	14.38	5.19	C	12.54	4.46	-
427	900	920	900	750	660/660	511	82	165	14.43	5.26	C	12.57	4.43	-
428	900	920	870	790	660/660	502	80	163	14.40	5.22	A	12.52	4.42	-
429	960	970	920	840	660/660	503	76	161	14.41	5.22	-	12.62	4.45	-
430	860	870	840	760	660/660	510	78	165	14.43	5.31	C	12.62	4.42	-
431	850	870	820	800	660/660	508	79	162	14.41	5.18	-	12.60	4.46	-
432	865	890	850	790	660/660	502	78	164	14.31	5.18	C	12.61	4.48	-
443	101	920	930	750	Air	492	77	165	14.39	5.19	-	12.61	4.45	-
444	900	790	840	760	Air	495	76	165	14.39	5.25	-	12.62	4.48	-

Table 5.4.2 Hot rolling and cooling conditions. mechanical properties and results of cold compression tests on prototype Lucchini steel wire-rods (C: Crack. F: Fracture. A: Askew).

Wire rods of Trials I and II														
No.	**Rolling**					**Tensile**		**Hardness**	**Compression Test**					
									as-rolled			machined		
	T1	T2	T3	T4	Heat treatment	UTS	**ROA**	HV30	h0	h1	Re.	h0	h1	Re.
	[°C]	[°C]	[°C]	[°C]	[s/°C]	[Mpa]	[%]	-	[mm]	[mm]	-	[mm]	[mm]	-
38Cr4 (SIDENOR)														
323	1015	1040	1010	840	660/660	740	70	-	14.42	5.26	**-**	12.61	4.57	**-**
325	1020	1040	980	770	660/660	720	69	-	14.42	5.22	**-**	12.61	4.53	**-**
331	840	870	850	800	660/660	750	70	-	14.39	5.14	**-**	12.61	4.52	**-**
332	870	900	850	760	660/660	738	70	-	14.39	5.26	**-**	12.63	4.52	**-**
344	1010	950	980	830	Air	991	48	-	14.45	5.37	**C**	12.64	4.62	**C**
347	950	860	910	800	Air	970	47	-	14.41	5.38	**-**	12.65	4.59	**C**
348	960	880	920	760	Air	954	49	-	14.39	5.35	**C**	12.64	4.60	**-**
351	830	770	790	780	Air	857	69	-	14.41	5.34	**-**	12.63	4.61	**C**
545SiCrV6 (SIDENOR)														
021	1000	1010	990	750	660/660	1157	60	-	14.38	5.29	**-**	9.91	3.62	C
022	1000	1010	990	850	660/660	1088	60	-	14.41	5.39	**C**	12.63	3.57	**C**
024	870	900	860	770	660/660	1123	54	-	14.42	5.37	**-**	12.60	3.57	**C**
025	870	910	870	800	660/660	1125	61	-	14.41	5.23	**-**	12.61	3.33	**C**
042	960	850	910	820	Air	1713	5	-	14.45	5.56	**F**	12.62	3.59	**C**
044	1060	960	980	850	Air	1551	5	-	14.19	5.52	**F**	12.62	-	**F**
049	900	780	860	800	Air	1295	58	-	14.41	5.42	**C**	12.63	3.56	**C**
050	910	790	860	760	Air	1311	57	-	14.27	5.40	**C**	12.62	3.60	**C**

Table 5.4.3 Hot rolling and cooling conditions. mechanical properties and results of cold compression tests on prototype Sidenor steel wire-rods (C: Crack. F: Fracture. A: Askew).

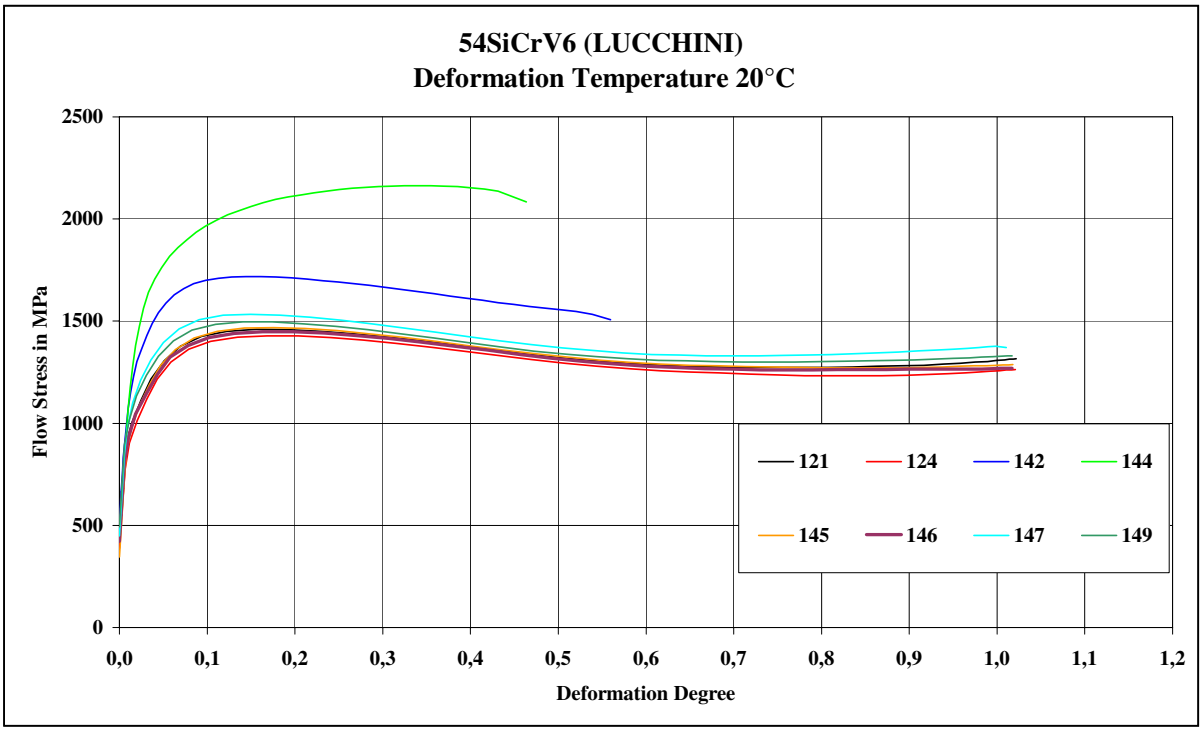


Fig.5.4.1 Cold compression flow curves of steel 54SiCrV6 (LUCCHINI).

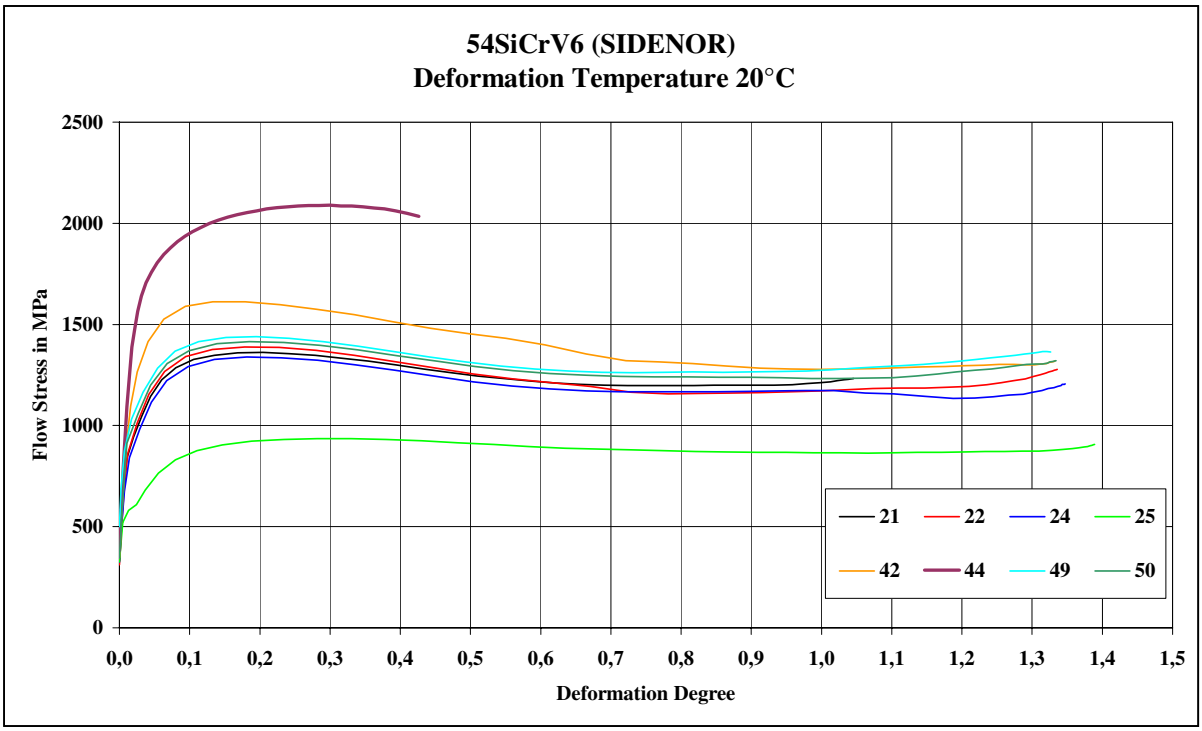


Fig.5.4.2 Cold compression flow curves of steel 54SiCrV6 (SIDENOR).

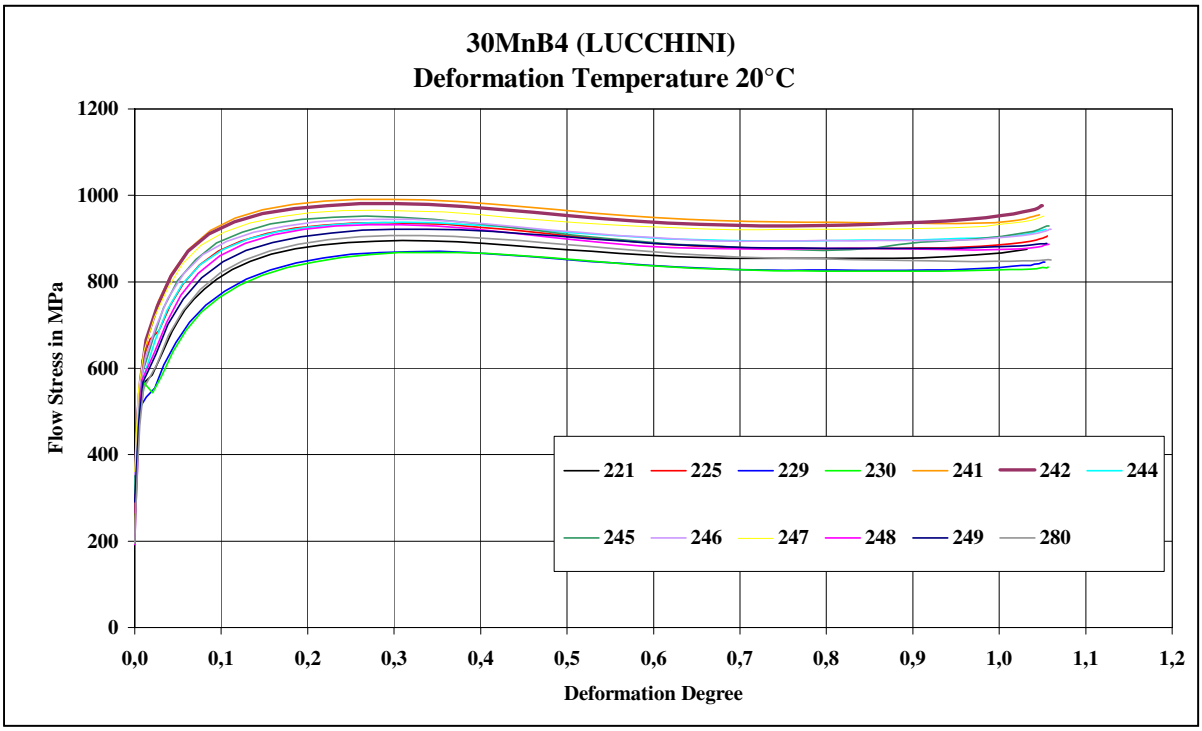


Fig.5.4.3 Cold compression flow curves of steel 30MnB4 (LUCCHINI).

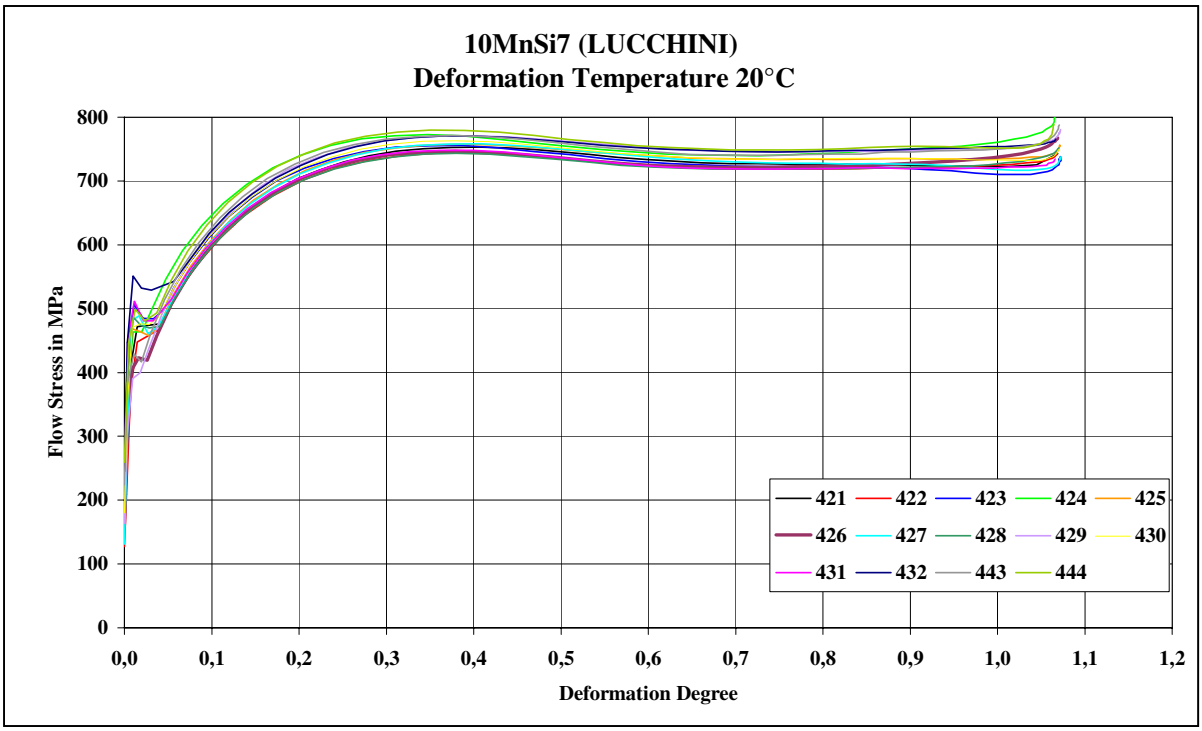


Fig.5.4.4 Cold compression flow curves of steel 10MnSi7 (LUCCHINI).

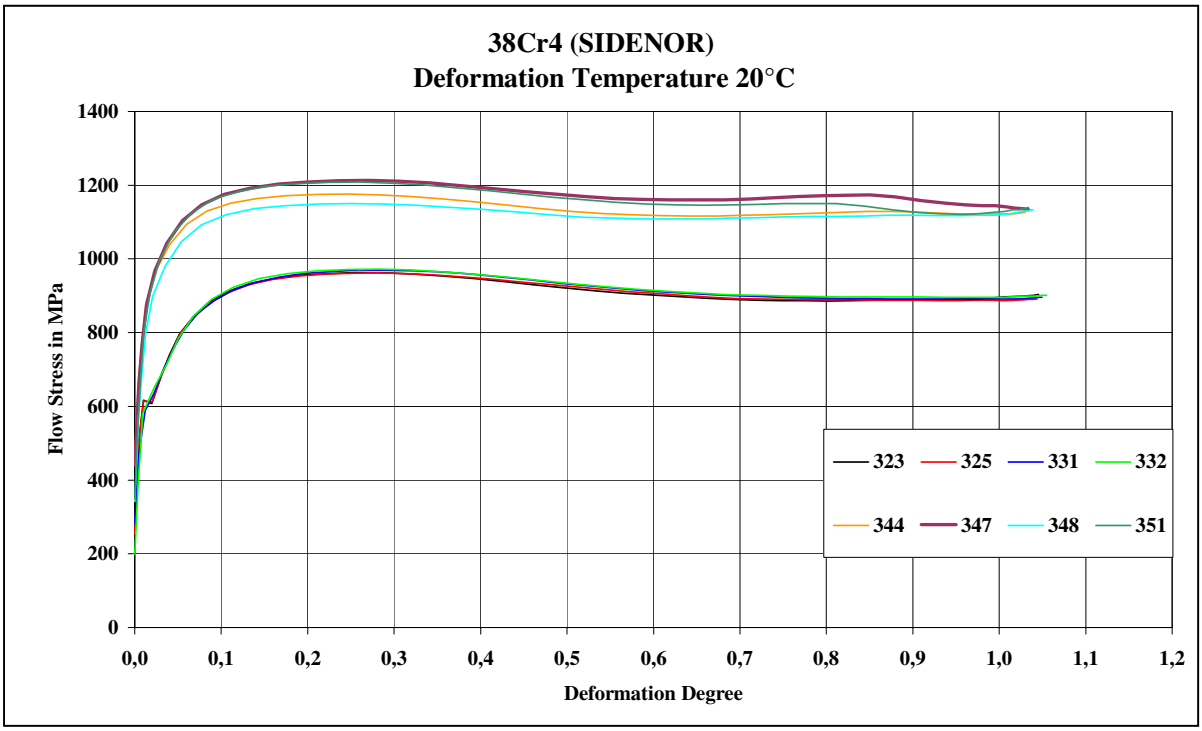


Fig.5.4.5 Cold compression flow curves of steel 38Cr4 (SIDENOR).

ANNEX 5.5

Cold drawing trials of selected
prototype wire-rods (WP5)

Experimental procedure

For this trial a drawing bench by Institute for Metal Forming was used. The cold drawing process was performed in two steps: ∅ 8.0 mm to ∅ 7.62 mm (ε = 0.097) and ∅ 7.62 mm to ∅ 7.18 mm (ε = 0.119), to obtain a total cold reduction of ε = 0.216 (Table 5.5.1).
Acid pickling was not possible, but the material was grinded to remove the typical scale formed during the rolling stage.

Steel	**Specimen Code**	**Cooling**	**Cold drawing** Final ∅: 7.18 mm	**Note**
54SiCrV6 (SID)	061	200°C / 600s	2 steps (∅7.62-∅7.18)	
54SiCrV6 (LUC)	176	500°C / 600s	2 steps (∅7.62-∅7.18)	Failure at the 2nd drawing step
30MnB4 (LUC)	268	Air	2 steps (∅7.62-∅7.18)	
38Cr4 (SID)	363	Air	2 steps (∅7.62-∅7.18)	
10MnSi7 (LUC)	485	600°C / 600s	2 steps (∅7.62-∅7.18)	

Table 5.5.1 Cold drawing trials of prototype wire-rods from the 3rd trial.

Steels 54SiCrV6 (SIDENOR), 30MnB4 (LUCCHINI), 38Cr4 (SIDENOR) and 10MnSi7 (LUCCHINI) exhibited good drawability. Steel 54SiCrV6 (LUCCHINI) failed during the 2nd drawing step.

Figs 5.5.1 to 5.5.10 show examples of the microstructures of the cold drawn wires, whilst the corresponding hardness values are reported in Table 5.5.2.

The wire from Lucchini steel 54SiCrV6, sample 176 (Figs 5.5.3 and 5.5.4), exhibited more evident bands due to segregated regions in the industrial heat, compared with Sidenor steel 54SiCrV6, sample 061 (Figs 5.5.1 and 5.5.2).

Steel	**Specimen No.**	**Average HV10**	**Central HV10**
54SiCrV6 (SIDENOR)	061	472	466
54SiCrV6 (LUCCHINI)	176	505	470
30MnB4	268	279	288
38Cr4	363	355	333
10MnSi7	485	230	234

Table 5.5.2 Hardness of as-drawn wires.

STEEL 54SiCrV6 (SIDENOR)

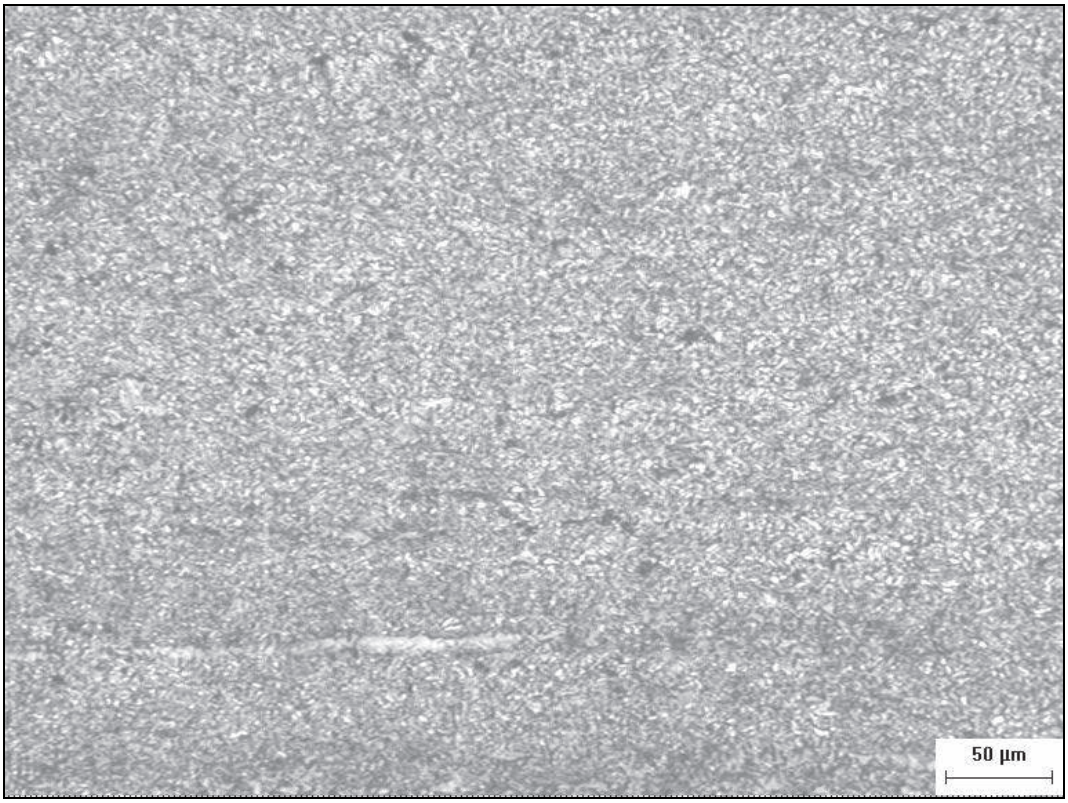


Fig. 5.5.1 Specimen 061 Long (200X).

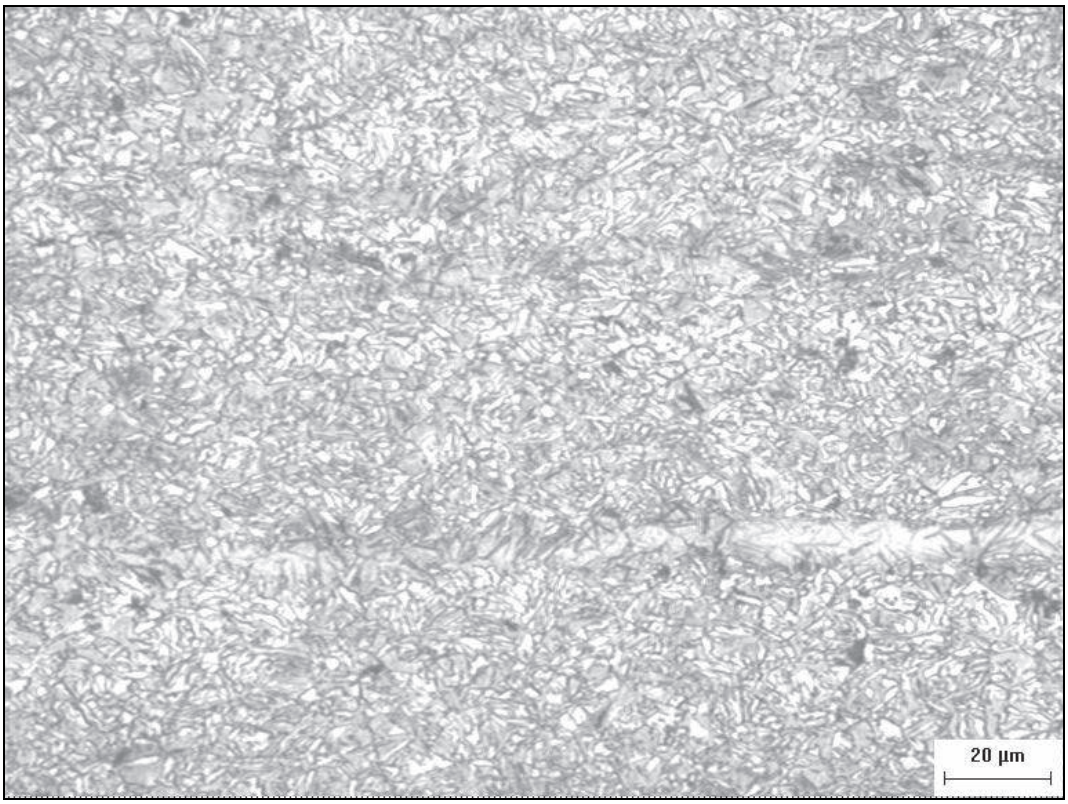


Fig. 5.5.2 Specimen 061 Long (500X).

STEEL 54SiCrV6 (LUCCHINI)

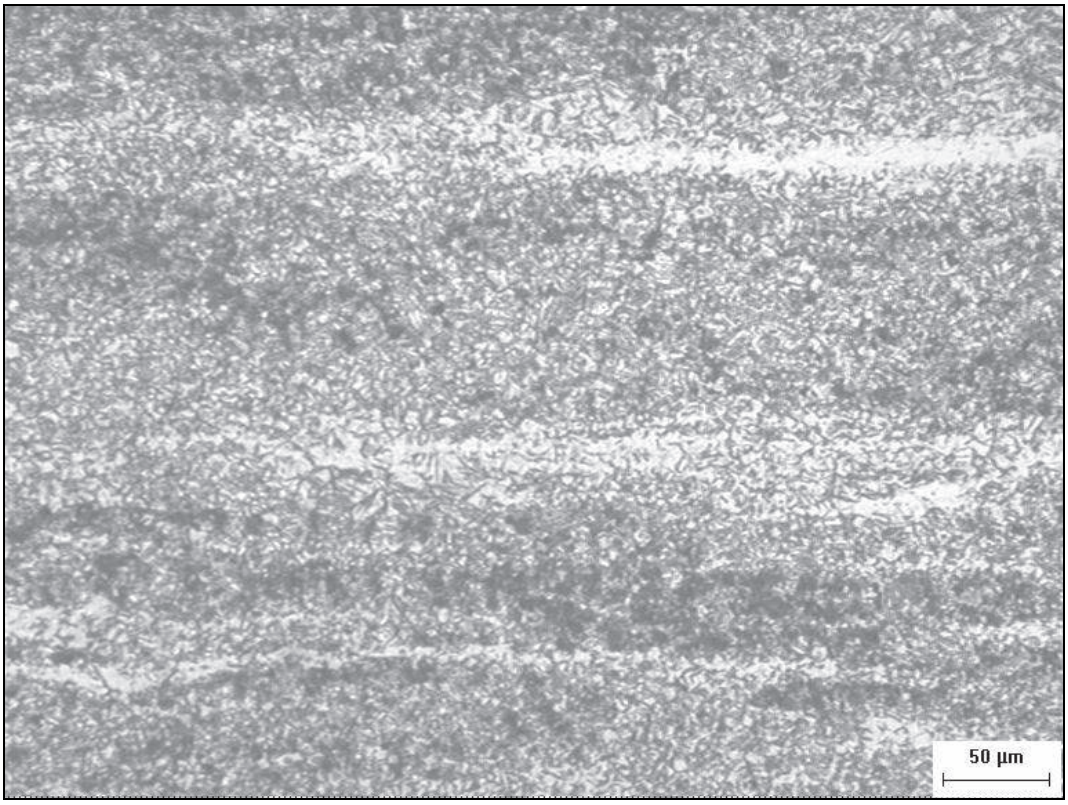


Fig. 5.5.3 Specimen 176 Long. (200X).

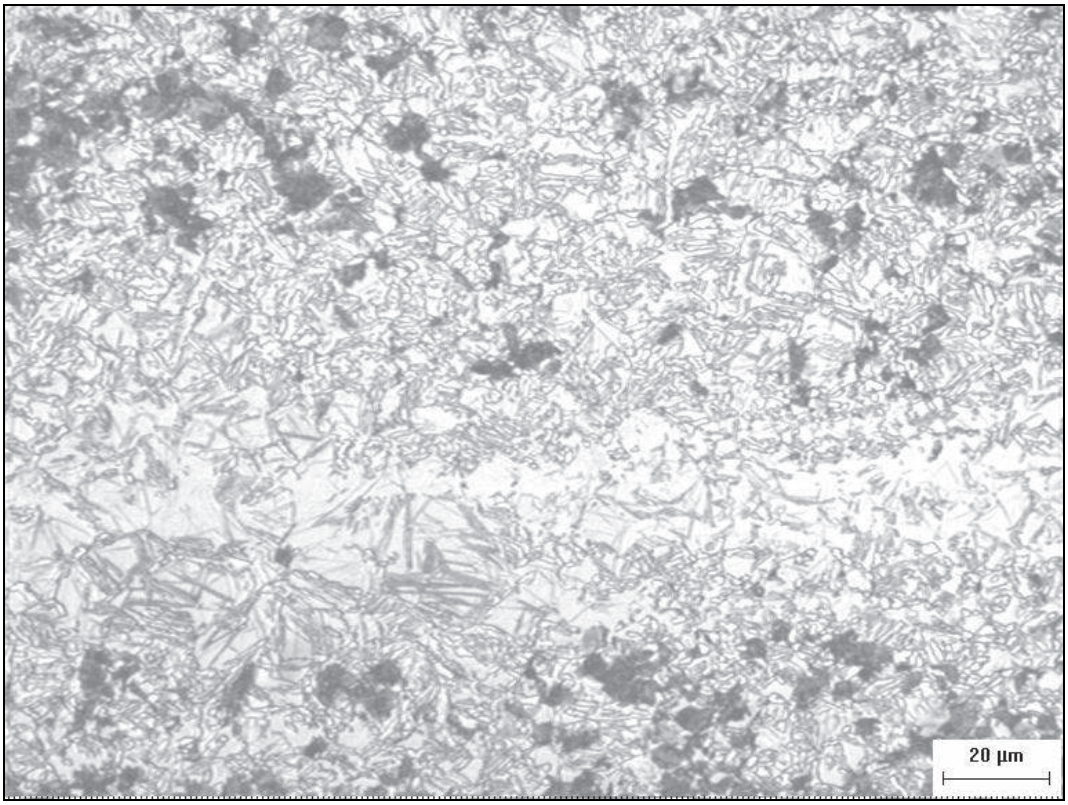


Fig. 5.5.4 Specimen 176 Long. (500X).

STEEL 30MnB4

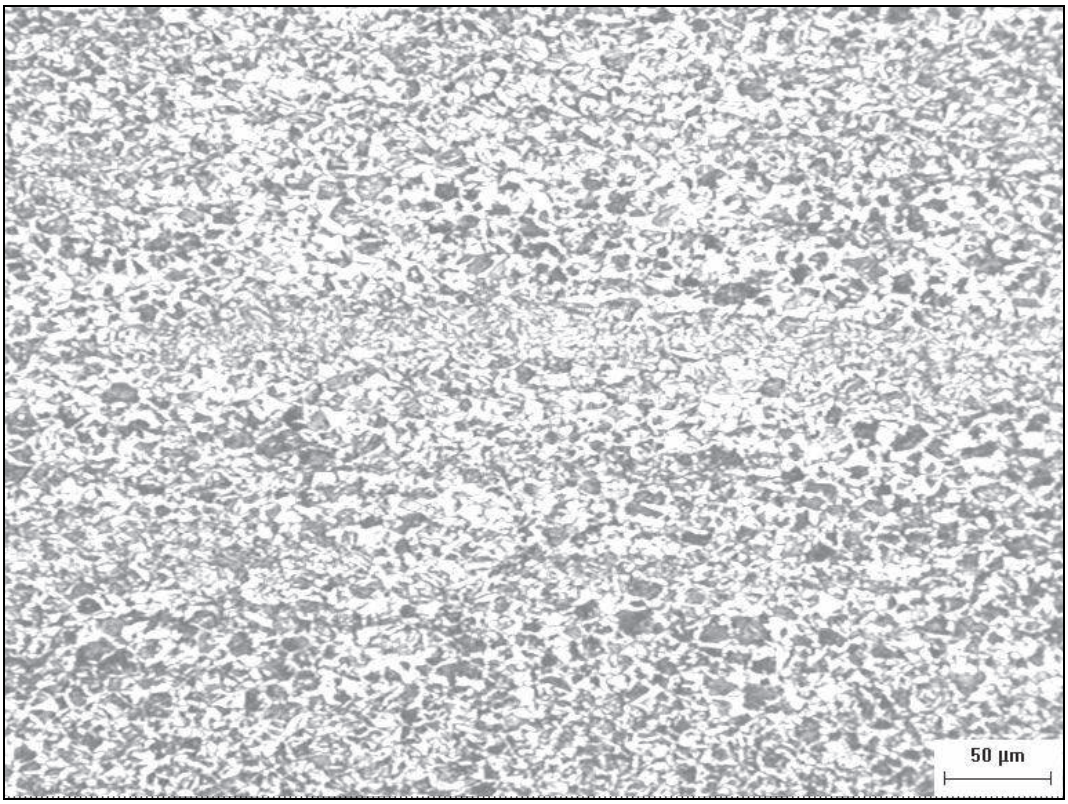


Fig. 5.5.5 Specimen 268 Long. (200X).

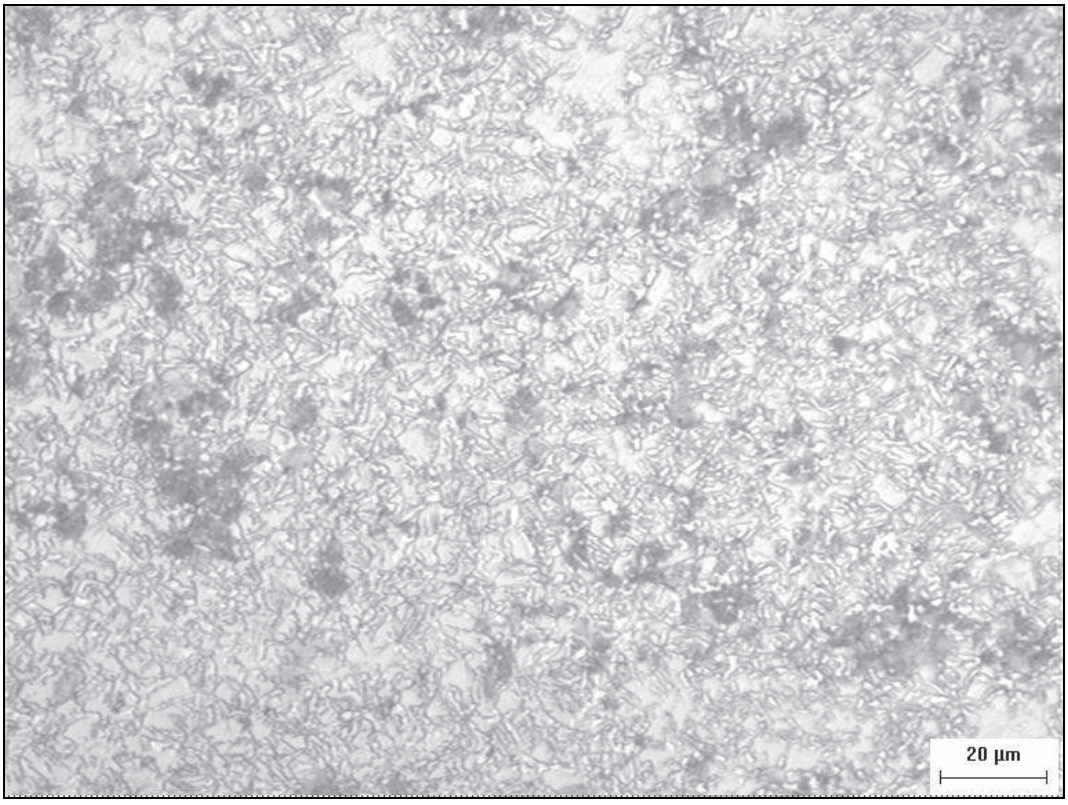


Fig. 5.5.6 Specimen 268 Long. (500X).

STEEL 38Cr4

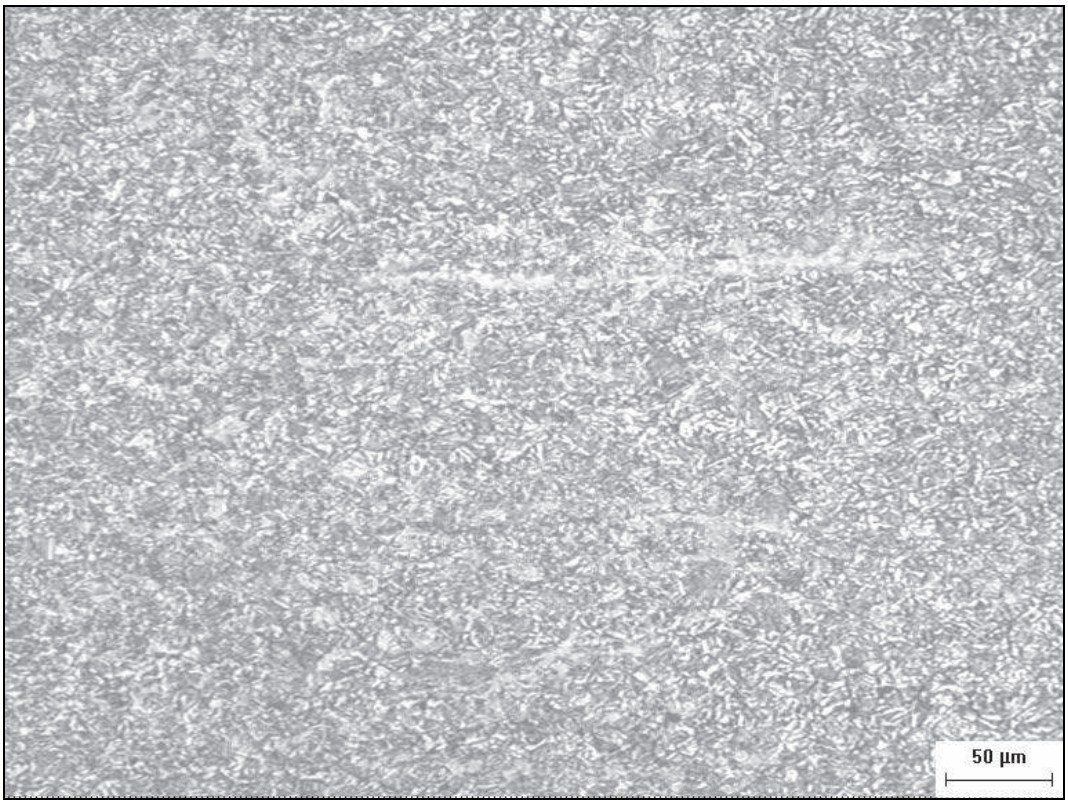


Fig. 5.5.7 Specimen 363 Long. (200X).

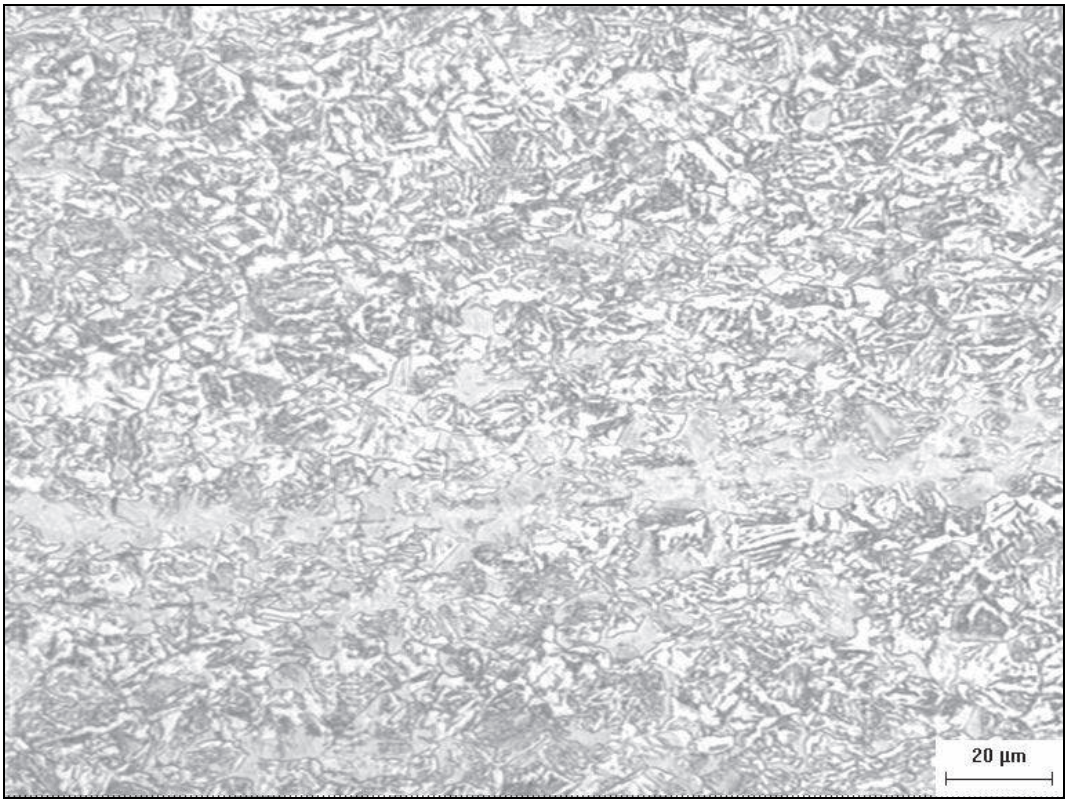


Fig. 5.5.8 Specimen 363 Long. (500X).

STEEL 10MnSi7

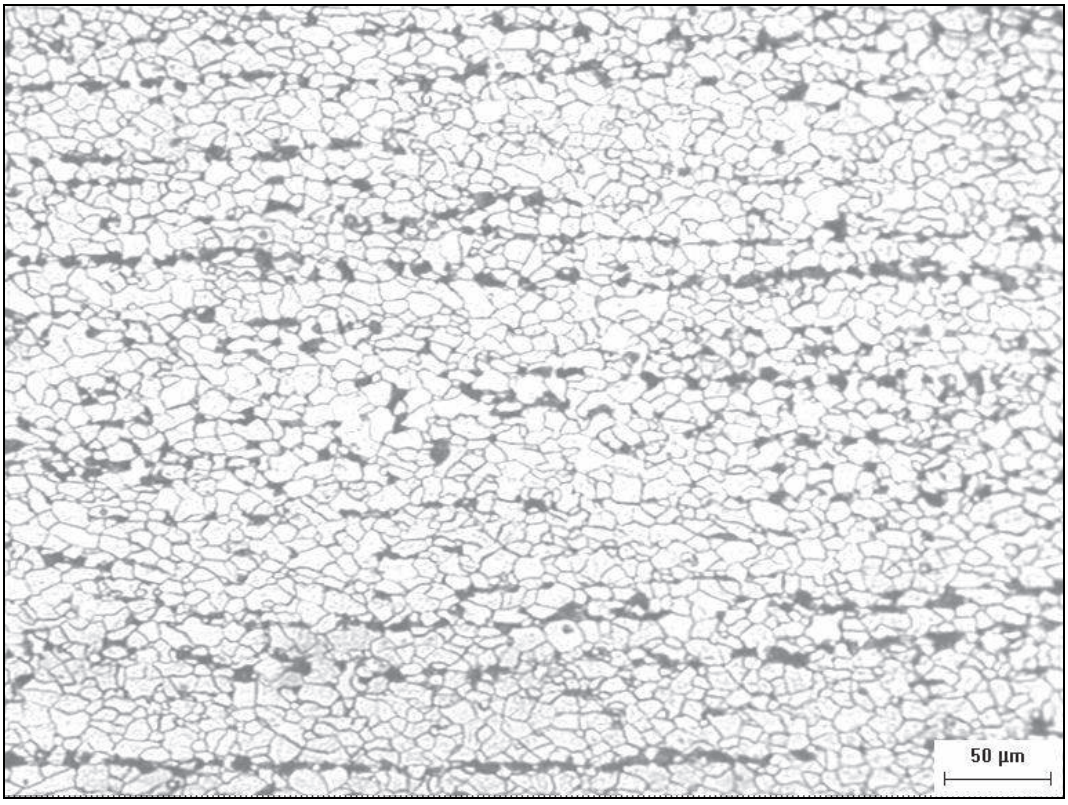


Fig. 5.5.9 Specimen 485 Long. (200X).

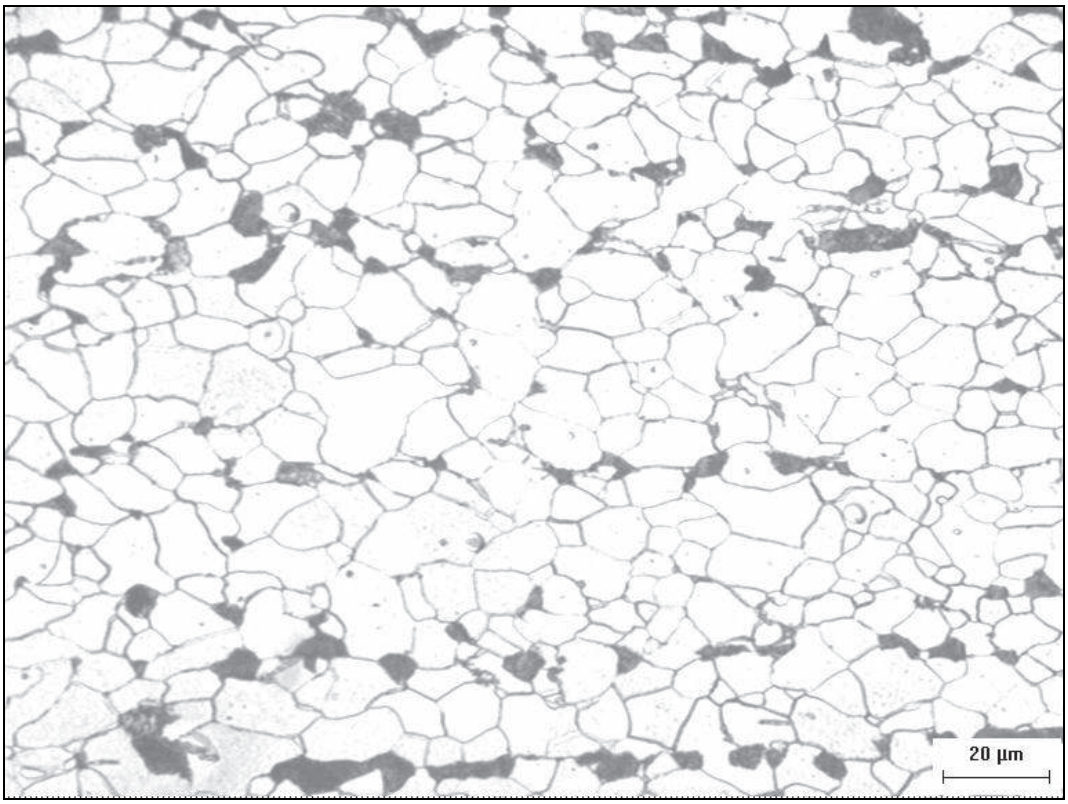


Fig. 5.5.10 Specimen 485 Long. (500X).

Results (Figs 5.5.11 to 5.5.18 and Table 5.5.3) confirm a good homogeneity in tensile properties.

Mechanical Properties		**Specimens**									
		061/1	**061/2**	**176/1**	**176/2**	**268/1**	**268/2**	**363/1**	**363/2**	**485/1**	**485/2**
Original diam.	mm	7.18	7.21	7.63*	7.63*	7.19	7.20	7.18	7.17	7.15	7.17
Original Cross Area	mm^2	40.47	40.81	45.70*	45.70*	40.58	40.69	40.47	40.36	40.13	40.36
YS 0.2%	MPa	1390	1403	1143	1182	869	857	1013	1078	654	712
UTS	MPa	1631	1556	1324	1362	967	906	1174	1220	691	740
ROA	%	12	N.A. **	55	56	55	59	25	26	78	76
* 54SiCrV6 (LUCCHINI) was tested only after 1st drawing step ** broken in the specimen holder											

Table 5.5.3 Summary of mechanical properties of cold drawn prototype wire rods.

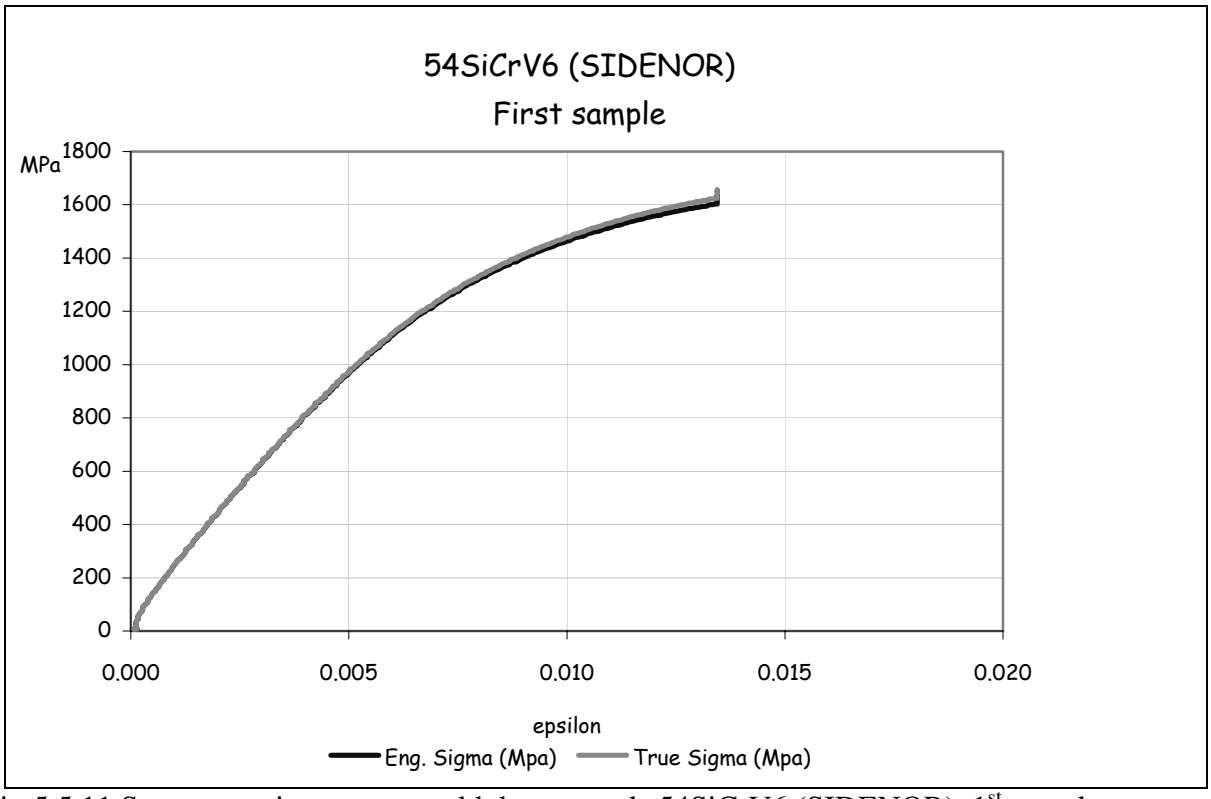


Fig.5.5.11 Stress – strain curve on cold drawn steels 54SiCrV6 (SIDENOR), 1st sample.

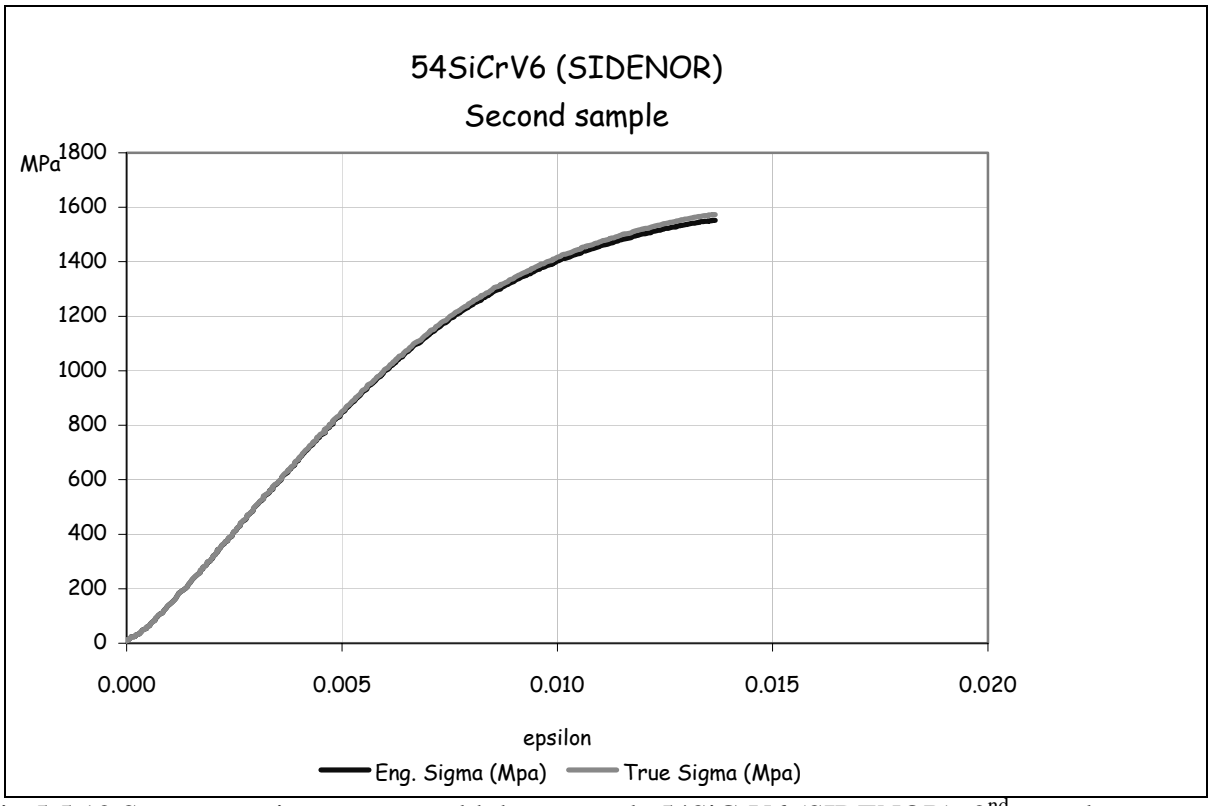


Fig.5.5.12 Stress – strain curve on cold drawn steels 54SiCrV6 (SIDENOR), 2nd sample.

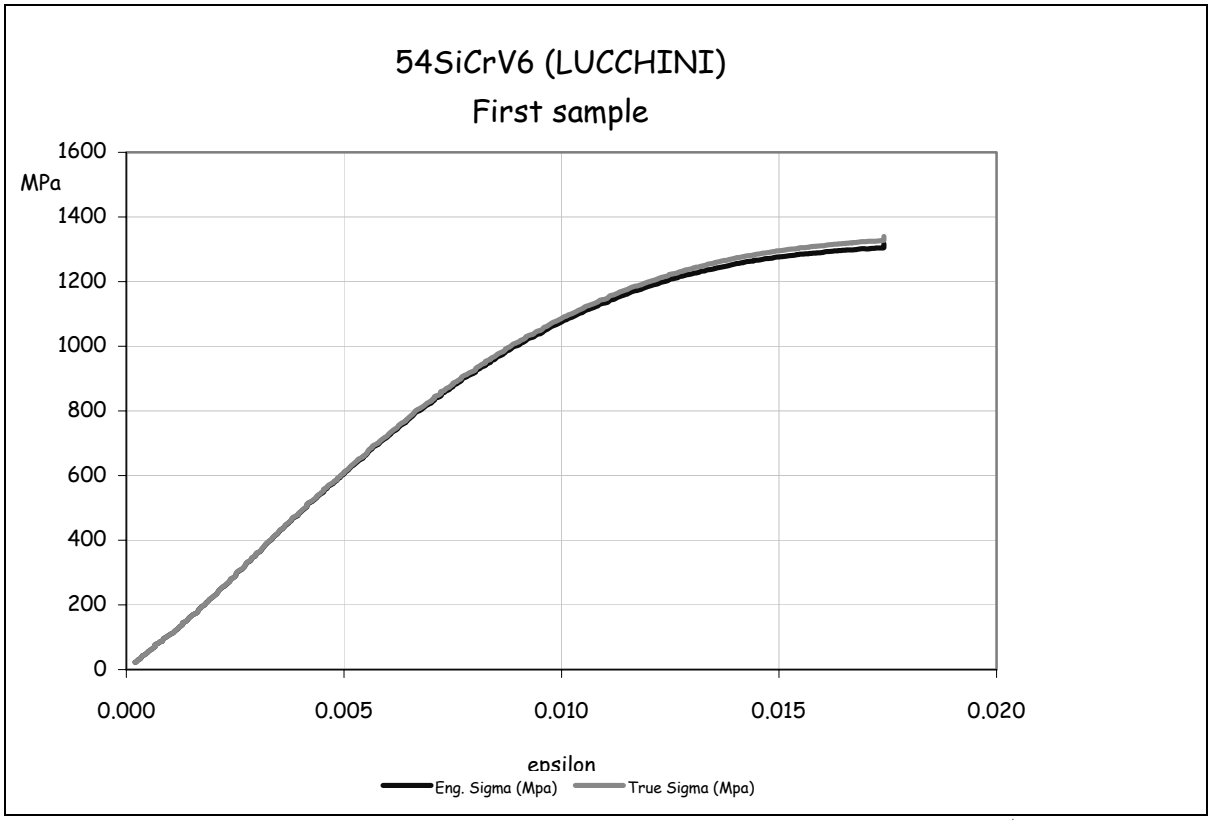


Fig.5.5.13 Stress – strain curve on cold drawn steels 54SiCrV6 (LUCCHINI), 1st sample.

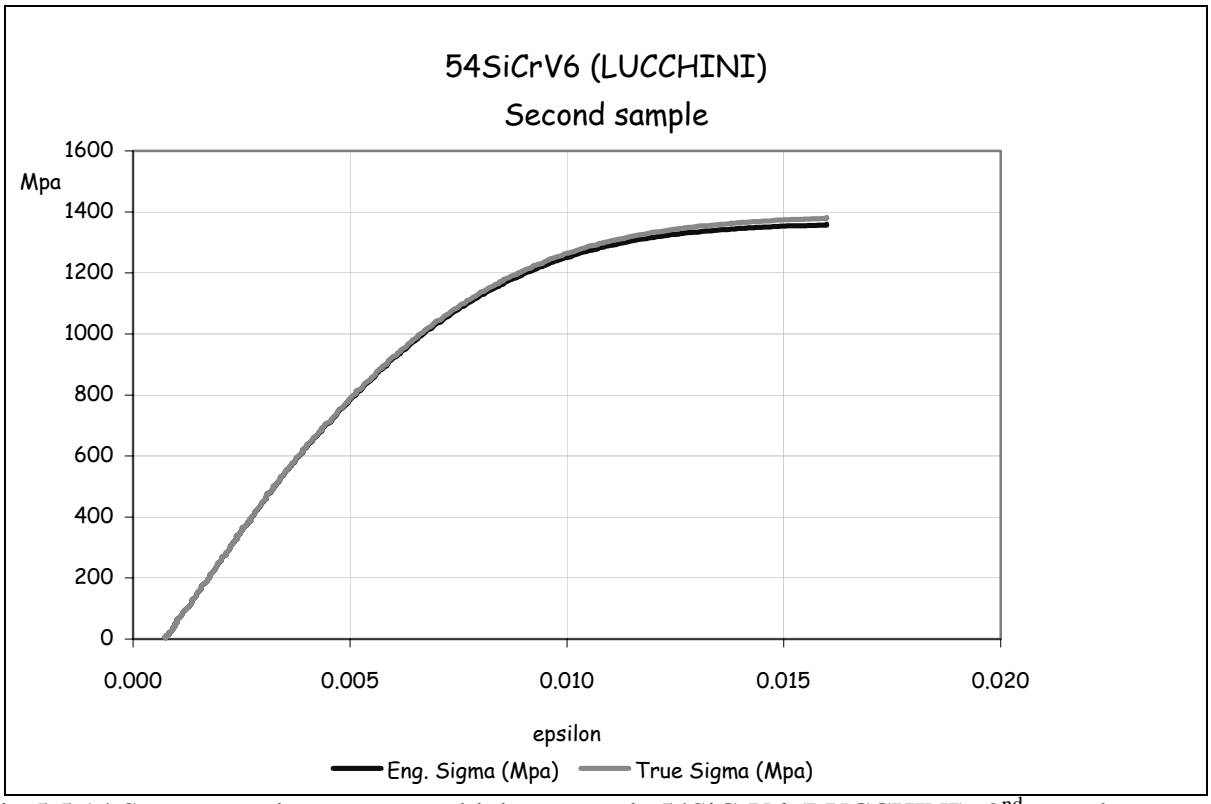


Fig.5.5.14 Stress – strain curve on cold drawn steels 54SiCrV6 (LUCCHINI), 2nd sample.

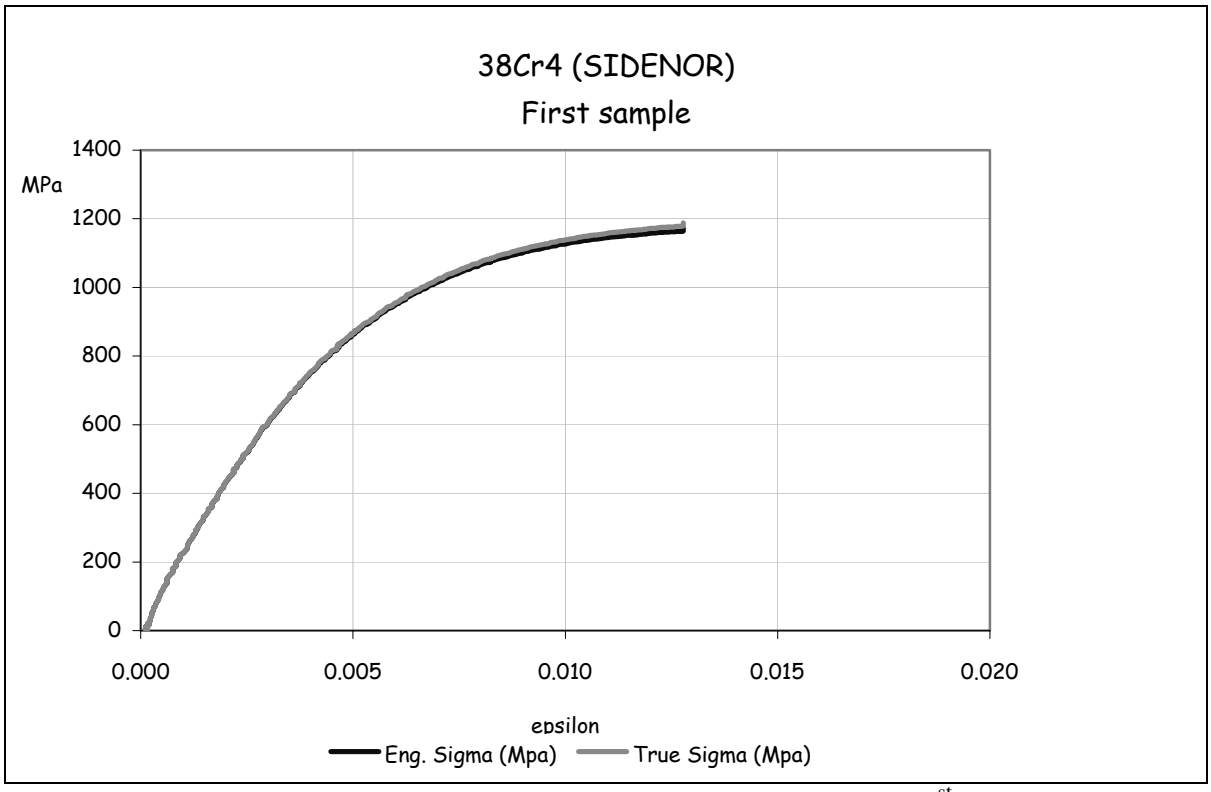


Fig.5.5.15 Stress – strain curve on cold drawn steels 38Cr4 (SIDENOR), 1st sample.

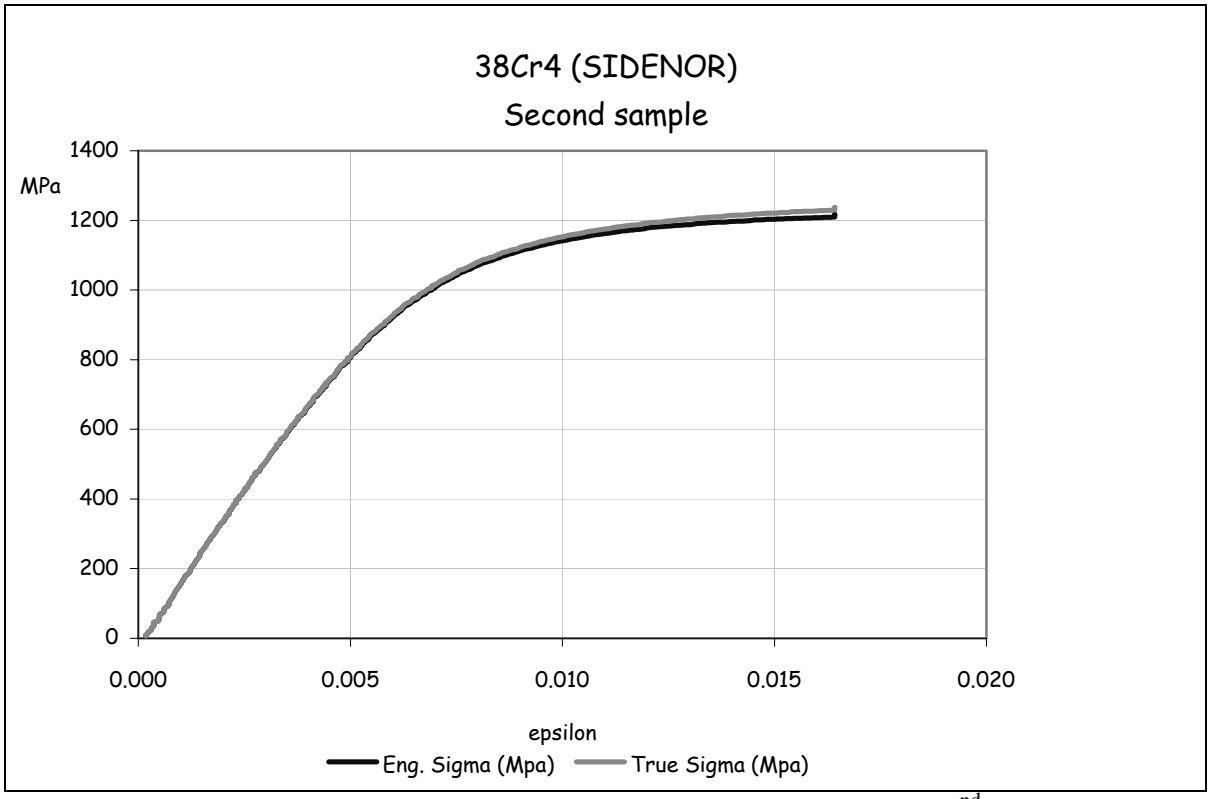


Fig.5.5.16 Stress – strain curve on cold drawn steels 38Cr4 (SIDENOR), 2nd sample.

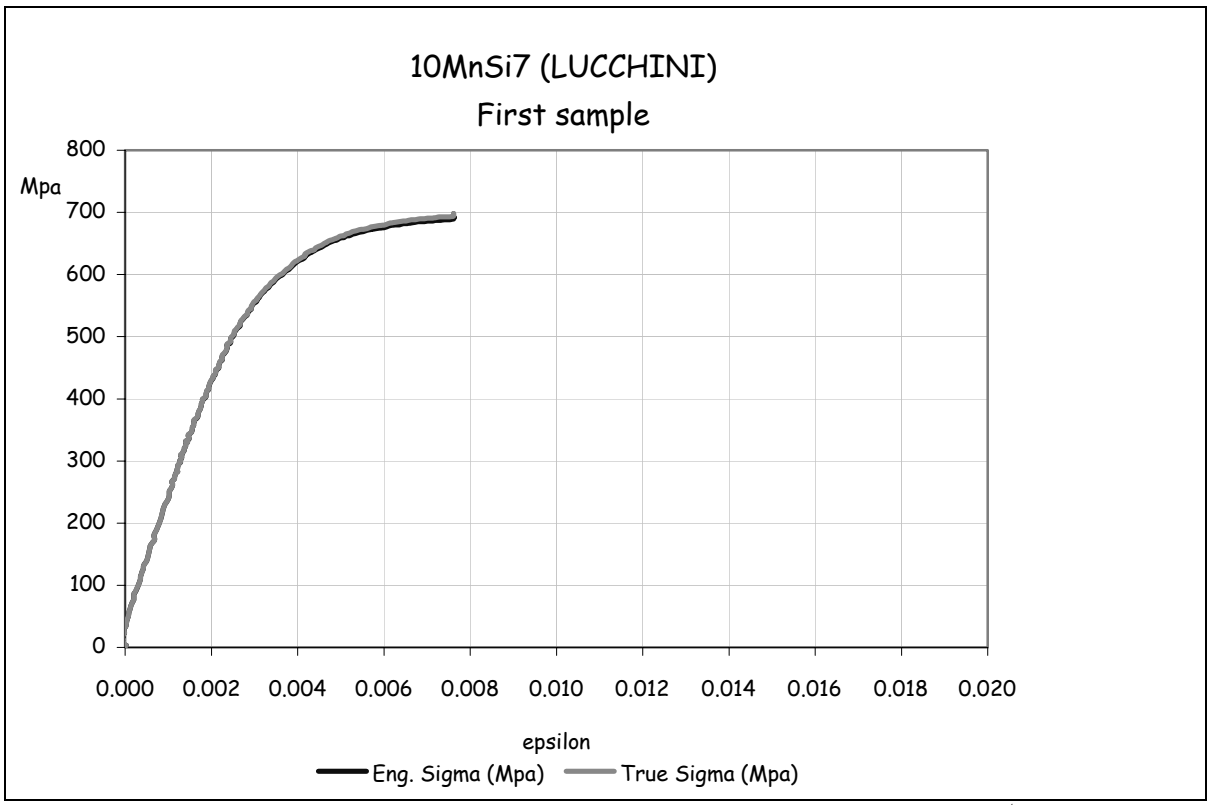


Fig.5.5.17 Stress – strain curve on cold drawn steel 10MnSi7 (LUCCHINI), 1st sample.

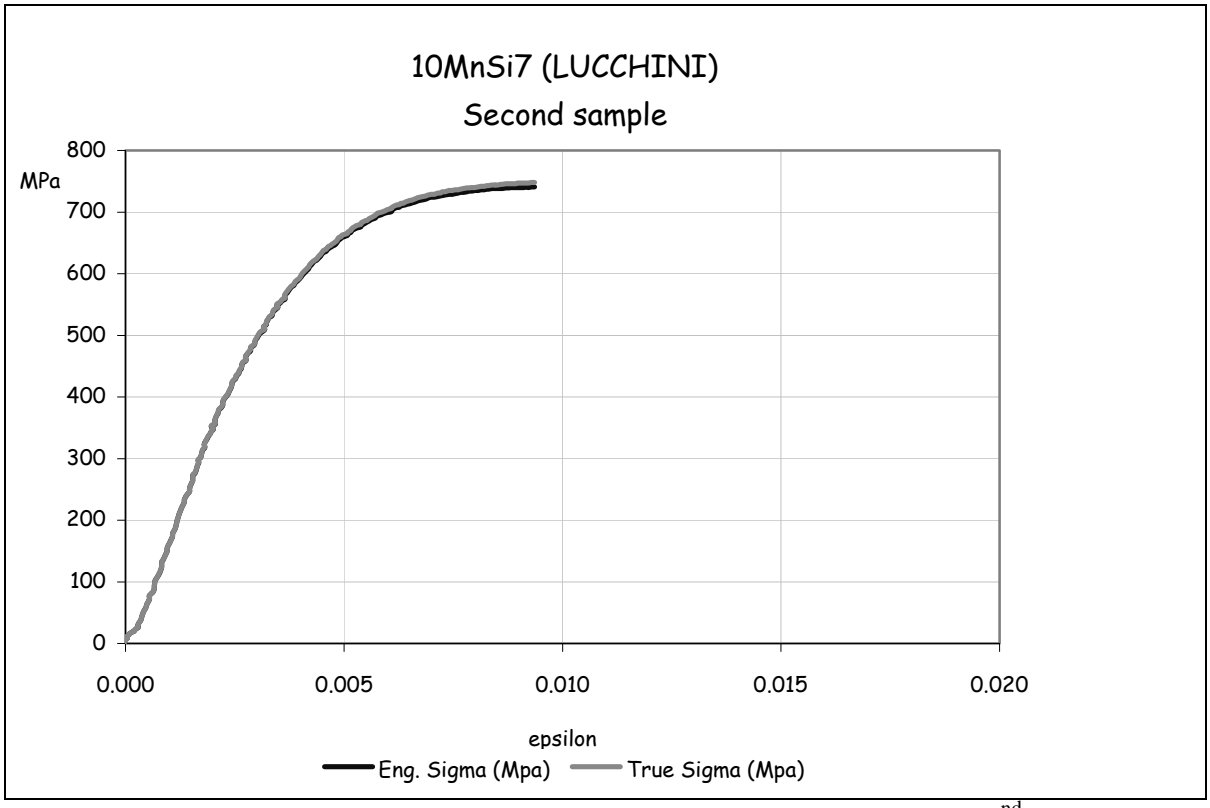


Fig.5.5.18 Stress – strain curve on cold drawn steel 10MnSi7 (LUCCHINI), 2nd sample.

European Commission

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